M. Eigen P. Schuster

The Hypercycle

A Principle of Natural Self-Organization

With 64 Figures

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Preface

This book originated from a series of papers which were published in „Die Naturwissenschaften" in 1977/78. Its division into three parts is the reflection of a logic structure, which may be abstracted in the form of three theses

A. Hypercycles are a principle of natural selforganization allowing an integration and coherent evolution of a set of functionally coupled self-replicative entities.

B. Hypercycles are a novel class of nonlinear reaction networks with unique properties, amenable to a unified mathematical treatment.

C. Hypercycles are able to originate in the mutant distribution of a single Darwinian quasi-species through stabilization of its diverging mutant genes. Once nucleated hypercycles evolve to higher complexity by a process analogous to gene duplication and specialization.

In order to outline the meaning of the first statement we may refer to another principle of material selforganization, namely to Darwin's principle of natural selection. This principle as we see it today represents the only understood means for creating information, be it the blue print for a complex living organism which evolved from less complex ancestral forms, or be it a meaningful sequence of letters the selection of which can be simulated by evolutionary model games.

Natural selection – and here the emphasis is on the word "natural" – is based on selfreproduction. Or: given a system of self-reproducing entities building up from a common source of material of limited supply, natural selection will result as an inevitable consequence. In the same way evolutionary behaviour governed by natural selection is based on noisy self-reproduction. These physical properties are sufficient to allow for the reproducible formation of highly complex systems, i.e. for the generation of such information as the blue print of a living organism. However, there are quantitative limitations in the extent of information to be gained, which are inherent to the Darwinian mechanism of natural selection. This is where the hypercycle comes onto the scene. The hypercycle, too, is a principle of selforganization – based on different prerequisites and hence yielding different consequences.

The theory of Darwinian systems as outlined in part A shows essentially two results:

a) Self-replicative entities compete for selection. This competition may be relaxed for unrelated species retreating to niches. It nevertheless has to be effective within each mutant distribution in order to keep the wild-type stable. Without such a competitive stabilization its information would melt away.

? (b) The information content of a stable wild-type is limited. In other words, the amount of information has to remain below a threshold, the magnitude of which is inversely proportional to the average error rate (per symbol). The threshold value furthermore depends on the logarithm of the superiority of the wild-type, which is the average selective advantage relative to the mutants of the total (stable) distribution. The distribution gets unstable whenever a mutant appears which violates this condition in being advantageous to the previously stable wild-type.

These properties are inherent to Darwinian systems. They guarantee evolutionary behaviour, characterized by selection *and stable* reproduction of the best adapted self-replicative entity and its replacement by any mutant that is still better adapted. On the other hand, the evolution of such a system is limited to a certain level of complexity defined by the threshold for maximum information content.

*The first self-replicative entities – owing to this limitation – must have been relatively short chains of nucleic acids. They were the only class of macromolecules fulfilling the condition of being inherently self-replicative. However, the specificity of physical forces on which the fidelity of selfreplication is based, is limited. Improvements of fidelity could result only from catalytic support, where the catalyst, in order to be subject to evolutionary adaptation had to be reproducible, too. Translation of information inherited by the reproductive material became a requirement at this stage of evolution. *

- degree of exactness of copying

The hurdle was immensely high. Evolution must have come almost to a standstill. Required was a machine, but in order to produce it this very machine had to be available right away. Even a primitive translation apparatus would have to involve a minimum of four adaptors assigning four different amino acids plus a corresponding number of enzymes and their messengers. The amount of information needed for such a system is comparable to that of a single stranded RNA-virus. However these particles can utilize the perfect translation apparatus of their host cell. They furthermore reproduce with the help of a highly adapted enzyme machinery, which represents a final – that is an optimal – product of evolution.

The genome of an RNA-phage hardly exceeds a few thousand nucleotides just enough to encode for a few (e.g. four) protein molecules. As is shown in part A, this limit is set by the fidelity which only could be reached with the help of a well adapted replication enzyme. Any further extension of the information content would require such sophisticated mechanisms as proofreading including exo-nuclease and ligase action, which is available only to the DNA-polymerases at quite avanced stages of evolution. How could even a primitive translation system originate, if reproduction fidelity was

based solely on the physical properties inherited by the nucleic acids not permitting the reproducible accumulation of more than fifty to hundred nucleotides in any individual nucleotide chain? The amount of information required for a translation system, without which no improvement of fidelity could be achieved, amounts to a multiple of what was available in the single self-reproducible chains.

The hypercycle is the tool for integrating length-restricted self-replicative entities into a new stable order, which is able to evolve coherently. No other kind of organization, such as mere compartmentation, or non-cyclic networks could fulfill simultaneously all three of the following conditions

- to maintain competition among the wild-type distribution of every self-replicative entity in order to preserve their information
- to allow for a coexistence of several (otherwise competitive) entities and their mutant distributions, and
- to unify these entities into a coherently evolving unit, where advantages of one individual can be utilized by all members and where this unit as a whole remains in sharp competition with any unit of alternative composition.

Our statement which comprises the results of part A represents a logical inference:

If we ask for a physical mechanism that guarantees the continuous evolution of a translation apparatus, hypercyclic organization is a minimum requirement. It is not sufficient - though necessary - that the information carriers involved are of a self-replicative nature. If we analyze the conditions of hypercyclic organization we immediately see their equivalence to the prerequisites of Darwinian selection. The latter is based on self-reproduction which is a kind of linear autocatalysis. The hypercycle is the next higher level in a hierarchy of autocatalytic systems (as shown in part A). It is made up of autocatalysts or reproduction cycles which are linked by cyclic catalysis, i.e. by another superimposed autocatalysis. Hence a hypercycle is based on non-linear (e.g. second or higher order) autocatalysis.

Hypercycles, because they show "regular" behaviour can be analyzed as a particular class of reaction networks. Such a general analysis is carried out in part B (cf. second statement). The fact that they show unique physical properties, which other types of couplings are devoid of, calls for a unified treatment of the "abstract hypercycle". Such a representation of the subject matter in itself justifies textbook representation.

On the other hand, hypercycles are by no means just abstract products of our mind. The principle is still retained in the process of RNA-phage infection, though there it applies to the closed world of the host cell. The phage genome upon translation provides a factor which acts as a subunit of the replicase complex, the other parts of which are recruited from host factors. This phage-encoded factor turns the enzyme into absolute phage specificity. In disregarding all RNAs from host origin the phage-specific replicase complex now represents a superimposed feedback loop for the autocatalytic amplification of the phage genome.

Our statement regarding the necessity of a hypercyclic organization of a primitive translation apparatus is of an "if-then" nature and does not yet refer to historical reality. There, unexpected singular events, fluctuations that do not represent any regularity of nature, might occur and then influence the historical route. If we want to show that historical evolution indeed took place under guidance of a particular physical principle, we have to look for witnesses of history, namely remnants of early organizational forms in present organisms. This is done in part C and our third statement refers to it.

Transfer RNAs as the key substances of translation provide some information about their origin. They seem to offer a natural way by which the difficulties of a start of the nonlinear network – the nucleation problem – can be solved. All members of the network are descendents of the same master copy, a t-RNA precursor. Mutants of the quasi-species distribution of this precursor could accumulate before the organization principle of a hypercycle came into effect. Being closely related mutants all adaptors and messengers as well as their translation products provide very similar functions (as targets and as executive factors), hence automatically "fall" into a highly cross linked organization including a cycle. As shown in part C this cycle can gradually stabilize itself through evolving specificities of the couplings, which all may be of the replicase-target type still utilized by RNS-phages. The realistic hypercycle is subject to experimental testing, which includes detailed studies of the present translation mechanism.

We hope this book may contribute to raise the right kinds of questions for a study of problems of evolution. There is no absolute value in any theory, if its inferences cannot be checked by experiments. On the other hand, theory has to offer more than just an explanation of experimental facts. As Einstein said: Only theory can tell us which experiments are to be meaningful.

In this sense the book is written not only for the physicist who seeks for the uniform application of physical laws to nature. It addresses the chemist, biochemist and biologist as well, to provoke him to carry out new experiments which may provide a deeper understanding of life as "regularity of nature" and of its origin.

Our work was greatly stimulated by discussions with FRANCIS CRICK, STANLEY MILLER, and LESLIE ORGEL; which for us meant some "selection pressure" to look for more continuity in molecular evolution. Especially helpful were suggestions and comments by CHRISTOPH BIEBRICHER, IRVING EPSTEIN, BERND GUTTE, DIETMAR PÖRSCHKE, KARL SIGMUND, PAUL WOOLEY, and ROBERT WOLFF. RUTHILD WINKLER-OSWATITSCH designed most of the illustrations and was always a patient and critical discussant.

Thanks to all for their help.

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MANFRED EIGEN
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A. Emergence of the Hypercycle

I. The Paradigm of Unity and Diversity in Evolution

Why do millions of species, plants and animals, exist, while there is only one basic molecular machinery of the cell: one universal genetic code and unique chiralities of the macromolecules?

The geneticists of our day would not hesitate to give an immediate answere to the first part of this question. Diversity of species is the outcome of the tremendous branching process of evolution with its myriads of single steps of reproduction and mutation. It in-

volves selection among competitors feeding on common sources, but also allows for isolation, or the escape into niches, or even for mutual tolerance and symbiosis in the presence of sufficiently mild selection constraints. Darwin's principle of natural selection represents a principle of guidance, providing the differential evaluation of a gene population with respect to an optimal adaptation to its environment. In a strict sense it is effective only under appropriate boundary conditions which may or may not be fulfilled in nature. In the work of the great schools of population genetics of Fisher, Haldane, and Wright the principle of natural selection was given an exact formulation demonstrating its capabilities and restrictions. As such, the principle is based on the prerequisites of living organisms, especially on their reproductive mechanisms. These involve a number of factors, which account for both genetic homogeneity and heterogeneity, and which have been established before the detailed molecular mechanisms of inheritance became known (Table 1).

Table 1. Factors of natural selection (according to S. Wright [1])

Factors of genetic homogeneity	Factors of genetic heterogeneity
Gene duplication	Gene mutation
Gene aggregation	Random division of aggregate
Mitosis	Chromosome aberration
Conjugation	Reduction (meiosis)
Linkage	Crossing over
Restriction of population size	Hybridization
Environmental pressure(s)	Individual adaptability
Crossbreeding among subgroups	Subdivision of group
Individual adaptability	Local environment of subgroups

Realizing this heterogeneity of the animate world there is, in fact, a problem to understand its homogeneity at the subcellular level. Many biologists simply sum up all the precellular evolutionary events and refer to it as 'the origin of life'. Indeed, if this had been one gigantic act of creation and if it – as a unique and singular event, beyond all statistical expectations of physics – had happened only once, we could satisfy ourselves with such an explanation. Any further attempt to understand the 'how' would be futile. Chance cannot be reduced to anything but chance.

Our knowledge about the molecular fine structure of even the simplest existing cells, however, does not lend any support to such an explanation. The regularities in the build up of this very complex structure leave no doubt, that the first living cell must itself have been the product of a protracted process of evolution which had to involve many single, but not necessarily singular, steps. In particular, the genetic code looks like the product of such a multiple step evolutionary process [2], which probably started with the unique assignment of only a few of the most abundant primordial amino acids [3]. Although the code does not show an entirely logical structure with respect to all the final assignments, it is anything but random and one cannot escape the impression that there was an optimization principle at work. One may call it a principle of least change, because the structure of the code is such that consequences of single point mutations are reduced to minimum changes at the amino acid level. Redundant codons, i.e., triplets coding for the same amino acids, appear in neighbored positions, while amino acids exhibiting similar kinds of interaction differ usually in only one of the three, preferentially the initial or the terminal position of the codon. Such an optimization, in order to become effective during the evolutionary process, requires by trial and error the testing of many alternatives including quite a number of degenerated assignments. Hence, precellular evolution should be characterized by a similar degree of branching as we find at the species level, provided that it was guided by a similar Darwinian mechanism of natural selection.

However, we do not encounter any alternative of the genetic code, not even in its fine structure. It is quite unsatisfactory to assume that it was always accidentally the optimal assignment which occurred just once and at the right moment, not admitting any of the alternatives which, undoubtedly, would have led to a branching of the code into different fine structures. On the other hand, it is just as unsatisfactory to invoke that the historical route of precellular evolution was uniquely fixed by deterministic physical events.

The results of our studies suggest, that the Darwinian evolution of species was preceded by an analogous stepwise process of molecular evolution leading to a unique cell machinery which uses a universal code. This code became finally established, not because it was the only alternative, but rather due to a peculiar 'once-forever'-selection mechanism, which could start from any random assignment. Once-forever selection is a consequence of hypercyclic organization [4]. A detailed analysis of macromolecular reproduction mechanism suggests that catalytic hypercycles are a minimums requirement for a macromolecular organization that is capable to accumulate, preserve, and process genetic information.

II. What Is a Hypercycle?

Consider a sequence of reactions in which, at each step, the products, with or without the help of addi-

tional reactants, undergo further transformation. If, in such a sequence, any product formed is identical with a reactant of a preceding step, the system resembles a *reaction cycle* and the cycle as a whole a catalyst. In the simplest case, the catalyst is represented by a single molecule, e.g., an enzyme, which turns a substrate into a product:

$$S \xrightarrow{E} P$$

The mechanism behind this formal scheme requires at least a three-membered cycle (Fig. 1). More involved reaction cycles, both fulfilling fundamental catalytic functions are presented in Figures 2 and 3. The Bethe-Weizsäcker cycle [5] (Fig. 2) contributes essentially to the high rate of energy production in massive stars. It, so to speak, keeps the sun shining and, hence, is one of the most important *external* prerequisites of life on earth. Of no less importance, although concerned with the *internal* mechanism of life, appears to be the Krebs- or citric acid cycle [6], shown in Figure 3. This cyclic reaction mediates and regulates the carbohydrate and fatty acid metabolism in the living cell, and has also fundamental functions in anabolic (or biosynthetic) processes. In both schemes, energy-rich matter is converted into energy-deficient products under conservation, i.e., cyclic restoration of the essential material intermediates. Historically, both cycles, though they are little related in their causes, were proposed at about the same time (1937/38).

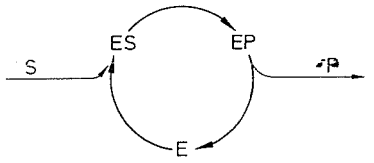


Fig. 1. *The common catalytic mechanism of an enzyme* according to Michaelis and Menten involves (at least) three intermediates: the free enzyme (E), the enzyme-substrate (ES) and the enzyme-product complex (EP). The scheme demonstrates the equivalence of catalytic action of the enzyme and cyclic restoration of the intermediates in the turnover of the substrate (S) to the product (P). Yet, it provides only a formal representation of the true mechanism which may involve a stepwise activation of the substrate as well as induced conformation changes of the enzyme.

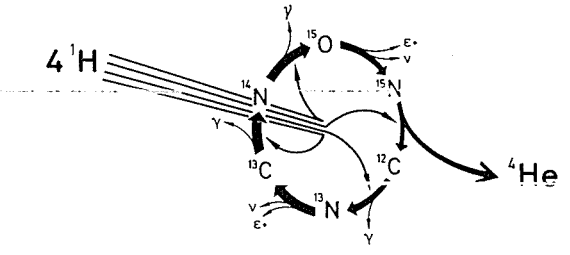


Fig. 2. *The carbon cycle*, proposed by Bethe and v. Weizsäcker, is responsible—at least in part—for the energy production of massive stars. The constituents: ^{12}C, ^{13}N, ^{13}C, ^{14}N, ^{15}O, and ^{15}N are steadily reconstituted by the cyclic reaction. The cyclic scheme as a whole represents a catalyst which converts four ^{1}H atoms to one ^{4}He atom, with the release of energy in the form of γ-quanta, positrons (ε^+) and neutrinos (ν).

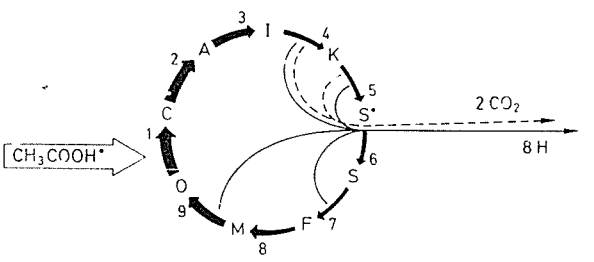


Fig. 3. *The tricarboxylic or citric acid cycle* is the common catalytic tool for biological oxidation of fuel molecules. The complete scheme was formulated by Krebs; fundamental contributions were also made by Szent-Györgyi, Martius, and Knoop. The major constituents of the cycle are: citrate (C), cis-aconitate (A), isocitrate (I), α-ketoglutarate (K), succinyl-CoA (S*), succinate (S), fumarate (F), l-malate (M) and oxaloacetate (O): The acetate enters in activated form as acetyl-CoA (step 1) and reacts with oxaloacetate and H_2O to form citrate (C) and CoA ($+H^+$). All transformations involve enzymes as well as co-factors such as CoA (steps 1, 5, 6), Fe^{2+} (steps 2, 3), NAD^+ (steps 4, 5, 9), TPP, lipoic acid (step 5) and FAD (step 7). The additional reactants: H_2O (steps 1, 3, 8), P, and GDP (step 6) and the reaction products: H_2O (step 2), H^+ (steps 1, 9), and GTP (step 6) are not explicitly mentioned. The net reaction consists of the complete oxidation of the two acetyl carbons to CO_2 (and H_2O). It generates twelve high-energy phosphate bonds, one formed in the cycle (GTP, step 6) and 11 from the oxidation of NADH and $FADH_2$ [3 pairs of electrons are transferred to NAD^+ (steps 4, 5, 9) and one pair to FAD (step 7)].

N.B.: The cycle as a whole acts as a *catalyst* due to the cyclic restoration of the substrate intermediates, yet it does *not* resemble a *catalytic cycle* as depicted in Figure 4. Though every step in this cycle is catalyzed by an enzyme, none of the enzymes is formed via the cycle

CoA = coenzyme A, NAD = nicotine amide adenine dinucleotide, GTP = guanosine triphosphate, FAD = flavine adenine dinucleotide, TPP = thiamine pyrophosphate, GDP = guanosine diphosphate, P = phosphate

Historically, both cycles, though they are little related in their causes, were proposed at about the same time (1937/38).

Unidirectional cyclic restoration of the intermediates, of course, presumes a system far from equilibrium and is always associated with an expenditure of energy, part of which is dissipated in the environment. On the other hand, equilibration occurring in a closed system will cause each individual step to be in detailed balance. Catalytic action in such a closed system will be microscopically reversible, i.e., it will be equally effective in both directions of flow.

Let us now, by a straight forward iteration procedure, build up hierarchies of reaction cycles and specify

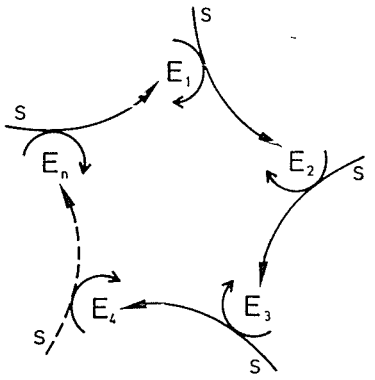


Fig. 4. *The catalytic cycle* represents a higher level of organization in the hierarchy of catalytic schemes. The constituents of the cycle $E_1 \rightarrow E_n$ are themselves catalysts which are formed from some energy-rich substrates (S), whereby each intermediate E_i is a catalyst for the formation of E_{i+1}. The catalytic cycle seen as an entity is equivalent to an autocatalyst, which instructs its own reproduction. To be a catalytic cycle it is sufficient, that only one of the intermediates formed is a catalyst for one of the subsequent reaction steps.

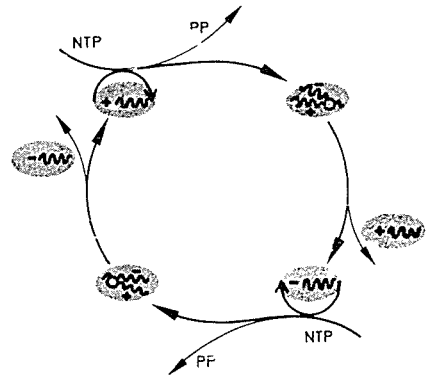


Fig. 5. *A catalytic cycle of biological importance* is provided by the replication mechanism of single-stranded RNA. It involves the plus and minus strand as template intermediates for their mutual reproduction. Template function is equivalent to discriminative catalysis. Nucleoside triphosphates (NTP) provide the energy-rich building material and pyrophosphate (PP) appears as a waste in the turnover. Complementary instruction, the mechanism of which will be discussed in connection with Figure 11, represents inherent autocatalytic, i.e., self-reproductive function

their particular properties. In the next step this means we consider a reaction cycle in which at least one, but possibly all of the intermediates themselves are catalysts. Notice that those intermediates, being catalysts, now remain individually unchanged during reaction. Each of them is formed from a flux of energy-rich building material using the catalytic halp of its preceding intermediate (Fig. 4). Such a system, comprising a larger number of intermediates, would have to be of a quite complex composition and, therefore, is hard to encounter in nature. The best known example is the four-member cycle associated with the template-directed replication of an RNA molecule (Fig. 5). *In vitro* studies of this kind of mechanism have been performed using a suitable reaction medium, buffered with the four nucleoside-triphosphates, as the energy-rich building material and a phage replicase present as a constant environmental factor [7, 8], (A more detailed description will be given by B.-O. Küppers [9]). Each of the two strands acts as a template instructing the synthesis of its complementary copy in analogy to a photographic reproduction process.

The simplest representative of this category of reaction systems is a single autocatalyst, or – in case of a whole class of information carrying entities I_i – the self-replicative unit. The process can be formally written as:

$$X \xrightarrow{I} I$$

Reactions of this type will be considered frequently in this paper, we characterize them by the symbol

(I)

Double-stranded DNA, in contrast to single-stranded RNA, is such a truely *self*-reproductive form, i.e., both strands are copied concomitantly by the polymerase [10] (cf. Fig. 6). The formal scheme applies to the prokaryotic cell, where inheritance is essentially limited to the individual cell line.

Both plain catalytic and autocatalytic systems share, at buffered substrate concentration, a rate term which is first order in the catalyst concentration. The growth curve, however, will clearly differentiate the two sys-

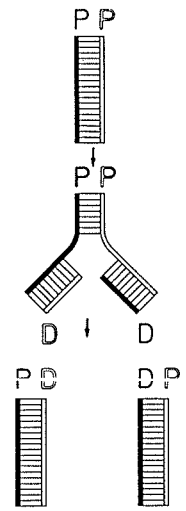


Fig. 6. *A true self-reproductive process* exemplified by a one-member catalytic cycle can be found with DNA replication. The mechanism which is quite involved (cf. Fig. 12), guarantees that each daughter strand (D) is associated with one of the parental strands (P)

tems. Under the stated conditions, the product of the plain catalytic process will grow linearly with time, while the autocatalytic system will show exponential growth.

In strict terminology, an autocatalytic system may already be called hypercyclic, in that it represents a cyclic arrangement of catalysts which themselves are cycles of reactions. We shall, however, restrict the use of this term to those ensembles which are hypercyclic with respect to the *catalytic function*. They are actually hypercycles of second or higher degree, since they refer to reactions which are at least of second order with respect to catalyst concentrations.

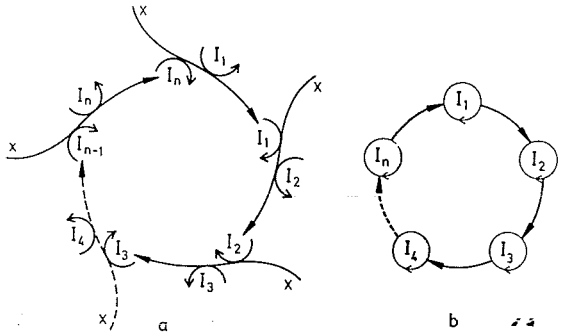


Fig. 7. *A catalytic hypercycle* consists of self-instructive units I_i with two-fold catalytic functions. As autocatalysts or – more generally – as catalytic cycles the intermediates I_i are able to instruct their own reproduction and, in addition, provide catalytic support for the reproduction of the subsequent intermediate (using the energy-rich building material X). The simplified graph (b) indicates the cyclic hierarchy

A catalytic hypercycle is a system which connects autocatalytic or self-replicative units through a cyclic linkage. Such a system is depicted in Figure 7. The intermediates I_1 to I_n, as self-replicative units, are themselves catalytic cycles, for instance, combinations of plus- and minus-strands of RNA molecules as shown in Figure 5. However, the replication process, as such, has to be directly or indirectly furthered, via additional specific couplings between the different replicative units. More realistically, such couplings may be effected by proteins being the translation products of the preceding RNA cycles (Fig. 8). These proteins may act as specific replicases or derepressors, or as specific protection factors against degradation. The couplings among the self-replicative cycles have to form a superimposed cycle, only then the total system resembles a hypercycle. Compared with the systems shown in Figures 4 and 5, the hypercycle is self-reproductive to a higher degree.

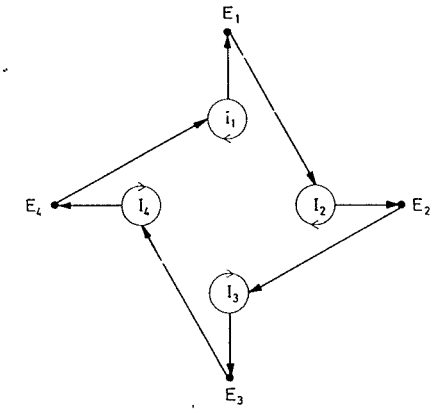


Fig. 8. *A realistic model of a hypercycle of second degree*, in which the information carriers I_i exhibit two kinds of instruction, one for their own reproduction and the other for the translation into a second type of intermediates E_i with optimal functional properties. Each enzyme E_i provides catalytic help for the reproduction of the subsequent information carrier I_{i+1}. It may as well comprise further catalytic abilities, relevant for the translation process, metabolism, etc. In such a case hypercyclic coupling is of a higher than second degree

The simplest representative in this category is, again, the (quasi)one-step system, i.e., the reinforced autocatalyst. We encounter such a system with RNA-phage infection (Fig. 9). If the phage RNA (+strand) is injected into a bacterial cell, its genotypic information is translated using the machinery of the host cell.

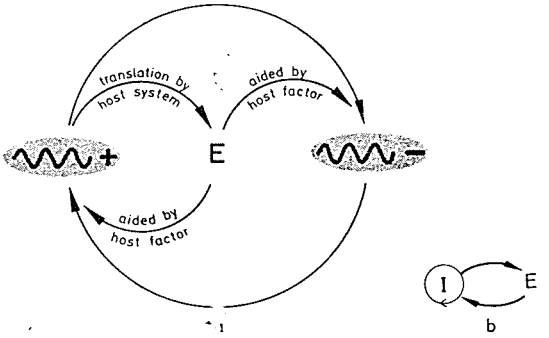


Fig. 9. *RNA-phage infection* of a bacterial cell involves a simple hypercyclic process. Using the translation machinery of the host cell, the infectious plus strand first instructs the synthesis of a protein subunit (E) which associates itself with other host proteins to form a phage-specific RNA-replicase. This replicase complex exclusively recognizes the phenotypic features of the phage-RNA, which are exhibited by both the plus and the minus strand due to a symmetry in special regions of the RNA chain. The result is a burst of phage-RNA production which – owing to the hypercyclic nature – follows a hyperbolic growth law (cf. part B, Fig. 17) until one of the intermediates becomes saturated or the metabolic supply of the host cell is exhausted. Graph (b) exemplifies that it is sufficient if one of the intermediates possesses autocatalytic or self-instructive function presuming that the other partners feed back on it via a closed cyclic link

One of the translation products then associates itself with certain host factors to form an active enzyme complex which specifically replicates the plus and minus strand of the phage RNA, both acting as templates in their mutual reproduction [11]. The replicase complex, however, does not multiply – to any considerable extent – the messenger RNA of the host cell. A result of infection is the onset of a *hyperbolic* growth of phage particles, which eventually becomes limited due to the finite resources of the host cell.

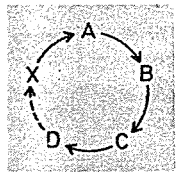
Another natural hypercycle may appear in Mendelian populations during the initial phase of speciation, as long as population numbers are low. The reproduction of genes requires the interaction between both alleles (M and F), i.e., the homologous regions in the male and female chromosome, which then appear in the offsprings in a rearranged combination. The fact that Mendelian population genetics [12] usually does not reflect the hypercyclic non-linearity in the rate equations (which leads to hyperbolic rather than exponential growth), is due to a saturation occurring at relatively low population numbers, where the birth rate (usually) becomes proportional to the population number of females only.

As is seen from the comparative schematic illustration in Figure 10, hypercycles represent a new level of organization. This fact is manifested in their unique properties. Non-coupled self-replicative units guarantee the conservation of a limited amount of information which can be passed on from generation to generation. This proves to be one of the necessary prerequisites of Darwinian behavior, i.e., of selection and evolution [13]. In a similar way, catalytic hypercycles are also selective, but, in addition, they have integrating properties, which allow for cooperation among otherwise competitive units. Yet, they compete even more violently than Darwinian species with any replicative entity not being part of their own. Furthermore, they have the ability of establishing global forms of organization as a consequence of their once-forever-selection behavior, which does not permit a coexistence with other hypercyclic systems, unless these are stabilized by higher-order linkages.

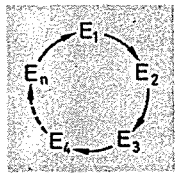
The simplest type of coupling within the hypercycle is represented by straight-forward promotion or derepression introducing second-order formation terms into the rate equations. Higher-order coupling terms may occur as well and, thus, define the degree p of the hypercyclic organization.

Individual hypercycles may also be linked together to build up hierarchies. However, this demands intercyclic coupling terms which depend critically on the degree of organization. Hypercycles H_1 and H_2, having the degrees p_1 and p_2 of internal organization, require intercyclic coupling terms of a degree $p_1 + p_2$, in order to establish a stable coexistence.

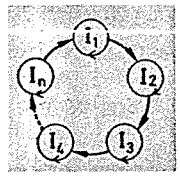
It is the object of this paper to present a detailed theoretical treatment of the category of reaction networks we have christened *hypercycles* [4] and to discuss their importance in biological self-organization, especially with respect to the origin of translation, which may be considered the most decisive step of precellular evolution.



Catalyst E



Autocatalyst (I)
(Self-replicative unit)



Catalytic
Hypercycle (H)

Fig. 10. *The hierarchy of cyclic reaction networks* is evident from this comparative representation (→ chemical transformation, ⇢ catalytic action)

III. Darwinian Systems

III. 1. The Principle of Natural Selection

In physics we know of principles which cannot be reduced to any more fundamental laws. As axioms, they are abstracted from experience, their predictions being consistent with any consequences that can be subjected to experimental test. Typical examples are the first and second law of thermodynamics.

Darwin's principle of natural selection does not fall into the category of first principles. As was shown in population genetics [14], natural selection is a consequence of obvious, basic properties of populations of living organisms subjected to defined external constraints. The principle then makes precise assertions about the meaning of the term fittest in relation to

environmental conditions, other than just uncovering the mere tautology of 'survival of the survivor.' Applied to natural populations with their variable and usually unknown boundary conditions, the principle still provides the clue for the fact of evolution and the phylogenetic interrelations among species. This was the main objective of Charles R. Darwin [15] and his contemporaty Alfred R. Wallace [16], namely, to provide a more satisfactory foundation of the tenet of descendence.

Actually, most of the work in population genetics nowadays is concerned with more practical problems regarding the spread of genetic information among Mendelian populations, leaving aside such academic questions as whether 'being alive' really is a *necessary* prerequisite of selective and evolutive behavior. The fact that obvious attributes of living organisms, such as metabolism, self-reproduction, and finite life span, as well as mutability, suffice to explain selective and evolutive behavior under appropriate constraints, has led many geneticists of our day to believe that these properties are *unique* to the phenomenon of life and cannot be found in the inanimate world [14]. Test-tube experiments [7], which clearly resemble the effects of natural selection and evolution *in vitro*, were interpreted as post-biological findings rather than as a demonstration of a typical and specific behavior of matter. Here one should state that even laser modes exhibit the phenomenon of natural selection and an analysis of their amplification mechanism reveals more than formal analogies. Yet, nobody would call a laser mode 'alive' by any standard of definition.
Wsuch questions may not have much appeal to those who are concerned with the properties of actual living organisms, they become of utmost importance in connection with the question of the *origin of life*. Here we must, indeed, ask for the *necessary* prerequisites in order to find those molecular systems which are eligible for an evolutionary self-organization. The underlying complexity we encounter at the level of macromolecular organization requires this process to be guided by similar principles of selection and evolution as those which apply to the animated worled.
Recent work (loc. cit.) both theoretical and experimental, has been concerned with these questions. In the following we shall give a brief account of some previous results concerning Darwinian systems.

III. 2. Necessary Prerequisites of Darwinian Systems

What is the molecular basis of selection and evolution? Obviously, such a behavior is not a global attribute of any arbitrary form of matter but rather is the consequence of peculiar properties which have to be specified.

The essential requirement for a system to be self-selective is that it has to stabilize certain structures at the expense of others. The criteria for such a stabilization are of a dynamic nature, because it is the distribution of competitors present at any instant that decides which species is to be selected. In other words, there is no static stability of any structure, once selected; it may become unstable as soon as other more 'favourable' structures appear, or in a new environment. The criteria for evaluation must involve some feedback property, which ensures the indentity of value and dynamic stability. An advantageous mutant, once produced as a consequence of some fluctuation must be able to amplify itself in the presence of a large excess of less advantageous competitors. Therefore, advantage must be indentical with at least some of those dynamic properties which are responsible for amplification. Only in this way can the system selectively organize itself in absence of an 'external selector'. The feedback property required is represented by inherent autocatalysis, i.e., self-reproductive behavior.
In a general analysis using game models [18], we have specified those properties of matter which are necessary to yield Darwinian behavior at the molecular level. They can be listed as follows:

1) *Metabolism.* Both formation and degradation of the molecular species have to be independent of each other and spontaneous, i.e., driven by positive affinities. This cannot be achieved in any equilibrated system, in which both processes are mutually related by microscopic reversibility yielding a stable distribution for all competitors once present in the system. Complexity, i.e., the huge multiplicity of alternative structures in combination with time and space limitations simply doesn't allow for such an equilibration, but rather requires a steady degradation and formation of new structures. Selection can become effective only for intermediate states which are formed from energy-rich precursors and which are degraded to some energy-deficient waste. The ability of the system to utilize the free energy and the matter required for this purpose is called metabolism. The necessity of maintaining the system far enough from equilibrium by a steady compensation of entropy production has been first clearly recognized by Erwin Schrödinger [19].

2) *Self-reproduction.* The competing molecular structures must have the inherent ability of instructing their own synthesis. Such an inherent autocatalytic function can be shown to be necessary for any mechanism of selection involving the destabilization of a

population in the presence of a single copy of a newly occurring advantageous mutant. Furthermore, self-copying is indispensable for the conservation of information thus far accumulated in the system. Steady degradation, a necessary prerequisite with respect to condition 1) and 3), would otherwise lead to a complete destruction of the information.

3) *Mutability*. The fidelity of any self-reproductive process at finite temperature is limited due to thermal noise. This is especially effective if copying is fast, more precisely, if it requires for each elementary step an energy of interaction not too far above the level of thermal energy. Hence mutability is always physically associated with self-reproducibility, but it is also (logically) *required* for evolution. Errors of copying provide the main source of new information. As will be seen, there is a threshold-relationship for the rate of mutation, at which evolution is fastest, but which must not be surpassed unless all the information thus far accumulated in the evolutionary process is to be lost.

Only those macromolecular systems which fulfil these three prerequisites are eligible as information carriers in a virtually unlimited evolutionary process. The properties mentioned have to be inherited by all members of the corresponding macromolecular class, i.e., by all possible alternatives or mutants of a given structure and, furthermore, they have to be effective within a wide range of concentration, i.e., from one single copy up to a macroscopically detectable abundance. The 'prerequisite of realizibility' excludes systems of a complicated composition and structure, in which the features mentioned would result from a particular coincidence of molecular interactions rather than from a general principle of physical interaction. As an example, consider the nucleic acids as compared with proteins. Reproduction in nucleic acids is a general property based on the physical forces associated with the unique complementarities among the four bases. Proteins, on the other hand, have a much larger functional capacity, including instructive and reproductive properties. Each individual function, however, is the consequence of a very specific folding of the polypeptide chain and cannot be attributed to the class of proteins in general. It might even be lost completely by a single mutation.

Systems of matter, in order to be eligible for selective self-organization, have to inherit physical properties which allow for metabolism, i.e., the turnover of energy-rich reactants to energy-deficient products, and for ('noisy') self-reproduction. These prerequisites are indispensible. Under suitable external conditions they also prove to be sufficient for selective and evolutive behavior.

III. 3. Dynamics of Selection

The simplest system in accordance with the quoted *necessary* prerequisites can be described by a system of differential equations of the following form [4] ($\dot{x}=\mathrm{d}x/\mathrm{d}t$; $t=$time):

$$\dot{x}_i=(A_iQ_i-D_i)x_i+\sum_{k\neq i} w_{ik}x_k+\Phi_i \qquad (1)$$

where i is a running index, attributed to all distinguishable self-reproductive molecular units, and hence characterizing their particular (genetic) information. By x_i we denote the respective population variable (or concentration). The physical meaning of the other parameters will become obvious from a discussion of this equation. The set of equations first of all involves those self-reproductive units i, which are present in the sample under consideration and which may be numbered 1 to N. It may be extended to include all possible mutants, part of which appear during the course of evolution.

In these equations describing an open system, *metabolism* is reflected by *spontaneous* formation ($A_iQ_ix_i$) and decomposition (D_ix_i) of the molecular species. 'Spontaneous' means that both reactions proceed with a positive affinity and hence are not mutually reversible. The term A_i always contains some stoichiometric function $f_i(m_1, m_2 \ldots m_\lambda)$ of the concentrations of energy-rich building material (λ classes) required for the synthesis of the molecular species i, the precise form of which depends on the particular mechanism of reaction. This energy-rich building material has to be steadily provided by an influx of matter as have the reaction products to be removed by a corresponding outflux (Φ_i). For a spontaneous decomposition, the D_i-term is linearly related to x_i reflecting a common first-order-rate law. In more complex systems both A_i and D_i may include further concentration functions if the corresponding reactions are enzymé-catalyzed or if further couplings among the reactions are present.

Self-reproduction, the second prerequisite, manifests itself in the x_i-dependence of the formation rate term. A straightforward linear dependence represents only the simplest form of inherent autocatalysis. Other more complicated, yet still linear mechanisms, such as complementary instruction or cyclic catalysis, can be treated in an analogous way as will be shown Nonlinear autocatalysis, on the other hand, is the main object of this paper.

Mutability is reflected by the quality factor Q_i, which may assume any value between zero and one. This factor denotes the fraction of reproductions that take place at a given template i and result in an *exact* copy of i. There is, of course, a complementary term related to imprecise reproduction of the template i; $A_i(1-Q_i)x_i$. It means the production of a large variety of 'error copies' which in most cases are quite closely related to the species i. The production of error copies of i will then show up with corresponding terms in the rate equations of each of its 'relatives' k. Correspondingly, the copy i will also receive contributions from those relatives due to errors in their replication. These are taken into account by the sum term; $\sum_{i\neq k} w_{ik}x_k$. The individual mutation rate parameter w_{ki}, will usually be small compared with the reproduction rate parameter A_iQ_i—the smaller the more distant the 'relative' k. If all species present and their possible mutants are taken into account by the indices i and k (running from 1 to N), the following conservation relation for the error copies holds:

$$\sum_i A_i(1-Q_i)x_i=\sum_i \sum_{k\neq i} w_{ik}x_k \qquad (2)$$

The individual flow or transport term Φ_i, finally, describes any supply or removal of species i other than by chemical reaction. It is required due to the metabolic turnover (cf. above). In most

cases each species contributes to the total flow Φ_t in proportion to its presence:

$$\Phi_i = \Phi_t \frac{x_i}{\sum_k x_k} \tag{3}$$

In evolution experiments the overall flow can be adjusted in order to provide reproducible global conditions, such as constant overall population densities:

$$\sum_k x_k = \text{const} \equiv c_n \tag{4}$$

In this case, the flow Φ_t has to be steadily regulated in order to compensate for the excess overall production, i.e.

$$\Phi_t = \sum_k A_k x_k - \sum_k D_k x_k \equiv \sum_k E_k x_k \tag{5}$$

where we call $E_i \equiv A_i - D_i$ the 'excess productivity' of the template i. Notice that the error production does not show up explicitly in this sum as a consequence of the conservation relation (2).

If, in addition, the individual fluxes of the energy-rich building material are also regulated, in order to provide for each of the λ classes a constantly buffered level ($m_1 \ldots m_\lambda$), the stoichiometric functions $f_i(m_1 \ldots m_\lambda)$ appearing in the rate parameters A_i are constant and as such do not have to be specified explicitly. We shall refer to this constraint, in which, via flux control, both the non-organized, as well as the total organized material is regulated to a constant level, as 'constant overall organization'. It is usually maintained in evolution experiments, e.g., in a flow reactor [9], or – on average – in a serial transfer experiment [7]. An alternative straightforward constraint would be that of 'constant fluxes'. In this case, the concentration levels are variable, adjusting to the turnover at given in- and outfluxes. Both constraints will cause the system to approach a steady state with sharp selection behavior. The quantitative results may show differences for both constraints, but the qualitative behavior turns out to be very similar [4]. It is, therefore, sufficient to consider here just one of both limiting cases. The constraints to be met in nature may vary with time and, hence, will usually not correspond to either of the simple extremes – just as little as weather conditions usually resemble simple thermodynamic constraints (e.g., constant pressure, temperature, etc.). However, the essential *principles* of natural selection can only be studied under controlled and reproducible experimental conditions.

For the constraint of constant overall organization the rate equations (1) in combination with the auxiliary conditions (2) to (5) reduce to:

$$\dot{x}_i = (W_{ii} - \bar{E}(t))\, x_i + \sum_{k \neq i} w_{ik} x_k \tag{6}$$

where

$$W_{ii} = A_i Q_i - D_i \tag{7}$$

may be called the (intrinsic) selective value and

$$\bar{E}(t) = \sum_k E_k x_k \Big/ \sum_k x_k \tag{8}$$

the average excess productivity, which is a function of time. Only when the population variables $x_i(t)$ become stationary, will $\bar{E}(t)$ reach a constant steady-state value which is metastable since it depends on the population of the spectrum of mutants. For constant (i.e., time-invariant) values of W_{ii} and w_{ik} the non-linear system of differential equations (6) can be solved in a closed form. Approximate solutions of the selection problem have been reported in earlier papers. In recent years, an exact solution has been worked out by C.J. Thompson and J.L. McBride [20] and independently by B.L. Jones, R.H. Enns, and S.S. Rangnekar [21, 22]. The explicit expressions obtained from the exact solutions by second-order perturbation theory are in agreement with the formerly [4] reported approximations*. The following discussion is based on the exact solutions given by B.L. Jones et al. [21] which offers an elegant quantitative representation of the selection problem.

III. 4. The Concept "Quasi-Species"

The single species is not an independent entity because of the presence of couplings. Conservation of the total population number forces all species into mutual competition, while mutations still allow for some cooperation, especially among closely related species (i.e., species i and k with non-vanishing w_{ik} and w_{ki} terms).

Let us, therefore, reorganize our system in the following way. Instead of subdividing the total population into N species we define a new set of N quasi-species, for which the population variables y_i are linear combinations of the original population variables x_i, whereby, of course, the total sums are conserved:

$$\sum_{k=1}^{N} x_k = \sum_{k=1}^{N} y_k \tag{9}$$

How to carry out this new subdivision is suggested by the structure of the differential equations (6). It corresponds actually to an affine transformation of the coordinate system, well known from the theory of linear differential equations. One obtains a new set of equations for the transformed population variables y_i which reads:

$$\dot{y}_i = (\lambda_i - \bar{E}(t))\, y_i \tag{10}$$

An application of this procedure to the non-linear equations (6) is possible because the term causing the non-linearity, $\bar{E}(t)$ according to Eq. (8), remains

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* Jones et al. [21] pointed out that a neglection of the backflow term $\sum_{k \neq i} w_{ik} x_k$ in [4] is not valid for the approach to the steady state because the term $W_{mm} - \bar{E}(t)$ becomes very small. They referred to Eq. II-49 in [4], where any error rate was deliberately neglected (i.e. $Q = 1$) for the purpose of demonstrating the nature of solutions typical for selection. They overlooked, however, our explicit statement (p. 482 in [4]) that such an assumption may apply approximately only to a dominant species with a well established selective advantage, while cell mutants owe their existence solely to the presence of the w_{ik} terms. The approximations obtained previously (Eq. II-33a; II-43; II-59; II-69; II-72 in [4]; cf. also [22]) indeed agree quantitatively with those following from the exact solution by application of perturbation theory (Eq. (21) and (22) in [21] and Eq. (13), (18) and (19) in this paper).

On the other hand, we should like to state that we appreciate very much the availability of the exact solutions, as obtained by Thompson and McBride [20] as well as by Jones et al. [21] which aid tremendously the presentation of a consistent picture of the quasi-species.

invariant in the transformation and can be expressed now as the average of the λ_i's.

$$\bar{E}(t)=\sum_k \lambda_k y_k / \sum_k y_k \qquad (11)$$

The λ_i's are the eigen-values of the linear dynamic system. They—as well as the eigen-vectors which correlate the x_i's whith the y_i's—can be obtained from the matrix, consisting of the coefficients W_{ii} and w_{ik}.

The solutions of the system (10) are physically obvious. Any quasi-species (characterized by an eigen-value λ_i and a population variable y_i), whose λ_i-value is below the threshold represented by the average, $\bar{E}(t)$, will die out. (Its rate is negative!) Correspondingly, each quasi-species with a λ_i above the threshold will grow. The threshold $\bar{E}(t)$, then, is a function of time, and—due to equation (11)—will increase, the more the system favours quasi-species associated with large eigen-values. This will continue until a steady state is reached:

$$\bar{E}(t)\rightarrow\lambda_{\max} \qquad (12)$$

i.e., the mean productivity will increase until it equals the maximum eigen-value. By then all quasi-species but one, namely the one associated with the maximum eigen-value—will have died out. Their population variables have become zero.

Darwinian selection and evolution can, thus, be characterized by an extremum principle. It defines a category of behavior of self-replicative entities under stated selection constraints.

Such a process, for instance, can be seen in analogy to equilibration, which represents a fundamental type of behavior of systems of matter under the constraint of isolation and which is characterized by a general extremum principle. The extremum principle (12) is related to the stability criteria of I. Prigogine and P. Glansdorff [23]. As an optimization principle it holds also for certain classes of non-linear dynamical systems [21]. Furthermore, the validity of the solutions of the quasi-linear system (10) is not restricted to the neighborhood of the steady state.

What is the physical meaning of a quasi-species?

In biology, a species is a class of individuals characterized by a certain phenotypic behavior. On the genotypic level, the individuals of a given species may differ somewhat, but, nevertheless, all species are represented by DNA-chains of a very uniform structure. What distinguishes them individually is the very sequence of their nucleotides. In dealing now with such molecules, being the replicative units, we just use these differences of their sequences in order to define the (molecular) species. The differences are, of course, expressed also by different phenotypic properties, such as replication rates, life times, error rate, etc.

The single (molecular) species, however, is not the true target of selection. Eq. (10) tells us that it is rather the quasi-species, i.e., an organized combination of species with a defined probability distribution which emerges via selection. As such it is selected against all other distributions. Under selection strains the populations numbers of all but one quasi-species really will disappear. The quasi-species is closely correlated to what is called the "wild-type" of a population.

The wild-type is often assumed to be the standard-genotype representing the optimally adapted phenotype within the mutant distribution. The fact that it is possible to determine a unique sequence for the genome of a phage supports this view of a dominant representation of the standard copy. Closer inspection of the wild-type distribution of phage Q_β (in the laboratory of Ch. Weissmann) [24], however, clearly demonstrated that only a small fraction of the sequences actually is exactly identical with that assigned to the wild-type, while the majority represents a distribution of single and multiple error copies whose *average* only resembles the wild-type sequence. In other words, the standard copies might be present to an extent of (sometimes much) less than a few percent of the total population. However, although the predominant part of the population consists of non-standard types, each individual mutant in this distribution is present to a very small extent (as compared with the standard copy). The total distribution, within the limits of detection, then exhibits an *average* sequence, which is exactly identical with the standard and, hence, defines the wild-type. The quasi-species, introduced above in precise terms, represents such an organized distribution, characterized by one (or more) *average* sequences. Typical examples of distributions (related to the RNA-phage Q_β) are given in Table 2. One unique (average) sequence is present only if the copy which *exactly* resembles the standard is clearly the dominant one, i.e., if it has the highest selective value within the distribution. Mutants, whose W_{ii} are very close to the maximum values, will on average be present in correspondingly high abundance (cf. Table 2). They will cause the wild-type sequence to be somewhat blurred at certain positions. If two closely related mutants actually have (almost) identical selective values, they may both appear in the quasi-species with (almost) equal statistical weights. How closely the W_{ii}-values have to resemble each

Table 2.
The abundance of the standard sequence in the wild-type distribution is determined by its quality function Q_m and its superiority σ_m. At given number of nucleotides ν_m the quality function can be calculated from the average digit quality $\bar{q}_m$ of the nucleotides referring to a particular enzymic read-off mechanism. Both $\bar{q}_m$ and σ_m also determine the maximum number of nucleotides ν_{max}, which a standard sequence must never exceed, otherwise the quasi-species distribution becomes unstable. The data refer to RNA sequences consisting of 4500 nucleotides (phage Q_β).

The values in the dark fields (a) show the relative abundance of the standard sequence within the wild-type distribution (in percent) according to Eq. (18) and (25). Negative values mean that the distribution is unstable. The light fields show the threshold numbers ν_{max} for given $\bar{q}_m$ and σ_m values. The data clearly demonstrate the sensitivity of ν_{max} towards the parameter $\bar{q}_m$. For a well-adapted species, $1-\bar{q}_m$ should be slightly larger than $1/\nu_m$ (e.g., $=1-\bar{q}_m=0.0005$ for $\nu_m=4500$ nucleotides requiring σ_m values $\geqq 10$).

correct copies / ν_{max}	$\bar{q}_m$ = 0.9980	0.9990	0.9995	0.9998
superiority σ_m = 2	unstable (<0) / 346	unstable (<0) / 693	unstable (<0) / 1386	unstable (<0) / 3466
20	unstable (<0) / 1498	unstable (<0) / 2996	5.8 % / 5991	37.6 % / 14979
200	unstable (<0) / 2649	0.6 % / 5298	10 % / 10597	40 % / 26492

a

In part b a more realistic example of a quasi-species distribution comprising $1\cdot10^9$ individuals is presented. A sequence of 4500 nucleotides would have 13500 one-error mutants, supposed that for each correct nucleotide (A, U, G, or C) there are three incorrect alternatives. Experiments with RNA-replicases, however, show that purine → purine and pyrimidine → pyrimidine substitutions are by far more frequent than any cross-type substitutions: purine ↔ pyrimidine. Hence – in order to be more realistic – we have assumed only one incorrect alternative for any position. Accordingly the multiplicity of any k-error copy is just $\binom{\nu}{k}$. The total of 4500 different one-error mutants have been subdivided with respect to their degenerate (average) selective values into five classes. There is one mutant in class M_{1a} which resembles the standard quite closely. Its selective value (W_{kk}) differs by only one percent. Class M_{1b} contains four degenerate mutants possessing selective values within 10% of that of the standard while 495 mutants of class M_{1c} show W_{kk} values 30% of W_{mm}. A bulk of 2000 mutants is by one order of magnitude lower in their W_{kk} values and an equal amount of mutants is not viable at all, i.e., they do not reproduce with any speed comparable to that of the standard. Furthermore all the copies with more than two errors have been assigned W_{kk} values $\ll W_{mm}$. Although this may not be an realistic assumption, it is of no serious consequence with respect to the population numbers of *individual* sequences which are extremely small because of the large multiplicity of different error copies. Despite this fact, the sum of all multiple error copies represents the largest group in this example, followed by the total of one-error copies. On the other hand, the standard is by far the most abundant individual in the quasi-species distribution.

mutant class	degeneracy of class	assumed relative selective value W_{kk}/W_{mm}	population number of individual mutant (× degeneracy)
error-free	1	1	8.9×10^7 (×1)
one-error			
M_{1a}	1	0.99	4×10^6 (×1)
M_{1b}	4	0.9	5×10^5 (×4)
M_{1c}	495	0.3	6.3×10^4 (×495)
M_{1d}	2000	0.1	4.9×10^4 (×2000)
M_{1e}	2000	~0	4.3×10^4 (×2000)
$\sum M_1$	4500		2.2×10^8 (×1)
multiple-error			
M_2	$\sim10^7$	~0	< 30 (×10^7)
$\sum M_{k>1}$	$\sim2^{4500}$	~0	6.88×10^8 (×1)

b

An alternative calculation has been made in which the relative selective value of the one-error mutant 1a has been raised from 0.99 to 0.9995. While the gros of the distribution changes only slightly, the population number of the particular mutant 1a rises to the value found for the standard type (i.e., both result to 8.4×10^7). This example shows the limitation of the approximations behind Eqs. (18) and (19) requiring $w_{km}\ll W_{kk}-W_{mm}$. A more rigorous numerical evaluation yields a population number of mutant 1a amounting to only about 60% of that of the standard. Even for small differences of selective values the standard clearly remains the dominant species. Only if a one-error mutant resembles the standard within limits of $W_{mm}-W_{kk}\ll 1/\nu_m$ it may be considered to be a degenerate and hence undistinguishable individual.

other for both mutants to become selectively indistinguishable, depends on their 'degree of affinity.' For distant relatives, the correspondence has to be much more precise than for one-error copies. A special class of 'reversible neutral' mutants, hence, can be quantitatively defined. There is, of course, a second wider class of neutral mutants which belong to different quasi-species being degenerate in their eigenvalues λ_i. Most of these neutral mutants die out after appearance but a minor part may spread through the population and coexist with, or displace, the formerly established quasi-species. This diffusional spread of neutral mutants can be understood only on the basis of stochastic theory (cf. below).

III. 5. Realistic Approximations

An explicit representation of the eigen-value of the selected quasi-species can be obtained with the help of perturbation theory. The result of second-order perturbation theory resembles the previously reported expression for $W_{\max}$ ([4], Eq. II-33a):

$$\lambda_{\max} \approx W_{mm} + \sum_{k \neq m} \frac{w_{mk} w_{km}}{W_{mm} - W_{kk}} \tag{13}$$

Here the index m refers to that (molecular) species which is distinguished by the largest selective value. The approximation holds only if no other W_{kk} approaches this value too closely and the dominant copy m can be considered as representative of the wild-type. Table 2 shows, how effective the approximation indeed is for any system of realistic importance. The larger the information content the smaller the individual w-values. The approximation thereby reveals a very important fact; selection (under strain) is extremely sharp with respect to distant relatives (the smaller the w_{mk}- and w_{km}-values the closer may W_{kk} resemble W_{mm} without being of any restriction to m). However, selection is smooth with respect to very close relatives. These are always present in the distribution if their selective value W_{kk} is much smaller than W_{mm} (or even zero). If the sum term in Eq. (13) can be numerically neglected (cf. values in Table 2), the extremum principle (Eq. (12)) can be expressed as:

$$W_{mm} > \bar{E}_{k \neq m} \tag{14}$$

or

$$Q_m > \sigma_m^{-1} \tag{15}$$

where

$$\bar{E}_{k \neq m} = \sum_{k \neq m} E_k x_k / \sum_{k \neq m} x_k \tag{16}$$

represents the average productivity of all competitors of the selected wild-type m and

$$\sigma_m = \frac{A_m}{D_m + \bar{E}_{k \neq m}} \tag{17}$$

is a superiority parameter of the dominant species.
With the same approximation the relative stationary population numbers can be calculated, yielding for the dominant copy

$$\bar{x}_m / \sum_{k=1}^{N} \bar{x}_k = \frac{W_{mm} - \bar{E}_{k \neq m}}{E_m - \bar{E}_{k \neq m}} = \frac{Q_m - \sigma_m^{-1}}{1 - \sigma_m^{-1}} \tag{18}$$

and for the one-error copy

$$\bar{x}_{1k} / \bar{x}_m = \frac{w_{km}}{W_{mm} - W_{kk}} \tag{19}$$

which is valid, as long as $w_{km} \ll W_{mm} - W_{kk}$ (cf. Table 2).
Higher approximations could be obtained using $\lambda_{\max}$ in which-ever form it is expressed. Eqs. (16)–(18) would change accordingly.

On the basis of these approximations we can quantitatively characterize Darwinian behavior of macromolecular systems. Eq. (13) shows to which extent the dynamics of selection is determined by the individual properties of the dominant (standard) species (m), while Eqs. (18) and (19) indicate the relative weights of representation of the standard species and its mutants (cf. also Table 2). It is surprising how small a fraction of the standard species is actually present within the wild-type distribution, despite the fact that its physical parameters almost entirely determine the dynamic behavior of the distribution. This provides a great adaptive and evolutive flexibility of the quasi-species and enables it to react quickly to environmental changes.

The approximations break down only in the case of the presence of two or more dominant species (cf. Table 2) which are (almost exactly) neutral mutants. However, those reversible neutral mutants can be combined to a selectively indistinguishable subclass of species and as such will determine the dynamic behavior like a single dominant species, according to the relations given above. It is interesting to note the fact that within the quasi-species there is no selection against reversible neutral mutants ('reversible' being defined as sufficiently large w_{ik} and w_{ki} to guarantee a reproducible representation). These reversible neutral mutants being part of the quasi-species have to be distinguished from unrelated neutral mutants, (w_{ik} and w_{ki} too small to provide a reproducible representation according to Eq. (18)). Those unrelated neutral mutants can still coexist, to a minor extent, as a consequence of random fluctuations [25]. The stochastic treatment shows that competition among those neutral species is a random drift phenomenon leading to extinction with an indeterminate 'survival of the survivor' as well as to a certain upgrowth and spread of new mutants, a phenomenon to which geneticists [26] refer as non-Darwinian behavior (i.e., survival without selective advantage). It should be realized that this stochastic behavior of unrelated neutral mutants, although it was not anticipated by Darwin and his followers, is not in contradiction with those properties which lead to the deterministic Darwinian behavior. The selection of reversible neutral mutants is even in accordance with the Darwinian principle, if this is interpreted in the correct way as a derivable physical principle, where it then applies to the concept of quasi-species.

III. 6. Generalizations

In the quantitative representation of Darwinian system by Eqs. (2) and (6) we have made a number of special assumptions in connection with the structural prerequisites and external constraints. In this section we want to find out how far we can generalize those assumptions without losing the characteristic features of Darwinian behavior.

1) *Rate Terms.* The linear rate terms for both autocatalytic formation and first-order decomposition may be substituted by more general expressions. Most common for enzymic mechanisms is the Michaelis-Menten form:

$$\text{rate} \sim \frac{x_i}{1 + a_i x_i} \quad \text{or} \quad \frac{x_i}{1 + \sum_k a_k x_k}$$

which replaces the simple x_i-dependence. It has been shown [4] that for autocatalytic mechanisms of this kind, selection remains effective for low population numbers and, hence, in the range which is critical for selection. Saturation will not prevent the upgrowth of advantageous mutants, but may allow for some coexistence due to a change from exponential to linear growth in the unconstrained mechanism.
In general, if the reaction order is defined by a term x_i^k, Darwinian behavior may be found for exponents:

$$0 < k \leqq 1$$

For $k = 0$ coexistence would result in a growth-limited system. Under the constraint of constant fluxes, such a situation may occur for self-reproductive species if their formation rate is limited by the constant rate of supply of energy-rich building material. Species which feed on different (mutually independent) sources will not compete with each other. The independent sources then provide

'niches' for coexistence. The multiple varieties of different species often owe their existence to such or similar devices, the consequence of which is in complete accord with the scope of Darwin's theory.
The behavior for exponents $k>1$ is analyzed in more detail in part B of this paper. It leads to extremely sharp selection with some consequences which were not compatible with Darwin's view, especially regarding descendence.

2) *Autocatalysis.* According to our discussion in section II, straightforward self-reproduction is only the simplest example among a whole class of linear autocatalytic mechanisms. Self-reproduction may generally be effected via a cyclic catalytic process. Single-stranded RNA-phages, for instance, reproduce via mutual instruction through the two complementary strands. The rate equations for such a two-membered catalytic cycle [4] yield two eigen-values for the dominant species:

$$\lambda_{1,2}=-\frac{D_{+}+D_{-}}{2}\pm\sqrt{A_{+}A_{-}Q_{+}Q_{-}+\tfrac{1}{4}(D_{+}-D_{-})^{2}} \tag{20}$$

These expressions are based on similar approximations as Eq. (13) neglecting the mutation terms. One of these eigen-values, if it is positive (i.e., $A_{+}A_{-}Q_{+}Q_{-}>D_{+}D_{-}$), replaces the W_{mm} referring to the *self*-replicative unit. Here, the kinetic parameters of both strands contribute with equal weight (geometric mean) to the selective value. They both have to be optimized in order to yield optimal performance. Wherever phenotypic properties of the RNA strands are of importance, equivalence can be most suitably reached by structural symmetry (cf. t-RNA, midi-variant of Q_{β}-RNA [27]). The second always negative eigen-value refers to an 'equilibration' between plus- and minus-strands, the concentrations of which then assume a fixed ratio. Whenever this equilibration is reached, plus- and minus-strand will act like one replicative ann competitive *unit.*
The general treatment of catalytic cycles has been shown [4, 20–22], to resemble the results for simple self-instructive systems. A n-membered cycle again is characterized by one positive eigen-value which corresponds to the selective value of the single self-reproductive unit. The catalytic properties of all members contribute to this eigen-value, in the simplest case, in the form of the geometric mean of their AQ-values, hence, requiring finite AQ-values for all members of the cycle. Furthermore, a n-membered cycle is characterized by $(n-1)$ negative eigen-values, which are representative of the internal equilibration of the concentration ratios of all members of the cycle.

3) *Mutations.* The main source of mutations, especially in the early stages of evolution, is mis-copying, i.e., the inclusion of a nucleotide with a non-complementary base during the process of replication. The experiments with Q_{β}-phage show that error rates for replacing a given purine or pyrimidine by its homologue differ considerably from those for exchanging a purine by one of the pyrimidines or *vice versa* [24]. In the formal treatment, using the parameters Q and w, no distinction has to be made regarding the kind of mutation, such as point mutation or frame shifts resulting from deletion or insertion. Those distinctions, of course, are important if the functional properties of the mutants are to be studied. The formal representation is also invariant to the causes of mutation, such as misreading in replication, chemically induced changes, or radiation damage. It may be necessary to correlate some of the mutations more closely with the decomposition term (cf. below) which, however, has no influence on the formal structure of the equations.

4) *Decomposition.* Mutations as a consequence of external influences (e.g. radiation) should be attributed to the decomposition term. In general, decomposition of the individual i may lead to species k belonging to a different class comprising only fragments of i. Again, those processes do not change the formal structure of the differential equations, if the corresponding conservation relations are appropriately taken into account.

5) *External Constraints.* The explicit form of the solutions of the selection equations depends on the constraints to be imposed. We have discussed in detail the case of constant overall organization. Similar, though quantitatively different, results are obtained for the constraint of constant fluxes [4, 28, 29]. The externally regulated parameters may, of course, involve temporal (e.g., any type of periodical) variations. This may lead to mechanistic advantages, but does not alter the essential prerequisites and consequences of Darwinian behavior. The extremum principle (12) then assumes the general form:

$$\lim_{\tau\to\infty}\frac{1}{\tau}\int_{0}^{\tau}\bar{E}(t)\,dt=\lambda_{m} \tag{21}$$

A further generalization of the extremum principle has been achieved by Jones et al. [21].
Selection of a quasi-species relative to its competitors may also be considered in a growing system. It will be shown that a simple normalization procedure is applicable, which allows a generalized treatment of non-steady-state systems.

6) *Stochastic Treatment.* A final generalization is of a more principal nature. Deterministic rate equations describe generally the average behavior of ensembles consisting of a large number of individuals. The elementary processes, however, can be represented only by reaction probabilities. Game models which have been developed together with R. Winkler-Oswatitsch [18] demonstrate clearly three basic types of behavior which can be treated by stochastic theory:
a) Internal self-control of fluctuations as found around stable steady states and, in particular, at thermodynamic equilibrium,
b) self-amplification of fluctuations characterizing an instability, and
c) indifference towards fluctuations yielding random drift behavior.
In the first case, fluctuations are of importance only for small population numbers and simply mean an uncertainty of any momentary microstate. For the macrostates, which are accessible to experimental tests, they yield expectation values within specifiable average fluctuation limits.
In the second case deterministic behavior is restricted to the response to a given fluctuation. In other words, this 'if-then' determinacy will predict what happens if a certain fluctuation occurs and the accuracy of the prediction will increase with the extent of the fluctuation. The occurrence of the fluctuation itself, however, is uncertain and this microscopic uncertainty is mapped macroscopically via the finally deterministic amplification process. This case is of particular relevance for Darwinian systems. Most mutations represent fluctuations of the first type, i.e., they are not of any selective advantage and do not endanger the stability of the wild-type. After occurrence, they cease deterministically as in the case of equilibrium. However, there are also mutations which bring along a selective advantage, and they have the tendency to amplify themselves, hence, causing an instability. Whether or not they are successful in reaching dominance depends on the magnitude of their selective advantage and the extent of the fluctuation. A single copy still has a fairly high chance of dying out before it is reproduced, especially if its W-value is only slightly larger than $\bar{E}(t)$. The stochastic theory shows that for small advantages, i.e. $(W_{m+1}-W_{m})\ll W_{m}$, the fluctuation has to reach a certain extent, i.e., a number of copies which corresponds to the magnitude of $W_{m}/(W_{m+1}-W_{m})$, before the probability of growing up becomes larger than $1-e^{-1}$. This is equivalent to saying that only mutants

identified by distinct advantages will *deterministically* influence the evolutionary behavior. Nearly neutral mutants behave stochastically almost like truly neutral mutants, and, hence, refer to the third category of games which resembles random drift behavior. Neutral mutants of high frequency, of course, are part of a given quasi-species and as such are somewhat stabilized due to a finite mutation rate. They thereby utilize a fluctuation response which was characteristic of the first category. Since such a coupling is rather weak, there may be quite large fluctuations in the relative representation of neutral relatives. Unrelated (i.e., very rare) neutral mutants, on the other hand, may be considered as different quasi-species whose eigen-values have the same magnitude as that of the wild-type. Most of those neutral quasi-species will have to die out, but if they happen to grow up they may become persistent and even replace the former wild-type ('survival of the survivor'). This kind of behavior can be deduced only from stochastic theory. Corresponding calculations have been carried out for the spread of genes in Mendelian populations, especially by M. Kimura [25] and his school.

In conclusion, evolution is a deterministic process with respect to its progressive character. There will always be favorable competition of the wild-type with less advantageous mutants, coexistence of neutral or nearly neutral, closely related mutants, and upgrowth of new clearly advantageous quasi-species. However, evolution is indeterminate with respect to the temporal sequence of the appearance of mutants, as well as with respect to genetic drift caused by unrelated neutral mutants. Only a small fraction of these neutral quasi-species actually can grow up. Evolution via rare neutral mutations may, therefore, be more important in the later than in the earlier stages of evolution, where many advantageous alterations are still possible and, hence, occur with relatively high frequency.

III. 7. Information Content of the Quasi-species

In our approach to molecular evolution we have not yet dealt explicitly with the concept of genetic information. We have defined the (molecular) species as the replicative unit with a distinct information content, represented by a particular arrangement of molecular symbols. We have also taken into account the similarity relations among different such symbol arrangements, which led to the concept of the quasi-species. In deriving the criteria of selection and evolution, it proved sufficient, just as in population genetics, to notice the individual differences in genotypic information and correlate them with their characteristic dynamic properties as expressed by the selective value W_{ii}.

On the other hand, questions such as, "How much information can be accumulated in a given quasi-species?" or, "Where is the limit of reproducibility and, hence, of the evolutionary power of a quasi-species?" would remain unanswered in such a treatment. We, therefore, now introduce more specifically the concept of information.

In the theory of communication, the information content of a message consisting of ν_k symbols can be expressed as

$$I_k = \nu_k i \tag{22}$$

where i is the average information content of a single symbol. According to Shannon i can be correlated with the probability distribution of the symbols [30, 31]:

$$i = -K \sum_j p_j \ln p_j \tag{23}$$

with

$$0 < p_j < 1 \quad \text{and} \quad \sum_j p_j = 1 \tag{24}$$

The constant K usually is taken to be $1/\ln 2$ in order to yield the unit 'bits/symbol'. The alphabet generally includes classes of symbols (e.g., $\lambda = 4$ for nucleic acids). The practical use of Eq. (23) is limited to those cases where the *a-priori* probabilities of all symbols are known and where the number of symbols in the message is large enough that the averages apply. In order to account for all cooperative effects or redundancies influencing the probability distribution, it might be necessary to know the probabilities of all λ^ν possible alternatives of symbol combinations.

In our case we are actually not so much interested in any statistical a-priori probability of a symbol, but rather in the probability that a given symbol is correctly reproduced by the genetic mechanism however sophisticated the machinery may be at the various levels of organization. This probability refers to the dynamical process of information transfer and, therefore, must be determined experimentally from kinetic data (wherever it is possible; cf. below). Let us call these probabilities for correct symbol reproduction q_j. A message consisting of ν_i (molecular) symbols then will be reproduced correctly with the probability or quality factor:

$$Q_i = \prod_{j=1}^{\nu_i} q_{ij} \equiv \bar{q}_i^{\nu_i} \tag{25}$$

Even if the symbols consist of only λ classes, the quality factor for each symbol of the message might be dependent on its particular environment so that the determination of the geometric mean $\bar{q}_i$ may require the consideration of various cooperative effects, specifically associated with the message 'i'. Nevertheless, this average $\bar{q}_i$ for any given message i can be determined and it turns out that for a particular enzymic reproduction machinery those averages apply quite universally, whenever the message is sufficiently long. Moreover, the individual q-values in general are so close to one that the geometric mean can be replaced by the arithmetic mean, i.e.,

$$\left(\prod_{j=1}^{\nu_i} q_{ij}\right)^{1/\nu_i} \approx \frac{\sum_{j=1}^{i} q_{ij}}{\nu_i} \quad \text{if } (1-q_{ij}) \ll 1. \tag{26}$$

Eq. (25) then comprises the information-theoretical aspect of reproduction where $\bar{q}_i$ however, refers to a dynamic rather than to a static probability. The numerical values of $\bar{q}$ may take into account all mechanistic features of symbol reproduction including any static redundancy which reduce the error rate of the copying process. Nature actually has invented ingenious copying devices ranging from complementary base recognition to sophisticated enzymic checking- and proof-reading mechanisms.

Genetic reproduction is a continuously self-repeating process, and as such differs from a simple transfer

of a message through a noisy channel. For each single transfer it requires more than just recovery of the meaning of the message, which, given some redundancy, would always allow a fraction of the symbols to be reproduced incorrectly. It is also necessary to prevent any further accumulation of mistakes in successive reproduction rounds. In other words, a fraction of precisely correct wild-type copies must be able to compete favorably with the total of their error copies. Only in this way can the wild-type be maintained in a stable distribution. Otherwise the information (now in a semantic sense, i.e., the copy with optimum W_{ii}-value) would slowly seep away until it finally is entirely lost.

The condition which guarantees the stable conservation of information is the selection criterion Eq. (14) or Eq. (15). This criterion can be expressed in a general form as:

$$Q_m > Q_{\min} = \sigma_m^{-1} \tag{27}$$

and as such applies to any reproduction mechanism, even if σ_m cannot be expressed in as simple a form as valid for the linear mechanism (cf. Eq. (17)). If we combine the selection criterion, as derived from dynamics theory, with the information aspect, as expressed by Eq. (25), we obtain the important threshold relationship for the maximum information content of a quasi-species:

$$\nu_{\max} = \frac{\ln \sigma_m}{1 - \bar{q}_m} \tag{28}$$

The number of molecular symbols of a self-reproducible unit is restricted, the limit being inversely proportional to the average error rate per symbol: $1-\bar{q}_m$

There is another way of formulating this important relationship. The expectation value of an error in a sequence of ν_m symbols; $\varepsilon_m = \nu_m(1-\bar{q}_m)$ must always remain below a sharply defined threshold:

$$e^{\varepsilon_m} < \sigma_m \tag{29}$$

otherwise, the information accumulated in the evolutionary process is lost due to an error catastrophe.

Table 2 contains examples which demonstrate the relevance of Eq. (28). The threshold is not sensitively dependent on the magnitude of the superiority function, but σ_m must be larger than one, i.e. $\ln \sigma_m > 0$, in order to guarantee finite values for $\nu_{\max}$. In practice (cf. below), $\ln \sigma_m$ is usually between one and ten. Relation (28) allows a quantitative estimate of the evolutionary potential, which any particular reproduction mechanisms can provide. It states, for instance, that an error rate of 1% (or a symbol-copying accuracy of 99%) is just sufficient to collect and maintain reproducibly an information content not larger than a few hundred symbols (depending on the value of $\ln \sigma_m$) or that the maintainance of the information content of the genome as large as that of *E. coli* requires an error rate not exceeding one in 10^6 to 10^7 nucleotides. It is a relation which lends itself to experimental testing, and we shall report corresponding measurements below. Eq. (28) also gives quantitative account of what has been called in population genetics the 'genetic load,' the importance of which has been stressed long ago.

The results of this section can be summarized as follows: Any mechanism of selective accumulation of information involves an upper limit for the amount of digits to be assembled in a particular order. If this limit is surpassed, the order, i.e., the equivalent of information, will fade away during successive reproductions. Stability of information is equivalent to internal stability of the quasi-species. It is as much based on competition as is selection of the quasi-species. However, two quasi-species can be brought to coexistence, while violation of the threshold relation will result in a total loss of genetic information. Hence internal stability of the quasi-species distribution, more than its 'struggle for existence' is the characteristic attribute of Darwinian behavior.

IV. Error Threshold and Evolution

IV. 1. Computer Test of the Error Catastrophe

The physical content of the threshold relation (Eq. 28) may be exemplified with a little computer game (cf. Table 3). The goal is to create a meaningful message from a more or less random sequence of letters. For this purpose the computer is initially given a set of N random sequences and programmed:

a) to remove each sequence from its memory after a defined (average) lifetime,

b) to reproduce each sequence which is present in the storage with a characteristic rate, and

c) to introduce random errors in the reproduced copies, again with a chosen average rate of substitutions per symbol.

The average rates for a) and b) are matched in such a way that a steady representation of N copies of sentences is maintained in the store, each sentence having a finite lifetime. Hence any information gained during the game can be preserved only via faithful reproduction of the sequences present. The gain of information, on the other hand, must result from a selective evaluation of the various mutant sequences

Table 3. Self-correction of sentences is the result of the evolution game, exemplified in this table.
The target sentence reads:

TAKE ADVANTAGE OF MISTAKE

It has been chosen because it provides 'selectively advantageous' information with respect to the mechanism of evolution. Its special form permits a cyclic closure, whenever functional links among the single words are introduced (as will be done in part B).
Using a code, in which each letter (and word spacing) is represented by a quintet of binary symbols, the information content amounts to $\nu_m = 125$ bits, allowing for about 4×10^{37} alternatives. This number excludes appearance of information by mere chance. The sequences of letters shown in this tables for given generations have been sampled as being representative for the total population of sequences in the computer store.

INITIAL SEQUENCE: BAK GEVLNT GUPIF LESTKKM

DIGIT QUALITY $\bar{q}_m$: 0.995

SELECTIVE ADVANTAGE PER BIT: 10

1. GENERATION:	RAK GEVNNT GUPQF KESTKKM
5. GENERATION:	NAK AEZ.NS GEPOF MESTMKU
10. GENERATION:	VAKF ADV!NT.GE OF MISD!KE
16. GENERATION: (GOAL REACHED)	TAKE ADVANTAGE OF MISTAKE

The first example demonstrates that evolution is very efficient near the critical value $1-\bar{q} \approx 1/\nu_m$, which with $\nu_m = 125$ amounts to $\bar{q}_m = 0.992$. Starting with a random sequence of letters the target sentence is usually reached within $20(\pm 6)$ generations for any value of $\bar{q}$ between 0.995 and 0.990. This efficiency near the threshold is even more evident if we compare the evolutionary progress at a given generation for various values of $\bar{q}_m$:

INITIAL SEQUENCE: BAK GEVLNT GUPIF LESTKKM
SELECTIVE ADVANTAGE PER BIT : 10

$\bar{q}_m$	BEST SEQUENCE AFTER 11 GENERATIONS	NUMBER OF MISTAKES
0.999	LAKD AEV.NTAGU AF KISTQKM	9
0.995	TAKE ADV!NT GE OF MISTAKE	2
0.990	TAKFRADV!NTAGE OF MISXAKE	3
0.985	VATA ADBKMDI DHOD ?CSYBKE	18

Analogous behavior is found for the disintegration of information for $\nu_m > \nu_{max}$. At an error rate $(1-\bar{q}_m) = 1.5\%$, a selective advantage per bit of 2.5 corresponds to a σ_m value of about 5. Under these conditions the information is not stable any more. For small selective values (as in this example), however, disintegration (or accumulation) of information is a comparatively slow process.

INITIAL SENTENCE: TAKE ADVANTAGE OF MISTAKE

DIGIT QUALITY $\bar{q}_m$: 0.985

SELECTIVE ADVANTAGE PER BIT: 2.5

NUMBER OF GENERATION	BEST SENTENCE	NUMBER OF MISTAKES
1	TAKE ADVANTAGE OF MISTAKE	0
5	TAKF !DVALTAGE OF MISTAKE	3
10	TALF ADVALTACE OF MISTAKI	5
20	DAKE ADUAVEAGE OF MJUTAKE	6
40	TAKE ADVONTQCU OF MFST!ME	7
71	TAKEB ?VALTAGI LV MIST!KE	8
71	GENERATION FOR $\bar{q}_m = 0.97$?AMEBADTIMOACFHQEBA!STBMF	18

Comparison of the last two rows (referring to generation 71) shows that disintegration is much faster at an error rate of 3% ($\bar{q}_m = 0.97$).
For the other example (selective advantage per bit = 10) the threshold is not yet passed at $\bar{q}_m = 0.985$; so that information is stable, as seen below, where the process starts with the correct sentence.

INITIAL SENTENCE: TAKE ADVANTAGE OF MISTAKE

DIGIT QUALITY $\bar{q}_m$: 0.985

SELECTIVE ADVANTAGE PER BIT: 10

71 GENERATION

OUTPRINT OF 8 REPRESENTATIVE SENTENCES	NUMBER OF MISTAKES
TAKE ADVANTAGE OF MISTAKE	0
TAKE ADVANTAGIPOF MISTAKE	2
TAKE ADVANTAGE OF MISTAKE	0
TBKE !DVANTAGE OF MISTAKE	2
SAKE ADVANTAGE OF MGSTAME	3
TAOE ADVANVAGE OF MISTAKE	2
TAKE ADVAVTAGE OF MISTAKE	1
TAKE .DVANTAGE OF MISTAKE	1

according to their meaning, or better, according to their more or less close relationship to any meaning. This evaluation is to be effected by intrinsic meaning-dependent properties of the sequences, which influence their rates of reproduction and (or) removal as expressed by the selective value.

In natural selection, the target of evolution is always the genotype which represents the phenotype with the optimum selective value. Evaluation then is effected via the phenotypic, i.e., physical and chemical properties of the particular individual which determine the reproduction rate and quality, as well as the lifetime of the genotype in relation to the average of its competitors present in the population. Likewise, in our psychic memory we can associate with any sequence of letters a reproductive value which is related to its meaning. In our game we could, thus, evaluate any sequence of letters according to its relation to meaning, using our imagination. The computer, of course, can achieve such a goal only via a particular program, in which the evolving sentences are compared with one or several possible target sentences.

Let us then assume, that each sequence is reproduced with a rate which depends on the number of symbols that coincide in their position with those of a (meaningful) target sentence. Using binary coding we may define, that with each bit closer to the target, we enhance the reproduction rate by a certain factor; with each bit more distant we slow it down correspondingly.

The information about possible target sentences which thereby has to be supplied to the computer in advance, however, is not used for any other purpose than to provide the computer with a value scheme. The evaluation procedure could easily be made more sophisticated, i.e., more closely resembling our mental evaluation of 'meaning,' up to a level that the particular target finally reached is not at all predetermined. However, these mechanistic details of simulation are not of so much importance for exemplifying the physical meaning of the threshold relation, after we can well define the intrinsic evaluation procedure of molecular evolution (an example is provided by a game model demonstrating the evolution of t-RNA molecules [18]). The results of the computer experiment, according to Table 3, can be summarized as follows:

At high symbol-reproduction quality, e.g., for a sentence of 100 bits with average error rates per symbol

$$1-\bar{q}_m \ll 10^{-2}$$

the evolutionary progress is very slow, even if large σ_m-values, i.e., large selective advantages are involved. Maximum progress is achieved if we choose average error rates $(1-\bar{q}_m)$ which are of the same order of magnitude as $1/\nu_m$ (e.g., for 100 bits: $\bar{q}\approx 0.99$).

For sufficiently large σ_m-values (>3) the target sentence is then obtained within a number of generations which corresponds to the order of magnitude of the evolutionary distance between the target and the initial (more or less random) sequence (e.g., 100 generations). However, as soon as the threshold for $1-\bar{q}_m = \ln\sigma_m/\nu_m$ is surpassed, no more information can be gained, regardless of how large a selective advantage per bit is chosen. If one starts out with a nearly correct sentence, the information disintegrates to a random mixture of letters, rather than to evolve to an error-free copy. The threshold is very sharp but the rate of disintegration varies near the threshold. There is only a weak dependence of the threshold value on the magnitude of σ_m, unless this parameter gets very close to unity. The superiority σ_m is calculated from the relative selective advantages, and, hence, some knowledge about the error distribution (relative to the respective optimal copy) is required. This distribution of course depends on the magnitudes of the selective advantages. The computer experiment closely resembles the expected error distribution, which near the critical value $1-\bar{q}_m \approx 1/\nu_m$ (with $\ln\sigma_m \approx 1$) yields an almost equal representation of the optimum copy, all one-error copies (relative to optimum), and the sum of all multiple-error copies (in which distribution the two-error copies again are dominantly represented, with strongly decreasing tendency for copies with more errors). For smaller selective advantages (e.g., $W_{mm}-W_{kk}<3$) this representation shifts in favor of the error copies and in disfavor of the (relative) optimum, which for $\ln\sigma_m=1$ is already present with less than 10% of the total.

IV. 2. Experimental Studies with RNA-Phages

As trivial as this game may appear—after one has rationalized its results—as relevant has it turned out in nature in determining the information gained at the various levels of precellular and cellular self-organization. An experiment resembling almost exactly the above game has been carried out with phage Q_β by Ch. Weissmann and his coworkers [32, 33].

An error copy of the phage genome has been produced by site-directed mutagenesis. The procedure consists of *in vitro* synthesis of the minus strand of the phage RNA containing at the position 39 from the 5′-end the mutagenic base analog N^4-hydroxy CMP, instead of the original nucleotide UMP. Using this strand as template with the polymerizing enzyme Q_β-replicase, an infectious plus strand could be obtained in which at position 40 from the 3′-end this

position corresponds to position 39 from the 5′-end in the minus strand and is located in an extra-cistronic region – an A-residue is substituted by G. *E. coli* spheroplasts then were infected with this mutant plus-strand yielding complete mutant phage particles, which could be recovered from single plaques. Serial transfer experiments *in vivo* (infection of *E. coli* with complete phage particles) as well as *in vitro* (rate studies with isolated RNA strands using Q_β-replicase) allowed for a determination of reproduction rate parameters for both the wild-type and the mutant-40 including their distributions of satellites. Combined fingerprint and sequence analysis, applied to successive generations, indicated changes in the mutant population due to the formation of revertants. Studies with different initial distributions of wild-type and mutant revealed the fact that natural selection involves the competition between one dominant individual and a distribution of mutants. The quantitative evaluation shows that the value depends on the particular selective advantage as well as on distribution parameters of the mutant population. The wild-type as compared with the particular mutant shows a selective advantage

$$W_{\text{wild-type}} - W_{\text{mutant}} \approx 2 \text{ to } 4$$

while the rate of substitution was estimated to be

$$1 - q \approx 3 \times 10^{4}.$$

The q-value is based on the rate of revertant formation and hence applies to the particular (complementary) substitutions

$$G \rightarrow A \quad \text{or} \quad C \rightarrow U, \text{ respectively.}$$

According to Eq. (20) the quality factors of both the plus and the minus strand contribute equivalently to the fidelity of reproduction. $G \rightarrow A$ and $C \rightarrow U$ substitutions are, therefore, equivalent. They may not differ too much from $A \rightarrow G$ and $U \rightarrow C$ replacements, the main cause being the similarity of wobbling for GU and UG interactions.

Since the replicating enzyme requires the template to unfold in order to bind to the active site, the q-values should not further depend on the secondary or tertiary structure of the template region. *In vitro* studies with a midi-variant of Q_β-RNA [27] yield rates for $C \rightarrow U$ substitutions which are consistent with the values reported above. Purine → pyrimidine and pyrimidine → purine substitutions seem to occur much less frequently and, hence, do not contribute materially to the magnitude of $\bar{q}$.

A determination of σ_m is more difficult, since it depends on the magnitude of $\bar{E}_{k \neq m}$. First of all, it is noteworthy that modification of an extracistronic region – which does not influence any protein encoded by the phage RNA – has such a considerable effect upon the replication rate. S. Spiegelmann was the first in stressing the importance of *phenotypic* properties of the phage RNA molecule with respect to the mechanism of replication and selection. The σ_m value reported above refers to a particular mutant and its satellites. Other mutants might influence the tertiary structure of Q_β-RNA in a different way and, hence, exhibit different replication rates. Moreover, mutations in intracistronic regions may be lethal and, therefore do not contribute to $\bar{E}_{k \neq m}$ at all. If we consider the measured value as being representative for the larger part of mutations we obtain for the maximum information content, then, a value only slightly larger than the actual size of the Q_β genome, which comprises about 4500 nucleotides. One might be somewhat suspicious with such a close agreement and we have mentioned our reservations. However, they refer mainly to the value of σ_m which enters only as a logarithmic term. Larger σ_m value would still yield acceptable limits for $\nu_{\max}$. Thus the value obtained may finally be not too far beyond reality.

There is another set of experiments, carried out by Ch. Weissmann and his coworkers [24], which indicates the presence of a relatively small fraction of standard phage in the wild-type distribution. These data suggest that $\sigma_m^{-1} \approx Q_m$ and that the actual number of nucleotides is indeed very close to the threshold value $\nu_{\max}$ (cf. Eq. 18).

The midi-variant, used in the evolution experiments of G. Mills and S. Spiegelmann et al. [27] consists of only 218 nucleotides and, hence, is not as well adapted to environmental changes as Q_β-RNA. It is, of course, optimally adapted to the special environment of the 'standard reaction mixture,' used in the test-tube experiments (which does not require the RNA particles to be infectious). However, its response to changes in the environment, e.g., to the addition of the replication inhibitor ethidium bromide, is fairly slow. The mutant obtained after twenty transfers, each allowing for a hundredthousand-fold amplification differs in only three positions from the wild-type of the midi-variant and shows a relative small selective advantage in the new environment. The reason for the slow response is that 218 nucleotides with an average single digit quality of 0.9995 yield Q values close to one and lead to wild-type sequences, that are very faithfully replicated, carrying along only a small fraction ($\lesssim 10\%$) of mutants in their error distribution.

The remarkable result of these studies in the light of theory is not the fact that the threshold relation as an inequality is fulfilled. Since its derivation is based on quite general logical inferences, any major disagreement would have indicated serious mis-

conceptions in our understanding of Darwinian systems. There is not the slightest reason for any such discrepancy, since we know quite well the molecular mechanism of self-replication in such a 'lucid' system as phage Q_β. The truly surprising result is that the actual value of ν not only remains below the threshold ν_{max} but, in fact, resembles it so closely. The number of nucleotides could easily have been restricted by factors other than symbol quality $\bar{q}$, thereby yielding ν values far below the threshold allowing for a reproducible accumulation of information.

Hence, we are forced to conclude that these RNA phages during their evolution, indeed, tried to accumulate as much information as was possible, utilizing also large extracistronic, but phenotypically active regions in their genome. This fact does not exclude that under other (e.g., artificial) conditions much smaller RNA molecules, such as the mentioned midi-variant, could win the competition, or that under different natural circumstances also much smaller viable phages exist.

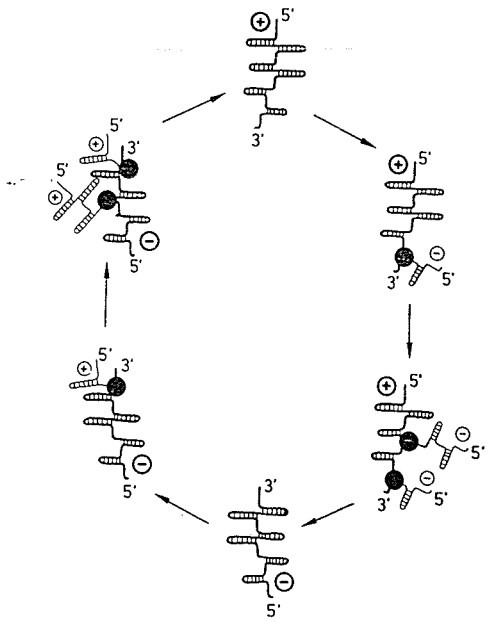


Fig. 11. *The replication cycle encountered in RNA phages* leads to single-stranded units of RNA with highly specific secondary and tertiary structures utilized as phenotypic targets [11, 34]. The formation of complete or partial duplices between plus and minus strands (the so-called Hofschneider or Franklin pairs, resp.) is prevented by an immediate internal folding of the newly synthesized strand. Using a midi-variant of the phage Q_β, S. Spiegelmann and coworkers [35] were able to demonstrate the existence of inhibitory effects due to duplex formation. Single-strand replication is based on an efficient interaction of the replicase with both the plus and the minus strand, requiring a certain symmetry in those regions of the tertiary structure which are phenotypically important.

Even more important is to realize that in nature no (single-stranded) RNA phage exists which comprises more than about 10000 nucleotides in its genome. This suggests that the enzymic mechanism of RNA replication, especially with respect to the discrimination of the bases A and G or U and C has reached its optimum and could not be improved further. It is not possible for any single-stranded RNA to maintain reproducibly more information than is equivalent to the order of magnitude of 1000 to 10000 nucleotides (the precise number depending on the value of σ_m).

Larger molecules could exist, of course, according to chemical criteria, but they would be of no evolutionary value. Moreover, the requirements for selective conservation of information must be fulfilled by both the plus and the minus strand, as prescribed by Eq. (20), although only one of the strands needs to carry the genetic information to be translated by the host mechanism. These conclusions apply only to RNA molecules which in their replication phase act as *single*-stranded templates and for which the mechanism depicted in Figure 11 has been proposed [11].

IV. 3. DNA-Replication

With double-stranded molecules, especially with DNA, we encounter a quite different situation. They generally reduplicate in the form of double-stranded units, i.e., they may be considered to be truly *self*-reproductive, at least in a phenomenological sense (even if the instruction conveyed is based again on the complementarity of the nucleotides).

Let us briefly summarize what is known [10] about reproduction of such double-stranded DNA molecules (cf. Fig. 12):

a) Replication is a semi-conservative process. Each of the two strands of the DNA duplex is copied during the reproduction phase, leading to two (essentially identical) duplices, each of which contains one parental strand.

b) Replication starts at a defined growing point and may proceed in both directions of the strand. The unwinding of the double strand is aided by so-called unwinding proteins, some of which have been isolated and identified. They enhance the rate of unwinding as much as thousand-fold, yielding a relatively fast movement of the replication fork. At the same time it is necessary to relieve the torque caused by the unwinding in some part of the molecule. It is suggested, that a chain break and repair mechanism, effected by endonucleases and ligases, may permit intermediate rotation of chain sections around a phosphodiester bond.

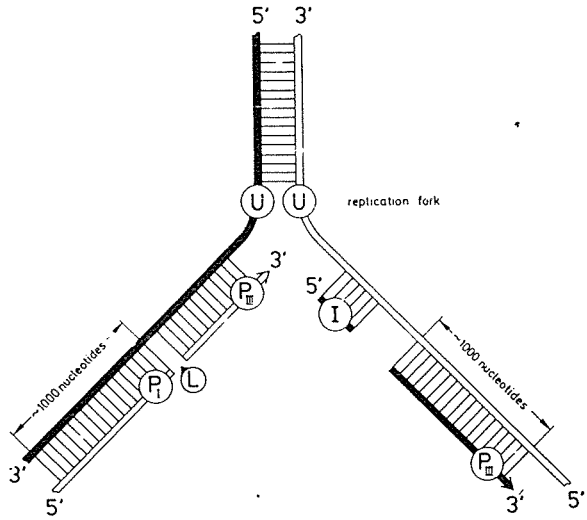


Fig. 12. *The semi-conservative replication of double-stranded DNA* is a highly sophisticated process including many steps of reaction and control of which the most important ones are indicated here [10]. Both daughter strands are polymerized in the 5′ → 3′ direction. The unwinding of the parental double helix is effected concomitantly by an unwinding protein (U). The synthesis of new fragments is initiated by RNA primers formed with the help of an enzymic system (I) and later hydrolyzed away presumably by the nuclease function of polymerase-I (P_I). Chain elongation up to fragments with 1000 to 2000 nucleotides is believed to be effected mainly by the polymerase-III complex (P_{III}). Those nascent so-called Okazaki fragments have to be linked together by a ligase (L). Mispaired nucleotides at the 3′-end of the fragments (and only those) are excised by the 3′ → 5′-exonuclease function, most likely of the polymerase-I complex (P_I), whose activity is predominantly associated with gap filling and repair. Other features, such as repair by 5′ → 3′-exonuclease action, through which whole fragments of DNA can be removed, are not included in this scheme since they may not be as important for the synthesis of new strands.

c) Replication of *both* strands proceeds by inclusion of nucleotide residues in the 5′-3′ direction. The DNA-polymerases can include the monomers only in this unique vectorial way, which cannot take place concomitantly at both strands. Electron microscopy with a resolution of about 100 Å has revealed the existence of single-stranded regions at only one side of the growing fork, suggesting that the other strand is completed only after a larger stretch, sufficient for a 5′-3′ progression, has been created.

d) Replication occurs in short, discontinuous pulses. In prokaryotes, the fragments produced in a single pulse are about 1000 to 2000 nucleotides long. They are initiated by the formation of primers, for which very short segments of RNA serve. The fragments of copied DNA occurring along both strands behind the replication fork are later sealed together by ligases.

e) The various functions required in DNA replication have been identified by isolation of the particular enzymes and by demonstration of their activity. In particular, several polymerase complexes (I, II and III) have been characterized, which comprise polymerizing as well as several chain-decomposing functions. Of special interest here is the 3′ → 5′ exonuclease activity of polymerase I. It allows for a preferential excision of a non-base-paired nucleotide at the 3′-terminus of the growing chain. Since chain growth occurs only in the 5′ → 3′ direction this exonuclease function allows for a proofreading of the newly synthetized chain fragments. Its optimal activity is about 2% of that of polymerizing functions. The 3′ → 5′ exonuclease is to be distinguished from a 5′ → 3′ exonuclease, which also is part of the DNA-polymerase I complex and probably involved in excision repair. It acts only at the 5′-terminus and cleaves the di-ester bond at a base-paired region, possibly up to 10 residues apart from the 5′-end. It, hence, can remove oligonucleotides, while the proofreading 3′ → 5′ enzyme only removes single non-base-paired nucleotides at the end of the growing chain.

We may conclude now an important difference between RNA and DNA replication, which is expressed in the average symbol quality factors for both mechanisms. In RNA replication the accuracy of information transfer has to be established in the continuous polymerization. However the RNA-replicase solves the problem, it achieves apparently a limiting value of $\bar{q}$ between 0.9990 and 0.9999. Approximately the same fidelity should be reached by any continuous DNA-polymerizing mechanism.

Mutant bacteriophage DNA-polymerases devoid of 3′ → 5′-exonuclease activity have been tested *in vitro* and shown to incorporate errors with the relatively high frequency of about one for every thousand nucleotides. Similar results have been reported for purified *avian myoblastosis virus* DNA-polymerase. However, also lower error rates have been found. For instance, studies with small eukaryotic DNA-polymerases, lacking proofreading exonuclease activity, yielded values one order of magnitude lower than in the cases mentioned above (i.e., one for every five- to tenthousand nucleotides) [36–39].

The observed DNA fragment of 1000 to 2000 nucleotides, appearing during DNA polymerization (in prokaryotic cells) may also directly refer to the limited fidelity of the polymerase function. Apparently, the polymerase cannot easily extend a mispaired terminus generated by itself ([10], p. 88), although this has been observed in the absence of exonucleases. On the other hand, 3′-5′-exonuclease if present will recognize the mismatch and excise the wrong nucleotide. There is

no reason to assume that the optimal resolving power at this step is much different from that in polymerization. Hence, proofreading may reduce the error rate (optimally) by another three orders of magnitude.

Correction of errors, on the other hand, cannot be postponed to any later stage, i.e., after both chains have been completed. Although repair systems using $5' \rightarrow 3'$-nuclease activities do exist, they cannot deter-

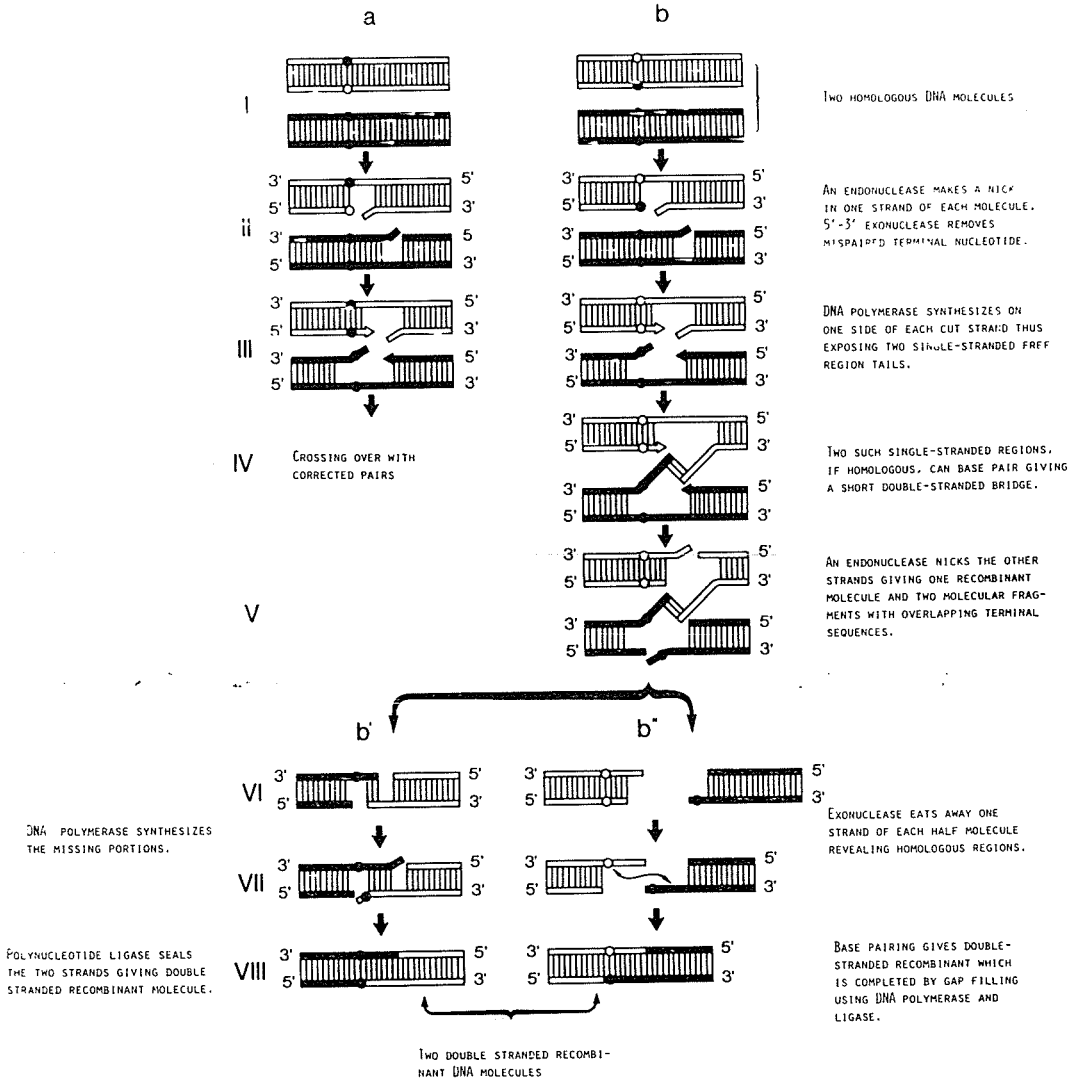


Fig. 13. *Genetic recombination allows for error detection* in completed double strands of DNA. This model was originally proposed to explain the mechanism of crossing over. It can be applied to error correction as well. The symbol ● designates a genetically correct, the symbol ○ an erroneous nucleotide. Accordingly ● ● always resembles the correct complementary, ● ○ the mismatched (non-complementary) and ○ ○ the complementary, but erroneous nucleotide pair, regardless of which of the four nucleotides is involved. Assume that strand nicking is triggered by a mismatched pair (stage II). Then $3' \rightarrow 5'$-exonuclease action will correct the error: ● ○ $\rightarrow$ ● ● in 50% of the cases (stage III), while in the other 50% the wrong nucleotide will complement itself. ○ ● $\rightarrow$ ○ ○. The incorrect (though complementary) pair, however, is not fixed as in a simple repair mechanism. Recombination with the correct copy (stage VI, VII) will restore the original situation (stage I and VIII) in which only one of four homologous positions is occupied by an incorrect nucleotide. Hence iteration of the procedure can lead to a steady reduction, rather than to a 50%-irreversible fixation of errors. This scheme points out that crossing over is associated with error checking in completed strands. The scheme (copied from [45]) can be understood on the basis of known functions of DNA-polymerase, which does not exclude the existence of other hypothetical and, even more efficient mechanisms. A complete understanding of the fidelity problem, which has to include a consideration of vegetative multiplication processes, requires a more detailed knowledge of the mechanism than is available hitherto

mine which of the two strands contains the mismatched member of the incorrect pair (cf. [36]). More detailed mechanisms of kinetic proofreading have been proposed [40] and experimentally tested [41, 42] (for a review cf. [43]).

We may therefore conclude: The optimum average symbol quality for DNA replication reaches values of 0.999999 or somewhat higher, thus allowing for an accumulation of information of up to an equivalent of one to ten million nucleotides (depending on the magnitude of σ_m). It is gratifying to notice that this number coincides with known sizes of genomes in prokaryotic cells (e.g. *E. coli*: 4×10^6 base pairs). Again there is no need at all for any individual to reach this limit. Other restrictions, such as packing requirements in the case of DNA phages, etc., may limit the actual size of a genome. As for RNA phages, any intermediate size below the threshold may, thus, be observed.

There is an upper limit for the genetic information content of a prokaryotic cell. Just as any extension beyond the single-strand information capacity of 10^4 nucleotides requires a new mechanism involving double-stranded templates and proofreading enzymes, the new limit of about 10^7 nucleotides set by the prokaryotic DNA-reproduction mechanism could not have been exceeded until another mechanism for further reduction of errors was available. Such a mechanism, namely genetic recombination was invented by nature at the prokaryotic level. However, it took about two to three billion (10^9) years before it reached perfection in order to give rise to another extension of the genetic information content of single individuals.

The process of genetic recombination utilized by all eukaryotic cells requires two alleles to be identified at their homologous positions. Since the error rate for the enzymic DNA reproduction is below 10^{-6} per nucleotide, uncorrected mistakes are very rare and cannot be present in more than one of the four equivalent sites of the two alleles. Hence, there is a further opportunity to identify and correct those errors in the recombinants, even if they occur in formerly completed duplices. A possible scheme is depicted in Figure 13. However, the mechanism of recombination is neither yet known in sufficient detail, nor is it clear how many steps finally are responsible for the further reduction of the error rate. The fact is that such a reduction *has* been achieved, as is revealed by an analysis of evolutionary trees, and that it is an important prerequisite for the expansion of genetic information capacity up to the level of man.

IV. 4. The First Replicative Units

For a discussion of the origin of biological information we have to start at the other end of the evolutionary scale and analyze those mechanisms which led to the first reproducible genetic structures. The physical properties inherent to the nucleotides effect a discrimination of complementary from non-complementary nucleotides with a quality factor $\bar{q}$ not exceeding a value of 0.90 to 0.99. The more detailed analysis based on rate and equilibrium studies of cooperative interactions among oligonucleotides has been presented elsewhere [4, 44]. In order to achieve a discrimination between complementary and non-complementary base pairs according to the known differences in free energies, the abundant presence of catalytically active, but otherwise uncommitted proteins as environmental factors might have helped. However, uncommitted protein precursors in some cases will favor the complementary, in other cases the non-complementary, interaction. Any preference of one over the other can only be limited to the difference of free energies of the various kinds of pair interaction. Any specific enhancement of the complementary pair interaction would require a convergent evolution of those particular enzymes which favor this kind of interaction. In order to achieve this goal they must themselves become part of the self-reproducing system which in turn requires the evolution of a translation mechanism.

The first self-reproductive nucleic acid structures with stable information content – given optimal $\bar{q}$ values of 0.90 to 0.99 – were t-RNA-like molecules. For any reproducible translation system, however, an information content larger by at least one order of magnitude would be required. As we know from the analysis of RNA-phage replication, such a requirement can be matched only by optimally adapted replicases, which could not have evolved without a perfect translation mechanism. The phages, we encounter today, are late products of evolution whose existence is based on the availability of such a mechanism, without which nature could not afford to accumulate as much information in one single nucleic acid molecule. Hence, there was a barrier for molecular evolution of nucleic acids at the level of t-RNA-like structures similar to those barriers we find at later stages of evolution, requiring some new kind of mechanism for enlarging the information capacity.

The t-RNA's or their precursors, then, seem to be the 'oldest' replicative units which started to accumulate information and were selected as a quasi-species, i.e., as variants of the same basic structure.

The first requirement was stability towards hydrolysis. It has been shown by a game model, similar to the one described in IV. 1., that the presently known secondary (and tertiary) structure of t-RNA (cf. Figs. 14a and b) is a direct evolutionary consequence of this requirement. The symmetry of this structure, furthermore, reflects the optimization of its replicative mechanism which, according to Eq. (20), supposes equivalent behavior of the plus and the minus strand

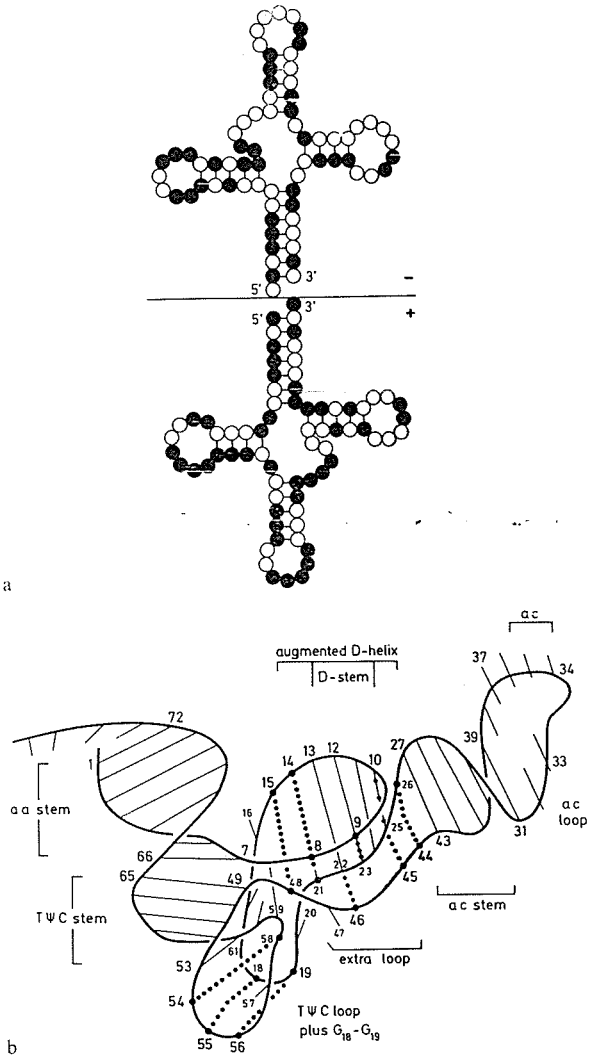


Fig. 14. *Symmetry of functional RNA molecules,* as exemplified with t-RNA^{phe}, aids single-strand replication by specifically adapted enzymes. The plus and minus strands of the symmetrical structure are distinguished by common phenotypic features. Although t-RNA's in present organisms are genotypically encoded, their symmetry might still reflect the ancestrial mechanism of single-stranded RNA reproduction, for which plus and minus strand are equally important. The symmetry is most obvious in the secondary structure (a), but shows up accordingly also in the tertiary structure (b) (reproduced from [46])

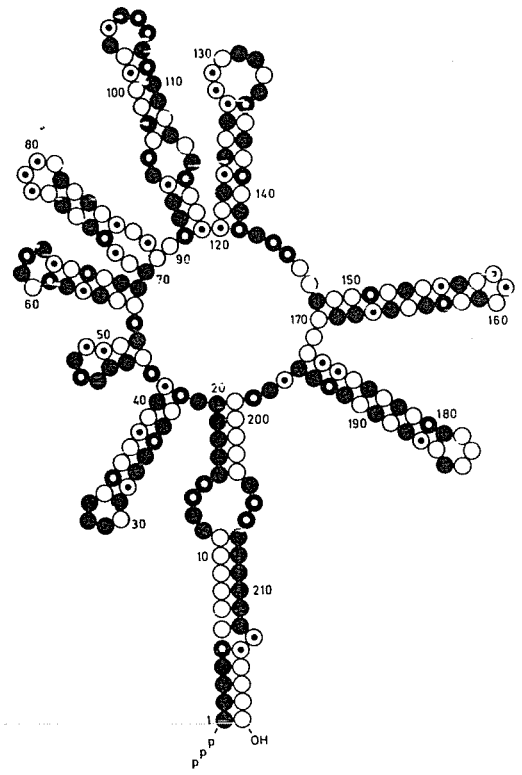


Fig. 15. *'Flower' model of Spiegelmann's midi-variant of Q_β-RNA (plus strand).* Symmetry requirements are less important where the information is mapped in genotypes which are reproduced via standardized polymerization mechanisms. The midi-variant of phage Q_β is selected *solely* for its phenotypic information, in that it exhibits an optimal target structure for recognition by the enzyme Q_β-replicase. This property must be inherited by both the plus and the minus strand. The symmetry of the structure becomes most obvious in the 'flower' model, although this arrangement probably does not represent the natural structure of the active molecule. According to the mechanistic conditions of single-strand replication shown in Figure 11, a model admitting immediate chain folding during synthesis [27] should be advantageous

in order to yield optimal performance. This symmetry can also be found at the level of RNA phages, especially for variants which are selected for being phenotypically most efficient with respect to *in vitro* replication, but otherwise not carrying genetic information (Fig. 15).

IV. 5. The Need for Hypercycles

It is the object of this paper to show, first that the breakthrough in molecular evolution must have been brought about by an integration of several self-reproducing units to a cooperative system and, second that a mechanism capable of such an integration can be provided only by the class of hypercycles. This conclu-

sion again can be drawn from logical inferences, based on the following arguments:

The information content of the first reproductive units was limited to $\nu_{max} \lesssim 100$ nucleotides. Several of those units representing similar functions but different specificities were required to build up a translation system. Such a system might have emerged from one quasi-species, but the equivalent partners had to evolve simultaneously. This is neither possible by linking them up into a larger self-reproductive unit (because of the error threshold), nor could it result from compartmentation, because of the strong competition among the equivalent self-reproductive units within the compartment. Such a process rather requires functional linkages among all self-reproductive units, to be distinguished by the following qualities:

a) The linkage must still permit competition of each self-reproductive unit with its error copies, otherwise these units cannot maintain their information.

b) The linkage must 'switch off' competition among those self-reproductive units which should be integrated to a new functional system and allow for their cooperation.

c) The integrated functional system then must be able to compete favorably with any other less efficient system or unit.

These three requirements can be fulfilled only by a cyclic linkage among self-reproductive units or, in other words, the functional linkage among autonomous self-reproductive units itself has to be of a self-enhancing cyclic nature, otherwise their total information content cannot be maintained reproducibly. Hypercyclic organization, thus, appears to be a necessary prerequisite for the nucleation of integrated self-reproductive systems of larger information content, as were required for the origin of translation. This statement is the conclusion of what is to be shown in the subsequent parts by a more detailed analysis of linked systems.

If we are asked, "What is particular to hypercycles?", our answer is, "They are the analogue of Darwinian systems at the next higher level of organization." Darwinian behavior was recognized to be the basis of generation of information. Its prerequisite is integration of self-reproductive symbols into self-reproductive units which are able to stabilize themselves against the accumulation of errors. The same requirement holds for the integration of self-reproductive and selectively stable units into the next higher form of organization, in order to yield again selectively stable behavior. Only the cyclic linkage—as an equivalent of autocatalytic reinforcement at this level (cf. II.)—is able to achieve this goal.

Table 4. The essential stages of information storage in Darwinian systems

Digit error rate $1-\bar{q}_m$	Superiority σ_m	Maximum digit content ν_{max}	Molecular mechanism and example in biology
5×10^{-2}	2	14	enzyme-free RNA replication[a]
	20	60	t-RNA precursor, $\nu=80$
	200	106	
5×10^{-4}	2	1386	single-stranded RNA replication via specific replicases
	20	5991	phage Q_β, $\nu=4500$
	200	10597	
1×10^{-6}	2	0.7×10^{6}	DNA replication via polymerases including proofreading by exonuclease
	20	3.0×10^{6}	*E. coli*, $\nu=4\times10^{6}$
	200	5.3×10^{6}	
1×10^{-9}	2	0.7×10^{9} [a]	DNA replication and recombination in eukaryotic cells
	20	3.0×10^{9}	vertebrates (man), $\nu=3\times10^{9}$
	200	5.3×10^{9}	

[a] Uncatalyzed replication of RNA never has been observed to any satisfactory extent; however, catalysis at surfaces or via not specifically adapted proteinoids (as proposed by S.W. Fox) may involve error rates corresponding to the values quoted.

The results of section IV are summed up in Table 4, showing the essential stages of information storage in Darwinian systems, which could be facilitated by various storage mechanisms of reproduction.

This table will be useful for a discussion of a model of continuous evolution from single molecules to integrated cellular systems, as presented in part C.

B. The Abstract Hypercycle

Topologic methods are used to characterize a particular class of self-replicative reaction networks: the hypercycles. The results show that the properties of hypercycles are sufficient for a stable integration of the information contained in several self-replicative units. Among the catalytic networks studied, hypercyclic organization proves to be a necessary prerequisite for maintaining the stability of information and for promoting its further evolution. The techniques used in this paper, though familiar to mathematicians, are introduced in detail in order to make the logical arguments accessible to the nonmathematician.

V. The Concrete Problem

In Part A of this trilogy on hypercycles we have arrived at some essential conclusions about Darwinian systems at the molecular level, which may be summarized as follows:

1. The target of selection and evolution is the quasispecies, which consists of a distribution of (genotypically) closely related replicative units, centered around the copy (or a degenerate set of copies) corresponding to the phenotype of maximum selective value.
2. The information content of this master copy – expressed as the number ν of symbols (nucleotides) per replicative unit – is limited to $\nu_m < \frac{\ln \sigma_m}{1-\bar{q}_m}$, where $\sigma_m (>1)$ is the superiority of the master copy, i.e., an average selective advantage over the rest of the distribution, and $\bar{q}_m$, the average quality of symbol copying. Exceeding this threshold of information content will cause an error catastrophe, i.e., a disintegration of information due to a steady accumulation of errors.
3. A highly evolved enzymic replication machinery is necessary to reach a stable information content of a few thousand nucleotides. Such an amount would be just sufficient to code for a few protein molecules, as we find in present RNA phages. The physical properties inherent in the nucleic acids allow for a reproducible accumulation of information of no more than 50 to 100 nucleotides.

The last of these three statements may be questioned on the basis of the argument that environmental factors – such as suitable catalytic surfaces or even protein-like enzyme precursors [47] – may cause a considerable shift of those numbers. In fact, the figures given were derived from equilibrium data, namely, from the free energies for (cooperative) complementary versus noncomplementary nucleotide interactions. Nevertheless, we still consider them upper limits which in nature may actually be reached only in the presence of suitable catalysts or via annealing procedures. Laboratory experiments on enzyme-free template-induced polymerization lead to considerably lower numbers. On the other hand, environmental catalysts cannot yield fidelities of symbol recognition exceeding the equilibrium figures, unless they themselves become part of the selectively optimizing system. There is no way of systematically favoring the functionally advantageous over the nonadvantageous interactions, other than via a stepwise selective optimization. The phage genomes could evolve in the form of single-stranded RNA molecules, only because a quite advanced replication and translation machinery was provided by the host cell. They are postcellular rather than precellular evolution products. Something like the magnitude of the information content of their genomes is just what would be required at the beginning of translation, namely, the reproducible information for a set of enzymes that could start a primitive translation mechanism. Hence the essential conclusion from Part A is:

The start of translation requires an integration of several replicative units into a cooperative system, in order to provide a sufficient amount of information for the build-up of a translation and replication machinery. Only such an integrated machinery can bring about a further increase of fidelity and hence allow for a corresponding expansion of the information content.

How can one envisage an integration of competitive molecules, other than by ligation to one large replicative unit, which is prohibitive due to the threshold relation for ν_{max}. (Note that the units to be integrated have to remain competitive with respect to their mutants in order to evolve further and not to lose their specific information.) Let us briefly investigate three possible choices:

1. *Coexistence.* Stable mutual tolerance of self-replicative units in the absence of stabilizing interactions is possible only for individuals belonging to the same quasi-species. The quasi-species distribution could well provide favorable starting conditions for the evolution of a cooperative system. However, it does not favor the evolution of functional features. The coupling stabilizing the quasi-species is solely dictated by the genotypic kinship relations, which usually do not coincide with functional needs. Required is a set of selectively equivalent genotypes that complement each other at the *phenotypic* level. The quasi-species distributions as such does not meet these selection criteria.

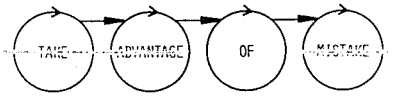
2. *Compartmentation.* Enclosure of a Darwinian system in a compartment will not provide a solution of this problem either. The main consequence of compartmentation is an enhancement of competition due to the restriction of living space and metabolic supply. Hence a compartment will only stabilize further a given selectively advantageous quasi-species; it will not favor the evolution of equivalent partners according to functional criteria, which requires the cooperating partners to diverge genotypically. A compartment, however, may offer advantages for a system that has already established a stable cooperation via functional linkages (cf. Part C). More sophisticated compartments such as present living cells, which comprise only one (or a few) copies of each replicative subunit together with a machinery for reproduction of the whole compartment require, of course, a symbol quality $\bar{q}_m$ which is adapted to the total information content according to the relationship for ν_{max}. In other words: They are subject to the same limitations as a fully ligated unit.

3. *Functional linkages.* Selection of functionally cooperating partners may be effected via the functional linkages, which provide either a mutual catalytic enhancement of reproduction or a structural stabilization. A closer inspection of such linkages is the main topic of this paper.

Let us aid our intuition again by playing another version of the computer game introduced in Part A. In the first part of the game the objective was to demonstrate the need for adapting the symbol-reproduction quality to the information content of the sentence to be reproduced. In the second part we assume now that the average quality factor $\bar{q}_m$ is not sufficient for a stable reproduction of the whole sentence in the form of a replicative unit, but suffices for copying units as small as single words. It refers to a natural situation in early evolution, where the physical forces inherent in the nucleotides may have been sufficient for an evolution of stable t-RNA-like molecules (=single words), but did not admit the build-up of an – even primitive – translation apparatus (=a whole 'meaningful' sentence). Accordingly, the computer is programmed just to reproduce single words using error rates sufficient to guarantee their stability against accumulation of errors.

As a first variant of the game let us try to establish a plain coexistence of the four words. For this purpose we attribute to all correct *words* in the sentence the same selective value, while a mistake in any word is of disadvantage with respect to the correct word by a given factor (per bit). As before, the words are allowed to reproduce, the total number being limited to N copies. This variant, however, differs from the original game in that the individual words now behave as independent replicative units. Table 5 shows some typical results: Despite the fact that all words have the same selective value and are able to compete favorably with their error copies, the sentence as a whole is unstable. Only one of the four words can win the competition, but it cannot be predicted by any means which of the four words actually wins. One may characterize this situation by the tautology: 'survival of the survivor'. The term 'fittest' means nothing but the mere result of the contest.

In the next variant of the game we introduce a functional linkage between related words: A given word provides catalytic help for the reproduction of the next word whenever it forms a meaningful sequence:



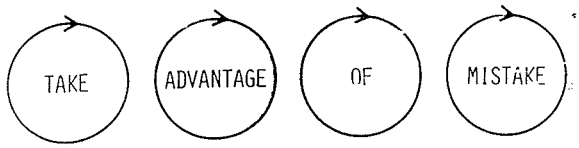
The coupling is proportional to the population number of the catalyst (i.e., to the representation of the particular word in the computer store). In other words, reproduction is facilitated according to a rate law:

$k_1 x_1$ and

$k_i x_i + k_i' x_i x_{i-1}$ for $i = 2, 3, 4$

Table 5. A game representing the competition of selectively equivalent replicative units

The aim of this game is to preserve the information of the sentence:



Each word symbolizes a replicative unit. All words have exactly the same selective value. The selective advantage per bit is 2.7. Each letter consists of 5 binary digits.

	Digit mutation probability $(1-\bar{q})$ [%]	Digit number ν	Word quality factor $Q=q^{\nu}$	Error expectation value ε
TAKE	3.15	20	0.53	0.63
ADVANTAGE	1.4	45	0.53	0.63
OF	6.3	10	0.53	0.63
MISTAKE	1.8	35	0.53	0.63

Since there is no coupling among the words every game ends with the *selection of one word*. All words are degenerate with respect to their selective values; therefore, each of the four words has an equal chance to be the survivor. Due to the high *average* error probability ($\sim 2\%$ per bit) the sentence as a whole (125 bits) is not a stable replicative unit.

Below typical results of ten games are listed. The 'X' indicates which word is selected, while all the others die out. The number denotes the generation after which selection is complete.

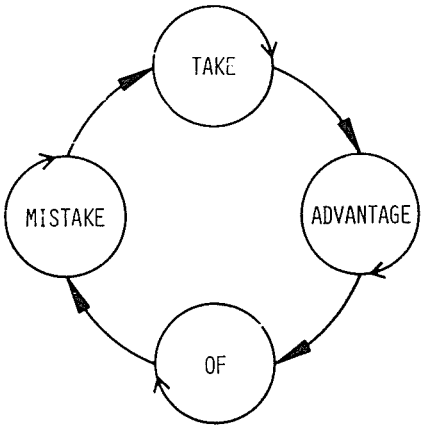
Game	TAKE	ADVANTAGE	OF	MISTAKE	Generation
1	X				12
2			X		15
3		X			19
4		X			23
5	X				10
6				X	20
7			X		9
8		X			13
9			X		22
10		X			26

Error distribution of the selected word ADVANTAGE. The solid line resembles the Poisson distribution $\frac{\varepsilon^k e^{-\varepsilon}}{k}$; where $\varepsilon=\nu(1-\bar{q})$ is the expectation value for an error in the word ($\nu=45$ bits).
(The errors refer to one single digit. All wrong letters differ from the correct ones in only one of their five digits.)

0 errors	1 error	2 errors	3 errors
ADVANTAGE			
ADVANTAGE			
ADVANTAGE			
ADVANTAGE	ADVANTAG!		
ADVANTAGE	ADVANDAGE		
ADVANTAGE	ADVARTAGE		
ADVANTAGE	ADVINTAGE		
ADVANTAGE	ADFANTAGE		
ADVANTAGE	ADZANTAGE		
ADVANTAGE	AHVANTAGE		
ADVANTAGE	AHVANTAGE	ADVANDACE	
ADVANTAGE	AHVANTAGE	ADVANDAIE	
ADVANTAGE	AFVANTAGE	ADVARXAGE	
ADVANTAGE	AFVANTAGE	ADVINTBGE	ITVANTAGU

x_i or x_{i-1}, resp. being population numbers, in this case referring to the words in the computer store. The result of this game variant is usually fixation of the last word of the sentence, i.e., 'mistake', while all other words die out. Only if the coupling is relatively weak and a particular k_i value is chosen large enough do we find that the corresponding word (i) may outgrow the others, representing selection among (essentially) independent competitors. The result that the last word in the sequence receives all the benefit of coupling (whenever the coupling terms are predominant) may be astonishing. One would expect that there must at least exist a range of stability for the whole sentence. This is certainly true for a certain magnitude of the population numbers, if the values of the rate parameters obey a certain order with respect to the position of the words in the sequence. However, fixed-point analysis as carried out in Section VII will show that, even under those special conditions, only the last member in the chain will grow in proportion to the total population, while all other members assume essentially constant population numbers, irrespective of the size of the total population. Hence, in a growing population, the relative abundance of the last member changes drastically until the system again reaches a range where only the abundant member remains stable. In the process of molecular evolution population numbers of individuals usually show those drastic changes, e.g., from one single mutant up to a detectable magnitude of (more than) billions of copies. Thus the result obtained in our game turns out to be quite representative of what actually would happen in nature.

The fact that linear coupling – if it works at all – feeds all the advantage forward to the last member in the sequence provides a strong hint for a possible solution of the problem: The couplings should form a closed loop:



Then the enhancement due to coupling will cyclically fluctuate through all words of the sequence. Our sentence actually was chosen as to provide automatically such a cyclic overlap through the word 'mis*take*'. Since each word is a catalytic cycle (i.e., a self-replicative unit) the system represents a hypercycle of second degree according to our definitions introduced in Part A. The result of the game is represented in Figure 16. All four words show a stable steady-state representation with a periodic variation of their population numbers. The selective values of different words do not have to be the same – which would seem very improbable for any realistic system. Each word, furthermore, is represented by a stable distribution of mutants. Unless one of the words is wiped out by a fluctuation catastrophe (which becomes very improbable at a sufficiently large number of copies) the population numbers will continue to oscillate. In other words: The information of the whole sentence is stable.

VI. General Classification of Dynamic Systems

VI.1. Definitions

In the following sections we shall carry out a more rigorous mathematical analysis of dynamic systems, especially of those which are of importance in precellular self-organization. To determine which systems are relevant we shall have to inspect different classes of reaction networks including both noncyclic and cyclic. Evolutionary processes can be described phenomenologically by systems of differential equations, as has been shown for a particular case in Part A. The term dynamical system then refers to the complete manifold of solution curves of a given set of differential equations.

Let us consider a general dynamic system that is described by n ordinary, first-order, and autonomous differential equations;

$$\frac{dx_i}{dt} \equiv \dot{x}_i = A_i(x_1 \ldots x_n, k_1 \ldots k_m; B); \qquad i = 1, 2, \ldots, n \quad (30)$$

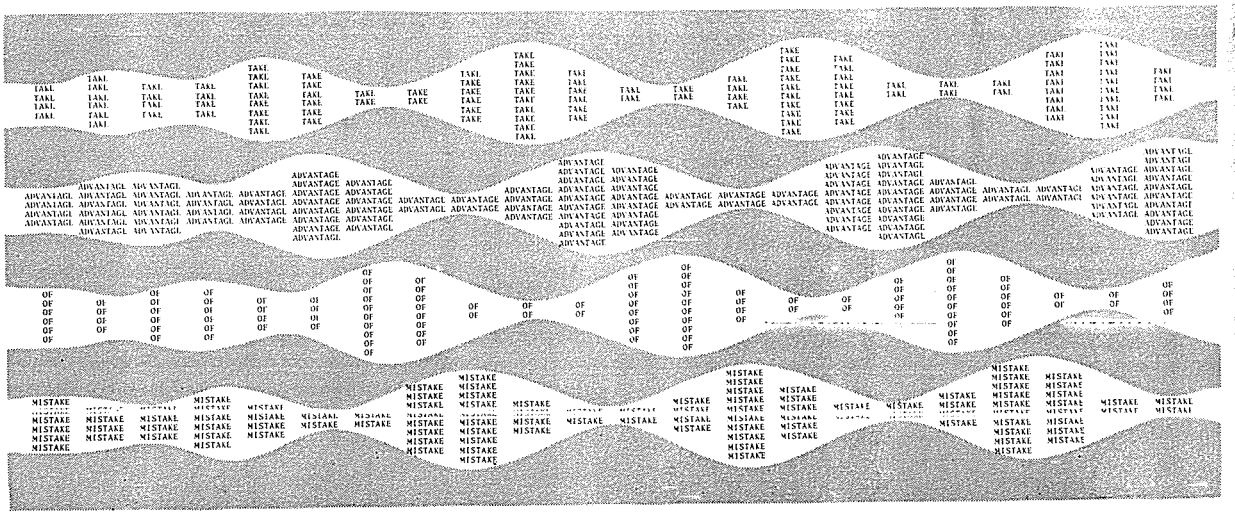


Fig. 16. Each word of the sentence TAKE ADVANTAGE OF MISTAKE represents a self-reproducing unit. The information of the sentence is stabilized by hypercyclic coupling among the words. In the graph, each printed word is representative of 10 copies in the computer store. All words in a vertical row refer to the same time, the intervals of which change discontinuously along the horizontal axis. The oscillation builds up from an initial equipartition of words and maintains a phase shift from word to word. The error distribution for each of the four words is stable and of the same kind as shown in Table 5

Later on we shall extend our analysis also to some nonautonomous systems for which $A_i = A_i(t)$.

As before, the x_i represent population variables that usually will refer to self-replicating macromolecular assemblies. The constants $k_i (i = 1, 2, \ldots, m)$ enter as parameters and may be composed of rate constants of elementary processes, of equilibrium constants for reversible and rapidly established reaction steps and of concentrations of those molecules that serve as the (energy-rich) building material for the synthesis of the macromolecules, assuming that these concentrations are buffered and hence can be included as time-independent values. Both the sets of x and k values can be represented as column vectors in a concentration space, or in a parameter space, respectively.

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \mathbf{k} = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_m \end{pmatrix}$$

By B we denote the initial conditions for a given set of solution curves, which in our case are represented by the set of initial concentrations $\mathbf{x}_0$.

According to the procedure employed in part A we split the functions A_i into three terms:

$$A_i = \mathrm{A}_i - \Delta_i - \phi_i \tag{31}$$

The A_is comprise all positive contributions to the chemical rate, representing an 'amplification' of the x_i variables, while the Δ_is include all negative rate terms resembling 'decomposition' of the macromolecular species. ϕ_i finally refers to a flux which may effect either dilution or buffering of the component i, depending on the external constraints applied to the system. The difference $\mathrm{A}_i - \Delta_i$ may be called a net growth function Γ_i. Referring to the Darwinian system (cf. part A), Γ_i, in particular, is given by $W_{ii}x_i + \sum_{k \neq i} w_{ik}x_k$, and if summed over all species $k = 1$ to n, it resembles the excess growth function $E = \sum_{k=1}^{n} E_k x_k$.

VI.2. *Unlimited Growth*

Removal of selection constraints leads to a new system of differential equations

$$\dot{x}_i = \Gamma_i(\mathbf{x}, \mathbf{k}, B) \tag{32}$$

describing a situation which in the following is called 'unlimited growth'. This terminology is representative for the system as a whole; for individual members it may also include decay or stationary behavior.

Assume we can express Γ_i as a polynomial in the various concentrations x_i (which as an approximation may also apply to irrational expressions or ratios of polynomials), then it will usually be possible to find leading terms in Γ_i which dominate in certain ranges of concentration. These leading terms usually are simple monomials of a given power of x_i. As such they determine the dynamic behavior of the system.

The simple case $\dot{x} = kx^p$ is illustrated in Figure 17. The textbook solutions have been normalized to $x(0) = 1$ and $\dot{x}(0) = 1$. As outlined in the Figure's legend, the whole family of solution curves can be subdivided into three classes, which are restricted to different regions of the concentration time diagram. Let us consider three representative examples, which will be of particular interest in our forthcoming discussion (cf. Table 6).

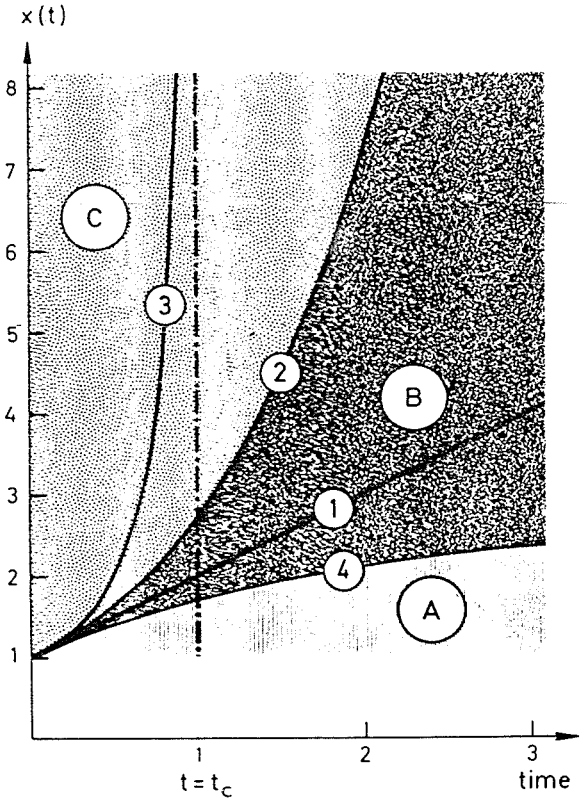


Fig. 17. Different categories of growth can be related to single-term growth functions $\Gamma(x) = \mathrm{d}x/\mathrm{d}t$ (normalized to $\Gamma = 1$ and $x = 1$ for $t = 0$). *Region A* does not include any growth function which could be represented by a simple monomial $\Gamma = x^p$. In this region all population numbers $x(t)$ remain finite at infinite time. The borderline between region A and B is given by the growth function $\Gamma(x) = e^{1-x}$ (curve 4). *Region B* is spanned by all monomials $\Gamma(x) = x^p$ with $-\infty < p < 1$. Curve 1 in this region exemplifies constant growth rate ($p = 0$) equivalent to linear growth. Exponential growth ($p = 1$) provides the borderline between regions B and C (curve 2). While population numbers in region B reach infinity only after infinite time, they show singularities at finite times in region C. As an example, hyperbolic growth ($p = 2$, singularity at $t = t_c = 1$) is shown by curve 3

i) Solution curve **1** represents a system with constant (positive) growth rate. The population variable $x(t)$ increases linearly with time. The solution curve also represents an example for the family of curves in region **B** of Figure 17, which grow to infinity at infinite time. An irreversible formation reaction with totally buffered concentrations of reactants may serve as the most common example. Self-reproducing species in ecologic niches, feeding on independent sources, may adjust their growth rates to the constant influx or production rate of food and then constitute another example for a growth behavior which is independent of the population size.

ii) Solution curve **2** results from growth rates linear in the population variable and exhibits an exponential increase of x with t, typical of Darwinian behavior, as was shown in Part A. The curve **2** furthermore establishes the borderline between regions **B** and **C**, i.e., between functions that reach infinity at infinite and at finite time.

iii) Solution curve **3** finally represents an example for functions with a singularity at finite time $[t_c=(kx_0)^{-1}]$. In this particular case, the growth rate was assumed to be proportional to the square of the population variable.

The whole range **C** may be characterized as 'hyperbolic growth'. Of course, in any real and finite world a population can never grow to infinity, because the available resources are finite and hence constraints will always take care of growth limitations. The phenomena giving rise to the hypothetical existence of a singularity will still cause a behavior quite different from that encountered in Darwinian systems.

At this point we may define the 'degree' p of the growth functions in a more general way, which will turn out to be useful for classification. As before, p_i is the power of the leading term in the growth function Γ_i. An n-dimensional dynamic system then may be characterized by a set of p_i values: $(p_1\, p_2 \ldots p_n)$. When we have a uniform distribution of powers p_i,

$$p_i=p_2=\cdots=p_n\equiv p, \tag{33}$$

we shall call the system 'pure'. Otherwise we are dealing with 'mixed' systems, which may be classified by their distributions of p_i values. Obviously, 'pure' systems can be analyzed much more easily than 'mixed' systems.

VI.3. Constrained Growth and Selection

In reality we shall always encounter constraints which provide certain limits for growth. For experimental studies, we must insure reproducible conditions. It is therefore necessary to formalize these conditions and include them in the theoretical treatment.

In irreversible thermodynamics we would prefer selection constraints that facilitate a thermodynamic description, e.g., constant generalized forces or constant generalized fluxes. For the analysis presented here, we have to adjust these to conditions for selection and evolution that can be materialized in nature. The constraints ϕ_i, as used in Equation (31), are too general for any straightforward analysis. In general we may distinguish between specific and nonspecific selection constraints. In the first case, the constraints act specifically on a single species or on a few species whereas the second case refers to regulation of a global flux ϕ. Then changes in all population variables are proportional to their actual values x_i:

$$\phi_i=\frac{x_i}{c}\,\phi \tag{34}$$

In practice, nonspecific selection constraints can be introduced into a dynamic system by the application of a continuous dilution flux. Thereby the total concentration, $c=\sum x_i$, can be controlled. The corresponding differential equation for c:

$$\dot{c}=\sum_{j=1}^{n}\dot{x}_j=\sum_{j=1}^{n}\Gamma_j(\mathbf{x})-\phi \tag{35}$$

fulfils the condition of stationarity: $\dot{c}=0$, when the flux is adjusted to compensate the net excess production:

$$\phi=\phi_0=\sum_{j=1}^{n}\Gamma_j(\mathbf{x}) \tag{36}$$

This selection constraint, referred to as 'constant organization', has been introduced previously and was also used in Part A. Condition (36) will be used frequently in the following sections to facilitate a general analysis of selection processes. Other constraints have been investigated as well [53]. As will be seen in the next section, the important features of selective and evolutive processes are fairly insensitive to the constraints applied. (These, of course, are always reflected in the quantitative results.)

The condition of constant organization leads to the following differential equations for the dynamic system:

$$\dot{x}_i=\Gamma_i(\mathbf{x})-\frac{x_i}{c_0}\sum_{j=1}^{n}\Gamma_j(\mathbf{x}),\qquad i=1,2,\ldots,n \tag{37}$$

Here c_0 denotes the stationary value of the total concentration which may be maintained by regulation of the flux to the value ϕ_0.

The individual selection behavior for the three simple growth functions: $p=0$, 1, and 2, as discussed in connection with Figure 17, is detailed in Table 6:

Table 6. Growth rates and selection behavior under the selection constraints of constant overall organization in the dynamic system $\dot{x}=\Gamma-\phi$

	p	Unlimited growth			Long-term behavior under constraints of constant organization	
		Growth rate $\Gamma(x)$	Solution curve	Type of growth	$\bar{x}$	Type of selection behavior
1	0	k	$x=x_0+kt$	linear	$\bar{x}_i=k_i c_0/\sum k_j$	Coexistence of species with no selection
2	1	kx	$x=x_0\cdot\exp(kt)$	exponential	$\bar{x}_k=c_0, \bar{x}_i=0$ $k_k-k_i>0, i\neq k$	Competition leading to selection of the globally 'fittest' species
3	2	kx^2	$x=x_0(1-kx_0t)^{-1}$	hyperbolic	$\bar{x}_k=c_0, \bar{x}_i=0$ $k=1,2,\ldots,n; i\neq k$	Competition aiming at local optimization 'once-for-ever' decision

i) Constant growth rates – corresponding to a linear increase of the population with time – yield under the constraint of constant organization a stable coexistence of all partners present in the system. Upgrowth of advantageous mutants shifts the stationarity ratios without causing the total system to become unstable.
ii) Linear growth rates, corresponding to an exponential increase of the population size, result in competition and selection of the 'fittest'. Advantageous mutants, upon appearance, destabilize and replace an established population.
iii) Nonlinear growth rates ($p>1$), characterized by hyperbolic growth, also lead to selection, more sharply than in the Darwinian system mentioned under ii). Mutants with advantageous rate parameters, however, in general will not be able to grow up and destabilize an established population, since the selective value is a function of the population number (e.g., for $p=2$, $W\sim x$). The advantage of any established population with finite x hence is so large that it can hardly be challenged by any single mutant copy. Selection then represents a 'once-for-ever' decision. Coexistence of several species here requires a very special form of cooperative coupling.

The examples mentioned are quite representative. We may classify systems according to their selection behavior as coexistent or competitive. In a given system we may encounter more than one type of behavior.

VI.4. Internal Equilibration in Growing Systems

While the condition of constant organization simplifies the analysis of dynamic system considerably, it is limited to systems with zero net growth. In this section we shall try to extend the range of applicability. The main problem is to find out in which way and under which conditions predictions on growing systems can be made, given the results obtained from an analysis of the corresponding stationary states. For this purpose we introduce nonspecific and time-dependent selection constraints (Eq. 34):

$$\dot{x}_i=\Gamma_i(\mathbf{x})-\frac{x_i}{c(t)}\,\phi(t) \tag{38}$$

Either $c(t)$ or $\phi(t)$ can be chosen freely. The other function, however, is determined then by the following differential or integral equation, respectively.

$$\phi(t)=\sum_{i=1}^{n}\Gamma_i(\mathbf{x})-\frac{\mathrm{d}c}{\mathrm{d}t} \quad \text{or} \tag{39}$$

$$c(t)=c_0+\int_0^t\left\{\sum_{i=1}^{n}\Gamma_i(\mathbf{x})-\phi(\tau)\right\}\mathrm{d}\tau \tag{40}$$

It is appropriate now to introduce normalized population variables $\xi=\frac{1}{c}\mathbf{x}$. The differential equations then can be brought into the form:

$$\dot{\xi}_i=\frac{1}{c(t)}\{\Gamma_i(\mathbf{x})-\xi_i\Sigma_j\Gamma_j(\mathbf{x})\} \tag{41}$$

As we see immediately, $\dot{\xi}_i$ does not depend explicitly on the selection constraint $\phi(t)$. There is, however, an implicit dependence through $c(t)$. We therefore push our general analysis one step further by considering some obvious examples: Let us assume that the net growth functions $\Gamma_i(\mathbf{x})$ are homogeneous of degree λ in x. Although this condition seems to be very restrictive we shall see that almost all our important model systems will correspond to it, at least under certain boundary conditions. Homogeneity in x leads to the same condition as the requirement of a defined degree $p(\lambda=p)$ in the unlimited growth system (see Sec. I.5). Now, the transformation of variables is rather trivial:

$$\Gamma_i(\mathbf{x})=\Gamma_i(c\,\xi)=c^{\lambda}\Gamma_i(\xi) \tag{42}$$

and we obtain for the rate equation:

$$\dot{\xi}_i=c^{\lambda-1}\left\{\Gamma_i(\xi)-\xi_i\sum_{j=1}^{n}\Gamma_j(\xi)\right\} \tag{43}$$

Two important conclusions can be drawn from a simple inspection of this equation: If $\lambda=p=1$, i.e., for a

Darwinian system as discussed in Part A, the dependence on c vanishes and not only the long-term behavior but also the solution curves are identical in growing and stationary systems as far as relative population variables ξ_i are concerned.
If $\lambda = p \neq 1$, the long term behavior of ξ will be identical with that for the stationary system at constant organization, provided $c(t)$ becomes neither zero/nor infinite. Thus for all realistic systems with homogeneous net growth functions f_i, the results of fixed-point analysis in ξ space, as obtained in the next section, will be valid also for the case of growing populations.
It is possible to generalize the latter result also to other classes of growth functions, as will be shown in the section on fixed-point analysis. Internal equilibration simplifies analysis of complex dynamic systems tremendously. In many cases the results become identical or similar to those for stationary conditions. If we analyze for the selective behavior of a system, these are the conditions which count. In the following section we shall inspect more closely various dynamic systems under these conditions.

VII. Fixed-Point Analysis of Self-Organizing Reaction Networks

VII.1. The Appropriate Method of Analysis

In analyzing various molecular processes of self-organization we are naturally more interested in the final outcome of selection than in a detailed resolution of the dynamic process. Accordingly, in this section we do not need all the information that is provided by the complete set of solution curves satisfying a system of differential equations. Fixed-point analysis, therefore, is our method of choice, because it serves best the purposes of a comparative analysis of selective behavior. Only in some cases shall we also consult more sophisticated techniques, such as the inspection of the complete vector fields.
Nowadays, fixed-point analysis is a routine technique for studying the long-term behavior of dynamic systems. It can be found in mathematical textbooks or treatises (see, e.g., [48]). Fixed-point analysis has also been applied to problems of economics and to ecologic models as well as to chemical reactions far from equilibrium [49]. A summary of the present stage of development was given recently in a progress report [50].

VII.2. Topologic Features

Let us imagine a mountainous country for which we have a map (cf. Fig. 18). The contour lines in the two-dimensional map provide us with a vague feeling for the three-dimensional scenery. It is this kind of problem that fixed-point analysis deals with. The landscape corresponds to a potential surface along which the dynamic system is moving. In most cases a complete knowledge of this potential surface is not required, and therefore a 'fixed-point map' will turn out to be much simpler than a survey map we use to orient ourselves in an unknown region. In general, the 'fixed-point map' shows exclusively the positions of locally highest and lowest points, such as mountain peaks, passes, and valleys, which are called sources, saddles, and sinks. Such special points are the fixed points of the potential field. Often it is necessary to include also the ridges as

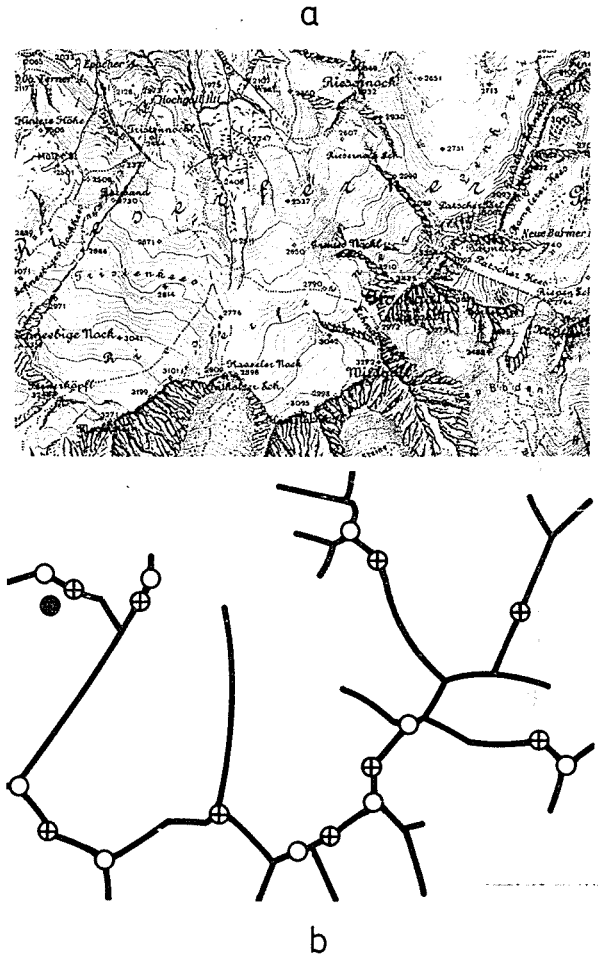


Fig. 18. a) A topographic map provides an abstract representation of a landscape. The lines drawn connect points of equal altitude. The picture shows a region in the Eastern Alps. (reprinted from „Österreichische Karte" 1:50000 Blatt Nr.177 (1962) by courtesy of Bundesamt für Eich- und Vermessungswesen Abt. Landesaufnahme). b) The fixed-point map is a further abstraction of the topographic map. The drawing records the fixed points of a): ○ sources or peaks, ⊕ saddle points, ● sinks; the solid lines here mark the separatices

lines which separate one valley from another (Fig. 18). Characteristically, they are called 'separatrices'. A fixed-point map including separatrices is sufficient to predict where a trajectory starting from a given point on the map will lead. Trajectories are the lines of steepest descent, which in a landscape will be followed by flowing water. The gravitational potential field on the surface of the earth, on the other hand, is less complicated than the fields we encounter in self-organizing dynamic systems. Whereas water flowing on earth always approaches a sink, such as a lake, self-organizing dynamic systems may show a more complex behavior. For instance, there are situations called limit cycles where – in the language of our illustration – water would never stop flowing at a certain point but rather would circulate forever along a closed line determined by the shape of the potential field. Even stranger situations have been described, to which mathematicians actually refer as 'strange attractors', representing something like nonperiodic orbits. Attractor is a more general expression than sink. It includes not only sinks, but also stable closed and nonperiodic orbits.

In a fixed-point map, the whole area under consideration can be separated into a number of regions usually called basins, belonging to individual attractors. The boundaries of these regions are the separatrices. Thus, from all points of a basin the water flows to the same attractor, which of course has to lie inside that region.

Now let us be more precise and characterize the quantities and expressions that are necessary for further discussion in mathematical terms. Fixed points or invariant points of dynamic systems are defined as those points at which all concentrations or population variables, x_i, are constant in time. Hence, the first time derivatives vanish

$$\dot{x}_i = 0, \qquad i = 1, 2, \ldots, n \tag{44}$$

thereby determining the positions of all fixed points belonging to a given dynamic system. When all random fluctuations in population variables are strictly suppressed, integration of the dynamic system starting at a fixed point trivially leads to time-independent, constant populations. The response of the system to small changes in concentration at given fixed points provides an excellent tool for a classification of these points. It can be described by a set of normal modes with reciprocal time constants ω_k, which are obtained as the eigenvalues of a system of linear differential equations representing the best fit of the nonlinear system around the point of reference (cf. VII.4). Accordingly we can distinguish four main classes of fixed points.

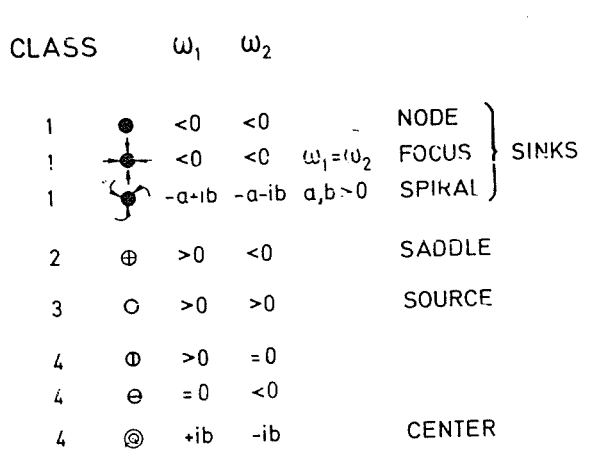


Fig. 19. Symbols are used to classify various fixed points: Class 1: stable fixed points or sinks; Class 2: saddle points; Class 3: sources; Class 4: unstable fixed points including eigenvalues ω with zero real parts. The examples refer to a two-dimensional dynamic system

(1) *Stable fixed points or sinks*, i.e., locally lowest points. All eigenvalues ω_k have negative real parts and hence fluctuations along all possible directions in concentration space are compensated by an internal counteracting force. In chemistry, sinks correspond to chemical equilibria in closed and to stable stationary states in open thermodynamic systems.

(2) *Saddle points* with at least one direction of instability. Here one ω_k value at least will have a positive real part. Consequently, a small perturbation or fluctuation in this direction introduces a force that tends to increase the fluctuation. As a result the dynamic system will move away from the saddle.

(3) *A source* representing a locally highest point. It differs from the saddle only by the fact that it is unstable with respect to *all* directions. All ω_k values have positive real parts.

(4) *Another class of fixed points*, which cannot be analyzed completely within the linear approach. Some of the frequencies ω_k show zero real parts and the nonlinear contributions may change their nature. An example is provided by centers that are defined by purely imaginary eigenvalues, and whose trajectorial representations are manifolds of concentric orbits. We shall encounter such situations in this paper.

After 'long enough' time – which means a period much longer than the largest time constant of the dynamic system – every realistic dynamic system (i.e., a system without external suppression of fluctuations) will approach an attractor. Thus the result of selection will always coincide with an attractor in concentration space.

The final result of a selection process corresponds either to a stable stationary state or to a continuously and periodically changing family of states. In some very rare situations nonperiodic changes within a defined set of states may occur as well. For all these stable or quasi-stable final situations a common, general expression is used in differential topology: the 'attractor' of the dynamic system, which includes the stable point, the closed orbit, and the aperiodic line. Within a given basin, the result of a selection process is approach to an attractor, which is independent of the particular initial conditions.

VII.3. An Appropriate Space: the Concentration Simplex

The concentration variables or population numbers span the n-dimensional open space $\mathbb{R}^n$: $\{x_1, x_2, \ldots, x_n; -\infty < x_i < \infty,\ i = 1, 2, \ldots, n\}$, only part of which is physically meaningful:

$$\mathbb{X}^n \subset \mathbb{R}^n;\ \mathbb{X}^n: \{x_1, x_2, \ldots, x_n;\ x_i \geqq 0,\ i = 1, 2, \ldots, n\} \quad (45)$$

All concentration variables can be summed to give a non-negative and finite total concentration c:

$$c = \sum_{i=1}^{n} x_i, \qquad 0 \leqq c < \infty$$

which is used for normalization:

$$\xi_i = \frac{x_i}{c}; \qquad 0 \leqq \xi \leqq 1; \qquad \sum_{i=1}^{n} \xi_i = 1 \quad (46)$$

According to the properties of the variables $\mathbb{X}^n$ can be mapped now isomorphically onto a unit simplex (S_n) for any given value of $c = c_0$. (The corresponding space will be denoted by $\mathbb{S}^n$):

$$c = c_0:\ \mathbb{X}^n \Leftrightarrow \mathbb{S}^n: \{\xi_1, \xi_2, \ldots, \xi_n\} \quad (47)$$

A unit simplex S_n is a regular polyhedron with n corners in the corresponding $(n-1)$-dimensional subspace defined by $\sum_{i=1}^{n} \xi_i = 1$. The edges of a simplex are of unit length and represent coordinate axes for the variables ξ_i. As an illustrative example S_3 is shown in Figure 20. Diagrams on S_3 are familiar to chemists from the representations of ternary systems.

As a consequence of Equation (46) the dynamic system on the unit simplex has lost one degree of freedom compared to $\mathbb{X}^n$. In other words: Due to normalization, the variables ξ_i always refer to a fixed value of $c = c_0$ and thereby one linear dependence is introduced among the variables.

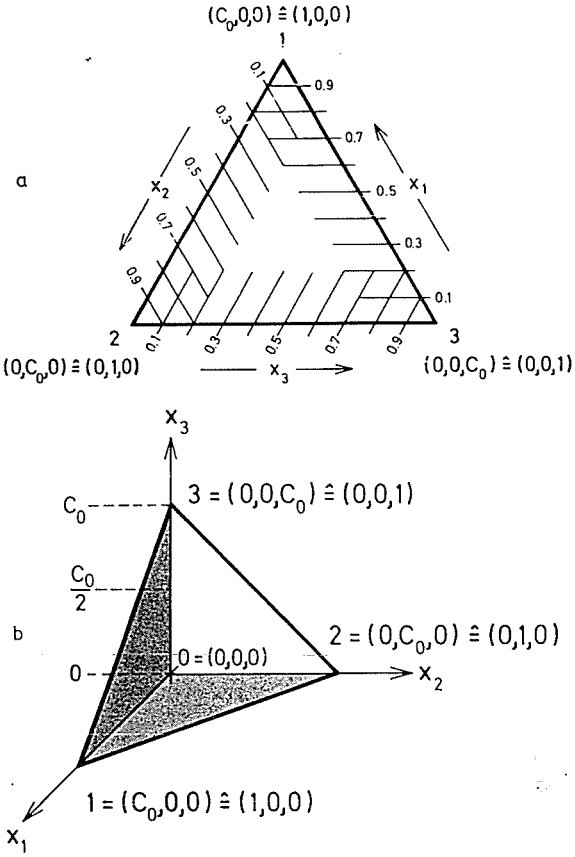


Fig. 20. Diagram a) illustrates the simplex S_3 while diagram b) shows how it is embedded in the physically accessible concentration space. For some of the points the total concentrations $c_0 = \sum_i x_i$, or the coordinates x_1, x_2, x_3 and ξ_1, ξ_2, ξ_3 are given in parenthesis

Finally, we would like to stress a difference between maps on $\mathbb{X}^n$ and $\mathbb{S}^n$ which becomes apparent when we compare results obtained for different values of c_0. Due to normalization, the size of the simplex S_n is fixed, whereas the physically accessible region of $\mathbb{X}^n$ varies with c_0. The positions and the normal modes of fixed points, in general, will depend on c_0 too. For a complete description of the long-term behavior of dynamic systems, it is necessary to evaluate fixed-point maps which themselves are 'functions' of the total concentration c_0. Many fixed points, as we shall see later, show a simple concentration dependence: Their coordinates are proportional to c_0. Upon changes of the total concentration c_0, these points move along lines passing through the origin of $\mathbb{X}^n$ (see Fig. 21) and consequently are mapped as single points on S_n. Accordingly the fixed-point map as a whole becomes much simpler. This formal dependence of fixed-point maps on the values of total concentration c_0 will be of particular importance in the analysis of growing systems.

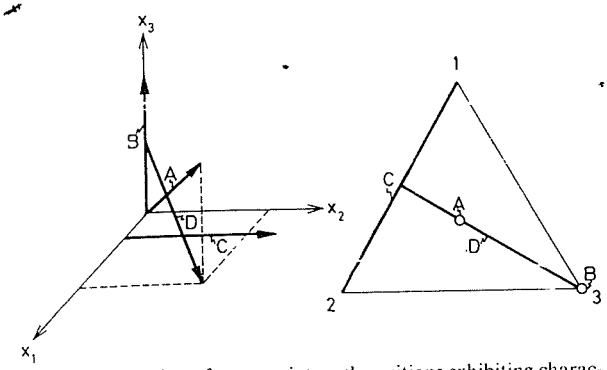


Fig. 21. Illustration of some points with positions exhibiting characteristic dependence on total concentration c_0 in concentration space $\mathbb{X}_3$ (a) and on the simplex S_3 (b). $\mathbf{A}=(c_0/3, c_0/3, c_0/3)$, $\mathbf{B}=(0, 0, c_0)$, $\mathbf{C}=(1, c_0-1, 0)$ and $\mathbf{D}=(c_0-1, c_0-1, 2-c_0)$. The arrows in a) indicate where the points migrate with increasing total concentration (note that those points for which all coordinates are proportional to c_0 like **A** and **B** are mapped into single points on S_3)

A highly symmetric part of a particular $(n-1)$-dimensional hyperplane embedded in n-dimensional concentration space is called the unit simplex. An illustration of a simplex, which can be described in three-dimensional space and therefore is easy to visualize, is given in Figure 20. The unit simplex includes the total physically meaningful range of concentrations and is best suited for a diagrammatic representation of selective processes.

VII.4. Normal Mode Analysis

Starting from the general system of nonlinear differential equations we first determine the fixed points according to $\dot{x}_i=0$. For a straightforward analysis of a dynamic system, it is important to know *all* the fixed points in the region under investigation. In general, however, this information will not be sufficient. Trajectories of the n-dimensional dynamic system will often end in sinks. However, there may be stable closed orbits or strange attractors, the existence of which can be guessed by a careful inspection of the nature of the fixed points in the surrounding region and an analysis of the vector fields. For instance, stable limit cycles in two dimensions can be identified by Poincaré maps. Information on the nature of the fixed points can be obtained by normal mode analysis.

For this purpose the dynamic system is linearized in the neighborhood of a given fixed point $\bar{\mathbf{x}}$:

$$\dot{z}_i=A_i(\bar{\mathbf{x}})+\sum_{j=1}^{n} A_{ij}z_j+O(|\mathbf{z}|^2) \tag{48}$$

The new variables z_i are defined by

$$z_i=x_i-\bar{x}_i \quad \text{or} \quad \mathbf{z}=\mathbf{x}-\bar{\mathbf{x}} \tag{49}$$

while the coefficients A_{ij} are the elements of an $n\times n$ Jacobian matrix **(A)** defined at the fixed point $\bar{\mathbf{x}}$:

$$A_{ij}=\left(\frac{\partial A_i}{\partial x_j}\right)_{\mathbf{x}=\bar{\mathbf{x}}} \tag{50}$$

Since $A_i(\bar{\mathbf{x}})=0$ by definition of the fixed point, the linearized system of differential equations is given by

$$\dot{\mathbf{z}}=\mathbf{A}\cdot\mathbf{z} \tag{51}$$

The reciprocal time constants referring to the normal modes are obtained now as eigenvalues of the matrix **A**. The eigenvectors ζ_i determine the corresponding linear combinations of concentration variables.

$$\mathbf{A}\cdot\zeta_j=\omega_j\cdot\zeta_j \tag{52}$$

The ω_j, in general, are complex quantities and determine the nature of the fixed points, the most important ones of which have been summarized already in Figure 19.

Provided the matrix **A** is not singular, a stable fixed point of the linearized system (51) corresponds to a stable fixed point of the nonlinear problem in almost all cases [51]. There are, however, some important exceptions ($\mathrm{Re}\,\omega_j=0$): A center in the linear system may appear as a spiral sink in the nonlinear case and *vice versa*. The famous Lotka-Volterra model system represents one example for this kind of behavior [52]. We shall encounter another one, the hypercycle of dimension $n=4$, in Section VIII.1.

If more than one stable fixed point, limit cycle, or other attractor is obtained for a given dynamic system, we would also like to know the basins for which the attractors represent the infinite time limits of the trajectories. Individual basins are separated by separatrices, which can be determined in principle by backward integration ($t\rightarrow -t$) starting from saddle points and following the lines of steepest descent. If all stable fixed points and other attractors for a given dynamic system as well as their basins are known, we can predict the result of a selection process starting from any point in the given concentration space.

In some cases we shall obtain $\mathrm{Re}\,\omega_j=0$. Linearization around the fixed point then does not provide enough information, and one has to go back to the nonlinear dynamic system for a complete characterization. Often, direct inspection of the vector field around the fixed point is not too involved and yields the desired results.

Determination of normal modes is an intrinsic part of fixed-point analysis. It represents an inspection of the trajectories of the dynamic system in the close vicinity of the fixed point. In most cases it is sufficient to characterize the stability properties of the fixed point. The linear approximations involved, however, may not always suffice to provide enough information, requiring more sophisticated methods of analysis.

VII.5. Growing Systems

From Equation (37) it is easy to deduce a differential equation for the total concentration c:

$$\dot{c}=\sum_{j=1}^{n} \Gamma_j(\mathbf{x})\left(1-\frac{c}{c_0}\right) \tag{53}$$

c_0 represents the stationary value of the total concentration which is controlled by the unspecific flux ϕ_0.

Apparently, this equation has a fixed point at $c=c_0$. The eigenvalue of the normal mode

$$\omega_c = \frac{1}{c_0}\left(\sum_{j=1}^{n} \Gamma_j(\mathbf{x})\right)_{c=c_0} \qquad (54)$$

is negative as long as the sum of all net growth terms Γ_i is positive. Thus we find a stable stationary state at $c=c_0$.

In certain systems the fixed-point maps referring to internal organization also depend on the values of the total concentrations c_0. Now, we are in a position to attribute some physical meaning to the former purely mathematical treatment. For this purpose we assume a nonstationary dynamic system, which starts to evolve at $t=t_0$ with a corresponding initial value of total concentration, $c(t_0)=c_0$. Selection constraints will be adjusted in such a way that the total concentration $c(t)$ changes slowly in comparison to the internal processes of the dynamic system, i.e., all changes due to external processes are much slower than changes due to internal organization. The system approaches a stable solution (i.e., a sink, a stable closed orbit or another kind of attractor) at every instant. When the preceding conditions are fulfilled the system comes closely enough to the long-term solution and the time-dependent process can be described as a sequence of stationary solutions with continuously changing total concentration. In more physical terms we may say, the dynamic system develops under established internal equilibrium. As expected, the analysis of a system can be simplified enormously if the condition of internal equilibration is fulfilled.

Internal equilibration in dynamic systems with *homogeneous* growth functions Γ_i is easy to analyze, because in this case the fixed-point maps S_n do not depend on the total concentrations c_0. From low to high values of c_0 no change in the selective behavior will occur. Moreover, in growing homogeneous system the long-term development does not depend on the degree of internal equilibration. The ultimate result of a selection process, thus, will be the same in systems of this type, independent of whether internal equilibrium has been established during the growth period or not. There are, however, situations to which the concept of internal equilibration must not be applied without careful analysis. At certain critical total concentrations, $c=c_{cr}$, discontinuous changes may occur in fixed-point maps, e.g., sinks may become unstable, stable limit cycles may disappear, etc. A well-known instability of this kind is represented by a 'Hopf bifurcation' [58]. An internally equilibrated dynamic system which reaches such a point from one side, e.g., a growing system approaching the critical concentration from lower values, is essentially off equilibrium after it passes the critical point. For the analysis of dynamic systems in the surroundings of such points, special care is needed. We shall encounter some such examples in Section VII. A very general study of similar situations has been pursued by R. Thom [59] in his catastrophe theory.

These complicated dynamic systems, of course, are more interesting from the biophysical point of view. In fact, the emergence of organized structures requires drastic changes like the discontinuities mentioned above in the fixed-point maps. Inevitably, dynamic systems describing transitions between different levels of organization have to pass through certain critical stages or periods. To be more concrete we shall consider one example representing an important problem in self-organization of biologic macromolecules: the transition from independent competitors to a functional unit consisting of cooperating polynucleotides and proteins. According to the definition given in Section I.4, only one species is selected in a competitive system and hence there is no stable attractor in the interior of S_n. Any cooperative system, on the other hand, has to have such an attractor, otherwise at least one of the cooperating macromolecules would die out after long enough time. Consequently, a dynamic system, which in principle is able to simulate the desired development from a more random to a more organized state, must contain a critical instability at certain values of its parameters.

VII.6. Analysis of Concrete Systems

a) Independent Competitors

As a lucid example for the application of the method of fixed-point analysis we consider the problem of selection of a quasi-species, treated in Part A. The mathematical framework is compiled in Table 7. The coordinates of the concentration space are given by the normal variables y_k; the eigenvalues λ_k are the growth parameters of the functions Γ_k. The analysis refers to a given distribution of mutants. Appearance of new mutants that provide any contribution to the selected quasi-species will change the meaning of the concentration coordinates y_k, i.e., their correlations with the true concentration variables x_k. The results in Table 7 are self-explanatory. We shall use them in the following for a comparative discussion of the three growth functions $\Gamma_i=k_i x_i^p$ which appear in Table 6, i.e.:

1. Constant growth rate: $p=0$
2. Linear growth rate: $p=1$
3. Quadratic growth rate: $p=2$

1. The first case yields one stable fixed point, a focal sink inside the unit simplex S_n:

$$\bar{\mathbf{x}} = \frac{c_0}{\sum_{j=1}^{n} k_j}\begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix} \qquad (55)$$

'Inside' the unit simplex means for all coordinates of $\bar{\mathbf{x}}$: $0<\bar{x}_i<c_0$. The (negative) eigenvalue of the Jacobian matrix is n-fold degenerate

$$\omega = -\frac{\sum_{j=1}^{n} k_j}{c_0} \qquad (56)$$

holding also for ω_c, which refers to the variation of the total concentration c.

The result is stable coexistence of all species.

2. The second case is treated in Table 7. As we recall there is only one stable fixed point. The fact that it is situated at the corner of the simplex indicates competitive behavior. Only one of the concentration coordinates of the nodal sink is positive ($=c_0$), all others being zero. As in the first case, the map does not depend on the overall concentration c_0, nor is the final result dependent on initial conditions.

Table 7. Fixed-point analysis of the selection of a quasi-species (cf. part A)

The rate equation reads:

$$\dot{y}_i = \lambda_i y_i - \frac{y_i}{c_0}\sum_{j=1}^{n}\lambda_j y_j; \qquad i = 1, 2, \ldots, n.$$

The long-term behavior is determined by n fixed points at the corners of the simplex S_n

$$\bar{y}_1 = \begin{pmatrix} c_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \bar{y}_2 = \begin{pmatrix} 0 \\ c_0 \\ \vdots \\ 0 \end{pmatrix} \quad \cdots \quad \bar{y}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ c_0 \end{pmatrix}$$

Normal mode analysis yields for every fixed point $\bar{y}_k$ a spectrum of n $\omega_j^{(k)}$ values.

$\omega_j^{(1)} = \lambda_j - \lambda_1$	$\omega_j^{(2)} = \lambda_j - \lambda_2$	$\omega_j^{(m)} = \lambda_j - \lambda_n$
$j = 2, 3, \ldots, n-1, n$	$j = 1, 3, \ldots n-1, n$	$\cdots \quad j = 1, 2, \ldots, n-2, n-1$
$\omega_c^{(1)} = -\lambda_1$	$\omega_c^{(2)} = -\lambda_2$	$\omega_c^{(m)} = -\lambda_n$

With respect to the degrees of freedom of the simplex S_n, each fixed point $\bar{y}_k$ has $n-1$ normal modes with the reciprocal time constants $\omega_j^{(k)}$ describing the process of internal organization of the distribution resulting from competition among different quasi-species. Furthermore the simplex S_n has one normal mode $\omega_c^{(k)}$ which corresponds to a variation of the total concentration c. All internal modes $\omega_j^{(k)}$ are represented by differences of eigenvalues λ. Hence there is only one stable fixed point for the largest eigenvalue: $\lambda_m > \lambda_j$, $j = 1, 2, \ldots, n$, $j \neq m$. It is a nodal sink, i.e., all $\omega_j^{(m)}$ values are different and negative. Accordingly, the quasi-species with the smallest eigenvalue is described by a source, owing to its positive ω_j values. The remaining $n-2$ fixed points then are saddle points, because they involve positive as well as negative normal-mode rate constants $\omega_j^{(k)}$.

3. The third case finally shows a total of $2^n - 1$ fixed points, which can be grouped in three classes.
The first class includes n focal sinks, one at each of the corners of S_n

$$\bar{x} = \begin{pmatrix} 0 \\ \vdots \\ x_k = c_0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{with} \quad \begin{aligned} \omega_j^{(k)} &= -k_k c_0 \qquad j = 1, 2, \ldots, n-1 \\ \omega_c^{(k)} &= -k_k c_0 \end{aligned} \tag{57}$$

These are the only stable fixed points. Being at the corners of the unit simplex they indicate again a competitive behavior, allowing for only one survivor, i.e., a pure state. In this case of nonlinear growth rates, however, the result of the competition depends on initial conditions, since there are n stable fixed points (in contrast to only one stable fixed point for linear autocatalysis). This means that each of the n competitors can decide the contest in its own favor, depending on initial population numbers. Once the winner is established, there is no easy way for any competitor to grow up and replace it. We therefore

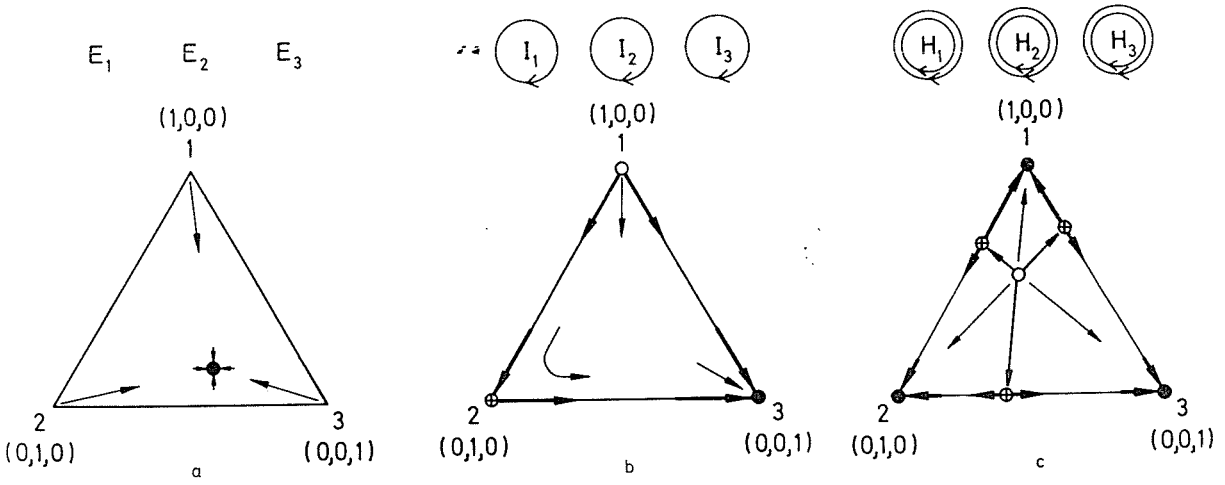


Fig. 22. Three-dimensional fixed-point maps for different types of independent competitors at constant organization. (The symbols E, Ⓘ and Ⓗ have been introduced in Fig. 10).

a) Constant growth rate ($p = 0$),

$$\dot{x}_i = k_i - \frac{x_i}{c_0}\sum_{j=1}^{3} k_j$$

$$k_1 = 1; \quad k_2 = 2; \quad k_3 = 3.$$

The map shows a focus inside the unit simplex S_3 which means stable and coexistent behavior of all three species. It is easy to visualize the whole manifold of trajectories approaching the stable focus along straight lines through every point in S_n.

b) Linear growth rate ($p = 1$),

$$\dot{x}_i = k_i x_i - \frac{x_i}{c_0}\sum_{j=1}^{3} k_j x_j \qquad k_1 = 1; \quad k_2 = 2; \quad k_3 = 3.$$

A pure state consisting of species 3 represents the only stable long-time-range solution of the system. With the exception of the two edges $\overline{12}$ and $\overline{23}$ all trajectories start in point 1 and end in point 3.
c) Quadratic growth rate ($p = 2$),

$$\dot{x}_i = k_i x_i^2 - \frac{x_i}{c_0}\sum_{j=1}^{n} k_j x_j^2 \qquad k_1 = 1; \quad k_2 = 2; \quad k_3 = 3.$$

The simplex S_3 is split into three regions, each being a basin for a stable fixed point. The size of the basin is correlated directly with the values of the corresponding rate constants. Since k_3 is largest the fixed point $\bar{x}_3$ has the largest basin

call this situation 'once-for-ever selection'. As in the two preceding cases the fixed-point map does not depend on the total concentration c_0.

The two other classes of fixed points include one source in the interior of the unit simplex (all coordinates being finite) and $2^n - n - 2$ saddle points, one on each edge and one in each face (including all possible hyperfaces) of S_n. Both classes of fixed points represent unstable behavior. We dispense with listing their coordinates and normal modes; they can be obtained by straightforward computations. Instead we illustrate the typical selection behavior of the different growth systems by showing some examples of unit simplices of dimension 3 (cf. Fig. 22).

The three relatively simple model cases have been chosen to exemplify the method of fixed-point analysis and to stress those properties we have to watch out for. The nature of the fixed point, especially whether it provides stable or unstable solutions, is of utmost importance for problems of selection and evolution. Of no less significance is the position of the fixed points in the unit simplex. Cooperative selection of a set of replicative units requires the fixed point to lie in the interior of the unit simplex S_k referring to a subspace $\mathbb{X}^k$ formed by the concentration coordinates of the k cooperating units. On the other hand, the position of a sink at one of the corners of S_k is representative of competition, leading to selection of only one component, while positions at edges, faces, or hyperfaces indicate partial competition and selection.

The build-up of a translation apparatus, for instance, requires the concomitant selection of several replicative units as precursors of different genes. None of the three systems discussed above fulfils the requirements for such a concomitant selection. The first system appears to be coexistent, but it is not selective and therefore cannot evolve to optimal function. The second system allows for coexistence only within narrow limits of the quasi-species distribution; it does not tolerate divergence of the genotypes, which is required to facilitate phenotypic diversification. The third system, finally, is strongly anticooperative, to such an extent that a species, once established, selects against any mutant, whether or not it provides a selective advantage.

Following up the suggestions which emerged from the comparative review in Section V we shall now investigate more closely ensembles with functional linkages. They will have to include replicative units for the purpose of conservation of the genetic information, at the same time they will have to be stabilized cooperatively via couplings, which will cause the growth function to be inherently nonlinear. The properties to be expected for the linked system hence will bear some relation to the third example of independent competitors.

b) Catalytic Chains

The most direct way of establishing a connection among all members of an ensemble is to build up a chain via reactive couplings, much as we link words into sentences in our language (Fig. 23).

The rate terms referring to these couplings will cause the net growth functions Γ_i for all but the first member to be nonhomogeneous:

$$\dot{x}_1 = k_1 x_1 - \frac{x_1}{c_0}\left[k_1 x_1 + \sum_{j=2}^{n}(k_j x_j + k'_j x_j x_{j-1})\right]$$

$$\dot{x}_i = k_i x_i + k'_i x_i x_{i-1} - \frac{x_i}{c_0}\left[k_1 x_1 + \sum_{j=2}^{n}(k_j x_j + k'_j x_j x_{j-1})\right]$$

for $i = 2, 3, \ldots, n$. (58)

Due to the lack of homogeneity the fixed-point maps will be of a more complicated form than in the cases discussed thus far.

To keep the procedure lucid we start with a three-dimensional system and extend our analysis later to higher dimensions. Table 8 contains a compilation of the pertinent relations of dimension three together with a brief characterization of the fixed-point map. According to this

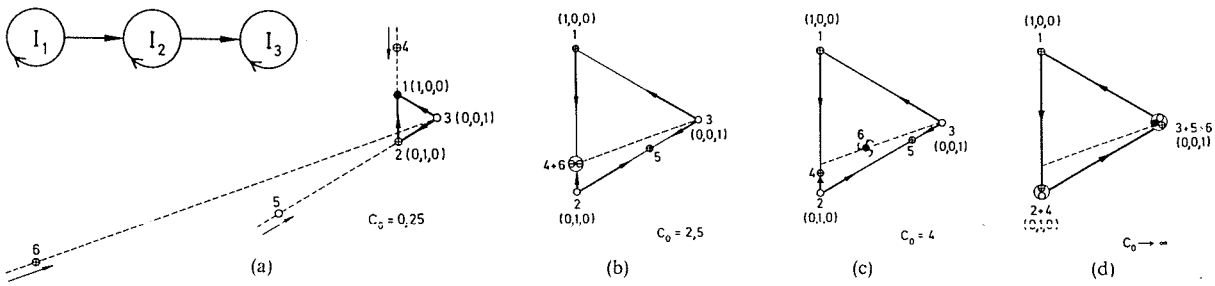


Fig. 23. Fixed-point maps of a catalytic chain of self-replicative units (I_i) under the constraint of constant organization:

$\Gamma_1 = k_1 x_1$;

$\Gamma_i = k_i x_i + k'_i x_i x_{i-1}$ (for $i = 2, 3$)

$k_1 = 3$; $k_2 = 2$; $k_3 = 1$; $k'_2 = 2$; $k'_3 = 1$

$1 = \bar{x}_1$; $2 = \bar{x}_2 \ldots 6 = \bar{x}_6$

At low concentrations (a) the stable solution corresponds to selection of species 1. If the two other species, however, have not yet been extinguished when the total concentration reaches a critical value, a new stationary state emerges, at which all three species become stable (b). With a further increase of the total concentration (c), only species 3 is favored so that the final situation (d) is equivalent to a selection of species 3. The underlying mechanism, however, differs from that for independent competitors

Table 8. Fixed-point analysis of catalytic chains of dimension three

The fixed-point map shows six fixed points with the coordinates and normal modes as given below:

$\bar{\mathbf{x}}_1 = \begin{pmatrix} c_0 \\ 0 \\ 0 \end{pmatrix}$	$\bar{\mathbf{x}}_2 = \begin{pmatrix} 0 \\ c_0 \\ 0 \end{pmatrix}$	$\bar{\mathbf{x}}_3 = \begin{pmatrix} 0 \\ 0 \\ c_0 \end{pmatrix}$
$\omega_1^{(1)} = k_3 - k_1$	$\omega_1^{(2)} = k_1 - k_2$	$\omega_1^{(3)} = k_1 - k_3$
$\omega_2^{(1)} = k_2' c_0 - k_1 + k_2$	$\omega_2^{(2)} = k_3' c_0 - k_2 + k_3$	$\omega_2^{(3)} \quad k_2 - k_3$
$\bar{\mathbf{x}}_4 \begin{pmatrix} \frac{k_1 - k_2}{k_2'} \\ \frac{k_2' c_0 - k_1 + k_2}{k_2'} \\ 0 \end{pmatrix}$	$\bar{\mathbf{x}}_5 \begin{pmatrix} 0 \\ \frac{k_2 - k_3}{k_3'} \\ \frac{k_3' c_0 - k_2 + k_3}{k_3'} \end{pmatrix}$	$\bar{\mathbf{x}}_6 \begin{pmatrix} \frac{k_1 - k_2}{k_2'} \\ \frac{k_1 - k_3}{k_3'} \\ \kappa \end{pmatrix}$
$\omega_1^{(4)} = \frac{(k_2' c_0 - k_1 + k_2)(k_2 - k_1)}{k_2' c_0}$	$\omega_1^{(5)} = k_1 - k_2$	$\kappa = \frac{k_2' k_3' c_0 - k_3'(k_1 - k_2) - k_2'(k_1 - k_3)}{k_2' k_3'}$
$\omega_2^{(4)} = \frac{1}{k_2'} \{k_2' k_3' c_0 - k_3'(k_1 - k_2) - k_2'(k_1 - k_3)\}$	$\omega_2^{(5)} = \frac{(k_3' c_0 - k_2 + k_3)(k_3 - k_2)}{k_3' c_0}$	$\omega_1^{(6)}$ and $\omega_2^{(6)}$ are the eigenvalues of the Jacobian matrix $\mathbf{A}(\mathbf{x} = \bar{\mathbf{x}}_6)$.

The three fixed points $\bar{\mathbf{x}}_1$, $\bar{\mathbf{x}}_2$, and $\bar{\mathbf{x}}_3$ coincide with the corners of the unit simplex S_3 (cf. Fig. 23) and hence signify competitive behavior – irrespective of their nature. The positions of the three other fixed points depend (linearly) on the total concentration c_0. The two fixed points $\bar{\mathbf{x}}_4$ and $\bar{\mathbf{x}}_5$ move along the edges $\overline{12}$ and $\overline{23}$ of the simplex, thereby showing partial competition. Solely the fixed point $\bar{\mathbf{x}}_6$ may pass through the interior of S_3 yielding cooperative selection of all chain members.

At low total concentration:

$$c_0 < (k_1 - k_2)/k_2', \quad (k_2 - k_3)/k_3' \quad \text{or} \quad (k_1 - k_3)/k_3'$$

the position of the fixed point $\bar{\mathbf{x}}_4$, $\bar{\mathbf{x}}_5$, or $\bar{\mathbf{x}}_6$, respectively, lies outside the simplex S_3, which means outside a physically meaningful region of the concentration space. (At least one concentration coordinate is negative.) For $c_0 \to 0$ the positions of these fixed points even approach infinity. The dynamic system becomes asymptotically identical with the system of exponentially growing (noncoupled) competitors, characterized by the fixed points $\bar{\mathbf{x}}_1$, $\bar{\mathbf{x}}_2$, and $\bar{\mathbf{x}}_3$.

If $k_1 > k_2$, k_3 and c_0 is above a threshold given by the sum of $[(k_1 - k_2)/k_2'] + [(k_1 - k_3)/k_3']$, the fixed point $\bar{\mathbf{x}}_6$, indicating cooperative behavior, enters the unit simplex. However, it does not approach any point in the interior of S_3, but rather migrates toward the corner 3.

analysis, the three members (I_1 to I_3) of the linear chain of self-reproductive units can be selected concomitantly only under very special conditions, namely

$$k_1 > k_2, k_3 \tag{59}$$

and

$$c_0 > \frac{k_1 - k_2}{k_2'} + \frac{k_1 - k_3}{k_3'} \tag{60}$$

It seems very unlikely that partners which happen to fulfill condition (59) can maintain it over long phases of evolution (which means that mutations that change relation (59) must never occur). Even if they are able to do so, the system will then develop in a highly asymmetric manner, whereby – at least under selection constraints – only the population number of the last member in the chain increases with c_0. Being aware that this soon means a divergence of population numbers by orders of magnitude, we may conclude that such a system will not be able to stabilize a joint function, since it cannot control the relative values of population numbers over a large range of total concentrations.

This behavior is illustrated with some examples in Figure 23, presenting some snapshots of a continuous process in a system growing in a stage close to internal equilibrium. For concentrations c_0 below the critical value given by Equation (60), the three fixed points $\bar{\mathbf{x}}_4$, $\bar{\mathbf{x}}_5$, and $\bar{\mathbf{x}}_6$ lie outside the unit simplex (Fig. 23a). If c_0 equals the critical value, the fixed point $\bar{\mathbf{x}}_6$ reaches the boundary of the simplex (Fig. 23b) and, with increasing c_0, migrates through its interior. At the same time it has changed its nature, now representing a stable fixed point (Fig. 23c), which in this particular case is a spiral sink. (A more detailed presentation of fixed-point analysis with inhomogeneous growth functions will be the subject of a forthcoming paper [53]). Figure 23d indicates the final fate of this stable fixed point, namely, migration to the corner 3. The system thereby approaches the pure state $\bar{x}_3 = c_0$.

The relevant results obtained for three dimensions can be generalized easily for the n-dimensional system. The role of species 3 is now taken by species $\mathbf{n}$. Instead of six we find $2n$ fixed points. The most interesting fixed point is $\bar{\mathbf{x}}_{2n}$. Its position can easily be determined:

$$\bar{\mathbf{x}}_{2n} = \begin{pmatrix} \frac{k_1 - k_2}{k_2'} \\ \frac{k_1 - k_3}{k_3'} \\ \vdots \\ c_0 - \sum_{j=2}^{n} \frac{k_1 - k_j}{k_j'} \end{pmatrix} \tag{61}$$

If and only if the rate constants fulfil the relations $k_1 > k_j, j = 2, 3, \ldots, n$ and the total concentration exceeds the critical value:

$$c_{c_1} = \sum_{j=2}^{n} \frac{k_1 - k_j}{k'_j} \tag{62}$$

the fixed point $\bar{\mathbf{x}}_{2n}$ lies inside the simplex S_n. Then, $\bar{\mathbf{x}}_{2n}$ corresponds to a stable stationary state. All concentrations besides $\bar{x}_n$ are constant at this state and hence the system approaches the pure state $\bar{x}_n = c_0$ at large total concentrations.

We may summarize the behavior of catalytic chains.
1) Stable sationary states exist only when the rate constants and the total concentration fulfil certain relations:

$$k_1 > k_j; \quad j = 2, 3, \ldots, n; \quad c_0 > \sum_{j=2}^{n} \frac{k_1 - k_j}{k'_j}$$

The system must be subject to selection constraints in order to select against other nonfunctional units, and the required order of rate constants must not be changed by selection of favorable mutants.
2) Provided the conditions given in (1) are fulfilled, the concentrations of individual species are of a comparable magnitude only in a rather small range of total concentrations. With increasing values of c_0 the last member of the chain, I_n, grows under (quasi-)stationary conditions and finally will dominate exclusively.
The catalytic chain therefore is not likely to be useful as an information-integrating system.

c) Branched systems

Wherever coupled systems evolve, branching of couplings as depicted in Figure 24 will be inevitable. Fixed-point analysis of such branched networks does not reveal any unexpected, new features. At very low total concentrations the three species behave like independent competitors. There are now two critical values of c_0, at which either I_1 and I_2 or I_1 and I_3 become coexistent. This finally depends on whether I_2 or I_3 is more efficiently favored by I_1. One of the two fixed points turns out to be a stable node, the other a saddle point. At higher total concentration the stable fixed point again migrates toward one of the corners 2 or 3, respectively. An illustration of this behavior is given in Figure 24.
The three-dimensional system investigated here can be generalized in two ways:
1. More than two branches may start out from a given point.
2. The individual branches may contain more than one member. The results of fixed-point analysis of these many-dimensional systems are essentially the same as those obtained in three dimensions and can be summarized as follows:

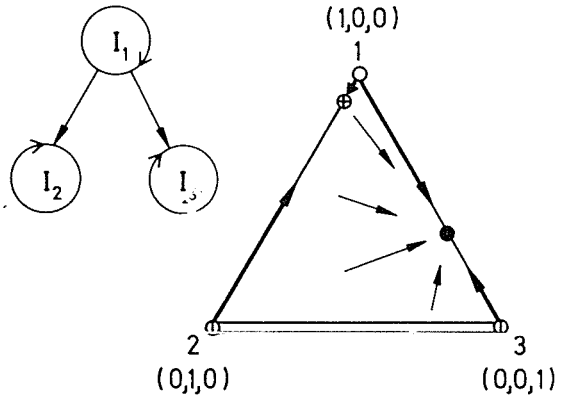


Fig. 24. Fixed-point map of a dynamic system representing a branching point in a catalytic network of self-replicative units (I_i) under the constraints of constant organization

$\Gamma_1 = k_1 x_1; \quad \Gamma_i = k_i x_i + k'_i x_i x_1$ (for $i = 2, 3$)

$k_1 = 3; \quad k_2 = k_3 = 0; \quad k'_2 = 1; \quad k'_3 = 2; \quad c_0 = 3.5$

Branched systems of self-replicative units are not stable over long time spans. The branch which is most efficient in growing will be selected while other branches will disappear. What remains finally is the most efficient linear chain and thereby the whole problem is reduced to a dynamic system of type (58), which we have discussed in the previous section.

VII.7. Fixed-Point Analysis of Hypercycles

a) Classification

Ring closure in dynamic systems leads to entirely new properties of the system as a whole, as we have seen in Part A. The set of molecules which are formed in a closed loop of chemical reactions represents a catalyst. A cycle of catalysts (Fig. 4) in turn has autocatalytic properties and may be regarded as a self-replicative system. After finding that straightforward or branching couplings among self-replicative units do not lead to a joint selection of the functionally linked system, we may ask whether any ring closure in the chain of couplings will bring about a change in the nature of the selective behavior of the total ensemble. We might argue that this must be the case, since we remember that in open chains the recipient of all the benefits of coupling was always the last member.
Hypercycles have been generally classified in Part A. The simplest constituents of this class of networks are obtained by a straightforward functional link among self-replicative units as shown in Figure 25.
This section on hypercycles will consist of three different parts. First we present some definitions and useful criteria for classification of this new kind of catalytic system. Then the results of fixed-point analysis for the most important 'pure' types of hypercycles are described. Finally, we shall study one example of a self-organizing system which represents a realistic catalytic hypercycle.

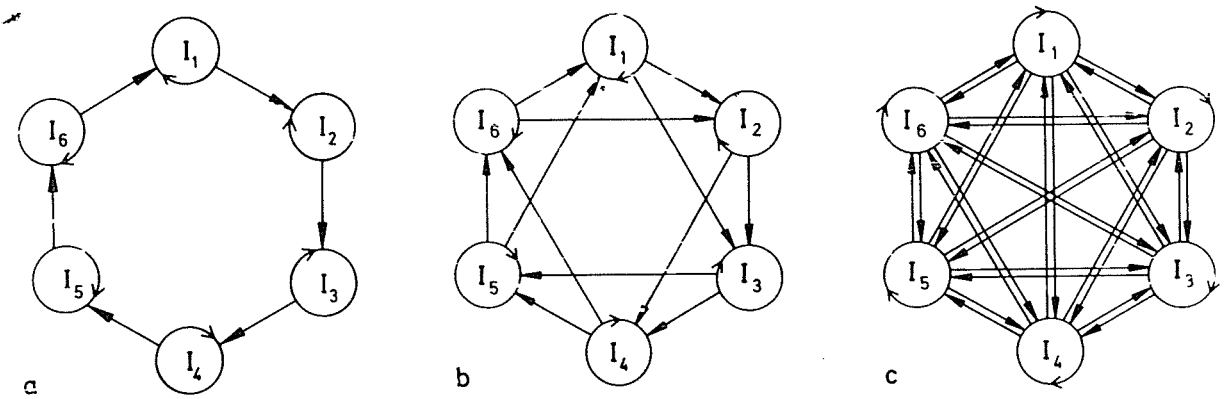


Fig. 25. Catalytic hypercycles, a) degree $p=2$. $n=6$, b) degree $p=3$, $n=6$, and c) degree $p=n=6$

Primarily, hypercycles differ from ordinary catalytic cycles by nonlinear terms in the growth rates. In simple cases, the functions I_i will be products of concentrations as shown in Equation (63). The exponents $p_{\lambda i}$ according to

$$\Gamma_i = k_i \prod_{\lambda=1}^{n} x^{p_{\lambda i}} \tag{63}$$

can be regarded as elements of a matrix **P**. The indices λ and i denote which population variable (x_λ) has to be raised to the power $p_{\lambda i}$ in the function Γ_i. The dynamic system is then determined completely by the matrix of exponents **P**, by the vector of rate constants $\mathbf{k}=(k_1 \dots k_n)$, and by a set of initial conditions. At first we shall study the 'pure' cases only, which are characterized by homogeneous growth terms Γ_i. The requirement of homogeneity leads to a first restriction for the exponents in the matrix **P**:

$$\sum_{\lambda=1}^{n} p_{\lambda i} = p; \qquad i=1,2,\dots,n \tag{64}$$

'p' is now common for all n differential equations and represents the degree of the growth functions introduced in Section V. In addition to the restriction of homogeneity we shall allow individual concentrations to appear only as first-order terms in Γ_i. Some important cases with higher-order dependences will be discussed below. Accordingly, the exponents $p_{\lambda i}$ are restricted to just two possible values: $p_{\lambda i}=\{0,1\}$.
Finally, we shall introduce cyclic symmetry into the net growth function:

$$\begin{aligned}
&\Gamma_i = k_i x_i x_j x_k x_l \dots x_r \\
&j=i-1+n(\delta_{i1}); \quad l=i-3+n(\delta_{i1}+\delta_{i2}+\delta_{i3}); \dots \\
&k=i-2+n(\delta_{i1}+\delta_{i2}); \quad r=i-p+n\sum_{\mu=1}^{p-1}\delta_{i\mu}
\end{aligned} \tag{65}$$

The assumption of cyclic symmetry is not essential for the most important features of the solutions. It is a reasonable assumption, however, if no further information on structural differences in the kinetic equations for the individual members of the cyclic system is available. The matrix **P** is of general and simple form now. A concrete example, the matrix **P** with $n=6$ and $p=3$ is shown below:

$$\mathbf{P}(n=6,p=3)=\begin{pmatrix} 1&0&0&0&1&1\\ 1&1&0&0&0&1\\ 1&1&1&0&0&0\\ 0&1&1&1&0&0\\ 0&0&1&1&1&0\\ 0&0&0&1&1&1 \end{pmatrix} \tag{66}$$

Hypercycles with cyclic symmetry and homogeneous growth functions Γ_i thus are completely determined by the values of n and p and the vector **k**.

Schematic diagrams for three examples of hypercycles with $n=6$ and $p=2,3$ and n are shown in Figure 25. Cases with $p=1$ must be excluded from the general class of catalytic systems called hypercycles, since they fall into the family of systems with linear growth rates Γ_i.

b) General Analysis

A summary of fixed-point analysis of hypercyclic systems is given in Table 9.
We discuss two cases which are most important in the context of this paper.
1. The simplest hypercycle possible with $p=2$
2. The hypercycle, which utilizes catalytic links among all members, i.e., $p_{\lambda i}=1$ for $i=1,\dots,n$ and $\lambda=1,\dots,n$, and hence, $p=n$
The first system we christen simply 'elementary hypercycle', the second – in accordance with its most common physical realization as a cooperatively behaving complex – 'compound hypercycle'.

c) The Elementary Hypercycle

Depending on the dimension of the dynamic system we observe interesting changes in the nature of the fixed point in the center of the simplex. For this purpose we inspect more closely the sets of eigenvalues obtained for different values of n, which are described appropriately as vectors, $\omega = \mathrm{Re}\,\omega\,\mathbf{e}_1 + i\,\mathrm{Im}\,\omega\,\mathbf{e}_2$ in the complex Gaussian plane (Fig. 26). The fixed point in the center is a focus for $n=2$, a spiral sink for $n=3$, a center for $n=4$. For $n\geqq 5$ we obtain saddle points with spiral components in some planes. These characteristic changes in the nature of a fixed point are reminiscent of a Hopf bifurcation despite the fact that our parameter is a

Table 9. The fixed-point map of a hypercycle

Subjecting the dynamic system (65) to the condition of constant organization we find:

$$\dot{x}_i = k_i x_i x_j \dots x_l - \frac{x_i}{c_0} \sum_{r=1}^{n} k_r x_r x_s \dots x_t$$

$$j = i - 1 + n\delta_{i1}, \dots, l = i - p + 1 + n \sum_{\mu=1}^{p-1} \delta_{i\mu}$$

$$s = r - 1 + n\delta_{r1}, \dots, t = r - p + 1 + n \sum_{\mu=1}^{p-1} \delta_{r\mu}; \quad p \leqq n \qquad \text{(T.9.1)}$$

Fixed-point analysis can be carried out *analytically* for any value of n if all the rate constants[a] are equal:

$$k_1 = k_2 = \dots = k_n = k \qquad \text{(T.9.2)}$$

The results obtained are:
1) One fixed point, which we denote by $\bar{\mathbf{x}}_0$, is always positioned in the center of the concentration simplex.
2) n additional fixed points occur at the corners of the simplex S_n: $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \dots, \bar{\mathbf{x}}_n$.

[a] Variations in the individual rate constants on the solutions will be discussed in Section VIII.

3) In many cases there are also one-, two-, three-, or even higher-dimensional manifolds of fixed points, e.g., fixed-point edges, triangles, tetrahedra, and higher-dimensional simplices [54]. These manifolds always occur in the boundaries of the corresponding simplices S_n, for example, fixed-point edges are found in the boundaries of S_n, $n \geqq 4$, fixed-point triangles in the boundaries of S_n, $n \geqq 6$, and fixed-point tetrahedra on S_n, $n \geqq 8$.
Normal-mode analysis around the central fixed point $\bar{\mathbf{x}}_0$, which is responsible for cooperative selection, yields:

$$\bar{\mathbf{x}}_0 = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \cdot \frac{c_0}{n}; \quad \omega_j^{(0)} = -\frac{1 - \gamma^{-(p-1)}}{1 - \gamma} \left(\frac{c_0}{n}\right)^{p-1} \cdot k$$

$$j = 1, 2, \dots, n-1; \quad \gamma = e^{\frac{2\pi i}{n} j}$$

$$\omega_n^{(0)} \equiv \omega_c^{(0)} = -\left(\frac{c_0}{n}\right)^{p-1} \cdot k \qquad \text{(T.9.3)}$$

For $p = 2$ there are n or $n-1$ distinct eigenvalues, while for $p = n$ all eigenvalues are equal to $\omega_c^{(0)}$. The first case is the usual situation. $n-1$ single and one double-degenerate eigenvalues $\omega_c^{(0)} = \omega_{j=n/2}^{(0)} = -k(c_0/n)^{p-1}$ are obtained for $p = 2$ and even n, whereas for odd n all the eigenvalues are distinct. The negative value of $\omega_c^{(0)}$ again indicates that the dynamic system on the simplex S_n is stable against fluctuations in total concentration c.

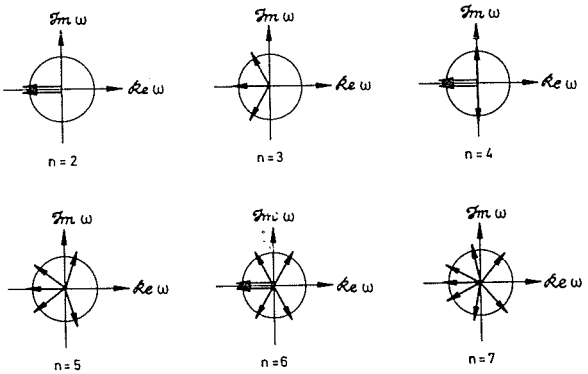


Fig. 26. Normal modes ω for the central fixed point ($\bar{\mathbf{x}}_0$) in hypercycles of type (65) with $p = 2$ and dimension n. Reω and Imω stand for imaginary and real part of the frequency ω, respectively

discretely varying quantity, the dimension n of the dynamic system. As we shall show by a more general analysis in Section VIII the central fixed point is asymptotically stable for $n = 2, 3$, *and* 4. In the case of higher dimensions $n \geqq 5$ we find a more complex attractor, namely, a stable closed orbit or limit cycle, which always remains inside the simplex and therefore never reaches the boundary. In the latter case, time averages of concentrations x_i:

$$X_i(t) = \frac{1}{t} \int_0^t x_i(\tau)\, d\tau, \qquad \bar{X}_i = \lim_{t \to \infty} x_i(t) \qquad (67)$$

rapidly approach c_0/n (for equal k_i values), which is just the same value as in the case of stable fixed points. For the fixed point at any corner k ($x_k = c_0$) of S_n we find one positive and $n-1$ zero $\omega_j^{(k)}$ values. In Section VIII we shall analyze the nonlinear contributions and identify these fixed points as saddle points. The corresponding long-term solutions hence do not contribute to the selection behavior.
The fixed-point map for a hypercycle with $p = 2$ and $n = 3$ is shown in Figure 27 for a concrete example.

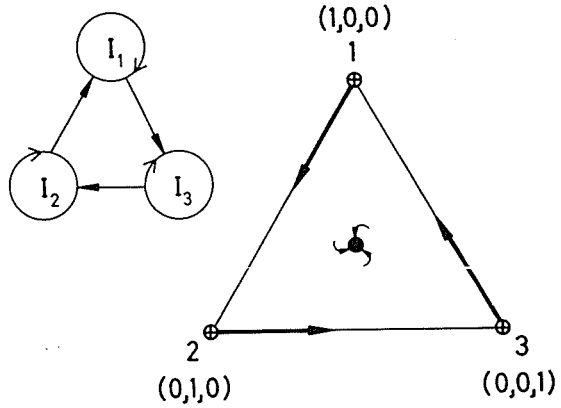


Fig. 27. Fixed-point map of a dynamic system (65) consisting of self-replicative units (I$_i$) forming a hypercycle under the constraint of constant organization

($\Gamma_i = k_i x_i x_j$; $j = i - 1 + n\delta_{i1}$; $n = 3$, $p = 2$, $\mathbf{k} = (1, 1, 1)$)

In general, the net growth functions for the individual self-replicative units which form the dynamic system of a hypercycle will contain not only catalytic terms but also first-order growth terms:

$$\Gamma_i = k_i x_i + k'_i x_i x_j \tag{68}$$

Subjecting a dynamic system with these growth functions to the constraints of constant organization we obtain:

$$\dot{x}_i = k_i x_i + k'_i x_i x_j - \frac{x_i}{c_0} \sum_k (k_k x_k + k'_k x_k x_l) \tag{69}$$

$$j = i - 1 + n\delta_{i1}, \quad l = k - 1 + n\delta_{k1}; \quad i = 1, 2, \ldots, n$$

From a mathematical point of view the catalytic chain (Fig. 23) differs from the hypercycle just by a single rate constant and results from the latter by putting $k'_1 = 0$. We may therefore expect some similarities between the two types of dynamic systems. Consistent with the inhomogeneity of the growth functions, the fixed-point maps depend on the total concentration. At the low concentration limit both systems become identical with the system of exponentially growing, indepedent competitors (Fig. 22). At high concentrations, on the other hand, the two systems will differ. The dynamic system (69) asymptotically resembles that of the corresponding elementary hypercycle ($p = 2$).

As a concrete example we consider again a system of dimension $n = 3$. There are seven fixed points: Three coincide with the corners of the simplex S_3, three other fixed points lie on the edges, and the seventh is in the interior of S_3.

Numerical results were calculated for a given set of parameters **k** and are shown in Figure 28. As for the catalytic chain in Figure 23 we present fixed-point maps for three different values of total concentration c_0, which represent low and high concentration limits (**a** and **c**) as well as the critical point

(**b**, $c_0 = c_{cr} = k_3(k_1'^{-1} + k_2'^{-1}) - k_1 k_1'^{-1} - k_2 k_2'^{-1}$).

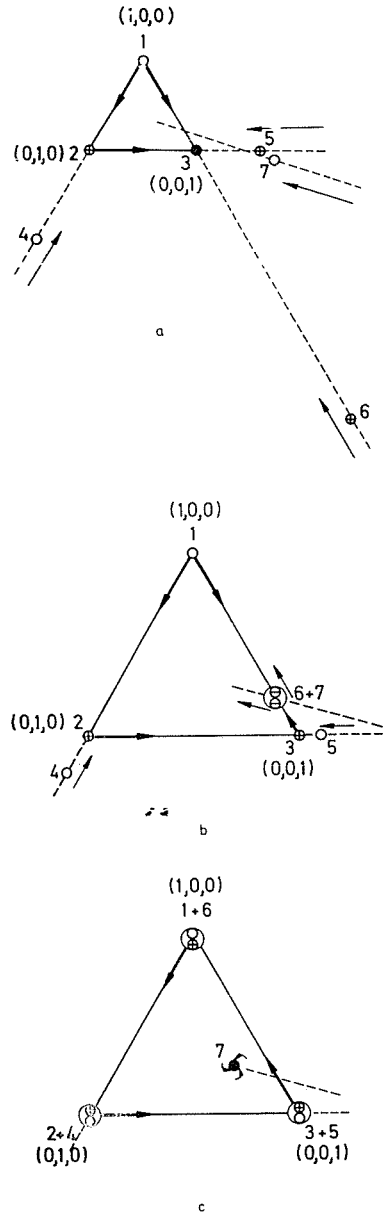


Fig. 28. Fixed-point map of a dynamic system (69) consisting of self-replicative units forming a catalytic hypercycle

($\Gamma_i = k_i x_i + k'_i x_i x_j$, $j = i - 1 + n\delta_{i1}$;
$n = 3$, $p = 2$, $\mathbf{k} = (1, 2, 3; 1, 2, 3)$;
a) $c_0 = 0.5$, b) $c_0 = 2.5$, c) $\lim c_0 \to \infty$;
$1 = \bar{x}_1$, $2 = \bar{x}_2$, $3 = \bar{x}_3$, $4 = \bar{x}_{12}$, $5 = \bar{x}_{23}$, $6 = x_{31}$, and $7 = x_0$)

Considering the development of the dynamic systems (58) and (69) close to internal equilibrium, we realize a very important difference between the cyclic and noncyclic systems: The cyclic system leads to an asymptotic high concentration limit which is characterized by constant relative concentrations of the individual species, whereas the open chain approaches the pure state ($x_n = c_0$) at high total concentration.

Summing up the whole development from low to high concentration limits we realize that a hypercycle, as described by the dynamic system (69), is an appropriate example of self-organization. Starting from competition among individual species the growing system approaches a final state with dynamically controlled net production of all members. This internal control leads to a stable stationary state or to a state with regular oscillation of population variables about a fixed point.

d) The Compound Hypercycle

The analysis of the case $p = n$ leads to a simple, general result: As in the other example treated before, there is one fixed point inside the simplex. The whole boundary of the simplex, however, consists of unstable fixed points, fixed-point edges, fixed-point planes, etc. Since the invariant point in the middle ($\bar{\mathbf{x}}_0$) is a focus for all values of n, all trajectories starting from the interior of the simplex, which is the whole physically meaningful domain, will approach this particular point after long enough time. All the eigenvalues $\omega_j^{(0)}$ associated with $\bar{\mathbf{x}}_0$

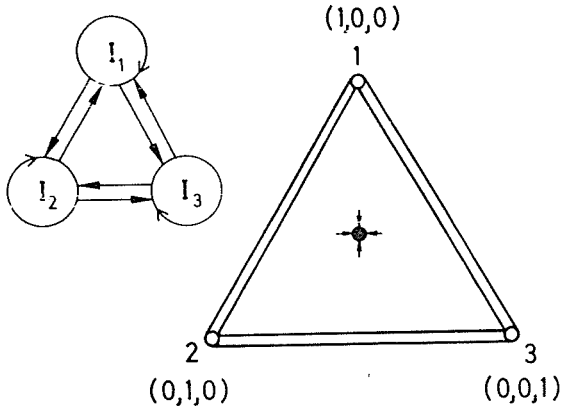


Fig. 29. Fixed-point map of a dynamic system (65) consisting of self-replicative units (I_i) forming a compound hypercycle under the constraint of constant organization

$(\dot{I}_i = k_i x_1 x_2 \ldots x_n;\ n=3,\ p=3,\ \mathbf{k}=(1, 1, 1))$

are the same for given values of k, c_0, and n. They follow from the expression (3) given in Table 9 if we set $p=n$. The fixed-point map for a compound hypercycle with $n=3$ is shown in Figure 29. The complexes, thus, represent excellent examples for the control of relative concentrations of their constituents.

e) Comparison of Various Hypercycles

Hypercycles have an intrinsic capability for integrating information. Indeed the simplest members of this class represent the least complicated dynamic structures that are able to prevent an ensemble of functionally linked self-replicative units from destroying information by elimination of some of their members as a result of selective competition. From the dynamic point of view, all kinds of hypercycles are equivalent with respect to this property. On the other hand, less sophisticated systems, such as simple catalytic cycles (Fig. 4), are not eligible as information-integrating systems since they lack the property of *inherent* self-reproducibility (cf. [4], p. 501 ff.).

A further discrimination within the hierarchy of hypercycles can be made according to their realizability in nature, which will be the subject of Part C. To present one example of such an argument, we compare here the simple type of hypercycle ($p=2$) and the complex ($p=n$) with respect to their physical materialization. Simple hypercycles require bimolecular encounters of macromolecules according to their growth law. These bimolecular terms can easily be provided by various mechanisms and also result from realistic assumptions about nucleic acid replication or messenger-instructed protein synthesis (see also Section IX and Part C). A compound hypercycle requires all partners to contribute to the rate of formation of each constituent. A mechanism leading to such a compound hypercycle then would need either a – highly improbable – multimolecular encounter or an intermediate formation of a complex of n different subunits, which is highly disfavored at low concentrations. Prebiotic conditions, on the other hand, are characterized by exceedingly low concentrations of *individual* macromolecules. For an efficient start of evolution via compound hypercycles one would have to assume extremely high association constants far outside the range of experimental experience and an inherent linkage between these constants and the functional efficiency of the single constituents. The compound hypercycle is thus likely to be of less importance for the nucleation of a translation system than any hypercycle of lower degree p. In more advanced phases of precellular evolution compound hypercycles might have had a chance to form.

Various systems consisting of catalytically active and self-reproductive units have been studied by fixed-point analysis. The results provide clear evidence for the necessity of hypercyclic coupling. Only catalytic hypercycles can fulfill the criteria for integration of information listed in Section IV.5.

1. Selective stability of each component due to favorable competition with error copies

2. Cooperative behavior of the components integrated into the new functional unit

3. Favorable competition of the functional unit with other less efficient systems

VIII. Dynamics of the Elementary Hypercycle

Hypercycles, being the relevant systems of prebiotic self-organization, deserve a more detailed analysis of their dynamic behavior. A complete qualitative description can be given for the class of elementary hypercycles ($p=2$) up to dimension $n=4$. For higher dimensions, as well as for hypercycles of a more complex structure, we have to aid the topologic analysis by numerical integration. We exemplify the methods with the class of elementary hypercycles, which reveal all the basic properties of hypercyclic self-organization.[1]

VIII.1. Qualitative Analysis

Since we are concerned with dynamic systems of cooperating constituents, the stable attractors in the interior of the physically accessible range of concentrations are of primary interest. More

[1] For a particular case of a hypercycle, in which the second-order term $x_k x_{k-1}$ of the growth function has been substituted by a term of the form $x_k \ln x_{k-1}$, an analytical solution could be provided [21].

specifically, we have to study the stability of those fixed points, for which some eigenvalues of the Jacobian matrix have zero real parts. In Section VII (Table 9) we encountered essentially two cases:

1. Zero eigenvalues ($\omega_j^{(i)}=0$, $j=2,3,\ldots,n$ and $i=1,2,\ldots,n$) for fixed points $\bar{\mathbf{x}}_i$ at the corners of the simplices S_n
2. Purely imaginary eigenvalues ($\omega_{2,4}^{(0)}=\pm i$) for the central fixed point of the four-membered hypercycle on S_4[2].

Before presenting a general proof for the stability of the central fixed points in hypercycles of low dimensions $n\leqq 4$, let us inspect the topology of these systems in more detail.

The dynamic systems corresponding to elementary hypercycles can be decomposed into several subsystems each of which is defined on a globally invariant subspace. A set of points or a subspace will be called 'globally invariant' with respect to a given dynamic system if and only if a trajectory that passes through any point of the subspace never leaves the subspace.

In particular we find that the dynamic systems on the simplices S_n can be subdivided into those on boundaries (BS_n) and those in the interiors (IS_n). The interior of a simplex, defined as before, is the region where no population variable vanishes: $0<\xi_i<1$, $i=1,2,\ldots,n$. Clearly, the dynamic systems on IS_n are most interesting since they describe the development of intact hypercycles. We denote them in the following by numbers: **2**, **3**, **4**, ... **N**.

On the boundary, one, two, or more population variables vanish. Consequently, the dynamic systems on BS_n can be subdivided into dynamic systems on simplices of lower dimension like edges, faces, and hyperfaces. To distinguish these systems from complete hypercycles we shall use **2A**, **2B**, **3A**, etc. as short-hand notations. All dynamic systems corresponding to elementary cycles of dimension $n\leqq 4$ are shown schematically in Figure 30. As a concrete example, the decomposition of the four-dimensional system into 11 subsystems is presented in Table 10.

Table 10. Globally invariant dynamic subsystems of the elementary hypercycle with dimension $n=4$

Symbol	Condition	Dynamic system
4	$\xi_1,\xi_2,\xi_3,\xi_4>0$	$\dot{\xi}_i=\xi_i\xi_j-\xi_i\phi$, $i=1,2,3,4$ $j=i-1+n\delta_{i1}$ and $\phi=\xi_1\xi_2+\xi_2\xi_3+\xi_3\xi_4+\xi_4\xi_1$
3A	$\xi_4=0$	$\dot{\xi}_1=-\xi_1\phi$ $\dot{\xi}_i=\xi_i\xi_j-\xi_i\phi$, $i=2,3$ $j=i-1$ and $\phi=\xi_1\xi_2+\xi_2\xi_3$
	$\xi_3=0$, $\xi_2=0$ or $\xi_1=0$	analogously
2A	$\xi_3=\xi_4=0$	$\dot{\xi}_1=-\xi_1\phi$, $\phi=\xi_1\xi_2$ $\dot{\xi}_2=\xi_1\xi_2-\xi_2\phi$
	$\xi_2=\xi_3=0$, $\xi_1=\xi_2=0$ or $\xi_4=\xi_1=0$	analogously
2B	$\xi_2=\xi_4=0$ $\xi_1=\xi_3=0$	$\dot{\xi}_i=0$, $i=1,2,3,4$

All the dynamic systems up to dimension $n=4$ can be analyzed by Lyapunov's method (Table 11). For the three systems **2**, **3**, and **4**, Lyapunov functions are given, and hence, the central fixed point represents a stable attractor. Moreover, the basin of this fixed point extends over the whole interior of the simplices. In more physical terms this means: Starting from any distribution of population variables we end up with the same stable set of stationary concentrations. The dynamic systems indeed are characterized by cooperative behavior of the constituents. This result is of particular importance for the four-dimensional system where the linear approximation, used in fixed-point analysis, yielded a center surrounded by a manifold of concentric closed orbits in the x, y-plane (see Fig. 31a), which does not allow definite conclusions about stability.

[2] Purely imaginary eigenvalues will occur also in elementary hypercycles of dimension $4k$, where k is an integer $\geqq 2$. In these higher-dimensional examples, however, the central fixed point is a saddle point independently of the nature of higher-order contributions to the purely imaginary eigenvalues.

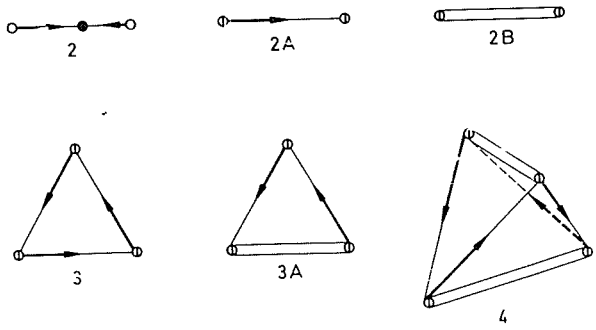


Fig. 30. Dynamic systems corresponding to elementary hypercycles of dimensions $n=2$, 3, and 4. The individual systems are mapped on the simplices S_n and can be decomposed into globally invariant dynamic subsystems (Table 10): The 'complete' subsystem **2**, **3**, and **4** are characterized by nonzero values of all population variables and thus describe the development in the interiors of the simplices S_n(IS_n: $0>x_i>c_0$, $i=1,2,\ldots,n$). On the boundaries of the simplices S_n(BS_n) one or more population variables vanish and dynamic subsystems of lower dimension like the 'flowing edge' **2A**, the 'fixed-point edge' **2B**, and the triangle of type **3A** are obtained (note that the dynamic system **2A** occurs in the boundaries surrounding **3**, **3A**, and **4**, **2B** in those surrounding **3A** and **4**, and **3A** finally occurs in the boundary around **4**)

The dynamic systems on the boundaries of the simplices (BS_n) determine the behavior of 'broken' hypercycles, i.e., catalytic hypercycles which are lacking at least one of their members. In reality, these systems describe the kinetics of hypercycle extinction. They are also of some importance in phases of hypercycle formation. On the boundaries of the complete dynamic systems up to dimension 4 we find two kinds of edges **2A** and **2B** as well as the face **3A** (Fig. 30). All three dynamic systems can be analyzed in a straightforward way.

The first kind of edge **2A** connects two consecutive pure states or corners, which we denote by 'i' and 'j' ($j=i+1-n\delta_{in}$). As shown in Figure 32 there is a steady driving force along the edge, always pointing in the direction $i\rightarrow j$. The only trajectory of this system thus leads from corner i to corner j. Accordingly, we shall call system **2A** a flowing edge. In approaching corner j, the driving force decreases parabolically (Fig. 32). Hence, the linear term of the Taylor expansion vanishes at the fixed point $\bar{\mathbf{x}}_j$, and fixed-point analysis cannot yield the desired prediction as to the nature of this fixed point.

In elementary hypercycles the corners of the simplices are saddle points: A corner (i) is stable with respect to fluctuations along the edge $\bar{ih}$ ($\delta x_h>0$, $h=i-1+n\delta_{i1}$) but unstable along the edge $\bar{ij}$ ($j=i+1-n\delta_{in}$). On the boundary of every complete dynamic system we thus find a closed loop, $\overline{12}\ \overline{23}\ \overline{34}\ldots\overline{n1}$, along which the system has a defined sense of rotation. This cycle is not a single trajectory. A particular kind of fluctuation is required at every corner to allow the system to proceed to the next pure state. The existence of this loop is equivalent to the cyclic symmetry of the total system. The asymmetry at each single corner reflects the irreversibility of biopolymer synthesis and degradation, assumed in our model.

The physically accessible range of variables in the dynamic system **3A** is circumscribed by two consecutive flowing edges, $\bar{ij}$ and $\bar{jk}$ ($j=i+1$

Table 11. Lyapunov functions [57] for basic hypercycles of dimension $n=2$, 3, and 4

To prove the stability of a certain fixed point $\bar{\mathbf{x}}$ of a dynamic system $\dot{\mathbf{x}}=A(\mathbf{x})$, we must find an arbitrary function $V(\mathbf{x})$ which fulfils the following two criteria:

$$(1)\quad V(\bar{\mathbf{x}})=0 \quad \text{and} \quad V(\mathbf{x})>0,\ \mathbf{x}\in U, \tag{T.11.1}$$

i.e., the function vanishes at the fixed point and is positive in its neighborhood U. Thus $V(\mathbf{x})$ has a local minimum at the fixed point.

$$(2)\quad \dot{V}(\mathbf{x})\equiv\frac{dV}{dt}=\sum_{j=1}^{n}\left(\frac{\partial V}{\partial x_j}\right)\frac{dx_j}{dt}<0, \qquad \mathbf{x}\in U \tag{T.11.2}$$

i.e., the time derivative of $V(\mathbf{x})$ is negative in the neighborhood of the fixed point. For trivial reasons $\dot{V}$ vanishes at $\bar{\mathbf{x}}$: $\dot{V}(\bar{\mathbf{x}})=0$. If it is possible to find such a function $V(\mathbf{x})$ for a given fixed point of a dynamic system, it is called a (strict) Lyapunov function and the point $\bar{\mathbf{x}}$ is (asymptotically) stable: All trajectories passing through a point in the neighborhood of $\bar{\mathbf{x}}$ will end in the fixed point $\bar{\mathbf{x}}$.
A Lyapunov function $V(\mathbf{x})$ may also be defined in a less strict way so as to fulfil the weaker criterion:

$$\dot{V}(\mathbf{x})\leqq 0 \tag{T.11.3}$$

Any trajectory entering the neighborhood U of $\bar{\mathbf{x}}$ will remain there. To give a concrete example; sinks are asymptotically stable in the stronger sense of Lyapunov, whereas centers are stable only with respect to the weaker criterion (T.11.3).
For the sake of convenience we use normalized variables ξ_i and assume the rate constants to be equal to 1, $k_1=k_2=\cdots=k_n=1$, when we apply Lyapunov's method to basic hypercycles.
The function

$$V=\left(\frac{1}{n}\right)^n-\xi_1\xi_2\ldots\xi_n \tag{T.11.4}$$

has a minimum and vanishes at the fixed point $\xi_i=\frac{1}{n}$ and thus meets condition (T.11.1). The time derivative of V can be obtained by straightforward differentiation

$$\dot{V}=-\xi_1\xi_2\ldots\xi_n(1-nr);$$
$$r=\sum_{k=1}^{n}\xi_k\xi_l; \qquad l=k-1+n\delta_{k1} \tag{T.11.5}$$

Now we have to check criterion (T.11.2) for systems with different values of n. In the interior of the simplex S_n the condition $\dot{V}<0$ becomes equivalent to the inequality

$$r(\xi)<\frac{1}{n}. \tag{T.11.6}$$

Clearly, we find $r(\bar{\xi}_0)=\frac{1}{n}$, which satisfies the equation $\dot{V}(\bar{\xi}_0)=0$. ($\bar{\xi}_0$ represents the central fixed point of the hypercycle.)
For the two-dimensional system ($n=2$), condition (T.11.6) can be easily verified:

$$\xi_1=\xi; \qquad \xi_2=1-\xi\rightarrow r=2\xi(1-\xi)\leqq\tfrac{1}{2} \tag{T.11.7}$$

The function r represents a parabola with the maximum at $\xi=\frac{1}{2}$. Thus $r(\xi)<\frac{1}{2}$ is fulfilled everywhere except at the fixed point $\xi=\frac{1}{2}$, where $r=\frac{1}{2}$. In this case V is a strict Lyapunov function and $\bar{\xi}_0$ is asymptotically stable.
For $n=3$ the situation is very similar. The inequality (T.11.3), $r<\frac{1}{3}$ is valid at every point in the interior of the simplex S_3, except at the fixed point $\bar{\xi}_0$ where $r=\frac{1}{3}$. V again represents a strict Lyapunov function and the central fixed point $\bar{\xi}_0$ is asymptotically stable.
In four dimensions the problem is more involved. We realize that condition (T.11.3) is verified almost everywhere on the simplex S_4:

$$r=(\xi_1+\xi_3)(\xi_2+\xi_4)=s(1-s), \qquad 0\leqq s\leqq 1 \tag{T.11.8}$$

In its interior we find $0\leqq r\leqq\frac{1}{4}$ with $r=\frac{1}{4}$ if and only if $s=\frac{1}{2}$. The equation $s=\frac{1}{2}$ determines the plane $\xi_1+\xi_3=\frac{1}{2}$ (see Figs. 31a and 34b). V apparently is a Lyapunov function solely in the weaker sense. This result suggests that the central fixed point at least is stable. To prove asymptotic stability we introduce new variables x, y, z

$$x=-2(\xi_2+\xi_3)+1 \qquad \dot{x}=-(1+z)(y-xz)$$
$$y=2(\xi_1+\xi_2)-1 \qquad \dot{y}=(1-z)(x-yz)$$
$$z=2(\xi_1+\xi_3)-1 \qquad \dot{z}=z^3-z+x^2-y^2$$

which shift the origin of the coordinate system into the center of the simplex S_4 and place the coordinate axes through the midpoints of the edges $\overline{23}$, $\overline{34}$, and $\overline{13}$, respectively, (cf. Fig. 31a). The fourth variable $\xi_4=1-\xi_1-\xi_2-\xi_3$ is eliminated. The z-axis thus points perpendicular to the critical plane $\xi_1+\xi_3=\frac{1}{2}$, which is spanned by the two variables x and y. In this plane the dynamic system simplifies to $\dot{x}=-y$, $\dot{y}=x$, and $\dot{z}=x^2-y^2$.
The time derivative of z vanishes only along the two lines $x=\pm y$ or $\xi_2=\xi_4$ and $\xi_1=\xi_3$, respectively. Consequently, there is no trajectory in this critical plane – except the fixed point $\bar{\xi}_0$ – and the system will pass through it in infinitely short time. Along any given orbit the condition $\dot{V}(\xi(t))<0$ is fulfilled at almost every moment, the only exceptions being the instances when the critical plane $\xi_1+\xi_3=\frac{1}{2}$ is passed. Along all trajectories, $V(\xi(t))$ is monotonically decreasing with t. V is a strict Lyapunov function, and thus the fixed point $\bar{\xi}_0$ is asymptotically stable.
In higher dimensions $n\geqq 5$, $V(\xi)$ is not a Lyapunov function and therefore no predictions on the stability of the central fixed point can be made by this method.

$-n\delta_{in}$ and $k=j+1-n\delta_{in}$) and one fixed-point edge $\overline{ik}$ ($k\neq i-1+n\delta_{in}$). The trajectories of this system are shown in Figure 31b. They start from some point of the fixed-point edge and end at the corner k, which thus represents the only stable attractor of the system. The species I_k consequently represents the survivor or remnant of this fragment of a hypercycle.
The investigation on the boundaries of basic hypercycles can be generalized to higher-dimensional systems. From the results obtained we can make the desired predictions about the long-term development of broken hypercycles. After one species in a hypercycle has been extinguished by some external event, the remaining dynamic system is unstable and approaches a pure state after long enough time. In all cases, a species will be selected which occurs just before the break in the hypercycle. In other words, species I_i will remain the last remnant of a hypercycle which has been destroyed by extinction of its constituent I_j, when i is the precursor of j ($j=i+1-\delta_{in}$). This behavior is not unexpected in light of the known properties of catalytic chains.

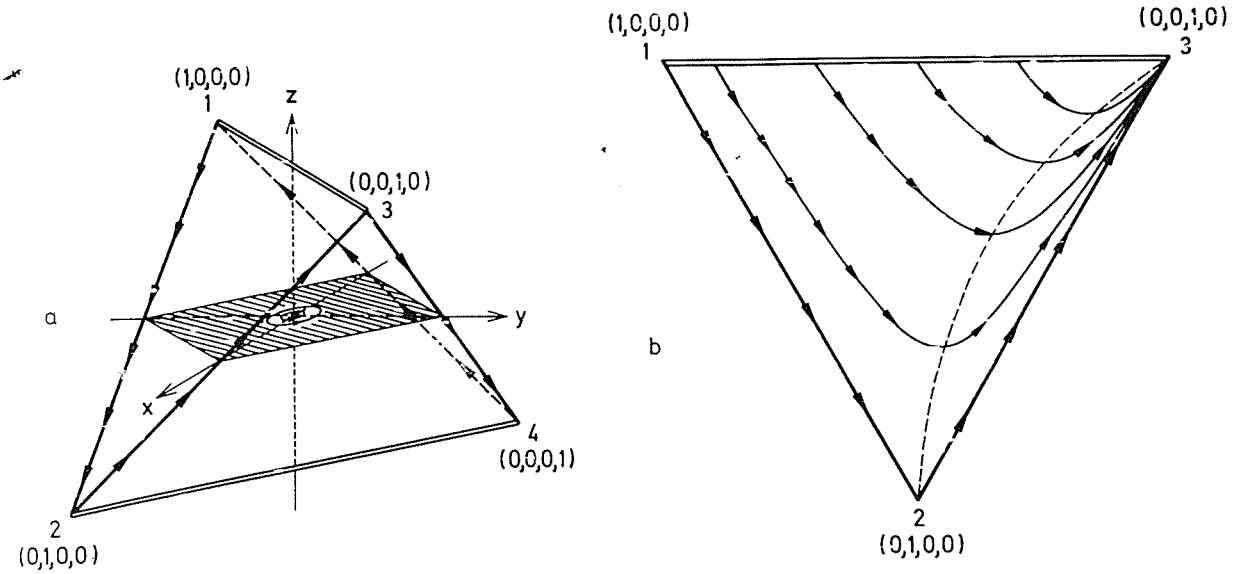


Fig. 31. Dynamic topology of the elementary hypercycle of dimension $n=4$. The dynamic system on the simplex consists of the system **4** on the interior IS_4 and four equivalent systems of type **3A** on equilateral triangles (S_3), each of which is circumscribed by two flowing edges **2A** and one fixed-point edge **2B**. a) The system on the interior is described appropriately in the variables x, y, and z (see Table 11). The manifold of closed concentric orbits belonging to the center of the linearized system lies in the x,y-plane (hatched region in the drawing). b) The dynamic system **3A**: All trajectories start out from some point on the fixed-point edge ($\overline{13}$) and end in the same corner (3). The dashed line connects all the points at which the trajectories are parallel to the fixed-point edge ($\overline{13}$)

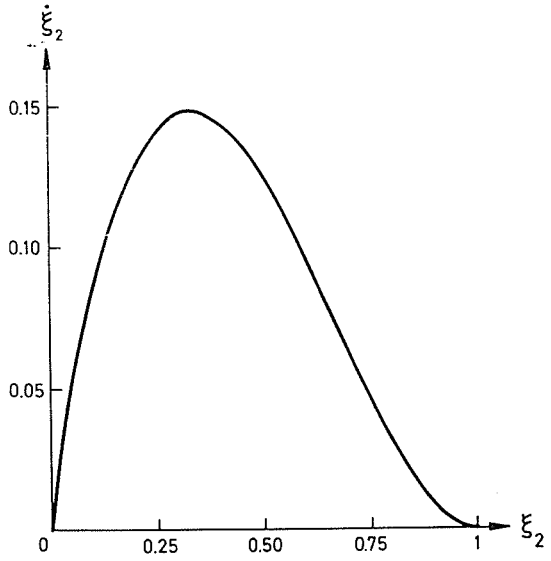


Fig. 32. The 'flowing edge' **2A**. The tangent vector $\dot{\xi}_2=\xi_2(1-\xi_2)^2$ is positive inside the whole physically allowed region ($0<\xi_2<1$) and vanishes at both ends, which represent the two fixed points of the system: $\bar{\xi}_1=(1,0)$ and $\bar{\xi}_2=(0,1)$. $\dot{\xi}_2>0$ means that ξ_2 is increasing during the next differential interval of time. Hence there is only one orbit on **2A** leading from $\bar{\xi}_1$ to $\bar{\xi}_2$, i.e., from corner 1 ($\xi_2=0$) toward corner 2 ($\xi_2=1$). The dynamic system 'flows' along this edge. Note that $d\dot{\xi}_2/d\xi_2$ vanishes at the fixed point $\bar{\xi}_2$ ($\xi_2=1$), leading to an eigenvalue $\omega=0$ in the linearized system. Consequently, fixed-point analysis fails to make a prediction about the stability of this point

VIII.2. Numerical Integration

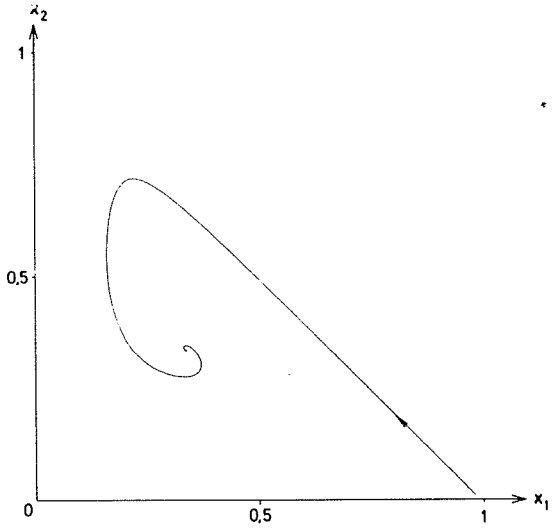
The systems of differential equations for elementary hypercycles with dimensions up to $n\leqq 12$ have been integrated by standard numerical techniques. The corresponding solution curves $\mathbf{x}(t)$ have been presented in a previous paper [4] and need not be repeated, since we are interested in a different aspect of the problem here. Our present purpose is to search for stable attractors in the interior of the simplices S_n that guarantee cooperative behavior of the constituents. For this goal an investigation of the manifold of trajectories is straightforward.

Differential equations for trajectories are obtained by elimination of the explicit time dependence in the original dynamic system:

$$\begin{aligned}
\frac{dx_2}{dx_1}&=\frac{A_2}{A_1}=f_2(x_1,x_2,\ldots,x_n)\\
\frac{dx_3}{dx_1}&=\frac{A_3}{A_1}=f_3(x_1,x_2,\ldots,x_n)\\
&\vdots\\
\frac{dx_n}{dx_1}&=\frac{A_n}{A_1}=f_n(x_1,x_2,\ldots,x_n)
\end{aligned}\qquad (70)$$

Integration of this new $(n-1)$-dimensional dynamic system yields the trajectories as solution curves:

$$\begin{aligned}
x_2&=g_2(x_1,x_2,\ldots,x_n)\\
x_3&=g_3(x_n,x_2,\ldots,x_n)\\
&\vdots\\
x_n&=g_n(x_1,x_2,\ldots,x_n)
\end{aligned}\qquad (71)$$



The trajectory thus is a curve in the n-dimensional concentration space. For graphical representation we shall use projections of these curves on the planes spanned by one selected coordinate x_k and by x_1. Trajectories for hypercycles of low dimension ($n=2$, 3, and 4) reflect the already known properties of these dynamic systems. The case $n=2$ is rather trivial: There are only two orbits which converge to the central stable focus (Fig. 30). The trajectories of the three-dimensional hypercycle ($n=3$) are spirals which rapidly approach the central fixed point (Fig. 33). This kind of trajectory corresponds to strongly damped oscillations of the solution curves $\mathbf{x}(t)$. The four-membered hypercycle deserves further consideration. Again, the trajectories spiral into the center of the simplex (Fig. 34a, b). In contrast to the three-dimensional example, the central force is much weaker than the rotational component. Accordingly, convergence

Fig. 33. A trajectory of the dynamic system (3) for the elementary hypercycle of dimension $n=3$ is shown as a projection on the plane (x_1, x_2). Initial conditions: $x_1(0)=0.98$, $x_2(0)=x_3(0)=0.01$

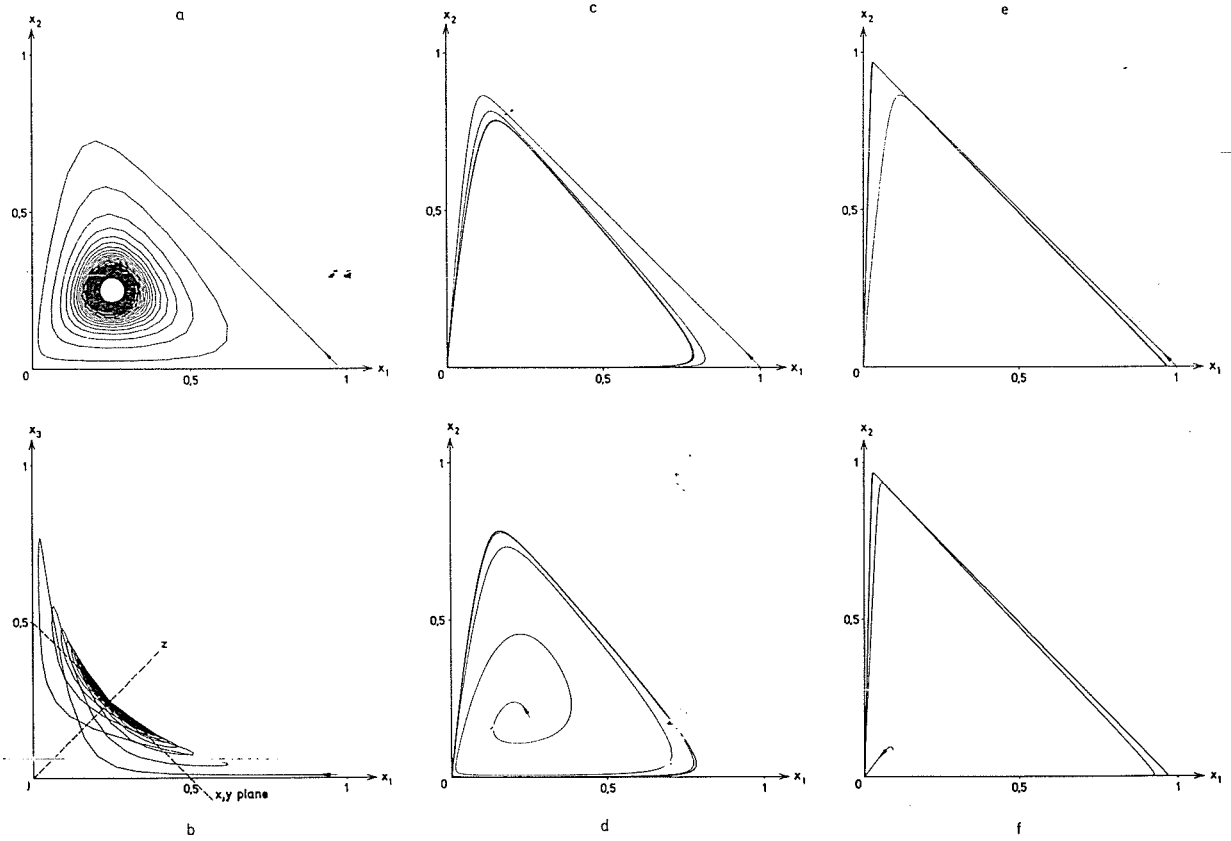


Fig. 34. Trajectories of the dynamic systems (4, 5, and 12) for elementary hypercycles of dimension $n=4$, 5, and 12, respectively. (1) $n=4$: initial conditions: $x_1(0)=0.97$, $x_2(0)=x_3(0)=x_4(0)=0.01$; two projections are shown. a) Projection of the trajectory on the plane (x_1, x_2). The trajectory spirals into the central fixed point with hardly damped oscillations. b) Projection of the trajectory on the plane (x_1, x_3). The plane of the center manifold (x, y-plane in Fig. 31) intersects the x_1, x_3-plane along the line $x_1+x_3=1/2$. Perpendicular to it we see the z-axis. Note that the trajectory crosses this plane (x, y) only at single points and does not stay therein for a longer period (Table 11). (2) $n=5$: projections on the plane (x_1, x_2). c) Initial conditions: $x_1(0)=0.9996$, $x_2(0)=x_3(0)=x_4(0)=x_5(0)=0.0001$. d) Initial conditions: $x_1(0)=0.2004$, $x_2(0)=x_3(0)=x_4(0)=x_5(0)=0.1999$. Note that the dynamic system approaches the same limit cycle from both sets of initial conditions. (3) $n=12$: projections on the plane x_1, x_2. e) Initial conditions: $x_1(0)=0.9989$, $x_2(0)=\ldots..x_{12}(0)=0.0001$. f) Initial conditions: $x_1(0)=0.0848$, $x_2(0)=\ldots..x_{12}(0)=0.0832$. Again both limit cycles are the same and come very close to the loop 12, 23 ... 11, 12, 12 1

toward the central fixed point is extremely slow. A projection of the trajectory on the x_1, x_3-plane nicely illustrates the previously derived result that there is no orbit in the plane $x_1 + x_3 = 1/2$, $x_2 + x_4 = 1/2$. Indeed, as one can see from Figure 34b, the trajectories follow closely a saddle-like bent surface.

For basic hypercycles of dimension $n \geqq 5$ the central fixed point represents an unstable saddle. There is no sink in the boundary and consequently one expects a stable closed orbit. The analytical techniques have not yet been developed to a sufficient extent to provide the proof of the existence of such an attractor in the interior of the simplices. Therefore, we have to rely on numerical results.

Numerical integration indeed provides strong evidence for a limit cycle or closed orbit. Starting from various points very close to the center, to a face, to an edge, or to a corner of the simplex we always arrive at the same limit cycle after long enough time. Two typical trajectories are shown in Figure 34c–f, for elementary hypercycles of dimensions $n=5$ and $n=12$, respectively. As we can see from a comparison of the two Figures, with increasing n the limit cycle approaches more closely the loop $\overline{12}, \overline{23}, \ldots, \overline{n1}$ mentioned in the previous section. Consequently the oscillations in the individual concentrations become more and more like rectangular pulses.

The use of numerical techniques also enables us to remove the assumption: $k_1 = k_2 \ldots = k_n$. Calculations with arbitrary k values have been performed for dynamic systems of dimensions $n=4$ and $n=5$. No change in the general nature of the solution curves is observed. Typical examples are shown in Figures 35 and 36. The individual concentrations in both systems oscillate. For $n=4$ the concentration waves are damped and the dynamic system approaches the central fixed point. Its coordinates are determined by the following equations:

$$\bar{x}_0: \; \bar{x}_i^0 = \frac{k_j^{-1}}{\sum_{l=1}^{n} k_l^{-1}} c_0; \quad j = i+1-n\delta_{in} \tag{72}$$

Five-membered hypercycles with unequal rate constants show the same kinds of undamped concentration pulses that we have observed in the system with equal k values. The size of the pulses is no longer the same for all subunits. Time-averaged concentrations [as defined by Equation (67)] fulfil the above equation (72), which determines the position of the (unstable) central fixed point. Accordingly, the pulses for those species which precede a step with a relatively small rate constant are broad, whereas those of species preceding a relatively fast reaction step are small in width and height. The system thus regulates the concentration of its constituents in such a way that the overall production rate is optimized.

Hypercycles of higher dimensions ($n \geqq 5$) do not exist in stable states with constant stationary concentrations but exhibit wave-like oscillations around an unstable fixed point in the center. Nevertheless, the constituents show cooperative behavior since their concentrations are controlled by the dynamics of the whole system and no population variable vanishes.

Dynamic systems corresponding to elementary hypercycles have one and only one attractor in the interior of the simplex, the basin of which is extended over the entire region of positive (nonzero) concentrations of all compounds. At low dimension ($n \leqq 4$) the attractor is an asymptotically stable fixed point, namely, a focus for $n=2$ and a spiral sink for $n=3$ and $n=4$. In systems of higher dimensions ($n \geqq 5$) numerical integration provides strong evidence for the existence of a stable limit cycle. All elementary hypercycles thus are characterized by cooperative behavior of their constituents.

Due to their dynamic features hypercycles of this type hide many yet unexplored potentialities for self-organization (dissipative structures, e.g., in case of superimposed transport). They may also play an important role in the self-organization of neural networks.

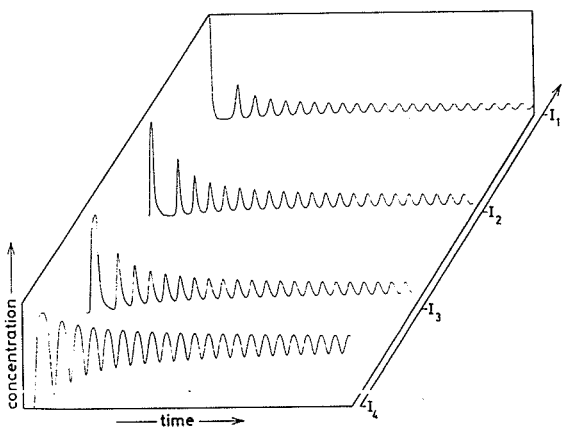


Fig. 35. Solution curves of the dynamic system for an elementary hypercycle with dimension $n=4$ and unequal rate constants ($k_1 = 0.25$, $k_2 = 1.75$, $k_3 = 1.25$, $k_4 = 0.75$; initial conditions: $x_1(0) = 0.9997$, $x_2(0) = x_3(0) = x_4(0) = 0.0001$; full concentration scale = 1 concentration unit, full time scale = 1000 time units). Note that the concentration of I_1 (the component preceding the fastest step) is smallest whereas that of I_4 (the component before the slowest step) is largest

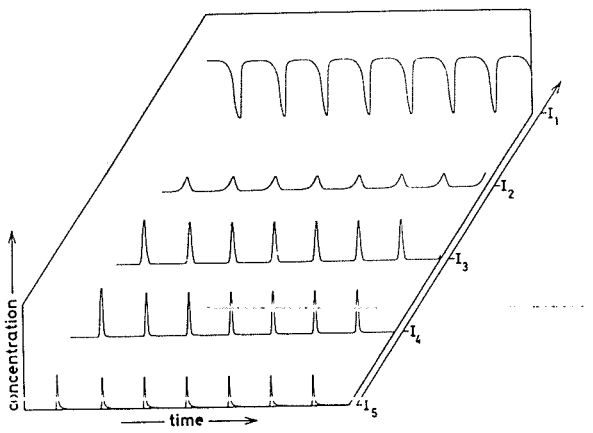


Fig. 36. Solution curves of the dynamic system for an elementary hypercycle with dimension $n=5$ and unequal rate constants ($k_1 = 25/13$, $k_2 = 1/13$, $k_3 = 19/13$, $k_4 = 1$, $k_5 = 7/13$; initial conditions: $x_1(0) = 0.9996$, $x_2(0) = x_3(0) = x_4(0) = x_5(0) = 0.0001$; full concentration scale = 1 concentration unit, full time scale = 1000 time units). Note that the concentration of I_5 (the component before the fastest step) is smallest, whereas that of I_1 (the component before the slowest step) is largest

IX. Hypercycles with Translation

IX.1. Ideal Boundary Conditions and General Simplifications

An appropriate set of boundary conditions can be realized in a flow reactor [4, 9, 55, 56]. The concentrations of all low-molecular-weight compounds (m_i, $i=1,2,\ldots,\lambda$) are buffered with the help of controlled flow devices, at the same time providing the energy supply for the system. The concentration variables x_i refer to the macromolecular species synthesized in the reactor, while all other compounds of the 'standard reaction mixture' do not show up explicitly in the differential equations, but appear implicitly in the effective rate constants of Equation 30.

Because of technical difficulties and also for heuristic reasons it is impossible to account explicitly for all elementary steps in the reaction mechanism. We rather have to apply simplified reaction schemes which lead to an appropriate 'over-all' kinetics. This strategy is a common procedure in chemical kinetics. Acid base reactions in aqueous solution for example are generally described by phenomenologic equations which do not account for individual proton jumps, but just reflect changes in protonation states of the molecules considered.

For the mechanism of template-directed polymerization and translation, the rate equations contain the population numbers of the complete macromolecules as the only variables. Hence chain initiation and propagation steps are not considered explicitly. A justification for these approximations can be taken from experiments. Actually, the kind of 'over-all' kinetics we are using here is well established (cf. Part C).

IX.2. The Kinetic Equations

The catalytic hypercycle shown schematically in Figure 37 consists of two sets of macromolecules: n polynucleotides and n polypeptides. The replication of polynucleotides (I_i) is catalyzed by the polypeptides (E_i) which, in turn, are the translation products of the former. The hypercyclic linkage is established by two types of dynamic correlations:

1. Each polynucleotide I_i is translated uniquely into a polypeptide E_i. The possibility of translation evidently requires the existence of an appropriate machinery which is composed of at least some of the translation products E_i and which uses a defined genetic code.
2. Polynucleotides and polypeptides form specific complexes that are also catalytically active in the synthesis of polynucleotide copies. The polypeptides may be specific replicases or specific cofactors of a common

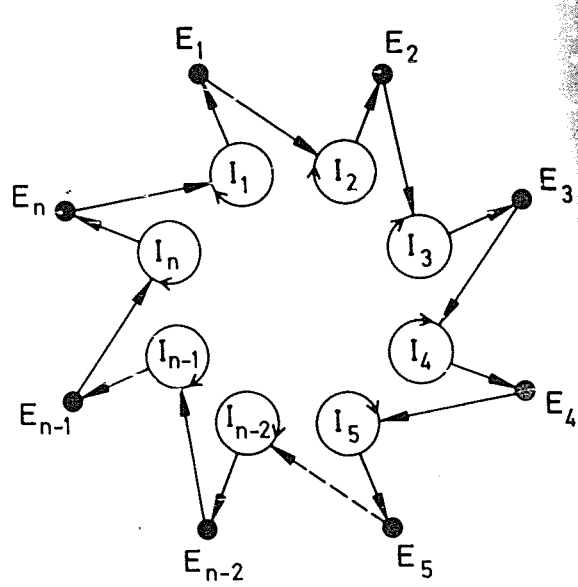


Fig. 37. Schematic diagram of a hypercycle with translation. Dimension: $2\times n$, i.e., n polynucleotides and n polypeptides

protein with polymerase activity. Altogether these primordial proteins provide at least two functions: specific replication and translation. How such a system can be envisaged is shown in Part C.

The couplings between the I_i and E_i have to be of a form which allows the closure of a feedback loop (Fig. 37). In mathematical terms cyclic symmetry is introduced by assuming specific complex formation between the enzyme E_j and the polynucleotide I_i, whereby $j=i+1-n\delta_{in}$.

The kinetics of polynucleotide synthesis follows a Michaelis-Menten-type reaction scheme, although we do not introduce the assumption of negligibly small complex concentrations.

$$I_i+E_j \underset{}{\overset{K_i}{\rightleftharpoons}} I_iE_j; \qquad I_iE_j+\sum \nu_\lambda^I M_\lambda^I \xrightarrow{f_i} I_i+I_iE_j \tag{73}$$

The four nucleoside triphosphates and their stoichiometric coefficients are denoted by M_λ^I and ν_λ^I; $\lambda=1,2,3,4$, respectively. Now we introduce z_i for the concentration of the complex I_iE_j and x_j^0, y_i^0 or x_j, y_i for the total or free concentrations of polypeptides (E_j) and polynucleotides (I_i). Mass conservation requires:

$$x_j^0=x_j+z_i \quad \text{and} \quad y_i^0=y_i+z_i \tag{74}$$

For fast equilibration of the complex the concentration z_i is related to the total concentrations x_j^0 and y_i^0 as:

$$z_i=\frac{y_i^0+x_j^0+K_i}{2}\left(1-\sqrt{1-\frac{4\,y_i^0\,x_j^0}{(y_i^0+x_j^0+K_i)^2}}\right) \tag{75}$$

Polypeptide synthesis is assumed to be unspecific, i.e., translation of the polynucleotide I_i occurs with the help of a common 'apparatus':

$$I_i+\sum_\lambda \nu_\lambda^E M_\lambda^E \xrightarrow{k_i} I_i+E_i \tag{76}$$

M_λ^E and ν_λ^E denote the activated amino acids and their stoichiometric coefficients, respectively. Selection constraints may be introduced properly by controlling total concentrations for both kinds of biopolymers (I and E) independently. By analogy with the constraints of constant organization we keep both sums of concentrations constant:

$$\sum_k y_k^0 = c_I^0; \quad \sum_k x_k^0 = c_E^0 \tag{77}$$

Under all these conditions our dynamic system consiting of $2n$ coupled differential equations reads:

$$\dot{y}_i^0 = f_i z_i - \frac{y_i^0}{c_I^0} \sum_{k=1}^{n} f_k z_k$$

$$\dot{x}_i^0 = k_i y_i - \frac{x_i^0}{c_E^0} \sum_{k=1}^{n} k_k y_k$$

$$i = 1, 2, \ldots, n. \tag{78}$$

For our purpose, it is sufficient to discuss two limiting cases:

1. The concentration z_0 of the complex $I_i E_j$ becomes proportional to the product of polynucleotide and polypeptide concentrations at sufficiently low concentrations:

$$y_i^0, x_j^0 \ll K_i; \quad z_i \sim \frac{1}{K_i} y_i^0 x_j^0; \quad z_i \ll y_i^0, x_j^0 \tag{79}$$

If we further assume the first-order translation process to be fast as compared to the second-order replication – an assumption that is well justified, at least for low concentrations of polynucleotides – the polypeptide concenfration will assume a stationary value that can be included in the rate parameters. The formation of polynucleotides then is described by a system of differential equations that is typical of an elementary hypercycle of dimension n.

2. At high concentrations, z_i becomes equal to the smaller of the two variables:[3]

$$z_i = \inf(y_i^0, x_j^0) \tag{80}$$

Accordingly, we approach two possible limiting situations

$$K_i \ll y_i^0 \ll x_j^0: \; z_i \sim y_i^0 \tag{81}$$

$$K_i \ll x_j^0 \ll y_i^0: \; z_i \sim x_j^0 \tag{82}$$

In the first of these two cases the polynucleotides behave like independent competitors, while the polypeptides – due to $y_i = y_i^0 - z_i \approx 0$ – remain stationary. Under natural conditions, where constraints like 'constant total concentrations' usually do not apply – at least not for the assumed small values of y – the resulting growth of polynucleotides would lead to a reversal of the concentration ratios $y{:}x$ and hence to an approach to condition (82). As a consequence, the approximations

$$z_i \approx x_j^0 \quad \text{and} \quad y_i \approx y_i^0 \tag{83}$$

become valid, leading to a $2n$-membered catalytic cycle, but not a hypercycle. Thus under saturation conditions, i.e., at high concentrations of the constituents, the hypercycle loses the behavior typical of nonlinear growth rates. As a unified system it simulates the properties of a simple catalytic cycle, which is equivalent to an autocatalyst or self-reproducing unit.

IX.3. Numerical Solutions

The differential equations derived for catalytic hypercycles with explicit consideration of complex formation between the polynucleotides and polypeptides are difficult to study by analytical methods, because of the irrational expressions involved. Numerical integration is time-consuming in these cases, but nevertheless, it represents the only source of information on the properties of these dynamic systems. To illustrate the dynamics of polynucleotide–polypeptide hypercycles we shall present computer graphs of solution curves as well as trajectories.

In comparison to elementary hypercycles the polynucleotide–polypeptide systems contain a new class of parameters, namely, the association constants of the complexes, K_i. As to be expected from the differences in kinetic behavior at the low and high concentration limits, the equilibrium constants exhibit a dominating influence on the dynamic properties of the system. For the sake of a systematic investigation we reduce the number of independent parameters. The assumptions made are essentially the same as those used for the elementary hypercycles: All rate constants for polynucleotide replication, $f_1 = f_2 = \cdots f_n = f$, for their translation into polypeptides $k_1 = k_2 = \cdots = k_n = k$, and all association constants, $K_1 = K_2 = \cdots = K_n = K$ are assumed equal. Then, we study the influence of K on the properties of the dynamic systems at fixed values of f and k and for a constant set of initial concentrations. For hypercycles of dimensions $n \leqq 4$ the solution curves approach a stable stationary state after long enough time. The individual concentrations may exhibit damped oscillations. The dynamics of these systems are essentially the same as for hypercycles with higher values of n and small equilibrium constants.

The dynamics of higher-dimensional hypercycles are more complicated. The long-term behavior of the system changes with increasing values of the equilibrium constant K. Below a certain critical value (K_{cr}) the system converges toward stable stationary states,

[3] inf = infimum is the mathematical term representing the smallest member of a set.

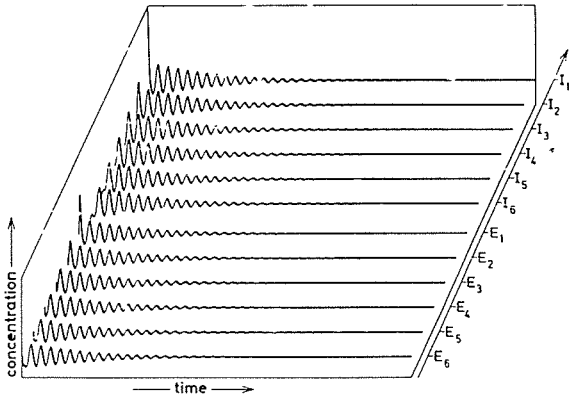


Fig. 38. Solution curves of the dynamic system for a hypercycle with translation. Dimension: 2×6, $k = 0.25$; initial conditions: $y_1(0) = 5.0$, $y_2(0) = \cdots = y_6(0) = 0.5$; $x_1(0) = \cdots = x_6(0) = 1.0$; full concentration scale = 5 concentration units, full time scale = 1000 time units. The value of the equilibrium constant K is below the critical value for the Hopf bifurcation and hence a damped oscillation is observed

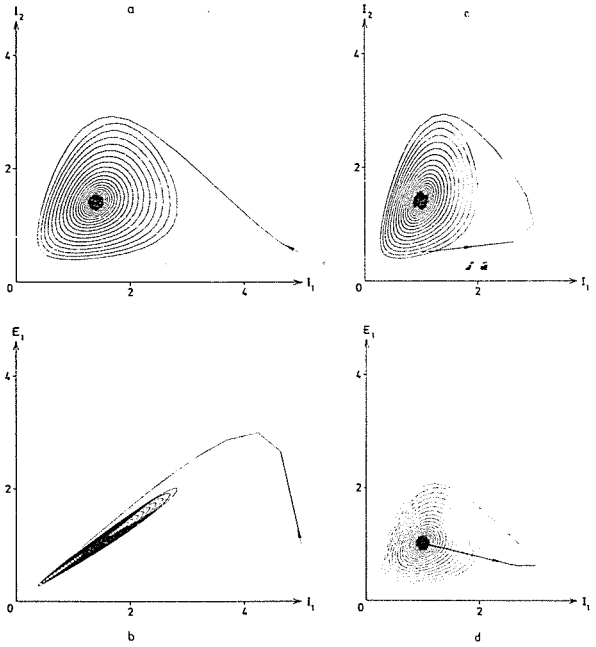


Fig. 39. A trajectory of the dynamic system for a hypercycle with translation. Dimension: 2×5, $k = 1.0$; initial conditions: $y_1(0) = 5.0$, $y_2(0) = y_3(0) = y_4(0) = y_5(0) = 0.5$; $x_1(0) = x_2(0) = x_3(0) = x_4(0) = x_5(0) = 1.0$. a) Projection of the trajectory on the plane (y_1, y_2) showing the concentrations of the polynucleotides I_1 and I_2. b) Projection on the plane (y_1, x_1) showing the concentrations of the polynucleotide I_1 and its translation product, the enzyme E_1. Note that the concentration of E_1 is roughly proportional to that of I_1 and thus the condition for simplifying the hypercycle with translation is fulfilled to a good approximation. c) Projection on the plane (x_1, y_2) showing the concentrations of the polypeptide E_1 and the polynucleotide I_2, the formation of which is catalyzed by the former. d) Projection on the plane (x_1, x_2) showing the concentrations of the polypeptides E_1 and E_2. Note that K again is below the critical value of the Hopf bifurcation and the trajectory converges to the central fixed point

whereas limit cycles are obtained for larger values of $K (K > K_{cr})$. According to the appearance of solution curves and trajectories we distinguish four different cases, arranged with respect to increasing values of the equilibrium constant K:

1. At small values of K the dynamic behavior is qualitatively the same as of hypercycles of lower dimensions. The solution curves exhibit strongly dampled oscillations (Fig. 38) and the trajectories spiral quickly into the center, which represents a stable stationary state (Fig. 39).

2. In principle we find the same general type of dynamic behavior as in case (1). The oscillations, however, are damped only slightly and the approach toward the stationary state is extremely slow (Fig. 40a, b). The situation is quite different from case (1), because the damping terms do not show up in normal mode anaiysis but require consideration of nonlinear contributions. Phenomenologically this fact reveals itself in the appearance of initially (almost) constant amplitudes of oscillation. This situation occurs at values of the equilibrium constant K that are slightly smaller than the critical value K_{cr}, i.e.: $K = K_{cr} - \delta K$.

3. At values of K that are slightly larger than the critical equilibrium constant ($K = K_{cr} + \delta K$), we observe an interesting phenomenon. The dynamic system first behaves much as in case (2). The individual concentrations oscillate with relatively small amplitudes. In contrast to case (2), the amplitudes increase slightly during the initial period. After this phase of sinusoidal oscillation, however, the concentration waves change abruptly in shape and frequency (Fig. 40c, d) and then resemble closely the rectangular pulses which we encountered in basic hypercycles of high dimension. Finally, the dynamic system approaches a limit cycle.

4. At large values of K the individual concentrations oscillate with increasing amplitude and the dynamic system steadily approaches the limit cycle (Fig. 40e, f). The kind of change in dynamic behavior with a continuously varying parameter as we have observed here is known in the literature as 'Hopf bifurcation' [58]. The characteristic retardation in convergence toward the long-term solution that we have found in cases (2) and (3) has been described also for other dynamic systems and is usually called the 'critical slowing down' at the Hopf bifurcation. In the case of hypercycles, the 'slowing down' around the critical value of K becomes more pronounced at larger values of n. In the five-membered cycle ($n = 5$) a situation corresponding to case (3) is hardly detectable. The catalytic hypercycle with $n = 10$, on the other hand, shows a much longer initial period, as referred to in case (3), than the six-membered system (Fig. 41). The initial

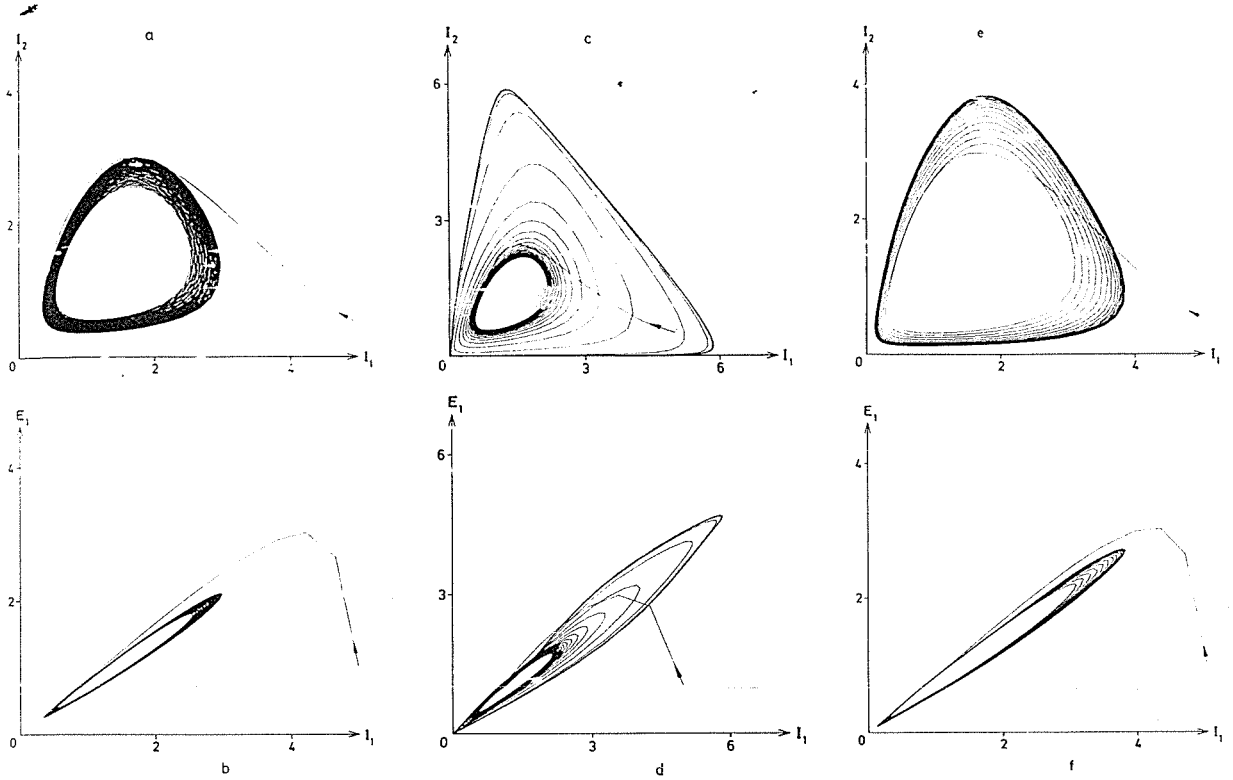


Fig. 40. Trajectories of the dynamic systems for hypercycles with translation. a and b) Dimension 2×5, $K = 1.1$, initial conditions: $y_1(0) = 5.0$, $y_2(0) = y_3(0) = y_4(0) = y_5(0) = 0.5$; $x_1(0) = x_2(0) = x_3(0) = x_4(0) = x_5(0) = 1.0$; projections are shown on the planes for the concentrations of I_1, I_2, and I_1, E_1, respectively; the value of the equilibrium constant K is slightly below the Hopf bifurcation and we observe very slow convergence toward the stable central fixed point. c and d) Dimension 2×6, $K = 0.2784$, initial conditions: $y_1(0) = 5.0$, $y_2(0) = \cdots = y_6(0) = 0.5$, $x_1(0) = \ldots x_6(0) = 1.0$; projections are shown on the planes for the concentrations of I_1, I_2, and I_1, E_1, respectively; the value of the equilibrium constant K is slightly above the Hopf bifurcation and we observe a metastable limit cycle before the system finally converges to the stable limit cycle. e and f) Dimension 2×5, $K = 1.2$, initial conditions: $y_1(0) = 5.0$, $y_2(0) = y_3(0) = y_4(0) = y_5(0) = 0.5$, $x_1(0) = x_2(0) = x_3(0) = x_4(0) = x_5(0) = 1.0$; projections are shown on the planes for the concentrations of I_1, I_2, and I_1, E_1, respectively; the value of the equilibrium constant K is above the Hopf bifurcation and the system converges steadily toward the stable limit cycle. Note that the proportionality between E_1 and I_1 is reasonably well fulfilled in all three cases (b, d and f)

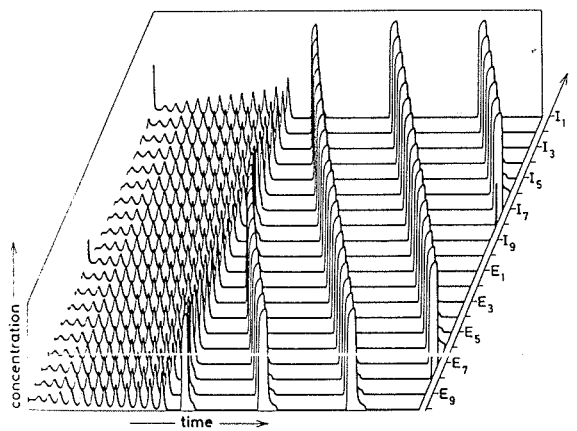


Fig. 41. Solution curves of the dynamic system for a hypercycle with translation. Dimension 2×10, $K = 0.026$; initial conditions: $y_1(0) = 5.0$, $y_2(0) = \cdots = y_{10}(0) = 0.5$; $x_1(0) = \cdots = x_{10}(0) = 1.0$; full concentration scale = 10 concentration units, full time scale = 1000 time units. The value of the equilibrium constant chosen is slightly above the critical value for the Hopf bifurcation. We observe a metastable oscillatory state which changes suddenly into the final limit cycle with the characteristic concentration waves

phase of sinusoidal oscillations resembles a metastable oscillatory state. The transition to the final limit cycle becomes sharper with increasing value of n and is quite pronounced for the ten-membered hypercycle.

All polynucleotide-polypeptide hypercycles studied so far have an attractor in the interior of the physically accessible range of concentrations. They are characterized by cooperative behavior of their constituents. Depending on the values of the product of total concentrations (c_0^N and c_0^P) and of association constants (K) as well as on the size of the hypercycles, we observe stable fixed points or limit cycles. Small K values then are complementary to high concentrations and *vice versa*. The long-term behavior at low and high concentration limits, obtained by numerical integration, agrees completely with the predictions based on the analysis given in the last section. One of the basic simplifications, which concerned quasi-stationarity of polypeptide synthesis, can be checked directly by an inspection of the projections of trajectories onto the E_1,

I_1 plane. For the stationary-state approximation we expect to find straight lines. As we can see from Figures 39b and 40b, d, f, proportionality of the two concentrations is roughly fulfilled and the simplified treatment appears to be well justified. It was, actually, the purpose of the numerical analysis of this complex reaction mechanism to verify the equivalence of complex and elementary hypercycles as far as their self-organizing properties are concerned. The conclusions reached with elementary systems therefore are relevant also for all kinds of realistic hypercycles of a more complex structure (cf. Part C).

X. Hypercyclic Networks

X.1. Internal Equilibration and Competition between Hypercycles

The concept of internal equilibration as introduced in Section VI seems to be very useful for a straightforward analysis of more complex networks since it permits a reduction of the number of independent variables.

At first we investigate the process of equilibration in elementary hypercycles. For that purpose we calculate time averages of the individual concentrations $X_i(t)$ – see Equation (67) and compare them with the corresponding solution curves $x_i(t)$ (Fig. 42). No matter whether the final state is stationarily inert or oscillating, the time averages $X_i(t)$ become practically constant after a few cycles. The assumption of established internal equilibrium therefore seems to be a well-justified approximation for hypercycles. Nevertheless, we shall check it in a few cases.

Using the concept of internal equilibration we can derive an equation for the net growth rate of entire hypercycles:

$$\dot{c}=\sum_{i=1}^{n}\dot{x}_i=\sum_{i=1}^{n}\Gamma_i(\mathbf{x})=\sum_{i=1}^{n}k_i x_i x_j$$

$$=\frac{1}{\sum_{i=1}^{n}k_i^{-1}}\,c^2\equiv\bar{k}c^2=\Gamma(c) \qquad (84)$$

$$j=i+1-n\delta_{in}$$

Hypercycles, thus, are characterized by quadratic growth rates and follow a hyperbolic growth law. They represent appropriate examples for the kind of non-Darwinian 'once-for-ever' selection discussed in Sections VI and VII.

According to the expression for $\bar{k}$ in Equation (84) the rate constant of an entire hypercycle will be of the same order of magnitude as that of its slowest single step.

Under the condition of unlimited growth, hypercycles grow hypothetically to infinity at a certain critical time (t_∞). In fully equilibrated systems these instabilities occur at

$$t^e_\infty=[k_i c(t=0)]^{-1} \qquad (85)$$

The results for equilibrated hypercycles calculated from Equation (85) are compared with the values obtained by numerical integration of systems far off internal equilibrium (t^n_∞) in Table 12. In fully equilibrated systems the instabilities always occur somewhat earlier, $t^e_\infty < t^n_\infty$. On the whole, these numerical differences are of minor importance only, the general behavior of the dynamic systems and the relative values of t_∞ being predicted correctly. The assumption of internal equilibration thus appears to be a good approximation for most nonequilibrated systems.

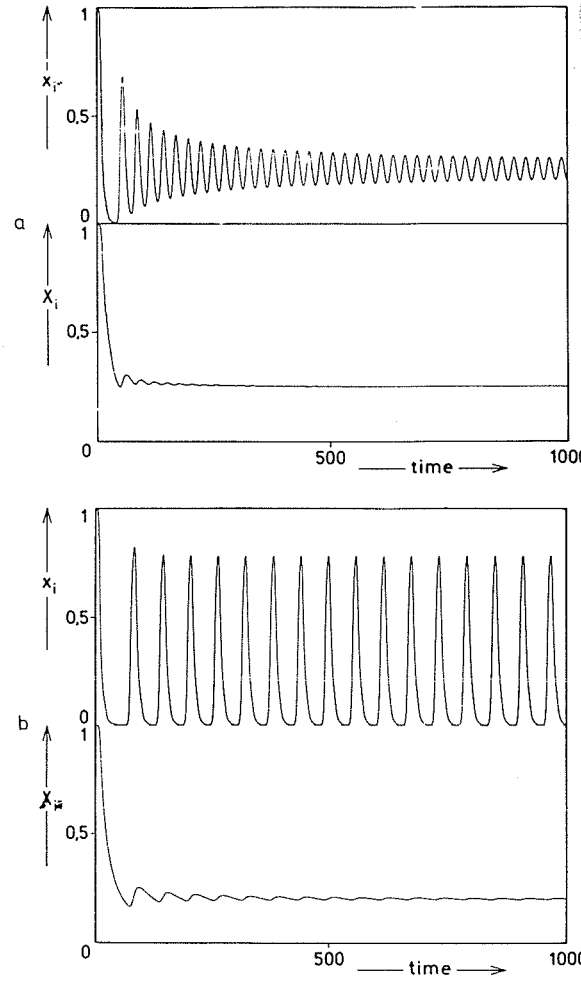


Fig. 42. Solution curves of the dynamic systems for elementary hypercycles of dimension $n=4$ and $n=5$ with equal rate constants and time-averaged concentrations $X(t)$; (a) $n=4$ and b) $n=5$. Note that $X(t)$ reaches $\bar{X}$ after a few oscillations, i.e., internal equilibrium is established relatively fast in both examples

Table 12. Instabilities in the dynamic systems for hypercycles under unlimited growth conditions

Dimension	Boundary and initial conditions			Critical time constants	
n	Rate constant k	Initial concentration $c(0)$	Initial distribution[a] $x(0)$	At equilibrium t^e_∞	Far off equilibrium t^n_α
2	$\frac{1}{2}$	0.55	(0.5, 0.05)	3.64	5.0
3	$\frac{1}{3}$	0.60	(0.5, 0.05, 0.05)	5.00	6.8
4	$\frac{1}{4}$	0.65	(0.5, 0.05, 0.05, 0.05)	6.15	7.3

[a] The distribution of initial concentrations applied in the numerical integration of the system far off equilibrium $\mathbf{x}(0)=(x_1(0), x_2(0)\ldots)$.

Selection among entire hypercycles as single entities can be studied generally under the assumption of internal equilibrium. The dynamic systems obtained thereby are, of course, identical with those describing independent competitors that are characterized by quadratic growth rates. Competition between non-equilibrated hypercycles is more difficult to investigate, since only numerical integration of the systems of differential equations is possible. An example has been treated elsewhere [53], demonstrating that the assumption of internal equilibration represents a powerful approximation.

As an example for competing systems we consider the two hypercycles H_A and H_B with n_A and n_B members subject to the constraints of constant organization.
If there is internal equilibration the system reduces to two competitors with quadratic growth-rate terms. From the results of fixed-point analysis we recall that hypercycle H_A will be selected when its relative initial concentration $c_A(0)$ exceeds a critical limit:

$$\lim_{t\to\infty} c_A(t)=c_0 \quad \text{if } c_A(0)>\frac{k_B}{k_A+k_B}c_0 \tag{86}$$

Otherwise hypercycle H_B wins the competition.
It seems illustrative to consider one more special case. We assume the individual rate constants to be very similar within a given hypercycle, i.e., $k_1\sim k_2\sim\cdots\sim k_n=\bar{k}_A$ and $k_{n+1}\sim k_{n+2}\sim\cdots\sim k_{n+m}=\bar{k}_B$. Then the rate constants for entire hypercycles are obtained as follows:

$$k_A=\frac{1}{n_A}\bar{k}_A \quad \text{and} \quad k_B=\frac{1}{n_B}\bar{k}_B \tag{87}$$

As we see, the constants are inversely proportional to the numbers of members in the hypercycle and consequently, smaller cycles seem to have a certain selective advantage. If we assume, however, all macromolecules to be present at roughly the same concentrations ($\bar{x}$), the disadvantage of the larger cycle is compensated exactly by a larger value of the total concentration, c:

$$\begin{aligned} &c_A(0)=n_A\cdot\bar{x}, \quad c_B(0)=n_B\cdot\bar{x} \quad \text{and} \quad c_0=(n_A+n_B)\bar{x} \\ &\lim_{t\to\infty} c_A(t)=c_0 \quad \text{if } \bar{k}_A>\bar{k}_B \end{aligned} \tag{88}$$

Therefore, the chance of survival is roughly the same for hypercycles of different sizes or dimension n, provided the initial concentrations of the individual members and the rate constants for the replication steps are equal.
The results obtained for two hypercycles can be generalized easily to N independent competitors.

X.2. Parasitic Coupling and Catalytic Networks

The cyclically closed catalytic link, which connects all active members $I_1 \ldots I_n$ of a hypercycle might include branching points and thereby provide a furthering of external species $I_{k\neq 1\ldots n}$, not being an intrinsic part of the cooperative unit. We call these external members parasites. To make an analytical treatment feasible we shall assume internal equilibrium within the cycle (Table 13). Two dynamic systems describing a hypercycle and a single-membered parasite have been investigated by the fixed-point method.
The first example is represented by a catalytic hypercycle and a parasite that is not capable of self-replication (Fig. 43a). Above a certain threshold value of total concentration ($k_A c_0 > k$), as we can see from Table 13, hypercycle and parasite are present with nonzero concentration at the stationary state. The equilibrium concentration of the hypercycle grows with increasing c_0, whereas the concentration of the parasite remains constant. At high enough concentration, consequently, the parasite will lose its importance for the dynamics of the cycle completely. At low total concentration ($k_A c_0 < k$) the system becomes unstable. Within the limits of the assumption of internal equilibrium the parasite destroys the hypercycle and finally represents the only remnant of the dynamic system.
The second case describes the development of a hypercycle with a self-replicative parasite attached to it (Fig. 43b). This dynamic system is characterized by sharp selection depending on the relative values of the rate constants k and k_A. For $k > k_A$ the parasite destroys the hypercycle whereas the inequality $k < k_A$ implies that the parasite dies out. It might be of some interest to consider the dynamic system explicitly on the level of individual polynucleotides. From Table 13 we obtain

$$k=k_x\frac{x_v}{c_A}=k_x\frac{k_{v+1}^{-1}}{\sum_i k_i^{-1}} \tag{89}$$

under the condition of established internal equilibrium. Using the previously derived expression

$$k_A=(\sum_i k_i^{-1})^{-1}$$

we find:

$$k<k_A \rightarrow \frac{k_x}{k_{v+1}} \cdots \frac{1}{\sum_i k_i^{-1}} < \frac{1}{\sum_i k_i^{-1}} \rightarrow k_x < k_{v+1} \qquad (90)$$

Thus the result of selection is determined completely by the ratio of the two rate constants for the reaction steps starting out at the branching point, no matter what the values of the other rate constants in the hypercycle are.

Numerical integration of dynamic systems of the types shown in Figure (43) has been performed in order to study the influence of deviations from the equilibrium distribution. In fact all the conclusions drawn here can also be verified for systems far off equilibrium.

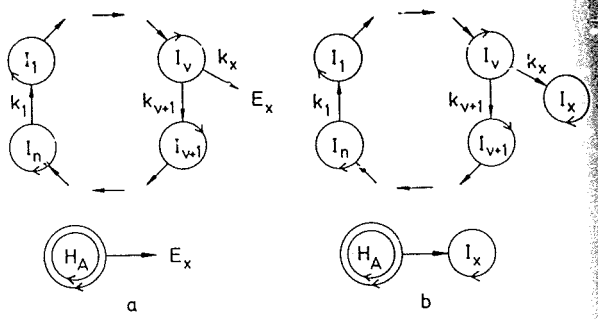


Fig. 43. Schematic diagrams for hypercycles with parasitic units, a) the parasite is not self-reproductive, b) the parasite is self-reproductive. The branching is assumed to occur at the component I_v

Table 13. Fixed-point analysis of hypercycles with parasitic couplings

A hypercycle with a parasitic unit I_x attached to it (Fig. 43) is described by $n+1$ differential equations, n of which are identical with those of the isolated hypercycle. For the $(n+1)$-th differential equation, which defines the concentration of the parasite, we obtain: $[I_x]=x$, $[I_v]=x_v$,

$$\dot{x}=k_x x_v-\frac{x}{c_0}\phi \qquad (T.13.1)$$

for the system depicted in Figure 43a and

$$\dot{x}=k_x x_v x-\frac{x}{c_0}\phi \qquad (T.13.2)$$

for the one exemplified in Figure 43b.
If an equilibrium is established within the hypercycle then the $n+1$ differential equations reduce to two. Here the introduction of new rate constants turns out to be appropriate:

$$k=k_x\cdot\frac{x_v}{c_A};\quad c_A=\sum_{j=1}^{n} x_j \qquad (T.13.3)$$

and k_A as defined in Equation (84).
(1) The parasite is not self-reproductive (cf. Fig. 43a):

$$\dot{c}_A=k_A c_A^2-\frac{c_A}{c_0}(k_A c_A^2+k c_A);\quad c_0=c_A+x \qquad (T.13.4)$$

$$\dot{x}=k c_A-\frac{x}{c_0}(k_A c_A^2+k c_A)$$

This dynamic system has two fixed points:

$$\bar{\mathbf{x}}_1:\ \bar{c}_A=\frac{k_A c_0-k}{k_A},\quad \bar{x}=\frac{k}{k_A} \qquad (T.13.5)$$

$$\omega^{(1)}=-\frac{(k_A c_0-k)^2}{k_A}$$

$$\bar{\mathbf{x}}_2:\ \bar{c}_A=0,\quad \bar{x}=c_0;\quad \omega^{(2)}=0 \qquad (T.13.6)$$

$\bar{\mathbf{x}}_1$ is stable unless the total concentration meets the critical condition $c_0=k/k_A$.
Stability analysis of $\bar{\mathbf{x}}_2$ requires a detailed inspection of the higher-order terms. For a point $x=c_0-\delta x$ we find

$$\dot{x}=\frac{(\delta x)^2}{c_0}(k-k_A c_0+k_A\delta x) \qquad (T.13.7)$$

which leads to

$\dot{x}>0$ for $c_0<k/k_A$

and

$\dot{x}<0$ for $c_0>k/k_A$.

Thus, the fixed point $\bar{\mathbf{x}}_2$ is stable at concentrations below the critical value and unstable above it. At low concentrations $\bar{\mathbf{x}}_1$ lies outside the physically accessible range, $\bar{x}=c_0$ is the only stable stationary state, and the hypercycle is destroyed by the parasite. At high concentrations the stable fixed point $\bar{\mathbf{x}}_1$ lies on the simplex S_2, which means that hypercycle and parasite are coexistent.
(2) The parasite is self-reproductive (Fig. 43b):

$$\dot{c}_A=k_A c_A^2-\frac{c_A}{c_0}(k_A c_A^2+k c_A x),\quad c_0=c_A+x$$

$$\dot{x}=k c_A x-\frac{x}{c_0}(k_A c_A^2+k c_A x) \qquad (T.13.8)$$

The system has two fixed points at the corners of S_2:

$$\bar{\mathbf{x}}_1:\ \bar{c}_A=c_0,\quad \bar{x}=0,\quad \omega^{(1)}=(k-k_A)c_0 \qquad (T.13.9)$$

$$\bar{\mathbf{x}}_2:\ \bar{c}_A=0,\quad \bar{x}=c_0,\quad \omega^{(2)}=0 \qquad (T.13.10)$$

The first fixed point is stable, provided that $k_A>k$. For the second fixed point it is again necessary to check the higher-order terms. At $x=c_0-\delta x$ we find

$$\dot{x}=\frac{k-k_A}{c_0}(\delta x)^2(c_0-\delta x) \qquad (T.13.11)$$

which now leads to

$\dot{x}>0$ for $k>k_A$

and

$\dot{x}<0$ for $k<k_A$.

Thus, $\bar{\mathbf{x}}_2$ is stable if the inequality $k>k_A$ holds. The system is competitive, which signifies that hypercycle and parastic unit cannot coexist except in the special situation where the rate constants are equal $(k=k_A)$.

The results obtained for single-membered parasites can be extended to arbitrary chains using the results derived in Section VII.6. In general, the fate of the entire parasite is strongly coupled to the development of the species attached to the cycle. The parasite will die out always when the concentration of the species after the branching point approaches zero. There is one interesting special case: $k_x = k_{v-1}$. The differential equations for I_{v+1} and I_x are identical and hence the ratio of the two species always remains constant at its initial value. Numerical integration of several dynamic systems of this type showed that in this special situation ($k_x = k_{v+1}$) all members of the parasite besides the species I_x will die out.
Chain-like parasites might fold back on the hypercycle, thereby leading toward a catalytic network with a branching point and a confluent. By numerical integration we found that systems of these types are unstable: The less efficient branch, i.e., the branch with the smaller values of the rate constants k dies out and a single, simple hypercycle remains.

Allowing for arbitrary assignment of catalytic coupling terms to a set of self-reproductive macromolecules we shall encounter highly branched systems or complicated networks much more frequently than regular hypercycles. It is of great importance, therefore, to know the further development of these systems in order to make an estimate of the probabilities of hypercycle formation. Analytical methods usually cannot be applied to this kind of system and hence we have to rely on the results of numerical techniques.
Some general results have been derived from a variety of solution curves obtained by numerical integration of the differential equations for various catalytic networks. As suggested by the previous examples, these systems are not stable and disintegrate to give smaller fragments. Apart from complicated dynamic structures, which owe their existence to accidental coincidence of the numerical values for different rate constants, the only possible remnants of catalytic networks of self-replicative units are independently growing species, catalytic chains, or catalytic hypercycles. Thus any catalytic network consisting of self-replicative units with uniform coupling terms will disintegrate either to yield a hypercycle which then is superior to the other fragments or to give competitive dynamic systems that are not suited for cooperative evolution.

X.3. Hierarchy of Coupling Between Hypercycles

In principle, hypercycles may be coupled to yield more highly organized systems by introduction of appropriate catalytic terms into the rate equation. We consider two basic hypercycles H_A and H_B and assume that the hypercycle H_A produces a catalytic growth factor for H_B and *vice versa*. Such a growth factor might be a constituent of the hypercycle or a substance produced by it. From our previous experience we would expect that mutual catalytic enhancement will lead to cooperative behavior.
To simplify a straightforward analysis, we assume internal equilibrium to be established in both hypercycles. The catalytic terms are of third order with respect to molecular concentrations ($k_A c_A^2 c_B$ and $k_B c_A c_B^2$, respectively, see also Table 14). Consequently, we may neglect the second-order growth functions of the uncatalyzed system at high enough concentrations. Fixed-point analysis is not sufficient to study the dynamic system obtained, since it yields zero eigenvalues for all normal modes. The vector field, however, can be investigated easily because the system has only one

Table 14. Fixed-point analysis of catalytically coupled hypercycles H_A and H_B

(1) Coupling terms of third order:

$$\dot{c}_A = k_A c_A^2 c_B - \frac{c_A}{c_0}\phi; \quad \phi = (k_A c_A + k_B c_B) c_A c_B$$

$$\dot{c}_B = k_B c_A c_B^2 - \frac{c_B}{c_0}\phi; \quad c_0 = c_A + c_B \qquad \text{(T.14.1)}$$

Fixed-point analysis of the dynamic system yields:

$$\bar{x}_1\colon\ \bar{c}_A = c_0, \quad \bar{c}_B = 0; \quad \omega = 0 \qquad \text{(T.14.2)}$$

$$\bar{x}_2\colon\ \bar{c}_A = 0, \quad \bar{c}_B = c_0; \quad \omega = 0 \qquad \text{(T.14.3)}$$

The system is competitive. Analysis of the higher-order terms (Fig. 44) reveals that $\bar{x}_1$ is stable for $k_A > k_B$. The condition $k_A < k_B$, on the other hand, leads to stability of $\bar{x}_2$.

(2) Coupling terms of fourth order:

$$\dot{c}_A = k_A c_A^2 c_B^2 - \frac{c_A}{c_0}\phi; \quad \phi = (k_A + k_B) c_A^2 c_B^2$$

$$\dot{c}_B = k_B c_A^2 c_B^2 - \frac{c_B}{c_0}\phi \quad c_0 = c_A + c_B \qquad \text{(T.14.4)}$$

This dynamic system is characterized by three fixed points:

$$\bar{x}_1\colon\ \bar{c}_A = 0, \quad \bar{c}_B = 0; \quad \omega = 0 \qquad \text{(T.14.5)}$$

$$\bar{x}_2\colon\ \bar{c}_A = 0, \quad \bar{c}_B = c_0; \quad \omega = 0 \qquad \text{(T.14.6)}$$

$$\bar{x}_3\colon\ \bar{c}_A = \frac{k_A}{k_A + k_B} c_0, \quad \bar{c}_B = \frac{k_B}{k_A + k_B} c_0;$$

$$\omega = \frac{k_A^2 k_B^2}{(k_A + k_B)^3} c_0^3 \qquad \text{(T.14.7)}$$

$\bar{x}_3$ thus represents a stable fixed point indicating cooperative behavior of the two coupled hypercycles H_A and H_B under all possible conditions. The vector field shown in Figure 44 reveals that the other two fixed points $\bar{x}_1$ and $\bar{x}_2$ are sources.

degree of freedom. As we see from Figure 44 the two hypercycles still compete despite the presence of the catalytic factors. The kind of catalytic coupling introduced, thus, was not sufficient to cause cooperative behavior.

If we increase the order of the catalytic terms by one, the dynamic system involves fourth-order growth-rate terms ($k_A c_A^2 c_B^2$, $k_B c_A^2 c_B^2$). Analyzing the vector field in the same way as before (Fig. 44), we find a stable fixed point at finite concentrations of both hypercycles (see also Table 14). Thus the quadratic coupling term is sufficient to cause cooperativity among catalytic hypercycles.

The physical realization of this type of catalytic coupling is difficult to visualize at the level of biologic macromolecules: The presence of a term like $k_A c_A^2 c_B^2$ or $k_B c_A^2 c_B^2$ in the overall rate equations requires either a complicated many-step mechanism or an encounter of more than two macromolecules, both of which are improbable*. One is tempted therefore to conclude that further development to more complex structures that consist of hierarchically coupled self-replicative units does not likely occur by introduction of higher-order catalytic terms into a system growing in homogeneous solution, but rather leads toward individualization of the already existing functional units. This can be achieved, for example, by spatial isolation of all members of a hypercycle in a compartment. Formation of prototypes of our present cells may serve as one possible mechanism leading to individualized hypercycles. After isolation is accomplished the individualized hypercycle may behave like a simple replicative unit. Hypercycles therefore are more likely to be intermediates of self-organization than final destinations.

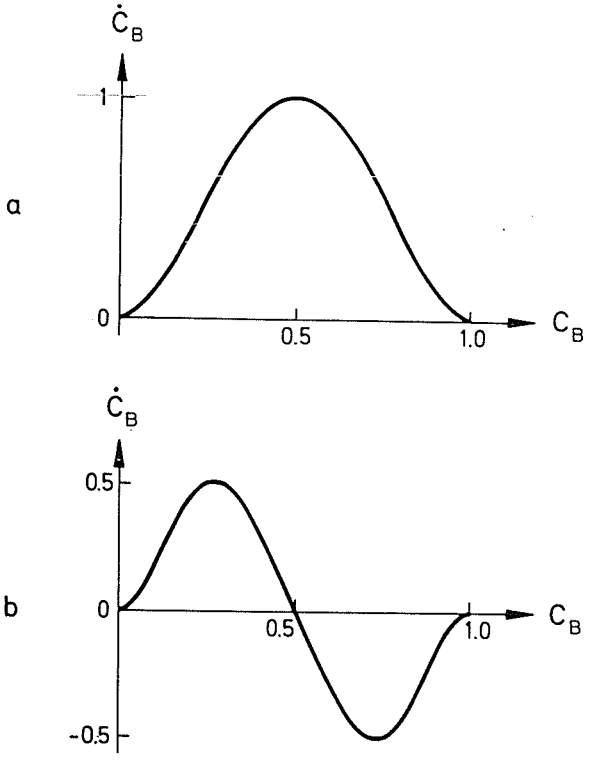


Fig. 44. Coupling between hypercycles. a) Catalytic coupling terms $k_A c_A^2 c_B$ and $k_B c_A c_B^2$, respectively. The tangent vector is positive inside the physically allowed region ($0 < c_B < 1$), except at the two fixed points; $k_B > k_A$ is assumed and consequently the hypercycle c_B is selected. The system is competitive despite the coupling term. b) Catalytic coupling terms $k_A c_A^2 c_B^2$ and $k_B c_A^2 c_B^2$, respectively. The system contains two unstable fixed points at the corners and a stable fixed point at the center ($\bar{c}_A = \bar{c}_B = 0.5$ because $k_A = k_B$). The system is cooperative

Conclusions

The main object of Part B is an abstract comparative study of various functional links in self-replicative systems. The methods used are common in differential topology. Complete analytical solutions – except in special cases – are usually not available, since the differential equations involved are inherently nonlinear. Self-reproduction always induces a dependence of production rates on population numbers of the respective species. Cooperation among different species via encoded functional linkages superimposes further concentration terms, which lead to higher-order dependences of rates on population variables.

A comparative analysis of selective and evolutive behavior does not require a knowledge of the complete solution curves. Usually it is sufficient to find their final destinations in order to decide whether or not stable coexistence of all partners of a functionally cooperative ensemble is possible. Fixed-point analysis, aided by Lyapounov's method and – in some cases – by a more detailed inspection of the complete vector field, serves the purpose quite well. The results of the combined analysis may be summarized as follows:

Functional integration of an ensemble consisting of several self-replicative units requires the introduction of catalytic links among all partners. These linkages, superimposed on the individual replication cycles of the subunits, must form a closed loop, in order to stabilize the ensemble via mutual control of all population variables. Independent competitors, which under certain spatial conditions and for limited time spans may coexist in 'niches', as well as catalytic chains or branched networks

* Artificial dynamic systems that are based on technical devices to introduce catalytic coupling terms like, e.g., electric networks may not encounter these difficulties.

are devoid of self-organizing properties, typical of hypercycles. Mere coexistence is not sufficient to yield coherent growth and evolution of all partners of an ensemble. In particular, the hypercycle is distinguished by the following properties:

1. It provides stable and controlled coexistence of all species connected via the cyclic linkage.

2. It allows for coherent growth of all its members.

3. The hypercycle competes with any single replicative unit not belonging to the cycle, irrespective of whether that entity is independent, or part of a different hypercycle, or even linked to the particular cycle by 'parasitic coupling'.

4. A hypercycle may enlarge or reduce its size, if this modification offers any selective advantage.

5. Hypercycles do not easily link up in networks of higher orders. Two hypercycles of degree p need coupling terms of degree 2p in order to stabilize each other.

6. The internal linkages and cooperative properties of a hypercycle can evolve to optimal function. 'Phenotypic' advantages, i.e., those variations which are of direct advantage to the mutant, are immediately stabilized. On the other hand, 'genotypic' advantages, which favor a subsequent product and hence only indirectly the replicative unit in which the mutation occurred, require spatial separation for competitive fixation.

7. Selection of a hypercycle is a 'once-for-ever' decision. In any common Darwinian system mutants offering a selective advantage can easily grow up and become established. Their growth properties are independent of the population size. For hypercycles, selective advantages are always functions of population numbers, due to the inherently nonlinear properties of hypercycles. Therefore a hypercycle, once established, can not easily be replaced by any newcomer, since new species always emerge as one copy (or a few).

All these properties make hypercycles a unique class of self-organizing chemical networks. This in itself justifies a more formal inspection of their properties – which has been the object of this Part B. Simple representatives of this class can be met in nature, as was shown in Part A. This type of functional organization may well be widely distributed and play some role in neural networks or in social systems. On the other hand, we do not wish to treat hypercycles as a fetish. Their role in molecular self-organization is limited. They permit an integration of information, as was required in the origin of translation.

However, the hypercycle may have disappeared as soon as an enzymic machinery with high reproduction fidelity was available, to individualize the integrated system in the form of the living cell. Individualized replicative systems have a much higher potential for further diversification and differentiation.

There are many forms of hypercyclic organization ranging from straightforward second-order coupling to the nth *order compound hypercycle in which cooperative action of all members is required for each reaction step. While we do not know of any form of organization simpler than a second-order hypercycle that could initiate a translation apparatus, we are well aware of the complexity of even this 'simplest possible' system. It will therefore be our task to show in Part C that realistic hypercycles indeed can emerge from simpler precursors present in sufficient abundance under primordial conditions.*

C. The Realistic Hypercycle

The proposed model for a 'realistic hypercycle' is closely associated with the molecular organization of a primitive replication and translation apparatus. Hypercyclic organization offers selective stabilization and evolutive adaptation for all geno- and phenotypic constituents of the functionally linked ensemble. It originates in a molecular quasi-species and evolves by way of mutation and gene duplication to greater complexity. Its early structure appears to be reflected in: the assignment of codons to amino acids, in sequence homologies of tRNAs, in dual enzymic functions of replication and translation, and in the structural and functional organization of the genome of the prokaryotic cell.

XI. How to Start Translation?

"The origin of protein synthesis is a notoriously difficult problem. We do not mean by this the formation of random polypeptides but the origin of the synthesis of polypeptides directed, however crudely, by a nucleic acid template and of such a nature that it could evolve by steps into the present genetic code, the expression of which now requires the elaborate machinery of activating enzymes, transfer RNAs, ribosomes, factors, etc."
Our subject could not be characterized more aptly than by these introductory phrases, quoted from a recent paper by F.H.C. Crick, S. Brenner, A. Klug and G. Pieczenik [3].

Let us for the time being assume that a crude replication and translation machinery, functioning with adequate precision, and adapted to a sufficiently rich alphabet of molecular symbols, has come into existence by some process not further specified, e.g., by self-organization or creation, in Nature or in the laboratory. Let us further suppose an environment which supplies all the activated, energy-rich material required for the synthesis of macromolecules such as nucleic acids and proteins, allowing both reproduction and translation to be spontaneous processes, i.e., driven by positive affinities. Would such an ensemble, however it came into existence, continue to evolve as a Darwinian system? In other words, would the system preserve indefinitely the information which it was given initially and improve it further until it reaches maximal functional efficiency?
In order to apply this question to a more concrete situation let us consider the model depicted in Figure 45. The plus strands of a given set of RNA molecules contain the information for a corresponding number of protein molecules. The products of translation can fulfill at least the following functions: (1) One protein acts as an RNA-polymerase similar to the specific replicases associated with various RNA phages. Its recognition site is adapted to a specific sequence or structure occurring in all plus and minus strands of the RNAs; in other words, it reproduces efficiently only those RNA molecules which identify themselves as members of the particular ensemble. (2) The other translation products function as activating enzymes, which assign and link various amino acids uniquely to their respective RNA adaptors, each of which carries a defined anticodon. The number of different amino acids and hence of adaptors is adjusted to match the variety of codons appearing in the messenger sequences, i.e., the plus strands of the RNAs,

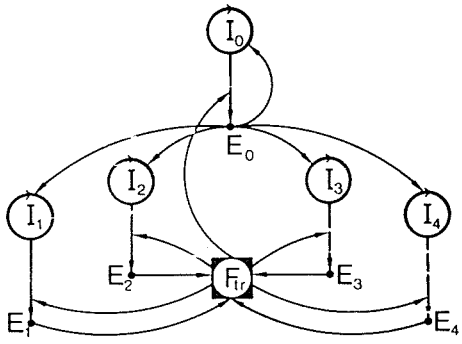


Fig. 45. A minimum model of primitive translation involves a messenger I_0 encoding a replicase E_0, which is adapted to recognize specifically the sequences I_0 to I_4. The plus strands of I_1 to I_4 encode four synthetase functions E_1 to E_4, while the minusstrands may represent the adapters (tRNAs) for four amino acids. Such a system, although it includes all functions required for translation and self-reproduction, is unstable due to internal competition. Coherent evolution is not possible, unless I_0 to I_4 are stabilized by a hypercyclic link

so as to yield a 'closed' translation system with a defined code. It does not necessarily comprise the complete genetic code, as it is known today, but rather may be confined to a – functionally sufficient – smaller number of amino acids (e.g., four), utilizing certain constraints on the codon structure in order to guarantee an unambiguous read-off. The adaptors may be represented by the minus strands of the RNA constituents, or, if this should be too restrictive a condition, they could be provided along with further machinery, such as ribosomes, in the form of constant environmental factors similar to the host factors assisting phage replication and translation inside the bacterial cell.

At first glance, we might find comfort in the thought that the system depicted in Figure 45 appears to be highly functionally interwoven; all I_i are supported catalytically by the replicase E_0, which in turn owes its existence to the joint function F_t of the translation enzymes E_1 to E_4 without which it could not be translated from I_0. The enzymes E_1 to E_4, of course, utilize this translation function for their own production too, but being the translation products of I_1 to I_4, they are finally dependent also upon E_0 or I_0 respectively.

However, a detailed analysis shows that the couplings present are not sufficient to guarantee a mutual stabilization of the different genotypic constituents I_i. The general replicase function exerted by E_0 and the general translation function F_{tr} are represented in all differential equations by the same term. The equations then reduce to those for uncoupled competitors, multiplied by a common time function $f(t)$. The system, which initially functions quite well, is predestined to deteriorate, owing to internal competition. A typical set of solution curves, obtained by numerical integration of the rate equations, is shown in Figure 46.

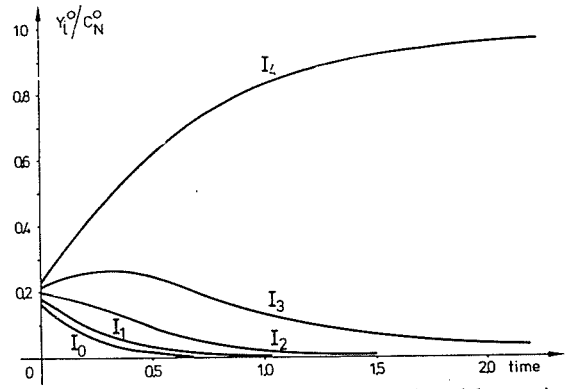


Fig. 46. Solution curves for a system of differential equations simulating the model represented in Figure 45. In this particular example, it is assumed that initial concentrations and autocatalytic-reproduction-rate constants increase linearly from I_0 to I_4, while the other parameters – such as translation-rate constants ($I_i \overset{k_i}{\rightarrow} E_i$), amino acid assignments (contribution of E_1, E_2, E_3 E_4 to F_{tr}) or enzyme-substrate-complex stabilities ($I_i + E_0 \overset{K_i}{\rightleftharpoons} I_i \cdot E_0$), etc. – are identical for all reaction partners. The time course of the relative population numbers (y_i^0/c_N^0) reflects the competitive behavior. The most efficiently growing template (I_4) will supersede all others and finally dominate ($y_4^0/c_N^0 \rightarrow 1$). However, since both replication (represented by E_0) and translation function (contributions of E_1, E_2 and E_3 to F_{tr}) disappear, I_4 will also die out. The total population is bound to deteriorate ($c_N^0 \rightarrow 0$)

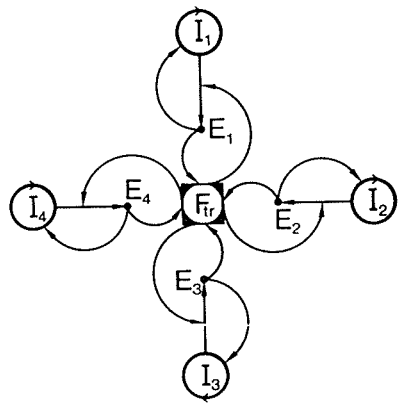


Fig. 47. In this alternative model for primitive replication and translation, the enzymes E_1 to E_4 are assumed to have dual functions, i.e., as specific replicases of their own messengers and as synthetases for four amino acid assignments. The fate of the system is the same as that of the system depicted in Figure 45, since the messengers are highly competitive

Another example of this kind is represented in Figure 47. Here all messengers produce their own specific replicases E_1 to E_4, which also provide synthetase functions (F_{tr}). Again, this coupling by means of a concomitant translation function does not suffice to stabilize the ensemble. The answer to our question, whether the mere presence of a system of messengers for replicase and translation functions and of translation products is sufficient for its continuous existence and evolution, is that unless a particular kind of coupling among the different replicative constituents I_i is introduced, such systems are not stable, despite the fact that they contain all required properties for replication and translation. Even if all partners were selectively equivalent (or nearly equivalent) and hence were to coexist for some time (depending on their population size), they could not evolve in a mutually controlled fashion and hence would never be able to optimize their functional interaction. Their final fate would always be deterioration, since an occasional selective equivalence cannot be coherently maintained over longer periods of evolution unless it is reinforced by particular couplings.

Knowing the results of part B, we are, of course, not surprised by this answer. A closer inspection of the particular linkages provided by the functions of replication and translation enzymes does not reveal any hypercyclic nature. Therefore these links cannot establish the mutual control of population numbers that is required for the interrelated evolution of members of an organized system. The couplings present in the two systems studied can be reduced to two common functions, which, like environmental factors, influence all partners in exactly the same way and hence do not offer any possibility of mutual control.

The above examples are typical of what we intend to demonstrate in this article, namely, that

1. In the early phases of evolution, characterized by low fidelities of replication and translation as well as by the initially low abundance of efficiently replicating units, hypercyclic organization offers large relative advantages over any other kind of (structural) organization (Sect. XV), and

2. That hypercyclic models *can* indeed be built to provide realistic precursors of the reproduction and translation apparatus found in present prokaryotic cells (Sect. XVI).

How could we envisage an origin of translation, given the possible existence of reproducible RNA molecules as large as tRNA and the prerequisites for the synthesis of proteins in a primitive form, utilizing a limited number of (sufficiently commonly occurring) amino acids?

XII. The Logic of Primordial Coding

XII. 1. The RRY Code

A most appealing speculative model for the origin of template-directed protein synthesis, recently proposed [3], is based on a number of logical inferences that are related to the problem of comma-free and coherent read-off. A primordial code must have a certain frame structure, otherwise a message cannot be read off consistently. Occasional phase slips would produce a frame-shifted translation of parts of the message and thereby destroy its meaning. The authors therefore propose a particular base sequence to which all codons have to adhere. Or, in other words, only those sequences of nucleotides that resemble the particular pattern could become eligible for messenger function. Uniformity of pattern could arise through instruction conferred by the exposed anticodon loop of tRNAs as well as by internal self-copying. Among the possible patterns that guarantee nonoverlapping read-off, the authors chose the base sequence purine-purine-pyrimidine, or, in the usual notation, RRY, to be common to all codons specifying a message. The particular choice was biased by a sequence regularity found in the anticodon loop of present tRNAs, which reads 3′NR$\alpha\beta\gamma$UY, $\alpha\beta\gamma$ being the anticodon, N any of the four nucleotides, and R and Y a purine and a pyrimidine, respectively. Another prerequisite of ribosome-free translation is the stability of the complex formed by the messenger and the growing polypeptide chain. A peptidyl-t-RNA must not fall off before the transfer to the subsequent aminoacyl-tRNA has been accomplished, that is, until the complete message is translated. Otherwise, only functionally inefficient protein fragments would be obtained. It is obvious from known base-pair stabilities that a simple codon-anticodon interaction does not guarantee the required stability of the messenger-tRNA complex. Therefore the model was based essentially on three auxiliary assumptions.

1. The structure of the anticodon loop of the adaptor (tRNA precursor) is such, that—given the particular and common codon pattern—an RNA can always form five base pairs with the messenger. The primitive tRNA is then assigned the general anticodon-loop sequence



where YYR is the anticodon.

2. The anticodon loop of each primitive tRNA can assume two different conformations, which are detailed in Figure 48. Both configurations had been described in an earlier paper by C. Woese [60] who

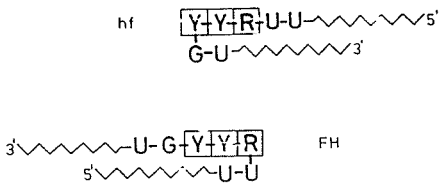


Fig. 48. Two possible configurations of the anticodon loop of tRNAs: FH according to Fuller and Hodgson [61] and hf according to Woese [60]. The anticodon pattern (framed) refers to the model of Crick et al. [3]

named them FH and hf. (FH refers to Fuller and Hodgson [61] who originally proposed that five bases at the 3′-end of the unpaired seven-base sequence in the anticodon loop are stacked on top of each other, while hf, according to Woese, designates a complementary configuration keeping the five bases at the 5′-end of the loop in a stacked position.) Woese assumed that the transition between both configurations plays an important role in ribosomal protein synthesis, but also referred to its possible significance in past, more primitive mechanisms.

3. Which of the two configurations is actually present depends on whether the tRNA is attached to an amino acid or to a peptide chain. By transferring an amino acid to the peptide chain, the tRNA flips from the hf to the FH configuration (cf. Fig. 49).

An additional, fourth postulate, not absolutely necessary as a prerequisite of the model, invokes an interaction between two adjacent tRNAs at the messenger, which assures that the required configration will be energetically the most favorable and contributes further to the stability of the polypeptide-messenger complex.

Figure 49 shows in more detail how polypeptide synthesis may be facilitated on the basis of the arguments given. The growing polypeptide chain is transported along the messenger, utilizing as 'fuel' the free energy of the transfer reaction. This reaction may be aided by a general nonspecific catalyst, but the mechanism does not require any sophisticated machinery, such as is nowadays provided by the ribosomes. Although interaction between codon and anticodon is stabilized by five base pairs, it is essential that the actual code make use of base triplets. It had been emphasized previously [62] that a primitive code utilizing anything but three bases is useless in explaining the present code.

The model explains well how in the absence of a sophisticated translation apparatus a message can be read off

spontaneously,
sequentially,
completely, i.e., unfragmented, and

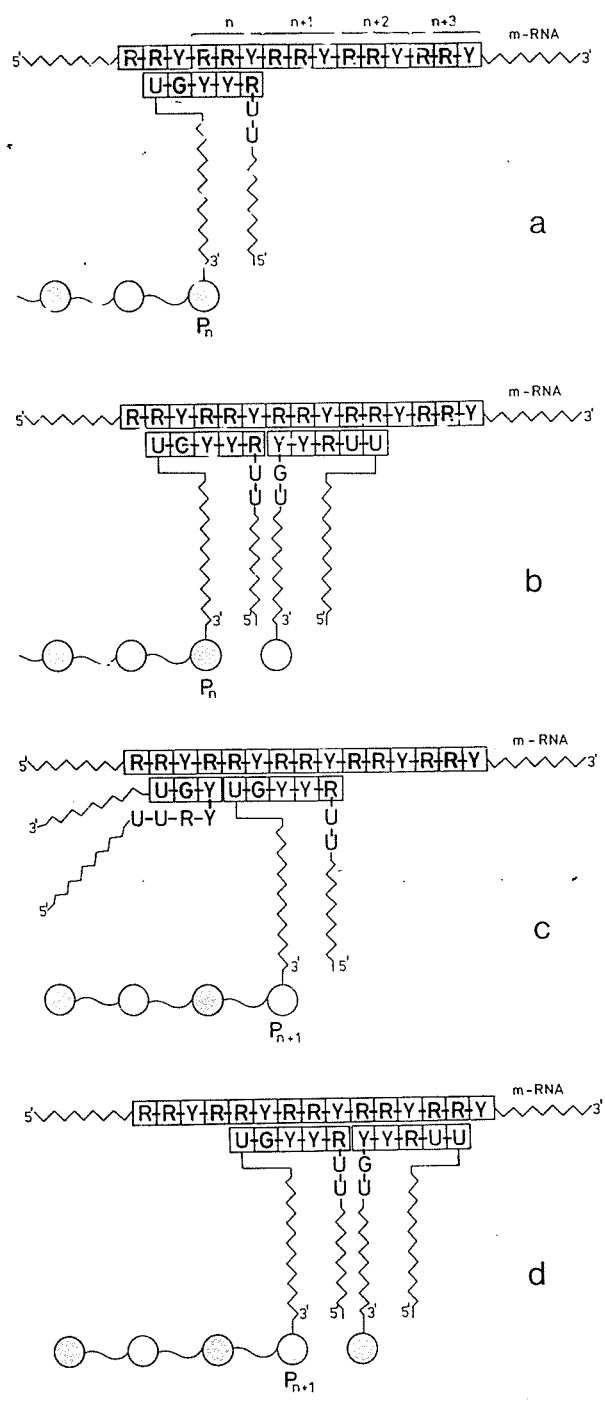


Fig. 49. The primitive translation mechanism requires 'sticky' interactions between the messenger and the peptidyl-tRNA. It thereby allows the growing peptide chain to remain in contact with the message until translation is completed. According to Crick et al. [3] the transport is effected by a flip mechanism involving conformational changes of the tRNA (FH⇌hf). The nascent peptide chain is always connected with the messengers via five base pairs with some additional stabilization by the adjacent aminoacyl-tRNA. The partial overlap of base pairing guarantees a consistent reading of a message encoded in base triplets

reproducibly, i.e., strictly maintaining a given codon frame.

The code is inherently related to certain structural features of the anticodon loop of present tRNAs, suggesting that these molecules are the descendents of the first functionally organized entities. The four amino acids assigned by this model are:

GG^{U}_{C} ; GA^{U}_{C} ; AG^{U}_{C} ; AA^{U}_{C} ;
glycine aspartic acid serine asparagine

Some of which were very abundant in the primitive soup [63].

On the other hand, the model also introduces some difficulties. A uniform RRY sequence has a large excess of purine over pyrimidine and therefore does not easily lend itself to stable internal folding.

As a consequence, such sequences

are quite labile toward hydrolysis
(if present as single strands),
have a greater tendency to form duplices
(which do not easily replicate by primitive mechanisms),
lack internal symmetry, and
produce minus strands of a different general code pattern (i.e., 5′RYY).

XII.2. The RNY Code

Before going into a more detailed discussion of the points listed above, let us now offer an alternative model that is free of these particular difficulties. The suggestion of using the general codon pattern RNY, where N stands for any of the four nucleotides A, U, G, C, is also to be credited to Crick et al. [1]. However, it was disfavored by its authors on the grounds of a disadvantage: If N represents a pyrimidine, then the anticodon loop, having the general sequence 3′ⓊGYNRUⓊ, can in some cases use only its five central nucleotides to form stable base pairs with the messenger. This argument may, however, be counteracted by the observation that an RNY code can assign eight amino acids, so that one may exclude certain combinations that do not fulfil the stability requirements for the messenger-peptidyl-tRNA complex (cf. below).

What are the advantages of a general code pattern RNY? First of all, the RNY code, like its RRY analog, is comma-free. Moreover, it is symmetric with respect to plus and minus strands. If read from 5′ to 3′, the general frame structure of both the plus and the minus strand is $R^{N}_{N'}Y$ where N and N′ are complementary, situated at mirror-image positions in the sequences of both strands. Similar symmetries can also develop internally within a single strand, allowing the formation of symmetric secondary folding structures. Typical examples of (almost) symmetric foldings are the present-day tRNAs. Single-stranded phage genomes and their derivatives (such as the midi-variant of Qβ-RNA) are also distinguished by such elements of symmetry. Here the selective advantage of a symmetric structure is obvious. If the molecules are to be reproduced by a polymerase, which recognizes specifically some structural feature, only a symmetric structure would allow the plus and the minus strands to be equally efficient templates. Such an equivalence of efficiency is required for selection. Thus the symmetry of tRNA may well be a relic from a time when these molecules still had to reproduce autonomously.

Internal folding also enhances the molcule's resistance to hydrolysis and offers an easy way of instructing the correct read-off of the message. In an open structure in which many nucleotides remain unpaired (e.g., for RRY sequences), replication and translation could start at any unpaired position of the sequence, leading to fragmentary products. In a completely paired structure, unmatched sticky ends are predestined to be the starting-points of replication and translation. In this way the complete message can be read off, requiring only transient partial unfoldings of the template structure, which may be enforced by interactions with the growing chain. Symmetry, although not absolutely required, would in this case again be of advantage (cf. Fig. 50).

From a logical point of view the RNY code seems to be more attractive than the RRY code, on the basis of three arguments.

1. Selective enhancement of RNA molecules must be effective for the plus as well as the minus strand. Symmetric RNY patterns can fulfil this requirement more easily than RRY sequences, which differ from their minus strands (RYY) and hence cannot both become equivalent targets for specific recognition by enzymes.

2. In view of the high complexity, there is little chance finding the very few sequences that offer useful properties for replication and translation. If these sequences, being symmetric structures, fulfil the requirements listed under (1), both the plus and the minus strands may be candidates for representing such functions.

3. Evolution of the translation apparatus with its various tRNAs and messengers requires a mutual stabilization of all replicative molecules. As will be shown below, hypercycles may emerge more easily from a quasi-species if this, on account of its symmetry, offers two complementary functions.

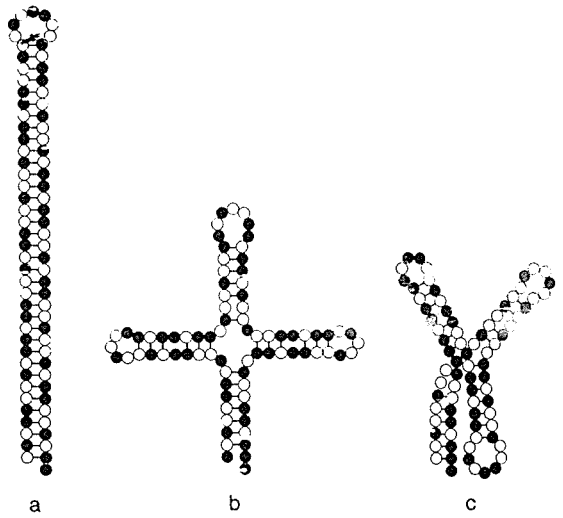


Fig. 50. Symmetric secondary structures of RNA require a corresponding internal complementarity (cf. Fig. 14 in Part A). Plus and minus strands will then exhibit similar foldings. RNY-codon patterns are able to produce such structures. A game has been devised, the rules of which take into account the physical interactions observed with oligonucleotides and tRNAs [17, 18]. It shows which of the structures is most likely to occur. Hairpins require two complementary halves of the molecule, e.g., a 5′-RRY sequence linked up to its minus strand involving a 5′RYY pattern

A system can accumulate information and eventually evolve to higher complexity only if it adopts the 'Darwinian logic' of selective self-organization. However, this logic must find its basis and its expression in material properties. All that the constituents can recognize at the beginning is natural abundance and strength of interaction. These are the properties with which we shall have to deal in order to understand the start of translation.

XIII. Physics of Primordial Coding

XIII.1. The Starting Conditions

Self-organization as a multimolecular process requires monomers as well as polymers to be present in sufficiently high concentrations. Its onset therefore must have been preceded by an extended phase of prebiotic synthesis, during which all the material necessary for yielding a 'highly enriched soup' was accumulated. We do not intend to dwell on these processes of primordial chemistry, nor do we quarrel about details of historical boundary conditions. Questions as to whether the 'soup' originated in the oceans, in a pond, or even in small puddles, or whether interfaces, coarse-grained, or porous surfaces were involved, may be important if absolute rates of the *historical* processes are to be estimated.

Here we simply start from the assumption that when self-organization began all kinds of energy-rich material were ubiquitous, including in particular:
amino acids in varying degrees of abundance,
nucleotides involving the four bases A, U, G, C,
polymers of both preceding classes, i.e., proteinoids as well as tRNA-like substances, having more or less random sequences.

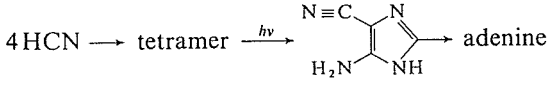
'Less random' in this context means the existence of nearest-neighbor and more complex folding interactions, leading to a preference of certain structures, while 'more random' refers to their primary unrelatedness to any functional destination, which—if initially present—could only be coincidental.

On the other hand, we definitely do not suppose the presence of any adapted protein machinery, such as specific polymerases,
adapted synthetases, or any of the ribosomal functions.

This does not exclude an involvement of non-instructed, poorly adapted protein catalysts, in facilitating the start of replication and translation. However, not being able to reproduce and improve selectively, those proteins must be subsumed together with other catalytic surfaces under 'constant environmental factors'.

XIII.2. Abundance of Nucleotides

Since nucleic-acid-like structures are the only ones that offer inherent self-reproductive properties, it is important to analyze first their abundances as well as their mutual interactions in more detail.
The monomeric nucleotides, especially their energy-rich oligophosphate forms, are more difficult to obtain (using possible prebiotic mechanisms) than are amino acids. Quantitative statements of relative abundance are therefore scarce. S. Miller and L. Orgel ([63], p. 104) emphasize the central role of adenine nucleotides both in genetic processes as well as in energy transfer and correlate it with the relative ease with which this substance is formed. J. Oró and his co-workers [64] found that adenine can be obtained in yields of 0.5% in concentrated aqueous solutions of ammonium cyanide, while Miller and Orgel ([63], p. 105) showed that even hydrogen cyanide alone, in a reaction catalyzed by sunlight, yields the important intermediate



The same intermediate can also react with cyanate, urea, or cyanogen to give guanine. Less well understood are the mechanisms of pyrimidine synthesis. A pathway could be demonstrated for the synthesis of cytosine, using cyanoacetylene – an electric-discharge product of mixtures of methane and nitrogen – in combination with cyanate. Uracil, on the other hand, appears to be a hydrolysis product of cytosine and may possible owe its primordial existence to this source.

Only little can be said about the primordial natural abundance of the purines and pyrimidines. The rate of template-instructed polymerization is proportional to the concentration of the monomer to be included. For complementary instruction, at least two nucleotides are required and they will have to be equivalently represented in informational sequences. Therefore the inclusion of the less abundant nucleotide will always be the rate-limiting step, at least for chain elongation. A possible large excess of A over U in the primordial monomer distribution would then have been of little help in favoring AU over GC copolymers, except in cases where the nucleation of oligo-A primers is the rate-limiting step. The capacity for replicative growth is limited to the template function of the less abundant member of the complementary nucleotide pair. If under primordial conditions the abundance of G and C was intermediate to that of A and U, then GC- and AU-rich copolymers may well have formed with comparable rates.

We therefore cannot maintain the earlier, speculative view that the first codons were recruited exclusively from a binary alphabet, made up of AU copolymers alone.

XIII.3. Stability of Complementary Structures

The stability of base pairing may provide clues that are more illuminating with respect to the question of the first codons. Stabilities and rates of base pairing have been studied using various nucleotide combinations. These results have been discussed in detail in earlier reviews [44, 4]. They reveal quantitatively the generally accepted view that GC pairs within a cooperative stack provide considerably more stability than AU pairs.

The stability constant of a continuous and homogeneous oligomeric sequence of n nucleotide pairs can be represented by the relation

$$K_n = \beta s^n \qquad (91)$$

which refers to a linear Ising model. The factor β is a cooperativity parameter, which for both the AU and GC pairs amounts to an order of magnitude of 10^{-3}*, while s is the stability constant of a single pair within the cooperative stack. For homogeneous sequences of AU pairs this parameter s is about one order of magnitude smaller than for homogeneous sequences of GC pairs, or in a rough quantitative representation:

$$s_{AU} \approx 10 \quad \text{while} \quad s_{GC} \approx 100.$$

Higher absolute stabilities than predicted by relation (91) are found if one of the complementary strands can assume the particular stacking configuration present in the anticodon loop of tRNA. Presumably the cooperativity parameter β is changed in this case. Yet O.C. Uhlenbeck, J. Batter and P. Doty [65, 66] found that tri- and tetranucleotides, complementary to the anticodon region of a tRNA and differing in one AU pair, exhibit a difference of one order of magnitude in their stability constants, quite in agreement with the figure quoted above. It is also reasonable that the largest absolute values of stability constants found are those for the interaction of two tRNAs that are complementary in their anticodons [67].

The data obtained with defined short sequences may serve at least for a comparison between various replication and translation models and for conclusions about their relative significance. It is obvious that single isolated AU or GC pairs are unstable under any realistic conditions of concentration. The start of replication, therefore, requires special help in the form of primer formation, and it is particularly this step that calls first for enzymic support. Present-day phage RNA replicases are also specifically adapted to a primary sequence pattern of the phage genome. For chain elongation, the incoming nucleotide is bound cooperatively on top of a stack of base pairs of the growing chain. Here the data suggest that the GC pair is about ten times more stable than the AU pair, resulting in a relatively higher fidelity q for G and C than for A and U. If the rate of replication is limited by the formation of the covalent link in the polynucleotide backbone (rather than by base-pair formation) the fidelity can be correlated with the monomer concentrations m_R and m_Y and the pair-stability constants K_{RY}, K_{RR}, and K_{YY}. The reproduction fidelity for any given nucleotide then may be obtained from the geometric mean of the fidelities for both complementary processes,

$\bar{R} \rightarrow \bar{Y}$; $\bar{Y} \rightarrow \bar{R}$:

$$q_{RY} = \frac{m_Y K_{RY}}{\sum_N m_N K_{RN}} \quad \text{and} \quad q_{YR} = \frac{m_R K_{YR}}{\sum_N m_N K_{YN}} \qquad (92)$$

* Such a relation is formally valid for both the internal base pairing within a given sequence and the binary association of two complementary sequences, where β has the dimension M^{-1}.

where $K_{RY} \approx K_{YR}$ and summation is extended over all N = A, U, G, C. Those q values are identical for A and U or G and C, respectively, since the error can appear either in the plus or in the minus strand. If the monomeric concentrations are of equal magnitudes, the stability constants determine what fidelities are obtainable. Then it follows that G and C reproduce considerably more accurately than A and U. The ratio of the error rates for GC and AU reproduction in mixed systems, however, does not exactly resemble the (inverse) ratio of the corresponding stability constants, owing mainly to the presence of GU wobble interactions, which are the main source of reproduction errors—even in present-day RNA phage replication [34].

We have made a guess of q values based on various sets of data for enzyme-free nucleotide interactions. They are summarized in Table 15. The first three sets refer to equal monomeric concentrations of A, U, G, and C. This assumption may be unrealistic and is therefore modified in the last three examples. One may object to the application of stability data that were obtained from studies with oligonucleotides. However, the inclusion of a single nucleotide in the replication process involves cooperative base-pair interactions and hence should resemble the relative orders found with oligonucleotides. All that is required for calculating the q values are *relative* rather than *absolute* stabilities.

The conclusion from the different estimates presented in Table 15 is: G and C reproduce with an appreciably higher fidelity than A and U. Depending on the superiority of the selected sequences (σ, cf. Eq. (28), Part A), the reproducible information content of GC-rich sequences in early replicative processes is limited to about twenty to one hundred nucleotides, i.e., to tRNA-like molecules, while that of AU-rich sequences can hardly exceed ten to twenty nucleotide residues per replicative unit. It should be emphasized at this point, that longer sequences of *any* composition may well have been present. However, they were *not* reproducible and therefore could not evolve according to any functional requirement.

From an analysis of experimental data for phage replication we concluded in Part A that even well-adapted RNA replicases would not allow the *reproducible* accumulation of more than 1000 to 10000 nucleotides per strand. This is equivalent to the actual gene content of the RNA phages.

We may now complete our statement regarding primordial replication mechanisms: A size as large as tRNA is *reproducibly* accessible only for GC-rich structures. Hence:

GC-rich sequences qualify as candidates for early tRNA adapters and for reproducible messengers, at least as long as replication is not aided by moderately adapted enzymes.

A similar conclusion can be drawn with respect to the start of *translation*. As was emphasized by Crick et al. [3], stability of the peptidyl-tRNA-messenger complex is critical for any primitive translation mo-

Table 15. Estimates of fidelities and error rates for G and C vs. A and U reproduction

Monomer concentrations	Stability constants of base pairs	Fidelity q		Error rate $1-q$	
		GC	AU	GC	AU
$m_A = m_G = m_C = m_U$	$K_{RR} = K_{YY} = 1$ $K_{AC} = 1$; $K_{GU} = 10$ $K_{AU} = 10$; $K_{GC} = 100$	0.93	0.59	0.07	0.41
$m_A = m_G = m_C = m_U$	$K_{RR} \approx K_{YY} \ll 1$ $K_{AC} = 1$; $K_{GU} = 10$ $K_{AU} = 10$; $K_{GC} = 100$	0.95	0.67	0.05	0.33
$m_A = m_G = m_C = m_U$	$K_{RR} = K_{YY} \ll 1$ $K_{AC} = 1$; $K_{GU} = 5$ $K_{AU} = 10$; $K_{GC} = 100$	0.97	0.78	0.03	0.22
$m_A = 10 m_G$ $m_G = m_C$ $m_C = 10 m_U$	$K_{RR} \approx K_{YY} \ll 1$ $K_{AC} = 1$; $K_{GU} = 5$ $K_{AU} = 10$; $K_{GC} = 100$	0.93	0.81	0.07	0.19
$m_A = 10\, m_G$ $m_G = m_C$ $m_C = 10\, m_U$	$K_{RR} \approx K_{YY} \ll 1$ $K_{AC} = 1$; $K_{GU} = 5$ $K_{AU} = 10$; $K_{GC} = 100$	0.95	0.69	0.05	0.31
$m_A = 10\, m_G$ $m_G = m_C$ $m_C = 10\, m_U$	$K_{RR} = K_{YY} = 1$ $K_{AC} = 2$; $K_{GU} = 10$ $K_{AU} = 10$; $K_{GC} = 100$	0.86	0.25	0.14	0.75

del. Applying the data quoted above, the stability constant of a complex including five GC pairs amounts to

$K_{5GC} \approx 10^7\ M^{-1}$

while that for five AU pairs is five orders of magnitude lower:

$K_{5AU} \approx 10^2\ M^{-1}$

Again these values must be seen as relative; they might actually be somewhat larger if the stacked-loop region or tRNA (as we know it today) were involved, which, however, would not invalidate the argument based on relative magnitudes.

We might also evaluate the models on the basis of lifetime data. The recombination-rate constants. as measured for complementary chains of oligonucleotides, were found consistently to lie in the order of magnitude of

$k_R \approx 10^6\ M^{-1}\ s^{-1}$.

Given the stability constants quoted above, the lifetimes of the respective complexes would amount to

$\tau_{5GC} \approx 10\ s$ and $\tau_{5AU} \approx 10^{-4}\ s$.

Again, these numbers might shift to larger values if stabilities turned out to be higher, and if two adjacent tRNAs are able to stabilize each other when attached to the messenger chain. Then lifetimes might just suffice for GC-rich sequences to start primitive translation. The lifetimes are certainly much too short if AU pairs are in excess. We see now that the slight disadvantage of the RNY relative to the RRY code resulting from stabilities can be balanced by utilizing primarily G and C at least for part of the R and Y positions. A four-membered GC structure is definitely more stable than any five-membered structure, including more than two AU pairs.

The conclusion is:

The start of translation is highly favored by GC-rich structures both of the tRNA precursors and of the messengers.

XIV. The GC-Frame Code

XIV.1. The First Two Codons

If we combine the conclusions drawn from stability data with the arguments produced by Crick et al., we can predict which codon assignments were probably the first ones.

The only sufficiently long sequences that are able to reproduce themselves faithfully must have been those in which G and C residues predominated. The first codons were then exclusively combinations of these two residues. The requirement for a comma-free read-off excludes the symmetric combinations GGG/CCC and GCG/CGC. This may be easily verified by writing down such sequences. Adaptors with the correct anticodon combinations can bind in various overlapping positions. This will have even more deleterious consequences, if further codon combinations, derived from symmetric precursors, are introduced. We are thus left with two complementary pairs of combinations, namely, GGC/GCC and CCG/CGG (all patterns to be read from 5′ to 3′). From the point of view of symmetry both appear completely equivalent. There is, however, a slight asymmetry based on wobble in the third position. Let us compare messenger sequences consisting exclusively of either CNG or GNC codons. In the first case the wobble base G is always situated in the third codon position, while in the second case it is restricted to the first position, in both the plus and the minus strands (if consistently read from 5′ to 3′). For replication the different codon positions are not distinguishable. Hence, wherever a wobble base occurs, an ambiguity may be introduced in the complementary strand, which, when it comes to translation, in one case affects the first, in the other case the third codon position:

5′----CNG CNG CNG----3′

↓

5′----$\overset{C}{U}$N′G$\overset{C}{U}$N′G$\overset{C}{U}$N′G----3′

and

5′----GNC GNC GNC----3′

↓

5′----GN$\overset{C}{U}$GN$\overset{C}{U}$GN$\overset{C}{U}$ ----3′

Only in the second case are the reproduced sequences correctly translated, i.e., if wobble interactions with the adapter occur preferentially in the third rather than in the first codon position.

In other words, an adapter with the anticodon 3′CNG can read both 5′GN′C and 5′GN′U, whereas an adapter with the anticodon 3′GNC will only read 5′CN′G, but reject 5′UN′G. The argument might be weak if five base pairs are required to keep the adapter bound to the messenger, since then wobble positions may not be as clearly distinct. Nevertheless, this asymmetric relationship between the first and third codon positions does exist and is obvious in the present genetic code.* Relatively small selective advantages are usually sufficient to bias the course of evolution. Crick et al.

* Our argument is aided by the fact that in the stationary distribution G is more persistent than C.

obviously preferred the RRY (or RNY) model on the basis of such arguments, too.
We are now able to make a unique assignment for the first two codons, namely

5'GGC and 5'GCC

which are complementary if aligned in an antiparallel fashion. This choice was dictated by four arguments, viz.,

stability of adapter-messenger interaction and
fidelity of replication, both favoring GC combinations to start with,
comma-free read-off in translation requiring an unsymmetric GC pattern, and
consistency of translation restricting *wobble* ambiguities to the third codon position.

We should like to emphasize that these arguments are based exclusively on the properties of nucleic acids. It is satisfying to notice that the two codons GGC and GCC in the present genetic code refer to the two simplest amino acids, glycine and alanine, which in experiments simulating primordial conditions indeed appear with by far the greatest abundance.
One may object that translation products consisting of these two residues only, will hardly represent catalysts of any sophistication. We shall return to this question in Section XVI. At the moment it suffices to note that translation at this stage is not yet a property required for the conservation of the underlying messengers. The first GC-rich strands are selected solely on the basis of structural stability and their ability to replicate faithfully. Many different GC sequences would serve this purpose equally well and hence may have become jointly selected as (more or less degenerate) partners of one quasi-species. Symmetric structures are greatly favored here, because they can fulfil the criteria of stability for the plus and for the minus strand simultaneously.
Among stable members, perhaps induced by template function of anticodon loops and subsequent pattern duplications, comma-free code sequences may have occurred and then started translation. If their translation products add any advantage to the stability or to the reproduction rates of their messengers, they will evolve further by a Darwinian mechanism and thereby change continuously the quasi-species distribution. Before we come back to such a stabilization by means of translation products, let us enlarge somewhat more on the question of stability of structure versus efficiency of replication, because it seems that both required properties are based on conflicting prerequisites.

XIV.2. The 'Aperiodic Linear GC Lattice'

The tRNA-like molecule with its internal folded structure strengthened by hydrogen bonds may be considered a microcrystallite. If it involves longer stretches of complementary GC pattern, the resulting internal structure may be quite inert. From melting curves of tRNA loops or corresponding oligonucleotides we know that an uninterrupted sequence of only four GC pairs is already quite stable. S. Coutts [68] studied an oligonucleotide corresponding to the extra loop of $tRNA^{Ser}_{1,2}$ from yeast, which contains 4 GC pairs (and was obtained by partial digestion of the tRNA molecule). He found a melting temperature of $84 + 1°$ C and a ΔH of 44 ± 4 kcal/mol. This is equivalent to a stability constant of about 2×10^5 at 25° C, in good agreement with the figures mentioned above. What is rated in the *selection* of a quasi-species is not merely the structural stability of the strands, but rather a favorable combination of structural stability with reproductive efficiency. Efficient template function requires a quick partial unwinding of a loop, a procedure for which excessively long stretches of GC pairs are prohibitive. However:

Natural sequences are not perfect anyway.

Given a high abundance of A-monomers and the limited fidelity of base pairing, the GC microcrystallites will always be highly doped with A-residues, acting like imperfections in the linear GC lattice. A priori, there may be any kind of sequence from high to low A, U, G, or C content. What is to be selected and then reproducibly mulitplied, will be a sequence rich in GC, but not perfect. If, for instance, every fifth position in such a sequence is substituted by an A or U residue, then base-paired regions, depending on internal complementarity, will involve on the average no more than four GC pairs (cf. present tRNA). Those structures can melt locallly with ease, especially if the replication process is aided by a protein, which then represents the most primitive form of a replicase.

Note: *A-U imperfections in the aperiodic GC lattice are selectively advantageous.*

As Thomas Mann* said: 'Life shrinks back from absolute perfection.'

XIV.3. From GNC to RNY

Given a certain abundance of A (and complementary U) imperfections in the GC-rich strands of the

* Th. Mann: Der Zauberberg (The Magic Mountain)

selected quasi-species, the next step in the evolution of a code seems to be preprogrammed. Mutations might occur in any of the three codon positions, but their consequences are quite different. Substitution of the middle base of a codon would enforce a complementary substitution of the middle base of the corresponding anticodon occurring in the minus strand and hence immediately introduce two new codons, GAC and GUC. Changes in the first or third position, on the other hand, would be complemented by changes in the third (or first) position, respectively, of the minus strand and – by wobble arguments – finally lead to only one further assignment. Moreover, the GC frame for comma-free reading would be perturbed.

Stability requirements do not initially allow for a substitution of more than one AU pair in the five-base-pair region of the messenger-tRNA complex. Hence the most likely codons to occur next are 5′GAC and 5′GUC. Being mutants of the pre-existing pair 5′GGC/5′GCC, they may be abundantly present as members of the selected GC-rich quasi-species.

However, if these mutants are assigned a function in translation, they have to become truly equivalent to the dominant 5′GGC/5′GCC species. It is at this stage that hypercyclic stabilization of the four codon adapters (and the messengers which encode the coupling factors) becomes an absolute requirement. Without such a link the different partners of the primary translation system may coexist for some time, but they would never be able to evolve or to optimize their cooperation in any coherent fashion.

The four codons allow four different amino-acid assignments, which can now offer a rich palette of functions. The resulting proteins therefore could become efficient coupling factors. Messengers and tRNAs, as members of the same quasi-species, might have emerged from complementary strands of the same RNA species, thus sharing both functions.

On the other hand, this may be too restrictive a constraint for their further evolution. We then have to assume that they were derived from a common precursor, but later on diverged into different seqences owing to their quite different structural and functional requirements.

The assignments for GAC and GUC, according to the present table of the genetic code, are aspartic acid and valine. Before we discuss the amino-acid aspect in more detail we may look briefly at some further steps in the evolution toward a more general code.

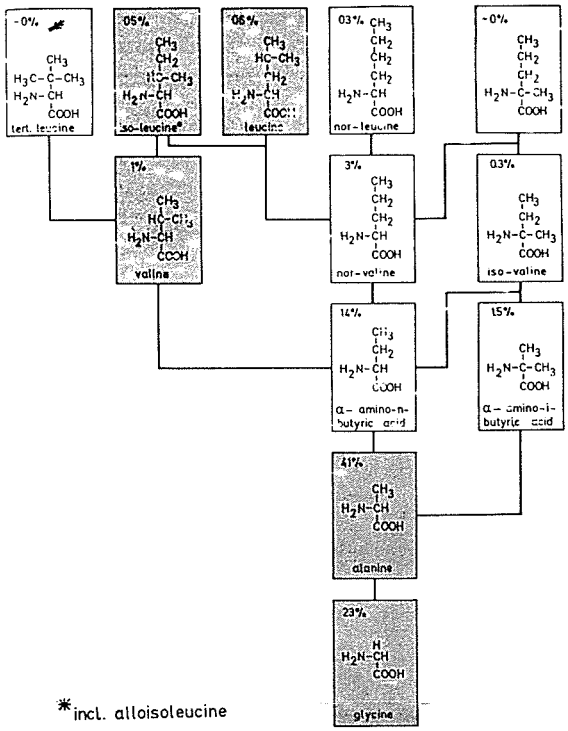
High stability of codon-anticodon interaction is required less and less as the translation products become better adapted. Wobble interactions are finally admitted and the GC frame code can evolve to the more general RY frame code. All together this brings four more amino acids onto the scene. The first substitution still occurs under the stability constraint, which forces the AU content to be as low as possible. Hence it introduces the two codons 5′AGC (=serine) and 5′ACC (=threonine). Their complementary sequences affect the third codon position, yielding 5′GCU and 5′GGU, which reproduce the assignments for alanine (GCC) and glycine (GGC). The degeneracy, accounting for the wobble interactions in the reproduction of these latter codons, may have been the primary cause of the appearance of AGC and ACC codons and their assignments.

If with the evolution of an enzymic machinery more than one AU pair is allowed in the codon region we arrive at two more new assignments, namely, AA^{U}_{C} (asparagine) and AU^{U}_{C} (isoleucine). This completes all possible assignments for an RNY code. The further evolution of the genetic code requires a relaxation of the constraint of nonoverlapping frames. Adaptation of ribosomal precursors is therefore now imperative.

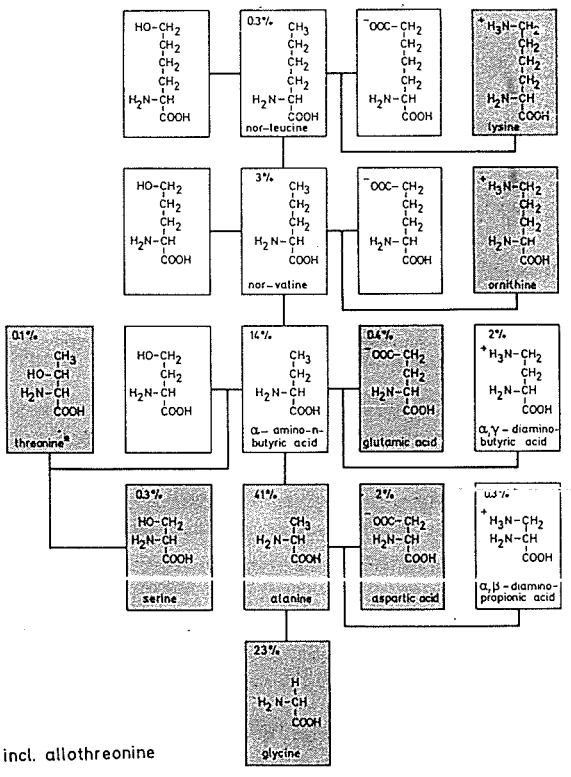
XIV.4. The Primary Alphabet of Amino Acids

Quite reliable estimates can be made for the *primordial abundance* of various amino acids. Structure and composition already provide the main clues for a guess about the likelihood of synthesis under primordial conditions. In Figure 51 the family tree of the first dozen nonpolar aliphatic amino acids as well as a few branches demonstrating the kinship relations for the simpler polar side chains are shown. Interesting questions concerning Nature's choice of the protein alphabet arise from this diagram.

The two simplest amino acids, glycine and alanine, are 'natural' representatives. It was apparently easier to fulfil requirements for hydrophobic interaction by adding some of the higher homolog, such as valine, leucine, and iso leucine. This specific choice may have been subject to chance; perhaps it was biased also by discriminative interactions with the adapters available. Among the polar side chains we find some obvious aliphatic carboxylic acids (aspartic and glutamic acid) as well as alcohols (serine and threonine), but not the corresponding amines (α, β-diaminopropionic acid and α,γ-diaminobutyric acid). Only the second next homolog (lysine) appears among the twenty 'natural' amino acids, while the intermediate (ornithine) still shows its traces. The reason may be that upon activation of the second amino group lactame formation or elimination occurs, which terminates the polymerization. Moreover, the second amino group may lead to a branching of the polypeptide chain (although



a



b

a similar argument may be raised for the carboxylic groups). Positively charged side chains may well have been dispensible in the first functional polypeptides. Even in present sea water the concentration of Mg^{2+} is high enough (~ 50 mM) to cause an appreciable complexation with carboxylic groups. Under reducing conditions even more divalent ions (such as Fe^{2+}) may have been dissolved in the oceans. Those metal ions, attached to carboxylic groups and still having free coordination sites, are especially important for close interactions between early proteins and (negatively charged) polynucleotides. From this point of view, side chains containing negatively charged ligands seem to be less dispensible than those containing positive charges.

The 'natural' amino acids not appearing in Figure 51 bear considerably more complex sidechains and were therefore present in the primordial soup at comparatively low concentrations.

These guesses from structure and composition are excellently confirmed by experiments simulating the prebiotic synthesis of amino acids, carried out by S. Miller and others (reviewed in [63]). The yields obtained for the natural amino acids (but also for other branches of the family tree) correspond grossly to the chemist's expectations (cf. numbers in Fig. 51), although many interesting items of detailed information are added by these experiments. The results are, furthermore, in good agreement with data obtained from meteorite analysis [69, 70] reflecting the occurrence of amino acids in interstellar space. Table 16 contains a compilation of data (taken from [63]), which are relevant for our discussion.

There is no doubt that the primordial soup was very rich in glycine and alanine. In Miller's experiments these amino acids appear to be about twenty times more frequent than any of the other 'natural' representatives. The next two positions in the abundance scale of natural amino acids are held by aspartic acid and valine with a clear gap between these and leucine, glutamic acid, serine, isoleucine, threonine, and proline.

There is every reason to assume that assignments of codons to amino acids actually followed the abundance scale. If glycine and alanine are by far the most abundant amino acids, why should they not have been assigned first, as soon as chemical mecha-

Fig. 51. The family tree of the first aliphatic amino acids and some branches for the simplest polar side chains. The number in the left upper corner of each plate refers to Miller's data of relative yields under primordial conditions [63] (i.e., molar yield of the particular amino acid divided by the sum of yields of all amino acids listed in their Table 7–2 on p. 87). The plates for the natural amino acids are shaded

Table 16. Abundance of natural amino acids in simulated prebiotic synthesis and in the Murchison meteorite. The first column refers to those amino acids which appear in the proteins. The data in the second column are typical results from S. Miller's experiments (as reviewed in [63]). They were obtained by sparking 336 mMol of methane in the presence of nitrogen and water. The total yield of amino acids (including those which do not appear in the proteins) based on carbon was 1.9%, the corresponding yields of glycine and alanine 0.26% and 0.71%, resp. The yields in the table refer to molar abundance. Similar data are obtained under different conditions, gly, ala, asp, val usually occurring as the most abundant natural amino acids. The meterorite data were reported by J. Oró et al. [69] and by K.A. Kvenvolden et al. [70], D-isomers appeared in all cases to be close to 50%. Further literature can be found in [63]

Compound	Yield [μM]	μg/g meteorite
Glycine	440	6
Alanine	790	3
Aspartic acid	34	2
Valine	19.5	2
Leucine	11.3	
Glutamic acid	7.7	3
Serine	5.0	
Isoleucine	4.8	
Threonine*	1.6	
Proline	1.5	1

* Including allothreonine

nisms of activation became available? The first primordial polypeptides—the instructed as well as the noninstructed—must then have been largely gly-ala copolymers with only occasional substitutions by other amino acids, probably including also those which finally did not become assigned.

The agreement between the abundances of natural amino acids and the order of the first four codon assignments is striking. It should be emphasized that our choice of codons is based exclusively on arguments that are related to the structure of nucleic acids. Not only do the first four GC frame codons coincide exactly with the order of abundance, but furthermore the four additional RNY codons are assigned to amino acids which—with the exception of asparagine—are well represented with appreciable yields in Miller's table. One may well ask whether the assignment of AA_C^U to asparagine is a primary one or—as suggested by the intimate neighborhood to the lysine codon—was originally related to any of the lower diamino acid homologs, which appear in Miller's table with a reversed order of yields as compared with aspartic acid (α,γ-diaminobutyric acid) and glutamic acid (α,β-diaminopropionic acid). Without further evidence, however, this might be a too far-reaching speculation.

Also of importance in this respect are some recent results from the protein-sequence analysis of nucleotide-binding enzymes, which are believed to have existed more than $3 \cdot 10^9$ years ago under precellular conditions [71, 72]. The data suggest the existence of a precursor sequence of the nucleotide-binding surface, which include the amino acids valine, aspartic (and glutamic) acid, alanine and glycine besides isoleucine, lysine and threonine (although these data actually refer to a later stage of precellular evolution than is discussed in this paper).

XV. Hypercyclic Organization of the Early Translation Apparatus

Any model for the evolution of an early code and translation apparatus will have to provide conditions that allow tRNA-like adapters as well as gene precursors (or messengers) for various enzymic factors not only to coexist, but also to grow coherently and to evolve to optimal function. In Parts A and B it was shown that such a self-organization requires cyclically closed reactive links among all individual partners, unless they all can be integrated structurally into one replicative unit. In this section we have to show how realistic code models are correlated with an hypercyclic organization, and how such systems can evolve.

An apparent difficulty of the hypercycle is how it is to originate. In plain language, the abundant presence of all members of the hypercycle seems to be a prerequisite for its origin. In more scientific terms, the hypercycle, being a higher-order-reaction network, has to 'nucleate' by some higher-order mechanism in order to come into existence.

Consider, for comparison, a simple replicative unit that grows according to a first-order autocatalytic rate law. In a solution buffered with energy-rich building material one copy is sufficient to start the multiplication process. Those experiments have been carried out with phage RNA or its noninfectious variants [7, 8, 32, 34, 73]. One template strand is sufficient to produce—within a few minutes—a large population of identical copies (cf. Part A).

A hypercycle never could start in this way. A single template copy would not multiply unless a sufficiently large number of its specific catalytic correspondents were present. These in turn are encoded by templates which themselves could not have multiplied without the help of their translation products. The growth of all templates in the system is dependent upon catalytic support, but the catalysts cannot grow unless the templates multiply. How large is the probability that nucleation will occur through some accidental fluctuation? Let us assume a test tube with 1 ml of sample solution. Diffusion-controlled reactions of

acromolecules may have rate constants of the order f magnitude of 10^8 M^{-1} s^{-1}. Hence at least 10^8 lentical copies of a given catalytic reaction partner ave to be present in order to start template multipli- ation with a half-time of about one day. There is o chance that correlated functions among several uch partners could result from coincidences of such iant fluctuations. It may, of course, be possible that he various templates multiply according to mixed- rder terms; in other words: that first-order (enzyme- ree) autocatalytic terms (representing template multi- lication without catalytic help by other members) re superimposed upon the second-order catalytic eplication terms. The hypercyclic link would then ecome effective only after concentrations have risen o a sufficiently high level. However, the system can- ot know in advance which of the many alternative equences multiplying according to a first-order auto- atalysis are the ones which provide the useful infor- nation for the catalysts required at the later stages f organization.

There is only one solution to this problem:

The hypercycle must have a precursor, present in high natural abundance, from which it originates gradually by a mechanism of mutation and selection.

Such a precursor, indeed, can be the quasi-species consisting of a distribution of GC-rich sequences. All members of a stable quasi-species will grow until they are present in high concentrations. As was shown in Section XIV, some GC-rich sequences may be able to start a translation by assigning amino acids to defined anticodons. At this stage the translation products are really not yet necessary for conserving the system, so translation can still be considered a game of trial and error. If, however, it happens that one of the translation products offers advantages for the reproduction of its own messenger, this messenger may become the dominant representative of the quasi-species distribution.

A single RNA species could at best assign a two-amino-acid alphabet, if both the plus and the minus strands act as adapters for two complementary codons (e.g., GGC and GCC). If adapter sequences are sufficiently abundant, there is also a finite chance that coexisting mutants assign the two or even four codons (including GAC and GUC for aspartic acid and valine), again possibly utilizing both plus and minus strands. All this may still happen during the quasi-species phase.

Such a system, however, can evolve only if the different RNA species stabilize each other with the help of their translation products. We defer a discussion of the details of assignments – e.g., as to whether plus and minus strands of a given RNA species can evolve concomitantly and thereby become two adapters for complementary codons, or whether the plus strand as messenger encodes for the coupling factor, while only the minus strand acts as an adapter – to Section XVI. Here we study the problem of how hypercyclic organization can gradually evolve out of a quasi-species distribution.

Figure 52 shows how such a process can be envisaged. Assume two abundant mutants of the quasi-species, whose plus and minus strands are able to act as adapters of (at most) two amino acid pairs (e.g., gly/ala and asp/val), and which at the same time may be translated into a protein made up of (at most) four classes of amino acids. If the translation products offer any catalytic function in favor of the reproduction of their messengers, one would probably encounter one of the situations represented in Figures 52 or 53.

Both messengers, being closely related mutants, encode for two proteins with closely related functions. If one is a specific replicase, the other will be too, both functions being self- as well as mutually enhanc-

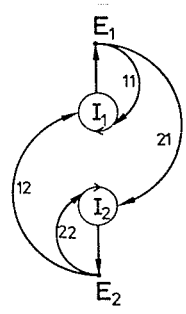


Fig. 52. *Two mutant genes I_1 and I_2*, encoding for their own replicases E_1 and E_2, may show equivalent couplings for self- [11, 22] and mutual [21, 12] enhancement due to their close kinship relation. Analogous behavior can be found in present RNA-phage replicases

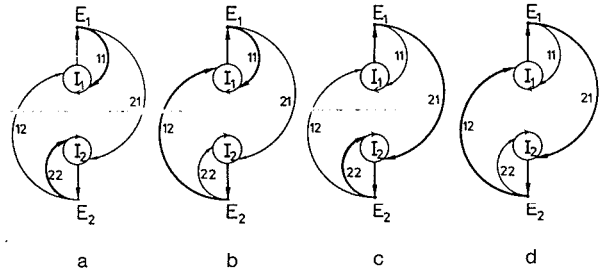


Fig. 53. *The evolution principle of hypercycles* is illustrated by the four possible situations arising from the couplings between two mutants shown in Figure 52. The thick lines indicate a preference in coupling (however small it may be). A stable two-membered hypercycle requires a preference for mutual enhancements as depicted in d)

Table 17. Fixed-point analysis of the two-member hypercycle

represented in Figure 52 has been carried out using the simplified rate equations

$$\dot{x}_i = \sum_{k=1,2} k_{ik} x_i x_k - \frac{x_i}{c} \sum_{l=1,2} \sum_{m=1,2} k_{lm} x_l x_m$$

$$i = 1,2; \quad x_1 + x_2 = c$$

yielding the three fixed points and their eigenvalues:

$$\bar{x}_1 = (c,0); \quad \omega^{(1)} = (k_{21} - k_{11})c$$

$$\bar{x}_2 = (0,c); \quad \omega^{(2)} = (k_{12} - k_{22})c$$

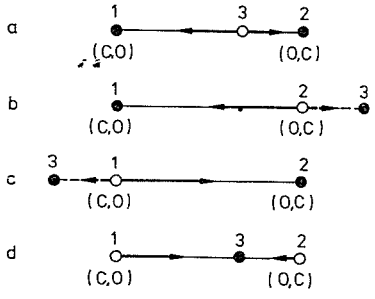
$$\bar{x}_3 = (k_{22} - k_{12}, k_{11} - k_{21}) \frac{c}{k_{11} - k_{21} + k_{22} - k_{12}};$$

$$\omega^{(3)} = \frac{(k_{11} - k_{21})(k_{22} - k_{12})}{k_{11} - k_{21} + k_{22} - k_{12}} c.$$

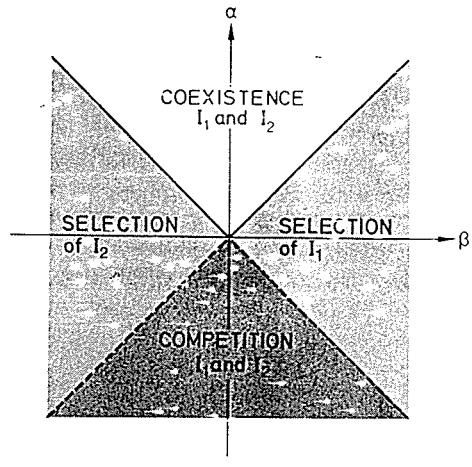
Four cases may be distinguished

a) $k_{11} > k_{21}$; $k_{22} > k_{12}$ yielding competition between I_1 and I_2
b) $k_{11} > k_{21}$; $k_{22} < k_{12}$ yielding selection of I_1
c) $k_{11} < k_{21}$; $k_{22} > k_{12}$ yielding selection of I_2
d) $k_{11} < k_{21}$; $k_{22} < k_{12}$ yielding hypercyclic stabilization of I_1 and I_2

The fixed-point diagrams of these four cases are (cf. Part B)



A unified representation can be achieved if the two coordinates: $\alpha = k_{12} + k_{21} - k_{11} - k_{22}$ and $\beta = k_{12} - k_{21} + k_{11} - k_{22}$ are introduced.

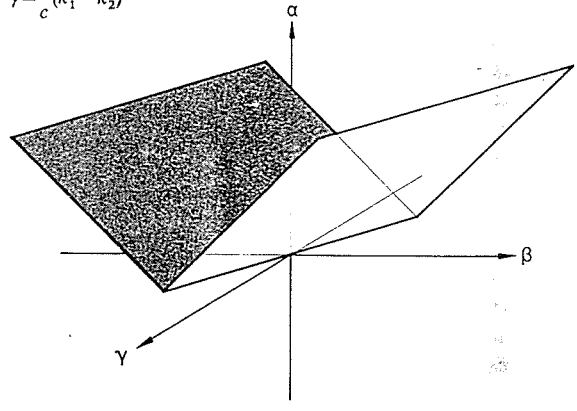


If in addition to the second-order term of the rate equations a linear autocatalytic term is introduced (yielding a growth function of the form $\Gamma_i = k_i x_i + \sum_{j=1,2} k_{ij} x_i x_j$), the region of stable hypercyclic coexistence of both species I_1 and I_2 is the space above the folded sheet in the three-dimensional parameter space with the coordinates:

$$\alpha = k_{12} + k_{21} - k_{11} - k_{22}$$

$$\beta = k_{12} - k_{21} + k_{11} - k_{22}$$

$$\gamma = \frac{2}{c}(k_1 - k_2)$$



ing. There may, however, be specificity, too, because both proteins do not necessarily recognize both sequences equally well, nor do they recognize unrelated sequences at all. The sequences provide a specific binding site for initiating replication. The differences in binding strength for the four possible interactions of E_1 and E_2 with I_1 and I_2 may be slight. These differences, indicated by the line strengths, however small they are, will have drastic consequences, as follows from an inspection of the corresponding fixed-point diagrams (Table 17). We may distinguish four cases:

(1) E_1 favors I_1 over I_2, and E_2 favors I_2 over I_1 (Fig. 53a).
Consequence: I_1 and I_2 both are hypercyclically enforced by their respective enzymes, leading to strong competition. Only one of the competitors can survive, even if they are selectively equivalent.
(2) E_1 favors I_1 over I_2, and E_2 also favors I_1 over I_2 (Fig. 53b).
Consequence: I_1 will win the competition and I_2 will die out.
(3) E_2 favors I_2 over I_1, and so does E_1 (Fig. 53c).
Consequence: I_2 is now the winner, while I_1 dies out.

(4) E_1 favors I_2 over I_1, and E_2 favors I_1 over I_2 (Fig. 53d).

Consequence: Here we obtain a mutual, hypercyclic stabilization of I_1 and I_2.

It is important to note that small differences suffice to produce the behavior outlined above. In this respect it is of interest to see what happens if both E_1 and E_2 are exactly equivalent in their treatment of I_1 and I_2. Here we have complete impartiality, regardless of how much the population numbers x_i or y_i differ. I_1 or I_2 can die out in consequence of a fluctuation catastrophe, since there is no mutual stabilization present as in case 4. On the other hand, the fluctuations do not amplify themselves, and if the population numbers are large enough, a fluctuation catastrophe will practically never occur. Quite different in this respect is case (1), mentioned above. Only for exactly equal population numbers of I_1, I_2, E_1, and E_2 is here the system in a dynamically balanced state. A small fluctuation may disturb the balance and then, through self-amplification, inevitably will lead to selection of one of the two species. The same is true for any ensemble in which each messenger provides help for its own replicase only (cf. Fig. 47). The coupling resulting from a common translation function – all replicases (utilizing their RNA-recognition sites) may function simultaneously as activating enzymes – is not sufficient to enforce a coexistence. As in the system shown in Figure 45, there will be only one survivor, and translation function will subsequently break down.

The exact criteria for hypercycle formation are derived in Table 17. The figures give a clear representation of the stability ranges in terms of generalized coordinates referring to the rate parameters.

We have thus obtained an evolution principle for hypercycles. This kind of organization can emerge from a single quasi-species distribution, as soon as means of reaction coupling develop. The prerequisites for coexistence of precursors can be met generally only by closely related mutants. Thus the emergence of hypercycles requires the pre-existence of a molecular Darwinian system, but it will then lead to quite new consequences. The evolution principle is effective even with very small differences in rate parameters and hence responds sensitively to small changes brought about by mutations. Given a quasi species distribution with developing interactions among the constituents, regardless, of how weak these interactions are, a hypercyclic organization will inevitably emerge whenever such interactions arise.

The hypercycle will also grow inevitably by way of mutations toward larger complexity (Fig. 54, 55). The evolution principle can be generalized by induction so as to apply to any n-membered hypercycle. A mutant I′ then may either replace I, die out, or enlarge the hypercycle to a size comprising $n+1$ members (cf. XVI. 10.). More general evolution criteria can be derived as indicated in Table 18.

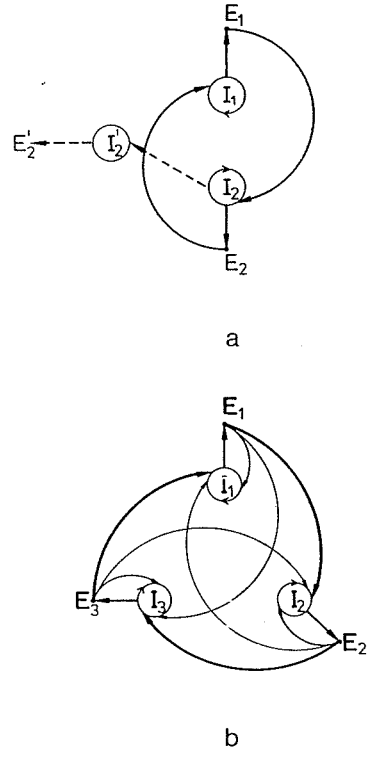


Fig. 54. *The generalization of the evolution principle* of hypercycles is illustrated in this diagram. The couplings have to fulfil the criteria derived in Tables 17 and 18, i.e., mutual enhancement has to prevail over self-enhancement (cf. thick lines). (a) A mutant of I_2 appears (I_2'). (b) The mutant (now I_3) is incorporated in the hypercycle

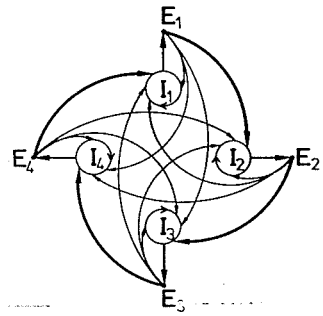


Fig. 55. *The 'realistic' four-membered hypercycle* assigns four messengers I_1 to I_4 (being mutants of a common precursor) to encode for four replicases E_1 to E_4 with common function, but slight preferences in specificity. The minus strands of I_1 to I_4 may concomitantly act as amino acid adapters

Table 18. Generalization of the evolution principle of hypercycles

is explained by the transition of a two-membered to a three-membered system as depicted in Figure 54.
The general rate equations are of the same form as in Table 17. Starting from a stable two-membered hypercycle, introduction of a third member I_3 (e.g., being a mutant of I_2) will frequently change the previously stable fixed point to a saddle point. This is most clearly seen if cyclic symmetry is assumed. Under these conditions, the notation can be simplified to:

$$k_{11}=k_{22}=k_{33}=k_D; \quad k_{13}=k_{21}=k_{32}=k_+; \quad k_{12}=k_{23}=k_{31}=k_-;$$

yielding the following matrix of rate coefficients.

$$\mathbf{K}=\begin{vmatrix} k_D & k_- & k_+ \\ k_+ & k_D & k_- \\ k_- & k_+ & k_D \end{vmatrix}$$

The fixed points and eigenvalues then are:

Corners: $\bar{x}_1=(c,0,0)$; $\omega_1^{(1)}=(k_- -k_D)c$, $\omega_2^{(1)}=(k_+ -k_D)c$

$\bar{x}_2, \bar{x}_3$ analogous

Edges: $\bar{x}_4=(0,k_D-k_-,k_D-k_+)\dfrac{c}{2k_D-k_+-k_-}$

$$\omega_1^{(4)}=\frac{k_-(k_D-k_-)+k_+(k_D-k_+)+k_+k_- -k_D^2}{2k_D-k_+-k_-}c$$

$$\omega_2^{(4)}=\frac{(k_D-k_+)(k_D-k_-)}{2k_D-k_+-k_-}c$$

$\bar{x}_5, \bar{x}_6$ analogous

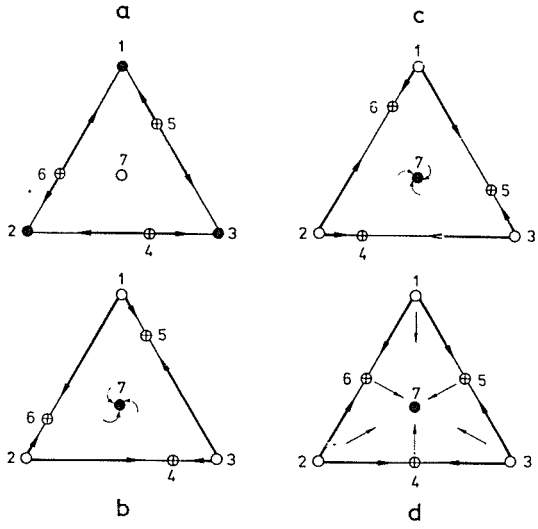
Interior: $\bar{x}_7=\left(\frac{c}{3},\frac{c}{3},\frac{c}{3}\right)$,

$$\omega_{1,2}^{(7)}=\{2k_D-k_+-k_- \pm i\sqrt{3}(k_+-k_-)\}\frac{c}{6}.$$

Again four cases are of special interest:

a) $k_D>k_+, k_-$
b) $k_+>k_->k_D$
c) $k_->k_+>k_D$
d) $k_+=k_-\gtrless k_D$

yielding the following fixed-point diagrams:



Large diagonal terms ($k_D \gg k_+, k_-$) lead to competition (diagram a). In the opposite situation, i.e., with large off-diagonal elements of **K**, the three species show cooperative behavior. The sense of rotation around the spiral sink in the center of the simplex is determined by the larger of the two constants k_+ and k_-. No rotational component is observed for equal constants $k_+=k_-$. The central fixed point is then a focus.
The example treated in this table provides a good illustration of the evolution to more complex hypercycles. In the absence of simplifying assumptions concerning the rate constants the analysis becomes quite involved. We refer to a more detailed representation [98], which includes a generalization to arbitrary dimensions.

XVI. Ten Questions

concerning our earliest molecular ancestors and the traces which they have left in the biosynthetic apparatus of present cells.

XVI.1. One RNA precursor?

This question is concerned with the complexity of the first molecules starting any reproducible function. A nucleotide chain of 100 residues corresponds to a complexity of about 10^{60} alternative sequences. If on grounds of stability we restrict ourselves to (AU-doped) GC copolymers only, we are still left with about 10^{30} possible arrangements. In order to achieve one or a few defined sequences, faithful self-reproduction is a necessary prerequisite. It will inevitably lead to Darwinian behavior, with selection of one defined quasi-species. The selected products are determined plainly by an optimal selective efficiency, but their structure depends on their historical route, which is strongly biased by self-copying of smaller oligonucleodtide patterns.

XVI.2. What Does Selective Advantage Mean to a Molecule?

Selective value is defined as an optimal combination of structural stability and efficiency of faithful replication. It can be expressed in quantitative terms related to the physical properties of a molecule in a given environment. Structural stability, resistance toward hydrolysis, and the development of cooperative properties call for elongation. Small oligonucleotides can-

not fold in any stable manner and may therefore easily be hydrolyzed. Furthermore, they do not offer sufficient adhesive strength for faithful copying or for translation. Length, on the other hand, is limited by replication rates and by copying fidelity. The properties of GC-rich sequences have been shown to be advantageous for forming stable copies with extended length. Whether these lengths resemble the sizes of present-day tRNA is uncertain. Sequence homologies have been found in tRNA [74], which indicate some self-copying of internal regions. This, however, may well have happened before codons became assigned. The onset of translation requires strong interactions between adapters and messengers, and these cannot be provided by molecules which are too small.

As soon as translation yields reproducible functions, selective value achieves a new dimension. It must, nevertheless, be expressed, for any given messenger, in terms of structural stability and efficiency of faithful reproduction. These properties now, however, also depend on the qualities (and concentrations) of the translation products. Specific coupling—as required for hypercyclic organization—is hence necessary for any system in which translation products are to be rated for selection and thereby become eligible for evolution. Such coupling is of a catalytic or protective nature.

XVI.3. Why Hypercyclic Organization of Single Mutant Genes Rather than One Steadily Growing Genome?

The answer to this question has been largely given in Part A. For a very primitive translation apparatus an amount of information would be required that corresponds to (or even exceeds) that of present RNA phages. The information of the phage genome can be preserved only with the help of a phage-specific enzyme complex, the availability of which is based on the efficiency of a complete translation machinery, provided by the host cell. If we accept the answers given to the first and second questions, the information needed to start translation must arise from cooperation among several mutants coexisting in the quasi-species distribution, rather than from a mere elongation of one sequence for which primarily no selection pressure would exist.

The hypercyclic stabilization of several coexisting mutants is equivalent to evolution by gene duplication. Originally, mutants appeared as single strands rather than as covalently linked duplicates. Fidelity restrictions would not allow for such an extension of length. Moreover, the probability of obtaining the required mutant combinations in one strand is very low. Sequences consisting of 100 G and C residues have

100 one-error mutants,
4950 two-error mutants
161700 three-error mutants, etc., or

$$N_k = \binom{100}{k}\ k\text{-error mutants}$$

The number of strands containing n mutant genes, each differing from the other in k specified positions (which may be necessary in order to qualify for a function) amounts to

$$\binom{N_k+n-1}{n} \approx \frac{N_k^n}{n!} \quad (\text{for } n \ll N_k),$$

e.g., for $n=4$ and $k=3$ to 3×10^{19} alternative sequences. Given even these small deviations in the multiplied genes, the chance of finding a copy with a favorable combination within one giant strand is almost nil for any reasonably sized population. Each of the *isolated* mutant genes containing three substitutions, however, would be abundantly present in any macroscopic population.

Last but not least, the tRNAs being the adapters for translation must have been present anyway as separate strands. Evolution of a unified genome would have required complicated transcription control right at the start.

The isolated RNA strands, on the other hand, have a natural origin in the quasi-species distribution. All sequences were similar and so must have been their translation products. Whenever one translation product provides coupling functions, all of them will do so, owing to their similarities. Cyclic coupling—as required for hypercyclic organization—may then occur as well. We might even say that hypercyclic organization is most naturally associated with any realistic primitive translation model.

Does the present genome organization, established in prokaryotic cells, offer any clue as to its early structure? Present genes are certainly much larger than the early messengers. Gene elongation, as well as duplication, provided an advantage whenever the steadily increasing fidelity of the enzymic machinery allowed for it. The translation products could gain in sophistication, and more complex multienzyme mechanism could evolve, utilizing differentiated enzymes which had descended from a common precursor. Recombinant mechanisms as utilized by present-day cells will not have been available in primitive systems. The present structure of the prokaryotic genome therefore may have been achieved through elongation of isolated genes, their duplication and triplication to operons and their final mapping onto DNA, which can utilize more advanced means of reproduction so

as to allow for the formation of a unified genome. The present operon sizes correspond well to those which can be handled by a sophisticated RNA replicase (e.g., 1000 to 10000 nucleotides).

XVI.4. Are tRNAs Necessary to Start With?

This question may be alternatively posed as: Why not small oligonucleotide adapters?

Adapters without messengers do not make much sense. Short nucleotide sequences do not qualify as messengers. Decapeptides are already equivalent to almost half a tRNA molecule. Furthermore, short oligonucleotides may be unstable since they lack any tertiary structure. The simplest symmetric structure, i.e., a single loop stabilized by four or five base pairs, requires as many as fifteen nucleotides. Enzyme-free specific recognition of an amino acid involving simultaneously the anticodon loop and the 3'-end of the adapter is possible only with more extended structures. The same is true for interactions between two adjacent adapters, necessary for the stabilization of the messenger-peptidyl-tRNA complex, or for the conformational change (e.g., from HF to fh) which may facilitate the transport of the growing peptide chain along the messenger. G. Maass et al. [99] recently reported such a conformational change in the anticodon loop of tRNA, which they recorded by observing a fluorescence change of the Y base. The effect appeared to be absent in the anticodon-loop fragment (i.e., a decanucleotide having the sequence of the anticodon loop). All of this suggests that insufficiently long RNA sequences do not qualify for adapter function.

We may then ask: What distinguishes an adapter from a messenger? They both require comparable minimum lengths. They both have to be specifically folded structures, through which they may become reproducibly recognizable by coupling factors.

Since each tRNA *and* each messenger require a coupling factor, e.g., a replicase, favoring their selective stabilization, dual functions of the RNA sequences are indispensable. Plus and minus strands of a given RNA sequence may thus be utilized jointly as messenger and adapter.

XVI.5. Do Present-day tRNAs Provide Clues about their Origin?

Similarities in structure might either be the consequence of adaptation to a common goal or, alternatively, indicate a common ancestor. Present tRNAs show many points of correspondence [75] in their structures. Are we able to infer a common ancestor from these analogies? According to an analysis carried out by T.H. Jukes [76], this question may be answered with a cautious 'yes'. Why one must be cautious may be illustrated with an example. One of the common features exhibited by all prokaryotic and eukaryotic tRNAs studied so far is the sequence TΨCG in the so-called T-loop, a common recognition site for ribosomal control. Recent studies of methanogenic bacteria [77] revealed that these microorganisms, which are thought to be the 'most ancient divergences yet encountered in the bacterial line', lack this common feature of tRNA, but rather contain a sequence ΨΨCG in one and UΨCG in another group. Although this finding does not call in question but rather underlines the close evolutionary relations of this class of microorganisms with other prokaryotes, it shows definitely that whole classes may concordantly adopt commont features. This is especially true for those molecules that are produced by a common machinery, such as the ribosome, which is the site of synthesis of all protein molecules.

Figure 56 shows an alignment of the sequences of four tRNAs from *E. coli*, which we think are the present representatives of early codon adpaters. Unfortunately, the sequence of the alanine-specific tRNA adapted to the codon GCC was not available. If we compare this species, which has the anticodon ^{5A}UGC, with its correspondent for valine, which has the anticodon ^{5A}UAC, we observe a better agreement

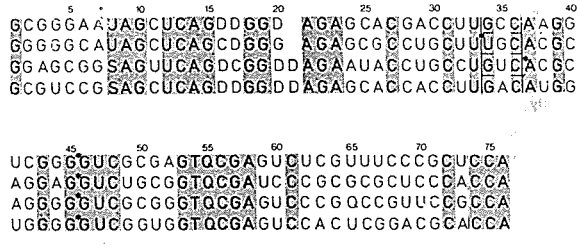


Fig. 56. *Alignment of the sequences of tRNAs* for the amino acids gly, ala, asp, and val. Unfortunately, the sequence referring to the codon GCC (for ala) is not yet available. Correspondences between gly- and ala-tRNAs are supposed to be closer for the correct sequence referring to the anticodon GCC (as suggested by the similarities between the two sequences for ala and val, referring to the anticodons *UGC and *UAC, resp.). The sequences show that base-paired regions consist predominantly of GC, and that close correspondences indicate the kinship between gly/ala and asp/val (cf. S in position 8 for asp and val instead of U for gly and ala, or the insertion of D between position 20 and 21 for asp and val). A = adenosine, *A = 2MA = N(2)-methyladenosine, C = cytidine, D = 5,6-dihydrouridine, G = guanosine, *G = 7MG = N(7)-methylguanosine, Q = Ψ = pseudouridine, S = thiouridine, T = ribosylthymine, U = uridine, *U = 5AU = 5-oxyacetyluridine

with the latter than with the one listed in the alignment (57 vis-à-vis 54 identical positions). Hence the correct alanine-tRNA with the anticodon GGC may have more coincidences with the gly-tRNA listed than for the 44 positions shown. Apart from this 'corporal defect' the data reveal

1. That all representatives agree in more than half of the positions (33 including the 'wrong' ala or 41 for gly, asp, and val),
2. That the subgroup gly/ala is distinguished from the subgroup asp/val by several features (thio-uridine 'S' instead of U in position 8, insertion of a 5,6-dihydro-U between position 20 and 21),
3. That all representatives have a pronounced excess of G and C over the A and U residues (or their derivatives), especially in the base-paired regions.

A comparison with other tRNA sequences, furthermore, indicates that these features—although certainly not uncommon among most tRNAs—are especially pronounced for this group. In particular, correspondences are as close as for different adapters of the same amino acid in the same organism.

One finding is particularly illuminating. If we compare the sequences of two adapters with complementary anticodons (e.g., asp and val) the coincidences between both plus strands of the tRNAs are much more pronounced than those between one plus strand (read from 3′ to 5′) and the other minus strand (read from 5′ to 3′). Actually, if we compare in this way the plus and minus strands of the same tRNA, the agreement is better. These coincidences are only the expression of the remarkable internal symmetry of tRNA, which places the anticodon almost exactly in the middle of the sequence and thereby allows for the formation of a symmetric two-dimensional pattern. We may rate this property as an indication of the early appearance of tRNA as an independent replicative unit. The requirement for the plus and minus strands to assume a similar pattern is important only if these represent independent replicative units rather than being structurally integrated into a large sequence of a genome, as they appear to be nowadays. We find a similar effect for phage RNAs or their variants, which have to multiply as single replicative units [78].

On the other hand, adaptation of tRNA to a common machinery must have brought about common deviations from symmetries originally required. The fact that the mirror images for plus and minus strands of the same tRNA show still more symmetric resemblance than those for plus and minus strands of tRNAs with complementary anticodons suggests that both tRNAs evolved as mutants of the same rather than of two complementary strands. We may then conclude that the present adapters for the codons GGC (gly), GCC (ala), GAC (asp), and GUC (val) derived from one quasi-species as single error mutants of a common ancestor. However, the original symmetry was not sufficient (why should it have been?) to allow adapter functions to derive from both the plus *and* the minus strand of a given RNA.

XVI.6. How Could Comma-free Messenger Patterns Arise?

The first messengers must have been identical with the first adapters (or their complementary strands). There is, indeed, a structural congruence behind adapter and messenger function. Whatever codon pattern occurs in the messenger sequence, it must have its complementary representation at the adapter. In primitive systems such a requirement could be most easily met by utilization of a common structural pattern for both types of molecules, such that the first adapters are the minus strands of the first messengers (if we define the plus strand as always being associated with a message) and that certain symmetries of structure allow both the plus and the minus strand to be recognized by the same replicase.

The first extended RNA molecules were rich in G and C, a consequence of selection based on criteria of structural stability and fidelity of copying. Molecules with a common codon pattern, such as GGC/GCC, require primer instruction (e.g., via catalysts or via exposed loops of RNAs present) with subsequent internal duplication. This will inevitably lead to structures that contain at least two codon patterns with internal complementarity, e.g., 5′GGC3′ and 3′CCG5′.

There is a good example for the efficiency of internal pattern duplication in the de-novo synthesis and amplification of RNA sequences by phage replicases. If Qβ replicase is severely deprived from any template, it starts to 'knit' its own primers, which it then duplicates and amplifies (selectively) until finally a uniform macroscopic population of RNA sequences—a few hundred nucleotides in length—appears. Under different environmental conditions, different (but uniform) sequence distributions are obtained [8]. S. Spiegelman, D. Mills and their co-workers have sequenced some of these 'midivariants,' all of which contain the specific recognition site for Qβ replicase [78]. Further experiments [73] have thrown light on the mechanism of this de-novo synthesis, showing that small pieces corresponding to sequences that are recognized by the enzyme are made as primers and then internally duplicated and selectively amplified. Earlier studies [22] have shown that, in particular, the sequences CCC(C) and UUCG can be recognized



Fig. 57. *Alignment of the sequence of Qβ-midivariant* (determined by S. Spiegelman et al. [78]) with an artificial sequence composed of CCC(C)- and UUCG-blocks, as well as their complements [GGG(G) and CGAA]. Agreement at 169 of 218 positions suggests that midivariant is a de-novo product made by the enzyme Qβ-replicase, which possesses recognition sites for CCC(C) and UUCG (EF Tu). The kinetics of de-novo synthesis indicates a tetramer formation at the enzymic recognition sites, followed by some internal self-copying with occasional substitutions. The specific midivariant usually wins the competition among all appearing sequences and hence seems to be the most efficient template. The process demonstrates how uniform patterns can arise in primitive copying mechanisms

by the enzyme. UUCG corresponds to the sequence TΨCG common to all tRNAs and known to interact specifically with the ribosomal elongation factor EF Tu, which acts as a subunit in the Qβ-replicase complex. An alignment of the midivariant with a sequence made up solely of the two oligonucleotides mentioned and their complementary segments—GGG(G) and CGAA—shows agreement in more than three-quarters of the positions, indicating the efficiency of internal copying of primer sequences (Fig. 57).

In a similar way we may think of the existence of primordial mechanisms of uniform pattern production. If among the many possible patterns 5′GGC/5′GCC and possibly also 5′GAC/5′GUC appeared, those messenger patterns could have started a reproducible translation according to the mechanism of Crick et al. [3] and have been capable of selective amplification with the help of their reproducible translation products.

XVI.7. What Did the First Functionally Active Proteins Look Like?

The simplest protein could be a homogeneous polypeptide, e.g., polyglycine. Does it offer any possible catalytic activity? This is a question that can and

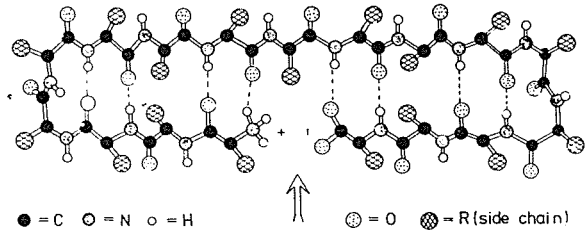


Fig. 58. *A simple enzyme precursor* is represented by a β-folded structure of some 15 to 25 amino acids (requiring messengers of 45 to 75 nucleotides). The active site includes a terminal amino group that is a very efficient proton donor (pK ~ 8), a terminal carboxylic group that acts as proton acceptor, and a catalytically active side chain (e.g., asp or ser). Many alternatives could be designed, only some of which have the correct pitch of the twisted chains to yield an efficient active site

should be answered with experiments. With mixed sequences, including a sufficiently large number of residues, say about fifteen to thirty, β-sheet structures may form with an active center, in which the terminal carboxylic group is placed in a defined position near the terminal amino group (Fig. 58). The proximal distance varies with the chain length, since the β-structure involves a twist among both antiparallel chains [79]. The pK of the terminal amino group is around eight, hence the catalytic site contains at least an efficient proton donor-acceptor system. Alternating gly-ala residues are very favorable for the formation of β-structures. However, there are serious solubility problems for chains consisting exclusively of gly and ala, which would restrict them to interfaces only.

The folding of β-sheets has been studied by P.Y. Chou and G.D. Fasman [80], who analyzed X-ray data for 29 proteins in order to elucidate 459 β-turns in regions of chain reversal. The three residues with the highest β-turn potential in all four positions of the bend include gly and asp, while in regions following the β-turn, hydrophobic residues are predominant.

An important prerequisite of catalytic efficiency is the defined spatial arrangement of the terminal groups. The utilization of two or more classes of amino acids may be necessary for stabilizing a reproducible folding. β-Sheets have long been known to be important building elements of protein structure. According to M. Levitt [81], they may be utilized in a very general manner to stabilize active conformations of proteins.

The large abundance of glycine and alanine might have determined in essence the appearance of the first proteins, but polar side chains are indispensible for the solubility of longer sequences. Four amino acid classes would of course offer much more flexibility. If aspartic acid and valine were the next two

candidates, globular structures might have formed, stabilized by hydrophobic interactions of the side chains of valine and alanine and solubilized by the carboxylic side chains of aspartic acid. This residue further offers many possibilities for forming specific catalytic sites with the participation of divalent metal ions.

Our imagination is taxed to estimate the vast number of possibilities. Experiments that are supposed to test various structures with respect to their efficiency in discriminating between RNA sequences and their structural features are under way. Results obtained with ribonucleases [82] encourage one to seek a 'minimum structure', able to recognize RNA sequences specifically.

XVI.8. Are Synthetases Necessary to Start With?

In the three-dimensional structure of present-day RNAs (cf. Part A, Fig. 14) the anticodon loop is fixed at a considerable distance from the amino acyl site. Such a structure is adapted to the functional needs of present tRNA molecules, imposed by the ribosomal mechanism and by the structure of synthetases. On the other hand, it is known that tRNA can undergo conformational changes that drastically alter its shape and dimensions. R. Rigler and his coworkers [83] studied conformational lifetimes as well as rotational relaxation times by fluorescence methods and concluded the existence of at least three different rapidly interconverting conformational states. Analogous results were obtained by T. Olson et al. [84], who used laser light-scattering techniques. The population of the different conformational states depends strongly on magnesium-ion concentration. It is important, again, to note that under conditions that correspond to those present in sea water (Mg^{2+}; ~50 m*M*), a conformer is present that differs in shape from the L-form found by crystallographic studies, being considerably more cylindric.

This point is stressed because it is most relevant to the question raised. Early enzymes were made of only a very limited number of amino acid residues and therefore cannot have been very bulky globular structures. In order to guarantee a unique assignment of an amino acid to an anticodon, either enzymes as sophisticated as present-day aminoacyl synthetases had to be available, or else the tRNA structure had to allow a much closer contact between the aminoacyl and anticodon sites than the L-form does, in order to admit a simultaneous checking of both sites. The high mutation rate at early stages would otherwise very soon have destroyed any unique coincidental correspondence between these two sites. On the other hand, the conformational transition is still required since the mechanism of peptide-bond formation (cf. Fig. 48) calls for a well-defined separation of the messenger and the growing peptide chain. The data quoted invite reflection about such possibilities. If, on the other hand, a structure similar to the pattern c) shown in Figure 49 is likely to arise, the first amino-acid assignments might even have been made without enzymic help. The tRNA structure as such certainly offers sufficient subtlety for specific recognition. It has been noted [85] that the fourth base from the 3'-end (i.e., the one following 3'ACC) is somehow related to the anticodon. The primary expectations regarding a unique correlation for all tRNAs finally did not materialize. However, such a correlation may have played a key role in the early specific recognition of amino acids by tRNAs. Referring to data from *E. coli* and T_4 phage, the nucleotides in the position following 3'ACC are: U for gly, A for ala, G for asp, and A for val. It was certainly important for early adapters to ensure unique assignment by sufficiently discriminative sites. This property might have been partially lost during later phases of evolution. This is admittedly a speculation and calls for experimental confirmation.

To conclude: Synthetases may have been dispensible at the very early stages, but tRNAs finally turned out to be an unsatisfactory attempt by Nature to make enzymes from nucleic acids. More efficient recognition may have evolved from the coupling factors, which were predestined to recognize tRNA-like structures specifically.

XVI.9. Which Were the First Enzymes?

If synthetases are not really necessary for a *start* of translation (and this is a big 'if!'), we are left with the coupling factors, probably replicases, as the only absolute primary requirements for a coherent evolution of translation. Via such a function, a selective advantage occurring in a translation product can be most efficiently fed back onto the messenger. Hence specific replicases (all belonging to one class of similar protein molecules) not only provide the prerequisites for hypercyclic coupling, but also turn out to be most important for the further evolution of proteins, since only they can tell the messenger what is phenotypically advantageous and how to select for it at the genotypic level, i.e., by enhanced synthesis of the particular messenger. As will be seen in the next paragraph, such a selective coupling between geno- and phenotypic levels works best in combination with spatial separation or compartmentation.

Next, of course, we have to look for catalytic support for the various translation functions. If replicases have established a defined relationship with tRNA-like messengers (including both the plus *and* the minus strands), their recognition properties may well be utilized for synthetase and translatase (i.e., preribosomal) functions. In other words, a gene duplicate of a replicase may well be the precursor of a synthetase messenger as well as of a translation factor such as EF Tu, the more so since the chemistry of replicase and transfer function is very similar and in present systems appears to be effected by similar residues.

Dual functions with gradual divergence may have been a very early requisite of replication and translation mechanisms, just as gene duplication was one of the main vehicles of evolution at later stages.

Those dual functions have clearly left their traces in present cell organelles, and viruses have utilized them as well for their postbiotic evolution in the host cell. The genome of the phage Qβ encodes for only one subunit of its replicase, but utilizes three more factors of the host cell, which have been identified as the ribosomal protein S_1 and the elongation factors EF Tu and EF Ts [87, 88].

Ch. Biebricher [89] has studied the properties of these factors and found that they are involved simultaneously in several functions of ribosomal control, utilizing their acquired property, namely, to recognize tRNA molecules. He argues that also the β-factor of Qβ replicase, which is encoded by the phage genome, has its precursor in the *E. coli* cell, and this seems, indeed, to be the case. Using immunologic techniques, he was able to identify a protein containing EF Tu and EF Ts that behaves like a precursor of the Qβ replicase in uinfected *E. coli* and that appears to be involved in an – as yet unspecified – RNA-synthesis function of *E. coli* [87]. Further, similar correspondences may yet be found with synthetases. It seems that once a certain function has been developed – such as the ability to recognize certain types of RNA – then Nature utilizes this function wherever else it is needed (e.g., specific replication, ribosomal transport and control, amino acid activation).

In some respects the formation of RNA phages may well have mimicked the evolution of early RNA messengers. Phages utilize as many host cell functions as possible except one, namely, specific recognition of their own genome (i.e., coupling via specific replication). Different phages (e.g., Qβ, Ms2, R17) inherit different recognition factors [9], although they all derive from a common ancestor in the host cell. In Part A it was also shown that the primary phase of RNA-phage infection is equvalent to a simple hypercyclic amplification process.

XVI.10. Why Finally Cells with Unified Genomes?

Hypercycles offer advantages for enlarging the information content by functional integration of a messenger system, in which the single replicative unit is limited in length due to a finite fidelity of copying. The increase in information content allows the build-up of a reproducible replication and translation apparatus, by which the translation products can evolve to higher efficiency. This will allow better fidelities, which in turn will increase the information content of each single replicative unit and thereby, again, enhance the quality of the enzymes. Simultaneously, as shown in Section XV, the hypercycle itself will evolve to higher complexity by integrating more differentiated mutant genes.

Increase of information content will not only produce better enzymes; it may also allow each replicative unit to inherit information for more than one enzyme. Dual functions can thereby be removed from the list of earlier evolutionary constraints, i.e., duplicated messengers may develop independently, according to the particular functional needs of their translation products. This may have been the origin of operon structures with control mechanisms for simultaneous replication of several structural genes. Replicases may thus have evolved to common polymerases associated with specific control factors for induction or repression.

After having realized the advantages of functional coupling, which seems to be a requirement for any start of translation, we should ask why functional coupling has finally been replaced by complete structural integration of all genes, the genomes of even the most primitive known cells being structural units. So where are the limitations of hypercyclic organization and what improvements can be made in it?

In a system controlled by functional links we have to distinguish two kinds of mutations. One class will primarily change the phenotypic properties of the messenger itself and thereby alter its target function with respect to a specific replicase or control factor. These mutations are especially important in the early phases of evolution owing to the important role of phenotypic properties of RNA structures. Those target mutations will immediately become selectively effective, advantageous mutations will be fixed, and disadvantageous ones will be dismissed.

The second kind of mutation – which may or may not be neutral with respect to the target function – refers to phenotypic changes in the translation products. The more precisely specified the messengers are, the more specifically a mutation may alter the function of the translation product.

Whether or not a mutant is specifically favored by

selection depends only on the target function, regardless of whether the translation product is altered in a favorable or an unfavorable sense or whether it remains neutral. For the later stages of precellular evolution the most common consequence of a mutation will be a phenotypic change in the translation product coupled with an unaffected target function. The mutant may then proliferate further, but it is not specifically selected against its former wild type, nor would the system select against the mutant, if its translation product proves to be unfavorable. What should really be achieved is a rating of the system as a whole. This may be accomplished by spatial separation of the messenger systems, by niches or—even more efficiently—by compartmentation. A messenger in a given compartment can enrich its environment with its own translation products and compete with other compartments using its efficiency of proliferation. To a limited extent this is also possible simply by spatial separation. However, a compartment without hypercyclic organization does not work at all. The enhanced competition among all messengers in the limited living space of the compartment would destroy any cooperative function.

A compartment could proliferate more efficiently by correlating its own reproduction with the re-duplication of its total gene content. This, of course, requires a fairly involved control mechanism, which could be facilitated by the integration of all genes into one giant replicative unit. Such an individualization of the total compartment requires high fidelity in the replication machinery. In Part A we compared the information content of various stages of life with their corresponding (and observed) replication fidelities (cf. Table 4).

The individualization of compartments is probably connected with the transition from RNA genes or operons to DNA genomes, since only the mechanism of DNA replication could guarantee sufficiently high fidelity. The new individualized unit was the integrated Proto-cell. The previous functional organization of genes and gene products has been superseded and amended by a coupled structural and functional organization. A closer study of the cyclic arrangement of genetic maps may still reveal some remnants of the origins of structural organization, although recombinative epigenetic effects may have covered many of the traces.

As a consequence of unification and individualization, the net growth of (asexual) multiplication of cells obeys a first-order autocatalytic law (in the absence of inhibitory effects). The Darwinian properties of such systems allow for selective evolution as well as for coexistence of a large variety of differentiated species. The integrated unit of the cell turns out to be superior to the more conservative form of hypercyclic organization.

On the other hand, the subsequent evolution of multicellular [90] organisms may again have utilized analogous or alternative forms of hypercyclic organization (nonlinear networks) applied to cells as the new subunits, and thereby have resembled in some respect the process of molecular self-organization.

XVII. Realistic Boundary Conditions

A discussion of the 'realistic hypercycle' would be incomplete without a digression on realistic boundary conditions. We shall be brief, not because we disregard their importance in the historical process of evolution—the occurrence of life on our planet is after all a historical event—but because we are aware of how little we really can say. While the *early stages of life*, owing to evolutionary coherence, have left at least some traces in present organisms, there are no corresponding remnants of the *early environment*. In our discussion so far we have done perhaps some injustice to experiments simulating primordial, template-free protein synthesis, which were carried out by S.W. Fox [91] and others (cf. the review by K. Dose and H. Rauchfuss [92]). It was the goal of our studies to understand the early forms of organization that allowed self-reproduction, selection, and evolutionary adaptation of the biosynthetic machinery, such as we encounter today in living cells. Proteins do not inherit the basic physical prerequisites for such an adaptive self-organization, at least not in any obvious manner as nucleic acids do. On the other hand, they do inherit a tremendous functional capacity, in which they are by far superior to the nucleic acids. Since proteins can form much more easily under primordial conditions, the presence of a large amount of various catalytic materials must have been an essential environmental quality. Research in this field has clearly demonstrated that quite efficient protein catalysis can be present under primordial conditions.

Interfaces deserve special recognition in this respect. If covered with catalytically active material they may have served as the most favorable sites of primordial synthesis. The restriction of molecular motion to the dimensions of a plane increases enormously the efficiency of encounters, especially if sequences of high-order reactions are involved.

L. Onsager [93] has emphasized that under primordial conditions the oceans must have been extensively covered with layers of deposited hydrophobic and hydrophilic material cf. also [94]). Those multilayers must have offered favorable conditions for a primordial preparative chemistry. In view of the obvious

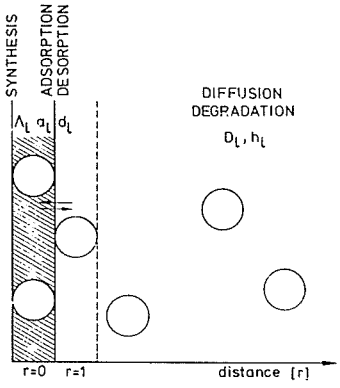


Fig. 59. Schematic representation of a heterogeneous reaction model including hypercyclic coupling. Three spatial regions are distinguished: $r=0$ bound to interface, $r=1$ transition layer at interface, $r>1$ bulk of solution phase. Diffusion to and from interface is superimposed on chemical reactions proceeding according to a hypercyclic scheme

advantages offered by interfaces we have examined the properties of hypercycles under corresponding environmental boundary conditions.

As a simple model we consider a system such as that depicted schematically in Figure 59. Polymer synthesis is restricted to a surface layer only ($r=0$), which has a finite binding capacity for templates and enzymes. The kinetic equations are similar to those applying to homogeneous solutions except that we have to account explicitly for diffusion. We distinguish a growth function that refers to the surface concentrations of replicative molecules and enzymes. Diffusion within the surface is assumed to be fast and not rate-determining. Adsorption and desorption of macromolecules is treated as an exchange reaction between the surface layer ($r=0$) and a solution layer next to the surface ($0<r\leqq 1$). Decomposition may occur at the interface and/or (only) in the bulk of the solution. Finally, transport to and from the interface is represented by a diffusion term.

Depending on the mechanism of synthesis assumed, it may be necessary to consider independent binding sites for both templates and enzymes. We used this model to obtain some clues about the behavior of hypercycles with translation (cf. Sect. IX in Part B). Numerical integration for several sets of rate parameters was performed according to a method described in the literature [95]. Three characteristic results – two of which are in complete analogy to the behavior of hypercycles in homogeneous solutions – can be distinguished:

(A) At very low concentrations of polynucleotides and polypeptides or large values of K_i [see Eqs. (73), (75), and (79) in Part B], the surface densities of polymers do not approach a steady-state value but decrease either monotonically or in damped oscillations. Consequently, the macromolecules die out after some time (Fig. 60).

(B) Above a certain threshold value of total concentration, we find limit cycle behavior in systems with $n<4$. The situation is analogous to the low-concentration limit in a homogeneous solution (Fig. 61).

(C) At sufficiently high concentrations we finally obtain a stationary state:

$$\lim_{t\to\infty}\frac{\partial x_i}{\partial t}=0, \qquad \lim_{t\to\infty}\frac{\partial y_i}{\partial t}=0$$

and $\bar{x}_i>0$, $\bar{y}_i>0$, $i=1,2\ldots n$ (Fig. 62), x_i and y_i being the concentrations of enzymes and messengers, respectively, $\bar{x}_i$ and $\bar{y}_i$ their final stationary values, and t the time.

In systems of lower dimensions ($n\leqq 4$) behavior of types (A) and (C) only was observed.

These model calculations were supplemented by several studies of closely related problems using stochastic computer-simulation techniques. The results again showed the close analogy of behavior of hypercycles at interfaces and in homogeneous solution (as described in detail in Part B).

Consideration of realistic boundary conditions is a point particularly stressed in papers by H. Kuhn [96]. We do not disagree with the assumption of a 'structured environment', nor do we know whether we can agree with the postulation of a very particular environment, unless experimental evidence can be presented that shows at least the usefulness of such postulates.

Our models are by no means confined to spatial uniformity (cf. the above calculations). In fact, the logical inferences behind the various models – namely, the existence of a vast number of structural alternatives requiring natural selection, the limitation of the information content of single replicative units due to restricted fidelities, or the need for functional coupling in order to allow the coherent evolution of a complete ensemble – apply to *any* realistic environment. Kuhn's conclusion that the kind of organization proposed is 'restricted to the particular case of spatial uniformity' is beside the point. Who would claim today, that life could *only* originate in porous material, or at interfaces, or within multilayers at the surface of oceans, or in the bulk of sea water? The models show that it may originate – with greater or lesser likelihood – under any of those boundary conditions, if – and only if – certain criteria are fulfilled. These criteria refer to the problem of generation and accumulation of information and do not differ qualitatively when different boundary conditions are applied.

Much the same can be said with respect to temporal uniformity or nonuniformity. It has been shown in Part B that selection criteria may assume an especially simple form if they apply to steady-state conditions. Since they refer to relative rather than to absolute reaction rates, they are qualitatively the same, regardless of whether the system is growing, oscillating, or in a stationary state.

It is true that annealing is a useful procedure for many problems related to phase separations. Whether, however, thermal fluctuations serve equally well for selection of longer polynucleotides, remains to be shown by experiments.

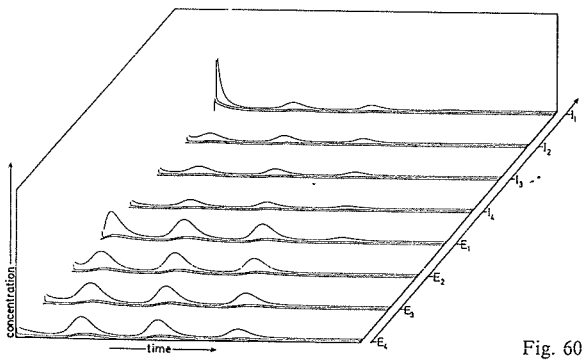


Fig. 60

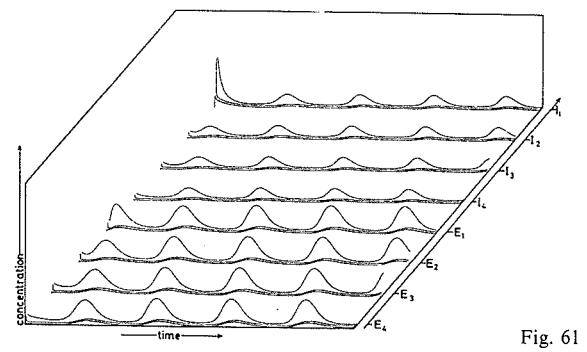


Fig. 61

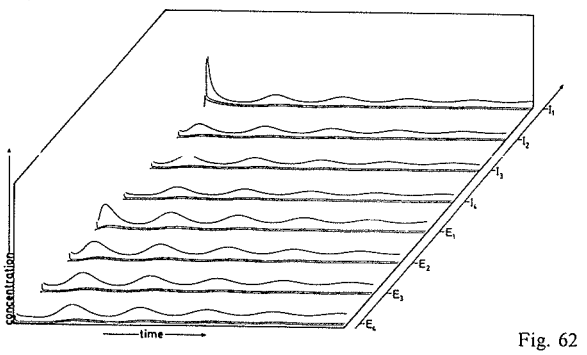


Fig. 62

In order to decide whether fluctuations of temperature improve the selection of strands with higher information content, one must analyze carefully the relative temperature coefficients of all processes involved. The tempererature coefficient of hydrolysis is likely to be the largest of all. *Instructed* replication is by no means generally enhanced at high temperatures. The incoming nucleotide has to bind cooperatively to its complementary base at the template, at the same time utilizing the stacking interaction with the top bases in the growing chain. This is not possible above the melting point of the templates. These considerations apply to any kind of environment, be it an aqueous bulk phase, a surface layer, or a compartment in a coarse-grained or porous material.

The important point for raising the information content is the relative strength of complementary vis-à-vis noncomplementary interactions. Discrimination generally works better at lower than at higher temperatures. S. Miller and L. Orgel ([63], p. 126) conclude from their experimental data:

"We do not know what the temperature was in the primitive ocean, but we can say that the instability of various organic compounds and polymers makes a compelling argument that life could not have arisen in the ocean unless the temperature was below 25° C. A temperature of 0° C would have helped greatly, and −21° C would have been even better. At such low temperatures, most of the water on the primitive earth would have been in the form of ice, with liquid sea water confined to the equatorial oceans.

There is another reason for believing that life evolved

Figs. 60 to 62. Solution curves, obtained by numerical integration, for a system of partial differential equations corresponding to the model depicted in Figure 59. The rate equations account for a growth function A_i as introduced in Part B, which refers to a four-membered hypercycle with translation (B. IX) and which has nonzero catalytic-rate terms only at the interface ($r=0$). The equations furthermore take care of adsorption and desorption (a_i, d_i describing the transition of particles between $r=0$ and $r=1$), hydrolysis (effective at $r \geqq 1$), and diffusion in the bulk of solution ($r>1$, i.e., to and from the transition layer at $r=1$). Each set of three curves refers to the three spatial positions $r=0$ (upper), $r=1$ (medium) and $r=2$ (lower). Figures 60 to 62 differ only in the assumption of different values for the stability constants of the catalytically active complexes $I_i \times E_{i-1}$, which are highest (0.16) in Figure 60, intermediate (0.06) in Figure 61, and lowest (0.04) in Figure 62. The balance between production and removal is sufficient to make the assumption of a dilution flux dispensable. As a consequence of the values chosen (relative to the uniform parameters f_i, k_i − according to B,IX − a_i, d_i and D) autocatalytic production cannot compete with removal by transport and decomposition in Figure 60, where all partners I_i and E_i die out. In both other cases a stable hypercyclic organization is established at the interface, where population numbers are either oscillatory (Fig. 61) or stationary (Fig. 62)

at low temperatures, whether in the oceans or lakes. All of the template-directed reactions that must have led to the emergence of biological organization take place only below the melting temperature of the appropriate organized polynucleotide structure. These temperatures range from 0° C, or lower, to perhaps 35° C, in the case of polynucleotide-mononucleotide helices.

The environment in which life arose is frequently referred to as a warm, dilute soup of organic compounds. We believe that a cold, concentrated soup would have provided a better environment for the origins of life."

XVIII. Continuity of Evolution

It has been the object of this final part of the trilogy to demonstrate that hypercycles may indeed represent realistic systems of matter rather than merely imaginary products of our mind.

Evolution is conservative and therefore appears to be an almost continuous process—apart from occasional drastic changes. Selection is in fact based on instabilities brought about by the appearance of advantageous mutants that cause formerly stable distributions to break down. The descendents, however, are usually so closely related to their immediate ancestors that changes emerge very gradually. Prebiotic evolution presents no exception to the rule.

Let us summarize briefly what we think are the essential stages in the transition from the nonliving to the living (cf. Fig. 63).

1. The first appearance of macromolecules is dictated by their structural stability as well as by the chemical abundances of their constituents. In the early phase, there must have been many undetermined protein-like substances and much fewer RNA-like polymers. The RNA-like polymers, however, inherit physically the property of reproducing themselves, and this is a necessary prerequisite for systematic evolution.

2. The composition of the first polynucleotides is also dictated by chemical abundance. Early nucleic acids are anything but a homogeneous class of macromolecules, including L- and D-compounds as well as various ester linkages, predominantly 2′–5′ besides 3′–5′. Reproducibility of sequences depends on faithfulness of copying. GC-rich compounds can form the longest reproducible sequences. On the other hand, AU substitutions are also necessary. They cause a certain structural flexibility that favors fast reproduction. Reproducible sequences form a quasi-species distribution, which exhibits Darwinian behavior.

3. Comma-free patterns in the quasi-species distribution qualify as messengers, while strands with exposed complementary patterns (possibly being the minus strands of messengers) represent suitable adapters. The first amino acids are assigned to adapters according to their availabilities. Translation products look monotonous, since they consist mainly of glycine and alanine residues. The same must be true for the bulk of noninstructed proteins.

4. If any of the possible translation products offers catalytic support for the replication of its own messenger, then this very messenger may become dominant in the quasi-species distribution and, together with

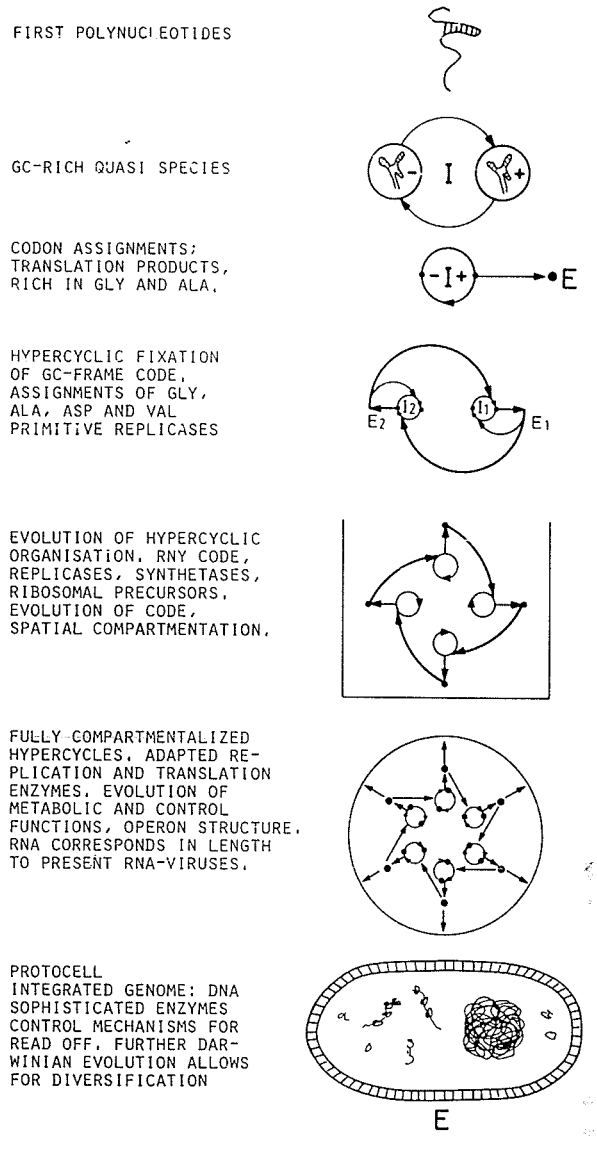


Fig. 63. Hypothetical scheme of evolution from single macromolecules to integrated cell structures

its closely related mutants, will be present in great abundance. The process may be triggered by some of the *noninstructed* environmental proteins, which in their composition reflect the relative abundance of amino acids and hence may mimic primitive *instructed* proteins in their properties.

5. The mutants of the dominant messenger – according to the criteria for hypercyclic evolution – may become integrated into the reproduction cycle, whenever they offer further advantages. Thus hypercyclic organization with several codon assignments can build up. Such a hypercyclic organization is a prerequisite for the coherent evolution of a translation apparatus. More and more mutants become integrated, and the steadily increasing fidelities will allow a prolongation of the sequences. Different enzymic functions (replicases, synthetases, ribosomal factors) may emerge from joint precursors by way of gene duplication and subsequently diverge. Units, including several structural genes, i.e., which are jointly controlled by one coupling factor.

6. The complex hypercyclic organization can only evolve further if it efficiently utilizes favorable phenotypic changes. In order to favor selectively the corresponding genotypes, spatial separation (either by compartmentation or by complex formation) becomes necessary and allows selection among alternative mutant combinations. Remnants of complex formation may be seen in the ribosomes.

We do not know at which stage such a system was able to integrate its information content completely into one giant genome molecule. For this a highly sophisticated enzymic machinery was required, and the role of information storage had to be gradually transferred to DNA (which might have happened at quite early stages).

These glimpses into the historical process of precellular evolution may suffice to show in which direction development, triggered by hypercyclic integration of self-replicative molecular units, may lead, and how the developing system may finally converge to give an organization as complex as the prokaryotic cell. We want to stress the speculative character of part C. The early phase of self-organization left traces, but no witnesses, so that many important steps still remain in the dark.

It was not even our intention to uncover historical truth. For a process so largely dependent upon chance – where indeterminate microscopic events, such as mutations, amplify and finally determine the course of macroscopic development – a complete reconstruction of history is not possible at all. Even in biology there is a 'poverty of historicism'. On the other hand, the principles governing the historical process of evolution – even in their finer details – may well be susceptible to our understanding. The traces of history in present systems may provide enough clues to allow one day the construction of 'those *n* equations for the *n* unknowns'.

All we wanted to show in this part is that the unique class of reaction networks, which we have termed hypercycles, is indeed the simplest realistic molecular organization that ensures the coexistence of functionally related self-replicative units. Self-replication is required for the conservation of information. Hence the hypercycle is the simplest system that can allow the evolution of reproducible functional links. It can originate from one self-replicating unit and its mutants, i.e., from a single (molecular) quasi-species. Its emergence was inevitable, whenever the conditions laid down by Nature allowed it. And yet:

"If anyone can name a more beautiful triangle underlying the composition of bodies, we will greet him not as an opponent but as a friend in the right."
(Plato, Timaios) [97]

This work was greatly stimulated by discussions with Francis Crick, Stanley Miller, and Leslie Orgel; which for us meant some 'selection pressure' to look for more continuity in molecular evolution. Especially helpful were suggestions and comments by Ch. Biebricher, I. Epstein, B. Gutte, D. Pörschke, K., Sigmund, P. Woolley, and R. Wolff.

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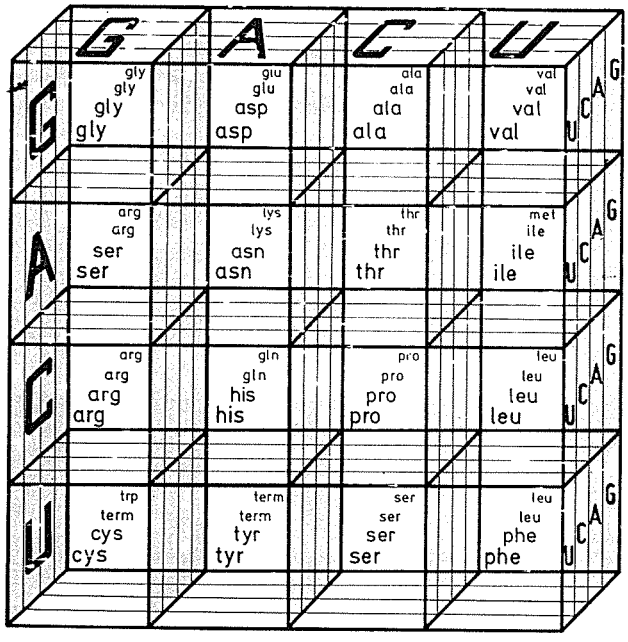


Fig. 64. The genetic code is the universal key for translation of genetic information from the legislative language of nucleic acids into the executive language of proteins. In this representation the three coordinates of geometric space are assigned to the positions of letters in the triplet codewords. The four letters of the nucleic acid alphabet are arranged in such a way that the codewords for the most abundant amino acids appear in the top layer of the cube

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12

Catalytic hypercycles

The hypercycle equation is a particular replicator equation whose interior rest point is unstable for higher dimensions. Nevertheless, the boundary is a repellor, so that the frequencies are bounded away from 0. We prove this permanence with the help of an average Lyapunov function.

12.1 The hypercycle equation

Let us consider the *hypercycle equation*

$$\dot{x}_i = x_i \left(k_i x_{i-1} - \sum_{j=1}^{n} k_j x_j x_{j-1} \right) \tag{12.1}$$

where the indices $i = 1, 2, \ldots, n$ are counted mod n (so that $x_0 = x_n$). This is just the replicator equation $\dot{x}_i = x_i((A\mathbf{x})_i - \mathbf{x} \cdot A\mathbf{x})$ on S_n, with

$$A = \begin{bmatrix} 0 & 0 & 0 & . & . & . & k_1 \\ k_2 & 0 & 0 & . & . & . & 0 \\ 0 & k_3 & 0 & . & . & . & 0 \\ . & . & . & . & . & . & . \\ 0 & 0 & 0 & . & . & k_n & 0 \end{bmatrix} . \tag{12.2}$$

This equation plays a crucial role in a theory of Manfred Eigen and Peter Schuster on prebiotic evolution: the x_i are the relative frequencies of RNA molecules that catalyse each other's replication in a closed feedback loop which provides a possible mechanism for the 'ganging up' of molecular information carriers.

The hypercycle equation (12.1) admits a unique rest point $\mathbf{p}$ in int S_n, given

by $k_1x_n = k_2x_1 = \cdots = k_nx_{n-1}$ and $x_1 + x_2 + \cdots + x_n = 1$. This yields

$$p_i = k_{i+1}^{-1}\left(\sum_j k_{j+1}^{-1}\right)^{-1} \qquad i = 1,\ldots,n\,. \tag{12.3}$$

The stability analysis of $\mathbf{p}$ is considerably simplified by a change in coordinates. Thus we set

$$y_i = \frac{k_{i+1}x_i}{\sum_{j=1}^n k_{j+1}x_j}\,. \tag{12.4}$$

The projective transformation $\mathbf{x} \to \mathbf{y}$ of S_n onto itself transforms (12.1) into

$$\dot{y}_i = y_i\left(y_{i-1} - \sum y_jy_{j-1}\right)\left(\sum_s k_{s+1}^{-1}y_s\right)^{-1}\,. \tag{12.5}$$

We may drop the last factor (see exercises 4.1.2, 4.1.3) and obtain

$$\dot{x}_i = x_i(x_{i-1} - \sum_j x_jx_{j-1}) \tag{12.6}$$

which is just (12.1) with $k_1 = k_2 = \cdots = k_n = 1$. Thus a change in coordinates followed by a change in velocity allows one to get rid of the coefficients k_i. The Jacobian of (12.6) at the unique rest point $\mathbf{m} = \frac{1}{n}\mathbf{1}$ in $\operatorname{int} S_n$ is a circulant matrix (see section 5.5). The elements in the first row are

$$-\frac{2}{n^2},\quad -\frac{2}{n^2},\quad \ldots,\quad -\frac{2}{n^2},\quad \frac{1}{n} - \frac{2}{n^2}$$

and the other rows are obtained by cyclic permutation. The eigenvalues are given by (5.20). This yields $\gamma_0 = -1/n$ and, for $j = 1,\ldots,n-1$,

$$\gamma_j = \sum_{k=0}^{n-1}\left(-\frac{2}{n^2}\right)\lambda^{kj} + \frac{1}{n}\lambda^{(n-1)j} = \frac{\lambda^{-j}}{n} \tag{12.7}$$

with $\lambda = e^{2\pi i/n}$. The eigenvalue γ_0 corresponds to the eigenvector $\mathbf{1}$ which is orthogonal to the simplex S_n. Since we are only interested in the restriction of (12.6) to S_n, we need not consider it any further. We see that for $n \geq 5$, the rest point $\mathbf{m}$ is unstable.

The function

$$P(\mathbf{x}) = x_1x_2\ldots x_n \tag{12.8}$$

vanishes on $\operatorname{bd} S_n$ (where $x_i = 0$ for at least one i) and is strictly positive in $\operatorname{int} S_n$. It attains its maximal value (in S_n) at the point $\mathbf{m}$. The time derivative

of $t \to \log P(\mathbf{x}(t))$ is given by

$$(\log P)^{\cdot} = \sum_{i=1}^n \frac{\dot{x}_i}{x_i} = 1 - n\sum_{j=1}^n x_jx_{j-1}\,. \tag{12.9}$$

For $n = 2$ and $n = 3$, this function is positive and vanishes only at $\mathbf{m}$.

Exercise 12.1.1 Show that for $n = 4$, the function $\dot{P}/P$ is still nonnegative, and vanishes on the set $\{\mathbf{x} \in S_4 : x_1 + x_3 = x_2 + x_4\}$, which contains only one invariant set, consisting of $\mathbf{m}$ only.

The following theorem is an immediate consequence.

Theorem 12.1.2 *For short hypercycles (i.e. $n = 2,3,4$) the inner rest point is globally stable.*

For $n \geq 5$, however, the interior rest point is unstable, and one is left with the question whether a permanent coexistence of the different types of polynucleotides is possible. This question will be addressed in the next section.

Exercise 12.1.3 An alternative way of proving the global stability of $\mathbf{m}$ for $n = 3$ would be to use other Lyapunov functions than (12.8). Try e.g. $x_1^{-p} + x_2^{-p} + x_3^{-p}$.

Exercise 12.1.4 Write down a global Lyapunov function for (12.1) if $n \leq 4$.

Exercise 12.1.5 Consider the hypercycle equation (12.1) with $n = 3$. For which values of k_i is the unique interior rest point evolutionarily stable? Show that for $n = 4$, the interior rest point (which, as we know, is globally stable) is never an ESS.

12.2 Permanence

As we have just seen, hypercycles do not reach a stable rest point for $n \geq 5$. They may nevertheless serve their purpose, which is to allow the coexistence of several types of self-reproducing macromolecules: whether the concentrations converge or oscillate in a regular or irregular way is of secondary importance. The main thing is that they do not vanish: more precisely, that there exists a *threshold value* $\delta > 0$ such that every solution of (12.6) in $\operatorname{int} S_n$ satisfies $x_i(t) > \delta$ for $i = 1,\ldots,n$ whenever t is large enough. This implies that if initially all species are present, even if only in very

small quantities, then after some time some sizeable amount of each will be present. No perturbation which is smaller than δ could wipe out a molecular species.

This property, which is important in many other contexts, deserves a name. A dynamical system defined on S_n is said to be *permanent* if there exists a $\delta > 0$ such that $x_i(0) > 0$ for $i = 1,\ldots,n$ implies

$$\liminf_{t\to+\infty} x_i(t) > \delta \tag{12.10}$$

for $i = 1,\ldots,n$.

Let us stress that δ does *not* depend on the initial values $x_i(0)$. Permanence means more than just that no component will vanish. If every state is a rest point, for example, the system is not permanent. Even if initially the concentrations were abundant, a sequence of tiny perturbations could lead to the extinction of a species. For a permanent system, on the other hand, perturbations which are sufficiently small and rare cannot lead to extinction. The boundary of the state space S_n acts as a *repellor*.

The proof that the hypercycle is permanent is not obvious. We shall start with a more general theorem giving conditions for permanence which will also be useful in many other situations.

Theorem 12.2.1 *Let us consider a dynamical system on S_n leaving the boundary invariant. Let $P : S_n \to \mathbb{R}$ be a differentiable function vanishing on* bd S_n *and strictly positive in* int S_n. *If there exists a continuous function Ψ on S_n such that the following two conditions hold:*

$$\text{for } \mathbf{x} \in \operatorname{int} S_n, \quad \frac{\dot{P}(\mathbf{x})}{P(\mathbf{x})} = \Psi(\mathbf{x}) \tag{12.11}$$

$$\text{for } \mathbf{x} \in \operatorname{bd} S_n, \quad \int_0^T \Psi(\mathbf{x}(t))dt > 0 \text{ for some } T > 0\,, \tag{12.12}$$

then the dynamical system is permanent.

The value $P(\mathbf{x})$ measures the distance from $\mathbf{x}$ to the boundary. If one had $\Psi > 0$ on bd S_n — a condition implying (12.12) — then $\dot{P}(\mathbf{x}) > 0$ for any $\mathbf{x} \in \operatorname{int} S_n$ near the boundary, and so P would increase, i.e. the orbit would be repelled from bd S_n. In that case, P would act like a Lyapunov function. Quite often, however, one cannot find a function P of this type. The weaker version defined above is said to be an *average Lyapunov function*: its time average acts like a Lyapunov function.

Proof The $T > 0$ in (12.12) can obviously be chosen as a locally continuous function $T(\mathbf{x})$. Its infimum τ is positive, since bd S_n is compact. For $h > 0$, we define

$$U_h = \left\{ \mathbf{x} \in S_n : \text{ there is a } T > \tau \text{ such that } \frac{1}{T}\int_0^T \Psi(\mathbf{x}(t))dt > h \right\}.$$

For $\mathbf{x} \in U_h$ we set

$$T_h(\mathbf{x}) = \inf \left\{ T > \tau : \frac{1}{T}\int_0^T \Psi(\mathbf{x}(t))dt > h \right\}.$$

We show first that U_h is open and T_h upper semicontinuous: in other words, if $\mathbf{x} \in U_h$ and $\alpha > 0$ are given, then for $\mathbf{y} \in S_n$ sufficiently close to $\mathbf{x}$, one has

$$\mathbf{y} \in U_h \quad \text{and} \quad T_h(\mathbf{y}) < T_h(\mathbf{x}) + \alpha\,. \tag{12.13}$$

Indeed, for α and $\mathbf{x}$ there is a $T \in [\tau, T_h(x) + \alpha[$ such that

$$\varepsilon = \frac{1}{T}\int_0^T \Psi(\mathbf{x}(t))dt - h > 0\,.$$

Since the solutions of ordinary differential equations depend continuously on the initial values, $\mathbf{x}(t)$ and $\mathbf{y}(t)$ are near each other, for all $t \in [0, T]$, if $\mathbf{x}$ and $\mathbf{y}$ are sufficiently close. The uniform continuity of Ψ implies $|\Psi(\mathbf{x}(t)) - \Psi(\mathbf{y}(t))| < \varepsilon$ for all times $t \in [0, T]$, and hence

$$\frac{1}{T}\int_0^T \Psi(\mathbf{y}(t))dt > \frac{1}{T}\int_0^T \Psi(\mathbf{x}(t))dt - \varepsilon = h$$

from which (12.13) follows.

By (12.12) the family of nested sets U_h (with $h > 0$) is an open covering of the compact set bd S_n. There exists one $h > 0$, then, such that U_h is an open neighbourhood of bd S_n (in S_n). Since $S_n \backslash U_h$ is also compact, P attains its minimum on this set. If we choose $p > 0$ smaller than this minimum, then the set

$$I(p) = \{\mathbf{x} \in S_n : 0 < P(\mathbf{x}) \leq p\}$$

is contained in U_h. $I(p)$ is a 'boundary layer' which is very thin if p is small. We shall show that if $\mathbf{x} \in I(p)$, then there exists a $t > 0$ such that $\mathbf{x}(t) \notin I(p)$. Indeed, otherwise $\mathbf{x}(t)$ would have to be in U_h for all $t > 0$. In this case

there exists a $T \geq \tau$ such that

$$\frac{1}{T}\int_t^{T+t} \Psi(\mathbf{x}(s))ds > h\ .$$

But since $\Psi = (\log P)^{\cdot}$ holds in $\operatorname{int} S_n$, this implies

$$h < \frac{1}{T}\int_t^{T+t} (\log P)^{\cdot}(\mathbf{x}(s))ds = \frac{1}{T}\left[\log P(\mathbf{x}(T+t)) - \log P(\mathbf{x}(t))\right]\ ,$$

that is,

$$P(x(T+t)) > P(\mathbf{x}(t))e^{hT} \geq P(\mathbf{x}(t))e^{h\tau}\ .$$

Hence there would exist a sequence t_n for which $P(\mathbf{x}(t_n))$ tends to $+\infty$, in contradiction to the boundedness of P.

Let us denote by $\bar{I}(p)$ the union of $I(p)$ with $\operatorname{bd} S_n$. All that remains to be shown is that there exists a $q \in]0, p[$ such that $\mathbf{x}(0) \notin \bar{I}(p)$ implies $\mathbf{x}(t) \notin I(q)$ for all $t \geq 0$.

The upper semicontinuous function T_h admits an upper bound $\bar{T}$ on the compact set $\bar{I}(p)$. Let t_0 be the first time when $\mathbf{x}(t)$ reaches $\bar{I}(p)$, i.e.

$$t_0 = \min\{t > 0 : \mathbf{x}(t) \in \bar{I}(p)\},$$

and let $\mathbf{x}(t_0) = \mathbf{y}$. Obviously $P(\mathbf{y}) = p$. Let m be the minimum of Ψ on S_n. For $m \geq 0$, all is clear, since P never decreases. In case $m < 0$ we set $q = pe^{m\bar{T}}$. For $t \in]0, \bar{T}[$,

$$\frac{1}{t}\int_0^t \Psi(\mathbf{y}(s))ds \geq m,$$

and hence, just as above

$$P(\mathbf{y}(t)) \geq P(\mathbf{y})e^{mt} > pe^{m\bar{T}} = q\ .$$

Hence the solution does not reach $I(q)$ for $t \in]0, \bar{T}[$. Furthermore, since $\mathbf{y} \in I(p)$, there is a time $T \in [\tau, \bar{T}[$ such that

$$P(\mathbf{y}(T)) \geq pe^{h\tau} \geq p\ .$$

At time $t + T$, thus, the orbit of $\mathbf{x}$ has left $I(p)$ without having reached $I(q)$. Repeating this argument, one sees that the orbit can never reach $I(q)$. □

Theorem 12.2.2 *It is sufficient to verify* (12.12) *for all* $\mathbf{x}$ *in the* ω*-limits of orbits on the boundary of* S_n.

Proof There exists, as before, an $h > 0$ such that U_h is an open neighbourhood of $\omega(\mathbf{x})$ (in S_n) for all $\mathbf{x} \in \operatorname{bd} S_n$. Since $\mathbf{x}(t)$ converges to $\omega(\mathbf{x})$, there is a t_1 such that $\mathbf{x}(t) \in U_h$ for all $t \geq t_1$. There is a $t_2 \geq t_1 + \tau$ such that

$$\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} \Psi(\mathbf{x}(t))dt > h\ ,$$

similarly a $t_3 \geq t_2 + \tau$ such that

$$\frac{1}{t_3 - t_2}\int_{t_2}^{t_3} \Psi(\mathbf{x}(t))dt > h,$$

etc. We obtain a sequence $t_1, t_2, t_3, \ldots$ satisfying

$$\frac{1}{t_k - t_1}\int_{t_1}^{t_k} \Psi(\mathbf{x}(t))dt > h\ .$$

If k is sufficiently large, the time average

$$\frac{1}{t_k}\int_0^{t_k} \Psi(\mathbf{x}(t))dt$$

is close to the previous expression and hence positive. □

Exercise 12.2.3 Show that theorems 12.2.1 and 12.2.2 are valid if Ψ is only assumed to be lower semicontinuous. This can always be achieved by defining, for $\mathbf{x} \in \operatorname{bd} S_n$,

$$\Psi(\mathbf{x}) = \liminf_{\mathbf{y}\to\mathbf{x}} \frac{\dot{P}}{P}(\mathbf{y})$$

where $\mathbf{y}$ belongs to $\operatorname{int} S_n$.

12.3 The permanence of the hypercycle

Theorem 12.3.1 *The hypercycle* (12.1) *is permanent.*

Proof We shall use theorem 12.2.1. As the average Lyapunov function $P(\mathbf{x})$ we choose the product $x_1x_2\ldots x_n$ which has already stood us in good stead in section 12.1. We have $\dot{P} = P\Psi$ with $\Psi = \sum k_i x_{i-1} - n\bar{f}$, where $\bar{f} = \sum k_j x_j x_{j-1}$. In order to show that P is an average Lyapunov function,

there exists a $T \geq \tau$ such that

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that is,

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Hence the solution does not reach $I(q)$ for $t \in]0, \bar{T}[$. Furthermore, since $\mathbf{y} \in I(p)$, there is a time $T \in [\tau, \bar{T}[$ such that

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it remains to verify condition (12.12). Thus we have to show that for every $\mathbf{x} \in \operatorname{bd} S_n$, there exists a $T > 0$ such that

$$\frac{1}{T}\int_0^T \sum_{i=1}^{n} (k_i x_{i-1} - n\bar{f})dt > 0 \tag{12.14}$$

i.e. such that

$$\frac{1}{T}\int_0^T \bar{f}(\mathbf{x}(t))dt < \frac{1}{nT}\int_0^T \sum k_i x_{i-1} dt\ . \tag{12.15}$$

Since $k := \min k_i > 0$ and $\sum k_i x_{i-1} \geq k$ for all $\mathbf{x} \in S_n$, the right hand side of (12.15) is not smaller than k/n. It is enough, therefore, to show that there is no $\mathbf{x} \in \operatorname{bd} S_n$ such that for all $T > 0$

$$\frac{1}{T}\int_0^T \bar{f}(\mathbf{x}(t))dt \geq \frac{k}{n}\ . \tag{12.16}$$

Let us proceed indirectly and assume that there is such an $\mathbf{x} \in S_n$. We shall show by induction that

$$\lim_{t\to+\infty} x_i(t) = 0 \tag{12.17}$$

for $i = 1, \ldots, n$. Since $\mathbf{x} \in \operatorname{bd} S_n$, there exists an index i_0 such that $x_{i_0}(t) \equiv 0$. Now if $x_i(t)$ converges to 0, then so does x_{i+1}; indeed, if $x_{i+1}(t) > 0$, one obtains from

$$(\log x_{i+1})^{\cdot} = \frac{\dot{x}_{i+1}}{x_{i+1}} = k_{i+1}x_i - \bar{f}$$

by integrating from 0 to T and dividing by T

$$\begin{aligned}\frac{\log x_{i+1}(T) - \log x_{i+1}(0)}{T} &= \frac{1}{T}\int_0^T (\log x_{i+1})^{\cdot} dt \\ &= \frac{1}{T}\int_0^T k_{i+1}x_i(t)dt - \frac{1}{T}\int_0^T \bar{f}(\mathbf{x}(t))dt\ .\end{aligned}$$

From $x_i(t) \to 0$ it follows that

$$\frac{1}{T}\int_0^T k_{i+1}x_i(t)dt < \frac{k}{2n}$$

for all sufficiently large T. Together with (12.16) this implies

$$\log x_{i+1}(T) - \log x_{i+1}(0) < -\frac{kT}{2n}$$

or

$$x_{i+1}(T) < x_{i+1}(0)\exp\left(-\frac{kT}{2n}\right)\ . \tag{12.18}$$

Hence $x_{i+1}(t) \to 0$ and (12.17) must hold. This contradicts the relation $\sum x_i = 1$. Hence P is an average Lyapunov function and (12.1) is permanent. □

Let us mention that the same proof shows that the considerably more general equation obtained by replacing the constants $k_i > 0$ in (12.1) by positive functions $F_i(\mathbf{x})$ is also permanent. Such equations describe the reaction kinetics for more realistic hypercycle models.

Exercise 12.3.2 Analyse the discrete time dynamics (7.23) for the hypercycle. Prove permanence for $C > 0$. Analyse the local stability for the interior fixed point $\mathbf{p}$. Prove that $\mathbf{p}$ is globally stable if $n \leq 3$ and $C > 0$. Analyse the asymptotic behaviour for $C = 0$.

12.4 The competition of disjoint hypercycles

Let us assume now that the 'primordial soup' contains n types of RNA molecules organized into *several* disjoint hypercycles. This can be described by a permutation π of the set $\{1, \ldots, n\}$. Every permutation can be decomposed into *elementary cycles* $\Gamma_1, \ldots \Gamma_s$: these correspond to hypercycles. The dynamics is given by

$$\dot{x}_i = x_i\left(k_i x_{\pi(i)} - \sum k_j x_j x_{\pi(j)}\right) \tag{12.19}$$

with $k_i > 0$ for $i = 1, \ldots, n$.

If π consists of a unique cycle, we obtain — up to a reordering of the indices — the familiar hypercycle equation (12.1). If the elementary cycle Γ_j consists of a unique element i (a fixed point of the permutation π), then M_i is an autocatalytic molecular type. As in section 12.1 we may perform a transformation

$$y_i = \frac{k_{\tau(i)}x_i}{\sum_j k_{\tau(j)}x_j} \tag{12.20}$$

(with $\tau = \pi^{-1}$) and get rid of the k_i. Again there is a unique rest point in $\operatorname{int} S_n$, namely the centre $\mathbf{m} = \frac{1}{n}\mathbf{1}$.

Exercise 12.4.1 Show that short hypercycles go to equilibrium: if the length $|\Gamma_k|$ of Γ_k is smaller than 5 and if i and j belong to Γ_k, then

$$\frac{x_i}{x_j} \to 1 \tag{12.21}$$

for $t \to +\infty$. (Hint: if Γ_k consists of the indices $1,\dots,m$ (with $m \leq 4$) consider $V = PS^{-m}$ with $P = x_1 \dots x_m$ and $S = x_1 + \dots + x_m$.)

Exercise 12.4.2 Show that as soon as there is more than one hypercycle in the reactor, the interior rest point **m** is unstable and the system is not permanent.

Exercise 12.4.3 For the catalytic chain described by (12.1) with $k_1 = 0$, $k_2 > 0,\dots,k_n > 0$, all x_i with $i \neq n$ will tend to extinction. (Hint: show by induction, for $1 \leq i \leq n-1$, that $x_i \to 0$ and x_i/x_{i+1} converges in an ultimately monotonic way to some limit. Alternatively, use Theorem 7.6.1.)

12.5 Notes

For the hypercycle theory see Eigen and Schuster (1979) and Eigen *et al.* (1981). Permanence was introduced in Schuster *et al.* (1979a). Permanence of a general 'hypercycle equation' was shown in Hofbauer *et al.* (1981). Theorem 12.2.1 on average Lyapunov functions and its application to hypercycles is from Hofbauer (1981b), while exercise 12.2.3 is due to Hutson (1984), who set it up in a considerably more general situation. A discrete time version of the hypercycle (see exercise 12.3.2) was analysed in Hofbauer (1984). In Hofbauer *et al.* (1991), it has been shown that (12.1) admits a stable periodic orbit for $n \geq 5$.

13

Criteria for permanence

Using topological arguments, we show that every permanent dynamical system admits an interior rest point. We introduce the notion of a saturated rest point, which corresponds to a Nash equilibrium. For replicator equations we derive a series of necessary and of sufficient conditions for permanence.

13.1 Permanence and persistence for replicator equations

For the replicator equation

$$\dot{x}_i = x_i(f_i(\mathbf{x}) - \bar{f}) \tag{13.1}$$

on S_n, with $\bar{f}(\mathbf{x}) = \sum x_j f_j(\mathbf{x})$, and for ecological equations of the type

$$\dot{x}_i = x_i f_i(\mathbf{x}) \tag{13.2}$$

on $\mathbb{R}^n_+$, we can rarely expect to obtain a full description of the attractors. For many purposes, however, the precise asymptotic behaviour is less important than the question of *extinction*, i.e. whether all species in the system can survive or not. There are several mathematical concepts dealing with this aspect. In particular, a system of type (13.1) or (13.2) is said to be *permanent* if there exists a compact set K in the interior of the state space such that all orbits in the interior end up in K. This means that the boundary is a *repellor* (for (13.2), we have to consider the points at infinity as part of the boundary of $\mathbb{R}^n_+$).

Equivalently, permanence means for (13.1) that there exists a $\delta > 0$ such that

$$\delta < \liminf_{t \to +\infty} x_i(t) \tag{13.3}$$

for all i, whenever $x_i(0) > 0$ for all i. (Thus we recover the definition of

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NL3516 **Catalytic hypercycle**

The concept of the "hypercycle" has been invented in the nineteen seventies in order to characterize a functional entity that integrates several autocatalytic elements into an organized unit (Eigen, 1971; Eigen & Schuster, 1977, 1978a,b). A catalytic hypercycle is defined as a cyclic network of autocatalytic reactions (figure 1). Autocatalysts, in general, compete when they are supported by the same source of energy or material. Hypercyclic coupling introduces mutual dependence of elements and suppresses competition. Consequently, the fate of all members of a hypercycle is identical with that of the entire system and, in other words, no element of a hypercycle dies out provided the hypercycle as such survives. The current view of biological evolution distinguishes periods of dominating Darwinian evolution based on variation, competition, and selection interrupted by rather short epochs of radical innovations often called *major transitions* (Maynard Smith & Szathmáry, 1995; Schuster, 1996). In the course of biological evolution major transitions introduce higher hierarchical levels. Examples are: (i) the origin of translation from nucleic acid sequences into proteins including the invention of the genetic code, (ii) the transition from independent replicating molecules to chromosomes and genomes, (iii) the transition from the prokaryotic to the eukaryotic cell, (iv) the transition from independent unicellular individuals to differentiated multicellular organisms, (v) the transition from solitary animals to animal societies, and (vi) presumably a series of successive transitions from animal societies to man. All major transitions introduce a previously unknown kind of cooperation into biology. The hypercycle is one of very few mechanisms that can deal with cooperation of otherwise competing individuals. It is used as a model system in prebiotic chemistry, in evolutionary biology, in theoretical economics as well as in cultural sciences.

The simplest example of a catalytic hypercycle is the *elementary hypercycle*. It is described by the dynamical system[1]

$$\frac{dx_i}{dt} = x_i \left(f_i x_{i-1} - \sum_{j=1}^{n} f_j x_{j-1} x_j \right); \; i,j = 1,2,\ldots,n; \; i,j = \bmod n\,. \quad (1)$$

The catalytic interactions within a hypercycle form a directed closed loop comprising all elements, often called Hamiltonian arc: $1 \to 2 \to 3 \to \ldots \to n \to 1$ (figure 1). Hypercycles are special cases of replicator equations of the class

$$\frac{dx_i}{dt} = x_i \left(\sum_{j=1}^{n} a_{ij} x_j - \sum_{j=1}^{n} \sum_{k=1}^{n} a_{jk} x_j x_k \right); \; i,j,k = 1,2,\ldots,n \quad (2)$$

with $a_{ij} = f_i \cdot \delta_{i-1,j}$; $i,j = \bmod n$.[1] For positive rate parameters and initial conditions inside the positive orthant[2]the trajectory of a hypercycle remains within this orthant:

[1]The '$\bmod n$' function implies a cyclic progression of integers, $1,\ldots,n-1,n,1,\ldots$. The symbol $\delta_{i,j}$ represents Kronecker's symbol: $\delta = 1$ for $i = j$ and $\delta = 0$ for $i \neq j$.

[2]The notion of *orthant* refers to the entire section of a Cartesian coordinate system in which the signs of variables don't change. In n-dimensions the positive orthant is defined by $\{x_i > 0 \,\forall\, i = 1,2,\ldots,n\}$.

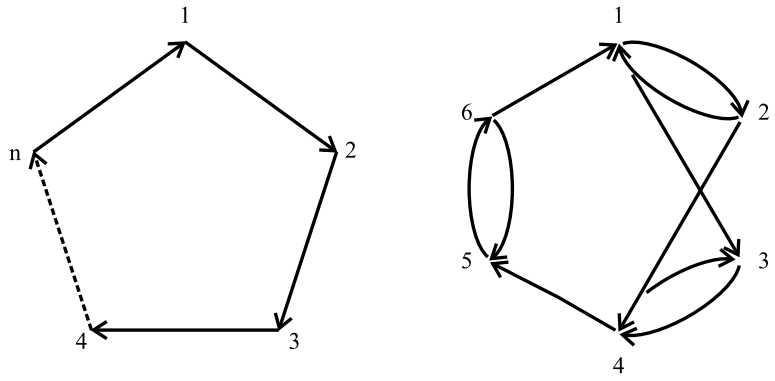


Figure 1. Definition of hypercycles. Replicator equations as described by the differential equation (2) can be symbolized by directed graphs: The individual species are denoted by nodes and two nodes are connected by an edge, $j \cdot \longrightarrow \cdot i$, if and only if $a_{ij} > 0$. The graphs of hypercycles consist of single Hamiltonian arcs as sketched on the left hand side of the figure. These dynamical systems are permanent independently of the choice of rate parameters f_i. For $n \leq 5$ they represent the only permanent systems but for $n \geq 6$ the existence of a single Hamiltonian arc is only a sufficient but not a necessary condition for permanence. The graph on the right hand side, for example, does not contain a Hamiltonian arc but the corresponding replicator equation is permanent for certain choices of rate parameters (Hofbauer & Sigmund, 1998).

$f_i > 0$, $x_i(0) > 0 \forall\, i = 1, 2, \ldots, n \implies x_i(t) > 0 \forall\, t \geq 0$. In other words, none of the variables is going to vanish and hence, the system is *permanent* in the sense that no member of a hypercycle dies out in the limit of long times, $\lim_{t\to\infty} x_i(t) \neq 0 \forall\, i = 1, \ldots, n$. The existence of a Hamiltonian arc, i.e. a closed loop of directed edges visiting all nodes once, is a sufficient condition for permanence (Hofbauer & Sigmund, 1998). It is also a necessary conditions for low-dimensional systems with $n \leq 5$, but there exist permanent dynamical systems for $n \geq 6$ without a Hamiltonian arc; one example is shown in figure 1.

The dynamics of equation (1) remains qualitatively unchanged when all rate parameters are set equal: $f_1 = f_2 = \ldots = f_n = f$, which is tantamount to a *barycentric* transformation of the differential equation (Hofbauer, 1981). The hypercycle is invariant with respect to a rotational change of variables, $x_i \implies x_{i+1}$ with $i = 1, 2, \ldots, n$; $i \bmod n$, it has one equilibrium point in the center and its dynamics depends exclusively on n. Some examples with small n are shown in figure 2. The systems with $n \leq 4$ converge towards stable equilibrium points, whereas the trajectories of the equation (1) with $n \geq 5$ approach limit cycles. Independently of n, elementary hypercycles do not sustain chaotic dynamics.

Hypercycles have two inherent instabilities, which are easily illustrated for molecular species: (i) The members of the cycle may also catalyze the formation of non-members that do not contribute to the growth of the hypercycle and thus hypercycles are vulnerable to parasitic exploitation (Eigen & Schuster, 1978a,b), and (ii) concentrations of individual species in oscillating hypercycles ($n \geq 5$) go through very small values and

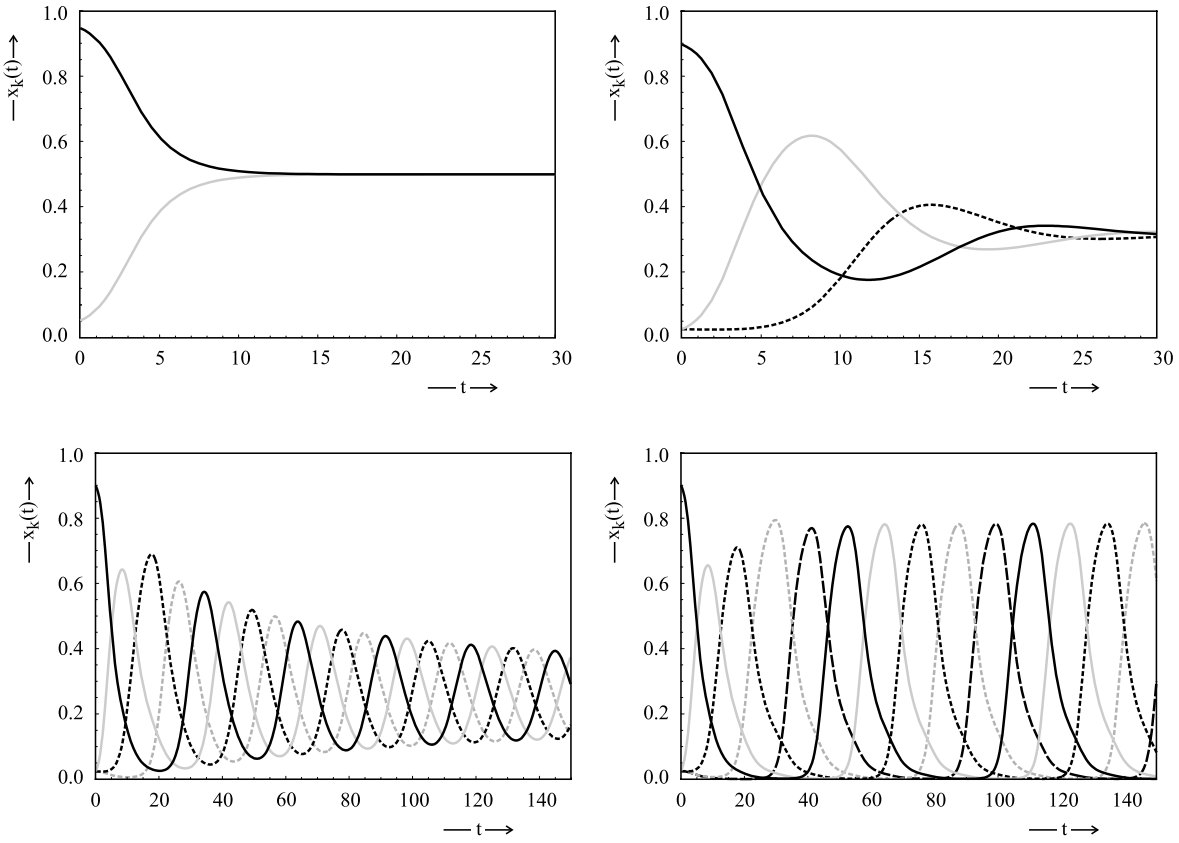


Figure 2. Solution curves of small elementary hypercycles. The figure shows the solution curves of equation (1) with $f_1 = f_2 = \ldots = f_n = 1$ for $n = 2$ (upper left picture), $n = 3$ (upper right picture), $n = 4$ (lower left picture), and $n = 5$ (lower right picture). The initial conditions were: $x_1(0) = 1 - (n-1) \cdot 0.025$ and $x_k(0) = 0.025 \; \forall \; k = 2, 3, \ldots, n$. The sequence of the curves $x_k(t)$ is: $k = 1$ full black line, $k = 2$ full grey line, $k = 3$ hatched black line, $k = 4$ hatched grey line, and $k = 5$ black line with long hatches. The cases $n = 2$, 3, and 4 have stable equilibrium points in the middle of the concentration space $c = (1/n, 1/n, \ldots, 1/n)$; equation (1) with equal rate parameters, $n = 4$, and linearized around the midpoint c exhibits a marginally stable 'center' and very slow convergence is caused by the non-linear term that becomes smaller as the system approaches c. For $n = 5$ the midpoint c is unstable and the trajectory converges towards a limit cycle (Hofbauer *et al.*, 1991).

these species might become extinct through random fluctuations. More elaborate kinetic mechanisms can stabilize the system in case (ii). Exploitation by parasites, case (i), can be avoided by compartmentalization. Competition between different hypercycles is characterized by a strong non-linearity in selection (Hofbauer, 2002): Once a hypercycle has been formed and established, it is very hard to replace it by another hypercycle. Epochs with hypercyclic dynamics provide explanations for 'once for ever' decisions or 'frozen accidents'.

See also Biological evolution; Optimization

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